ASCP CERTIFICATION PREPARATION

BOC STUDY GUIDE

Clinical Laboratory Certification Examinations

7th edition

MEDICAL LABORATORY TECHNICIAN
MLT(ASCP) & MLT(ASCP^i)

MEDICAL LABORATORY SCIENTIST
MLS(ASCP) & MLS(ASCP^i)

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The Importance of Certification, CMP, Licensure & Qualification

Seventy percent of medical decisions are based on results provided by the clinical laboratory. A highly skilled medical team of pathologists, specialists, medical laboratory scientists, technologists, technicians, and phlebotomists works together to determine the presence or absence of disease and provides valuable data needed to determine the course of treatment.

Today's laboratory uses many complex, precision instruments and automated processes. However, the success of the laboratory begins with the laboratorians' dedication to their profession and willingness to help others. Laboratorians must produce accurate and reliable test results, have an interest in science, and be able to recognize their responsibility for affecting human lives.

Role of the ASCP Board of Certification (ASCP BOC)

Founded in 1928 by the American Society of Clinical Pathologists (ASCP—now, the American Society for Clinical Pathology), the ASCP BOC is considered the preeminent certification agency in the US and abroad within the field of laboratory medicine. Composed of representatives of professional organizations and the public, the ASCP BOC mission is to:

"Provide excellence in certification of laboratory professionals on behalf of patients worldwide."

The ASCP BOC consists of more than 100 volunteer technologists, technicians, phlebotomists, laboratory scientists, physicians, and professional researchers. These volunteers contribute their time and expertise to the ASCP BOC Board of Governors and the Examination Committees. They allow the ASCP BOC to achieve the goal of excellence in credentialing medical laboratory personnel in the US and abroad.

The Board of Governors is the policymaking governing body for the ASCP BOC and is composed of 23 members. These 23 members include technologists, technicians, and pathologists nominated by the ASCP, representatives from the general public, and representatives from the following societies: the American Association for Clinical Chemistry, the AABB, the American Society for Microbiology, the American Society for Clinical Laboratory Science, the American Society of Cytopathology, the American Society of Hematology, the American Association of Pathologists' Assistants, the Association of Genetic Technologists, the National Society for Histotechnology, and the Clinical Laboratory Management Association.

The Examination Committees are responsible for the planning, development, and review of the examination databases; determining the accuracy and relevancy of the test items; confirming the standards for each examination and performing job or practice analyses.

Certification and Credential Maintenance Program (CMP)

www.ascp.org/cmp

Certification is the process by which a nongovernmental agency or association grants recognition of competency to an individual who has met certain predetermined qualifications, as specified by that agency or association. Certification affirms that an individual has demonstrated that he or she possesses the knowledge and skills to perform essential tasks in the medical laboratory. The ASCP BOC certifies those individuals who meet academic and clinical prerequisites and who achieve acceptable performance levels on examinations.

In 2004, the ASCP BOC implemented the Credential Maintenance Program (CMP), which mandates participation every three years for newly certified individuals in the US. The goal of this program is to demonstrate to the public that laboratory professionals are performing the appropriate and relevant activities to keep current in their practice. Additional information on ASCP's CMP program may be found on the ASCP website at www.ascp.org/cmp.

United States Certification

www.ascp.org/boc/us-certifications

Specific directions on how to apply for the Medical Laboratory Technician (MLT) and Medical Laboratory Scientist (MLS) examinations are posted on the Get Credentialed page of the ASCP BOC website. The steps are summarized below:

- Identify the examination you are applying for and determine your eligibility.
- Gather your required education and experience documentation.
- Apply for the examination online and pay by credit card. Pay by mail instructions will be available upon the completion of the online application process.
- Submit required documentation.

Upon notification of your eligibility, schedule an appointment to take your examination.

International Certification

www.ascp.org/boc/international-certifications

ASCP offers its gold standard credentials in the form of international certification (ASCPi) to eligible individuals. The ASCP^{i} credential certifies professional competency among new and practicing laboratory personnel in an effort to contribute globally to the highest standards of patient safety. Graduates of medical laboratory science programs outside the United States are challenged with content that mirrors the standards of excellence established by the US ASCP exams. The ASCP^{i} credential carries the weight of 90 years of expertise in clinical laboratory professional certification. Please visit the ASCP BOC website at to view the following:

- Current listing of international certifications.
- Eligibility guidelines.
- Step-by-step instructions to apply for international certification.

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State Licensure

www.ascp.org/boc/state-licensure

State Licensure is the process by which a state grants a license to an individual to practice their profession in the specified state. The individual must meet the state's licensing requirements, which may include examination and/or experience. It is important to identify the state and examination to determine your eligibility and view the steps for licensure and/or certification. For a list of states that require licensure, please go to the ASCP BOC website at www.ascp.org/boc/state-licensure and click on the specific state licensure of interest.

The ASCP BOC examinations have been approved for licensure purposes by the states of California and New York. The ASCP BOC examinations also meet the requirements for all other states that require licensure.

Qualification

www.ascp.org/boc/qualification

A qualification from the ASCP BOC recognizes the competence of individuals in specific technical areas. Qualifications are available in apheresis, immunohistochemistry, and laboratory safety. To receive this credential, candidates must meet the eligibility requirements and successfully complete an examination (QBRS, QDP, QIA, QIHC, QLS,). Candidates who complete the qualification process will receive a wall Certificate of Qualification, which is valid for 3 years. The qualification may be revalidated every 3 years upon completion of the Credential Maintenance Program (CMP) application and fee. (Documentation of acceptable continuing education may be requested.) Further information regarding requalification can be found at this link: www.ascp.org/cmp

Preparing for the ASCP BOC Certification Examination

Begin early to prepare for the Certification Examination. Because of the broad range of knowledge and skills tested by the examination, even applicants with college education and those completing formal laboratory education training programs will find that review is necessary, although the exact amount will vary from applicant to applicant. Generally, last minute cramming is the least effective method for preparing for the examination. The earlier you begin, the more time you will have to prepare, and the more you prepare, the better your chance of successfully passing the examination and scoring well.

Study for the Test

Make preparation for the exam part of your daily routine. Set aside a regular time and place to study. It is more beneficial to spend a short time studying every day than to spend several hours in one sitting every week or two. Be sure to allow enough time to cover all of the information specified in the Exam Content Guidelines. Also, be sure to devote some extra time to areas of weaknesses identified by your initial review.

Examination Content Guidelines

www.ascp.org/boc/mls
www.ascp.org/boc/mlt

The ASCP BOC has developed Examination Content Guidelines to delineate the content included in its tests. Current Content Guidelines for the MLS and MLT examinations, as well as other certification examinations offered by the ASCP BOC, are available on the website: **Indices of question numbers by content outline for both MLS and MLT examinations precede the questions in this text, and are located on pages xii-xv.**

Practice Analysis Reports

www.ascp.org/boc/practice-analysis

A practice analysis survey is a formal process for determining or verifying the responsibilities of individuals in the job/profession, the knowledge individuals must possess, and the skills necessary to perform the job at a minimally competent level. The results of the practice analysis inform the specifications and content of the ASCP BOC certification eexaminations. The practice analysis process ensures that the examinations are reflective of current practices. It also helps guarantee that individuals who become certified are up to date on the state of medical laboratory science practice and are competent to perform as certified laboratory professionals.

Study Guide

The questions in this study guide are in a format and style similar to the questions on the ASCP BOC examinations. The questions are in a multiple choice format with one best answer. Work through each chapter and answer all the questions as presented. Next, review your answers against the answer key. Review the answer explanation for any questions you responded to incorrectly or were uncertain about. Each question is referenced if you require further explanation.

Textbooks

www.ascp.org/content/docs/default-source/boc-pdfs/boc-us-reading-lists/mls_imls_reading_list.pdf
www.ascp.org/content/docs/default-source/boc-pdfs/boc-us-reading-lists/mlt_imlt_reading_list.pdf

The references cited in this study guide identify many useful textbooks. The most current Reading Lists for all examinations are available on the ASCP BOC website: www.ascp.org/certification. Textbooks tend to cover a broad range of knowledge in a given field and may provide expanded explanations, if needed.

Primary Reference Source Cited in this Study Guide

QCMLS [2021] Quick Compendium of Medical Laboratory Sciences. ISBN: 978-0891896616

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AABB [2020] Standards for Blood Banks and Transfusion Services, 32e. ISBN: 978 156395367-5
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Rifai N [2019] Tietz Fundamentals of Clinical Chemistry and Molecular Diagnostics 8e. ISBN: 978-8131258231

Some older citations still remain for historical interest.

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BOC Interactive Practice Exam

store.ascp.org/productlisting/productdetail?productId=121613463

Enables the user to build custom quizzes and timed practice test based on topics, difficulty and more, pulling from a library of over 2500 study questions. You can compare question-level results to see how your peers are answering. Practice Exam allows user to set a time limit to simulate a real exam scenario. MLT-only filter can be applied for those studying for the MLT Exam. Can be accessed anywhere on desktop or mobile.

Taking the Certification Examination

The ASCP BOC uses computer adaptive testing (CAT), which is criterion referenced. With CAT, when an examinee answers a question correctly, the next exam question has a slightly higher level of difficulty. The difficulty level of the questions presented to the examinee continues to increase until a question is answered incorrectly. Then a slightly easier question is presented. In this way, the test is tailored to the examinee's ability level.

Each question in the examination pool is calibrated for difficulty and categorized into a subtest area, which corresponds to the content guideline for a particular examination. The weight (value) given to each question is determined by the level of difficulty. All examinations (with the exception of Phlebotomy Technician [PBT] are scheduled for 2 hours and 30 minutes and have 100 questions. The PBT examination is scheduled for 2 hours and has 80 questions. Your preliminary test results (pass/fail) will appear on the computer screen immediately upon completion of your examination. A notification to view your examination scores will be emailed to you within four business days after the examination administration, provided all required documents have been received. Examination results cannot be released by telephone under any circumstances.

Your official detailed examination score report will indicate a "pass" or "fail" status and the specific scaled score on the total examination. A scaled score is mathematically derived (in part) from the raw score (number of correctly answered questions) and the difficulty level of the questions. Because each examinee has taken an individualized examination, scaled scores are used so that all examinations may be compared on the same scale. The minimum passing score is 400. The highest attainable score is 999.

If you were unsuccessful in passing the examination, your scaled scores on each of the subtests will be indicated on the report as well. These subtest scores cannot be calculated to obtain your total score. These scores are provided as a means of demonstrating your areas of strengths and weaknesses in comparison to the minimum pass score.

Tips for Preparing for the Examinations

The ASCP BOC MLS & MLT Study Guide contains over 2500 multiple choice questions and is designed to achieve 2 goals: (1) to assist students in preparing for clinical laboratory certification exams and (2) to provide clinical laboratory instructors with a resource of examination questions. To achieve the first goal, the ASCP BOC MLS & MLT Study Guide includes a comprehensive set of sample review questions, organized by chapter and further grouped by topic corresponding to the ASCP BOC Examination Content Outlines. Each chapter begins with a set of questions followed by an answer key. A brief explanation for each answer is provided followed by references which may be used to obtain additional information or expanded explanations. Several references were chosen to highlight information pertinent to the questions included in the ASCP BOC MLS & MLT Study Guide. Additionally, each question is indexed according to the ASCP BOC Examination Content Outlines for the Medical Laboratory Technician and International Medical Laboratory Technician, and/or the ASCP BOC Qualification in Donor Phlebotomy Exam.

To determine the set of questions included in the ASCP BOC MLS & MLT Study Guide, 2 criteria were used. First, the ASCP BOC Examination Content Outline for the Medical Laboratory Technician and International Medical Laboratory Technician and Medical Laboratory Scientist created a framework for content included in the ASCP BOC MLS & MLT Study Guide. A suggested approach for studying for a certification exam is outlined below using the medical laboratory technician certification exam as an example. The approach outlined below may be transferred to the Medical Laboratory Scientist (MLS) Exam using the MLS Exam Content outline. This guide addresses 7 categories of information that correlate to the following chapter headings: Blood Banking, Chemistry, Urinalysis & Body Fluids, Hematology, Hemostasis, Microbiology, Laboratory Operations, and Immunology.

Second, the weighted content of ASCP BOC Examination Content Outline for the Medical Laboratory Technician and International Medical Laboratory Technician provided direction for content emphasis. Over 2500 questions are included in the ASCP BOC MLS & MLT Study Guide. Content outline percentages for each certification examination follow below. The questions in this study guide track this percentage distribution.

ASCP BOC MLT Certification Exam

Content Category	Percentages
Blood Banking	15-20%
Chemistry	20-25%
Urinalysis & Body Fluids	5-10%
Hematology & Hemostasis	20-25%
Microbiology	15-20%
Laboratory Operations	5-10%
Immunology	5-10%

ASCP BOC MLS Certification Exam

Content Category	Percentages
Blood Banking	17-22%
Chemistry	17-22%
Urinalysis & Body Fluids	5-10%
Hematology & Hemostasis	17-22%
Microbiology	17-22%
Laboratory Operations	5-10%
Immunology	5-10%

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Tips for Preparing for the Examinations

In summary, the ASCP BOC MLS & MLT Study Guide is a comprehensive tool designed to assist phlebotomy students with preparing for phlebotomy certification exams, and secondly, to assist phlebotomy instructors in developing written competency assessments (tests). Questions with images will appear as they would on the certification examination. Laboratory results will be presented in both conventional and SI units. The practice questions are presented in a format and style similar to the questions included on the ASCP BOC examinations. The ASCP BOC Examination Content Outline for the Medical Laboratory Technician and International Medical Laboratory Technician and Medical Laboratory Scientist served as the framework for the ASCP BOC MLS & MLT Study Guide. Responses providing explanations to the answer key were based in a number of resources, including the Quick Compendium of Medical Laboratory Sciences.

Please note: None of these exact questions will appear on any ASCP BOC examination. This book is not a product of the ASCP BOC; rather, it is a product of the ASCP Press, the independent publishing arm of the American Society for Clinical Pathology. Use of this book does not ensure passing an examination. The ASCP BOC's evaluation and credentialing processes are entirely independent of this study guide; however, this book should significantly help students prepare to challenge the ASCP BOC examination.

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Preface

The 7th edition of the *Board of Certification Study Guide for Clinical Laboratory Certification Examinations* contains 2556 multiple choice questions, ordered to correspond to the latest Examination Content Guidelines. Questions are cross-referenced to exam content outlines following the introductory material (see pxii-xix). The questions in this edition are arranged in chapters which correspond to the major content areas on the examination. Within each chapter, the questions are further grouped by major topics. Unique to this study guide is the differentiation of questions appropriate for both the Medical Laboratory Technician and Medical Laboratory Scientist levels from questions that are appropriate for the Medical Laboratory Scientist level ***only*** (clearly marked MLS ONLY). Short answer explanations (with references) are provided for each practice question. Questions with images will appear as they would on the certification examination. Laboratory results are presented in both conventional and SI units. For this latest iteration of the Study Guide, a new typeface (one that approximates the one used on the examination itself) has been used to make the questions an even better approximation of the actual examination experience.

The practice questions are presented in a format and style similar to the questions included on the ASCP BOC certification examinations. **Please note: None of these exact questions will appear on any ASCP BOC examination.**

These practice questions were developed for this edition; some questions were compiled from previously published materials and submitted questions from recruited reviewers. (Note: neither editors nor reviewers currently serve on any reviewers do not currently serve on any Examination Committee.)

This book is not a product of the ASCP BOC; rather, it is a product of the ASCP Press, the independent publishing arm of the American Society for Clinical Pathology. Use of this book does not ensure passing of an examination. The ASCP BOC's evaluation and credentialing processes are entirely independent of this study guide; however, this book should significantly help you prepare for your ASCP BOC examination.

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CERTIFICATION

To the examinee,

I have had the honor to work on the ASCP BOC Study Guide throughout my career. I hope that this edition of this Study Guide provides a useful tool to prepare you for the ASCP BOC certification examination.

Certification is an important first step in your professional career. It is my hope that you find your career in laboratory medicine rewarding.

Good luck on your certification exam and on your future endeavors.

—Patricia A Tanabe, MPA, MLS(ASCP)^{CM}
Executive Director, Board of Certification (retired)

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Chapter 1

Blood Banking

The following items have been identified generally as appropriate for those preparing for both the MLS and MLT examinations. Items that are appropriate for the MLS examination **only** are marked with MLS ONLY.

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I. Blood Products

A. Donors

1. The minimum hemoglobin concentration in a fingerstick from a male blood donor is:

   a 12.0 g/dL (120 g/L)
   b 12.5 g/dL (125 g/L)
   c 13.0 g/dL (130 g/L)
   d 13.5 g/dL (135 g/L)

2. A cause for indefinite deferral from blood donation is:

   a a reactive test for *Babesia* species
   b residence in an endemic malaria region for 5 years
   c positive test for *Trypanosoma cruzi*
   d history of chicken pox vaccination

3. Which of the following prospective donors would be accepted for blood donation?

   a 62-year-old female with a blood pressure of 210/80
   b 18-year-old female who weighs 100 lb
   c 40-year-old male with a pulse of 115
   d 82-year-old male with a hemoglobin of 13.5 g/dL

4. Which one of the following constitutes permanent deferral status of a donor?

   a tattoo 5 months previously
   b recent close contact with a patient with viral hepatitis
   c 2 units of blood transfused 4 months previously
   d confirmed positive test for HBsAg 10 years previously

5. MLS ONLY A prospective donor with which of the following health histories would be accepted for blood donation?

   a hepatitis B immune globulin 2 months ago
   b HIV prevention drugs 6 months ago
   c blood transfusion 2 months ago
   d travel to malaria endemic country 1 month ago

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6. In order to be a plateletpheresis donor, the platelet count must be at least:

   a 150,000/µL
   b 200,000/µL
   c 250,000/µL
   d 300,000/µL

7. Prior to blood donation, the intended venipuncture site must be cleaned with a scrub solution containing:

   a hypochlorite
   b green soap
   c 10% acetone
   d povidone iodine

8. A donor who has just donated 2 units of Apheresis Red Blood Cells will be deferred from further blood donation for a minimum of how many weeks?

   a 8
   b 12
   c 16
   d 24

9. Which of the following infectious agents relies solely on donor questioning to avoid transmission from transfused blood products?

   a *Trypanosoma cruzi*
   b *Plasmodium falciparum*
   c HCV
   d CMV

10. Which of the following practices at the time of blood collection helps minimize bacterial contamination of platelet products?

    a use of 18-gauge needle
    b diversion pouch
    c green soap scrub
    d UV irradiation

11. If a unit of autologous Red Blood Cells is collected and transfused in the same facility, which of the following is not required?

    a ABO and Rh typing
    b a physician's order
    c infectious disease screening
    d evaluation for risk of bacteremia

12. According to AABB Standards, what is the minimum hemoglobin level for an autologous donor?

    a 11.0 g/dL
    b 12.0 g/dL
    c 12.5 g/dL
    d 13.0 g/dL

B. Processing

13. Which of the following must be included on the label of a unit of Red Blood Cells Leukocytes Reduced?

    a known leukocyte count for the unit
    b phlebotomist identification
    c unique collection facility identifier
    d date of blood collection

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14. All donor blood testing must include:

a complete Rh phenotyping
b anti-CMV testing
c direct antiglobulin test
d serological test for syphilis

15. Which of the following is not a useful strategy to reduce bacterial contamination in platelet components?

a using green soap scrub to disinfect venipuncture site
b diverting the first 10-40 mL of donor blood from the main blood bag
c performing culture-based testing on all platelet components
d applying pathogen reduction technology to platelet components

16. Which of the following practices has been useful in reducing the incidence of transfusion related acute lung injury (TRALI)?

a use of Fresh Frozen Plasma from male donors
b use of Fresh Frozen Plasma from female donors
c pathogen reduction treatment of Fresh Frozen Plasma
d leukocyte-reduced Fresh Frozen Plasma

17. In allogeneic blood donation, which of the following infectious agents must also include a nucleic acid testing (NAT) assay?

a HBV
b HTLV I/II
c syphilis
d cytomegalovirus

18. What is the primary reason that infectious agents can be transmitted following blood transfusion?

a pathogen reduction technology failure
b donor in the window period of early infection
c leukocyte-reduction failure
d donor history questionnaire not completed

19. Apheresis Platelets that will undergo pathogen reduction technology do not require:

a hepatitis B virus NAT
b culture-based testing for bacteria
c West Nile virus NAT
d ABO and Rh testing

20. The results in this table are obtained on a blood donor sample at immediate spin:

anti-A	anti-B	A_1 cells	B cells	anti-D	Rh control
0	0	4+	4+	0	0

Before labeling blood components from this donation, what additional testing must be completed?

a test donor RBCs with anti-A,B
b test donor RBCs with anti-H
c perform weak D testing on donor RBCs
d test donor serum with A_2 cells

21. Which of the following courses of action must be performed immediately if a sample from an Apheresis Platelet undergoing culture-based testing indicates bacterial growth?

a set up another culture to confirm positivity
b determine sensitivity to antibiotics
c retrieve the unit if issued for transfusion
d identify the organism

22. A sample of blood from each donation is tested for which of the following infectious disease agents?

a HIV-1/2, HTLV-I/II, WNV, HCV
b HBV, HCV, CMV, *T. cruzi*
c HBsAg, HIV-1/2, *P. falciparum*, *Babesia* species
d dengue virus, syphilis, HCV, HBV

C. Storage

23. The transport temperature for Red Blood Cells Leukocytes Reduced is:

a 1-6°C
b 1-10°C
c 18-20°C
d 20-24°C

24. The transport temperature for Apheresis Platelets is:

a 1-6°C
b 1-10°C
c 18-20°C
d 20-24°C

25. A unit of Red Blood Cells expiring in 35 days is split into 5 small aliquots using a sterile pediatric quad set and a sterile connecting device. Each aliquot must be labeled as expiring in:

a 6 hours
b 12 hours
c 5 days
d 35 days

26. When platelets are stored on a rotator set on an open bench top, the ambient air temperature must be recorded:

a once a day
b twice a day
c every 4 hours
d every hour

27. Which of the following is the correct storage temperature for the component listed?

a Cryoprecipitated AHF, 4°C
b Fresh Frozen Plasma (FFP), –20°C
c Red Blood Cells, Frozen, –40°C
d Platelets, 37°C

28. Six units of Red Blood Cells are issued to the OR at 9 AM in a cooler, validated to maintain a temperature of 1-10°C for 2 hours. The cooler containing the units of blood is returned to the blood bank 40 minutes later because surgery is cancelled. What should be done with these units?

a discard the units as they were issued to a specific patient
b inspect units and establish that appropriate temperature has been maintained
c continue to store in cooler since surgery is rescheduled for tomorrow
d put units back into inventory as only 40 minutes has elapsed since issue

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29. An acceptable storage temperature for Red Blood Cells, Frozen is:

a –80°C
b –20°C
c –12°C
d 4°C

30. Red Blood Cells, Leukocytes Reduced must be stored at:

a 1-6°C
b 1-10°C
c 1-20°C
d 20-24°C

31. If the seal is entered or broken on a unit of Red Blood Cells stored at 1-6°C, what is the maximum allowable storage period, in hours?

a 6
b 24
c 48
d 72

32. Cryoprecipitated AHF must be stored at:

a ≤–10°C
b ≤–18°C
c 1-6°C
d 1-10°C

33. Plasma Frozen Within 24 Hours After Phlebotomy (PF24) and thawed for transfusion has an expiration of:

a 6 hours
b 12 hours
c 24 hours
d 5 days

34. Apheresis Platelets must be stored at:

a 1-6°C
b 1-10°C
c 10-18°C
d 20-24°C

35. Cryoprecipitated AHF, if maintained in the frozen state at –18°C or below, has a shelf life of:

a 42 days
b 6 months
c 12 months
d 36 months

36. Thawed Plasma must be stored at:

a ≤–18°C
b 1-6°C
c 1-10°C
d 20-24°C

37. MLS ONLY During storage, the concentration of 2,3-diphosphoglycerate (2,3-DPG) decreases in a unit of:

a Platelets
b Fresh Frozen Plasma
c Red Blood Cells
d Cryoprecipitated AHF

38. Upon inspection, a unit of Apheresis Platelets is noted to have visible clots, but otherwise appears normal. The technologist should:

a issue without concern
b filter to remove the clots
c centrifuge to express off the clots
d quarantine for Gram stain and culture

39. Upon expiration, a unit of thawed Plasma Frozen Within 24 Hours (PF24) is converted to thawed Plasma. This thawed Plasma can be stored for an additional:

a 1 day
b 4 days
c 14 days
d 28 days

40. The transfusion service is preparing aliquots from a unit of Red Blood Cells Leukocytes Reduced with the aid of a sterile connecting device for a pediatric patient. When checking the weld for one of these aliquots, it is noted that the weld is incomplete and leaking. This unit is then resealed with an acceptable weld. What will the expiration date of this unit be?

a 6 hours
b 24 hours
c 3 days
d original expiration date

D. Blood Components

41. A patient with a platelet count of 10,000/µL receives one unit of Apheresis Platelets, Leukocytes Reduced. After transfusion, the patient's platelet count is 50,000/µL. These results indicate:

a The presence of HLA antibodies and refractoriness to transfusion
b The patient is actively bleeding but still producing platelets
c The platelet count is the expected posttransfusion increment
d The patient's pre-transfusion platelet count was incorrect

42. Red Blood Cells Leukocytes Reduced must be prepared by a method known to reduce the leukocyte count to:

a $<8.3 \times 10^5$
b $<5.0 \times 10^6$
c $<5.5 \times 10^{10}$
d $<3.0 \times 10^{11}$

43. A unit of Red Blood Cells that expires in 32 days has just been irradiated. The expiration date of this unit will:

a remain the same
b be reduced by 4 days
c be reduced by 14 days
d be increased by 2 days

44. Cryoprecipitated AHF must be transfused within what period of time following thawing and pooling without the use of a sterile connection device?

a 4 hours
b 8 hours
c 12 hours
d 24 hours

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45. According to AABB Standards, Fresh Frozen Plasma must be infused within what period of time following thawing?

a 24 hours
b 36 hours
c 48 hours
d 72 hours

46. Cryoprecipitated AHF:

a is indicated for fibrinogen deficiencies
b should be stored at 4°C prior to administration
c will not transmit hepatitis B virus
d is indicated for the treatment of hemophilia B

47. Which Apheresis Platelets product should be irradiated?

a autologous unit collected prior to surgery
b random stock unit going to a patient with DIC
c a directed donation given by a mother for her son
d a directed donation given by an unrelated family friend

48. Irradiation of a unit of Red Blood Cells is done to prevent the replication of donor:

a granulocytes
b lymphocytes
c red cells
d platelets

49. Plastic bag overwraps are recommended when thawing units of FFP in 37°C water baths because they prevent:

a the FFP bag from cracking when it contacts the warm water
b water from slowly dialyzing across the bag membrane
c the entry ports from becoming contaminated with water
d the label from peeling off as the water circulates in the bath

50. Which of the following blood components must be prepared within 8 hours after phlebotomy?

a Red Blood Cells
b Fresh Frozen Plasma
c Red Blood Cells, Frozen
d Cryoprecipitated AHF

51. Which of the following is proper procedure for preparation of Platelets from Whole Blood?

a light spin followed by a hard spin
b light spin followed by 2 hard spins
c 2 light spins
d hard spin followed by a light spin

52. Of the following blood components, which one should be used to prevent HLA alloimmunization of the recipient?

a Red Blood Cells
b Granulocytes
c Irradiated Red Blood Cells
d Leukocyte-Reduced Red Blood Cells

E. Blood Component Quality Control

53. Quality control of Apheresis Granulocytes must demonstrate which of the following granulocyte counts in 75% of units tested?

a 1.0×10^{10}
b 2.0×10^{10}
c 3.0×10^{10}
d 4.0×10^{10}

54. An important determinant of platelet viability during storage is:

a plasma potassium concentration
b plasma pH
c prothrombin time
d activated partial thromboplastin time

55. According to AABB Standards, Apheresis Platelets shall demonstrate with 95% confidence that >75% of units contain ≥ how many platelets?

a 5.5×10^{10}
b 6.5×10^{10}
c 3.0×10^{11}
d 5.0×10^{11}

56. According to AABB Standards, Platelets prepared from Whole Blood shall have at least:

a 5.5×10^{10} platelets per unit in at least 90% of the units tested
b 6.5×10^{10} platelets per unit in 90% of the units tested
c 7.5×10^{10} platelets per unit in 100% of the units tested
d 8.5×10^{10} platelets per unit in 95% of the units tested

57. According to AABB Standards, Apheresis Platelets shall demonstrate with 95% confidence that >95% of units have what minimum pH at the time of issue?

a 6.0
b 6.2
c 6.8
d 7.0

58. What percentage of red blood cells must be retained when preparing Red Blood Cells Leukocytes Reduced?

a 50%
b 70%
c 85%
d 100%

59. In a quality assurance program, Cryoprecipitated AHF must contain a minimum of how many international units of Factor VIII?

a 60
b 70
c 80
d 90

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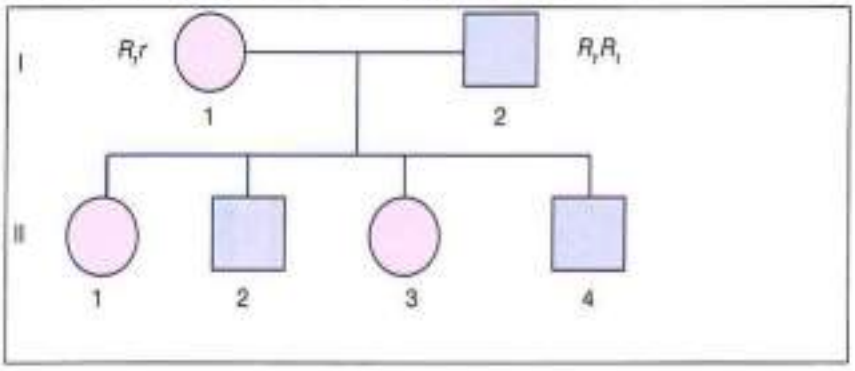
II. Blood Group Systems

A. Genetics

60. Given the most probable genotypes of the parents shown in this figure, which statement best describes the most probable Rh genotypes of the 4 children?



a 25% will be R_0r, 25% will be R_1r, and 50% will be R_1R_1
b 50% will be R_1r and 50% will be R_1R_1
c 100% will be R_1r
d 100% will be R_1R_1

61. The linked HLA genes on each chromosome constitute a(n):

a allele
b trait
c phenotype
d haplotype

62. Most blood group system genes and their resulting genetic traits display what type of inheritance?

a sex-linked dominant
b sex-linked recessive
c autosomal recessive
d autosomal codominant

63. The mating of an Xg(a+) man and an Xg(a–) woman will **only** produce:

a Xg(a–) sons and Xg(a–) daughters
b Xg(a+) sons and Xg(a+) daughters
c Xg(a–) sons and Xg(a+) daughters
d Xg(a+) sons and Xg(a–) daughters

64. MLS ONLY

The observed phenotypes in a particular population are:

phenotype	number of persons
Jk(a+b–)	122
Jk(a+b+)	194
Jk(a–b+)	84

What is the gene frequency of Jk^a in this population?

a 0.31
b 0.45
c 0.55
d 0.60

65. MLS ONLY In a random population, 16% of the people are Rh-negative (*rr*). What percentage of the Rh-positive population is heterozygous for *r*?

a 36%
b 48%
c 57%
d 66%

66. Which phenotype could *not* result from the mating of a Jk(a+b+) female and a Jk(a–b+) male?

a Jk(a+b–)
b Jk(a+b+)
c Jk(a–b+)
d Jk(a–b–)

67. Which of the following phenotypes is the result of homozygous inheritance of the corresponding genes?

a Le(a+b–)
b M+N+
c Fy(a–b+)
d Jk(a+b+)

68. What do the O_h (Classical Bombay), group O, and Lu(a–b–) phenotypes have in common?

a result from inheritance of identical sex-linked dominant genes
b result from inheritance of identical sex-linked recessive genes
c result from inheritance of identical autosomal dominant genes
d result from inheritance of identical autosomal recessive genes

B. Chemistry, Antigens

69. Which of the following antibodies is usually clinically insignificant?

a anti-P
b anti-P1
c anti-P^k
d anti-p

70. An individual's red blood cells give the reactions with Rh antisera shown in this table:

anti-D	anti-C	anti-E	anti-c	anti-e	Rh control
4+	3+	0	3+	3+	0

The individual's most probable genotype is:

a *DCe/DcE*
b *DcE/dce*
c *Dce/dce*
d *DCe/dce*

71. A blood donor has the genotype: *hh*, *AB*. Using anti-A and anti-B antisera, the donor's red cells will type as group:

a A
b B
c O
d AB

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72. An individual has been sensitized to the k antigen and has produced anti-k. What is the most probable KEL system genotype for this individual?

a *KK*
b *Kk*
c *kk*
d K_0K_0

73. MLS ONLY Anti-Fy3 will fail to react with which of the following enzyme-treated red cells?

a Fy(a+b–)
b Fy(a–b+)
c Fy(a–b–)
d Fy(a+b+)

74. A mother has the red cell phenotype D+C+E–c–e+ with anti-c (titer of 32 at AHG) in her serum. The father has the phenotype D+C+E–c+e+. The baby is Rh-negative and not affected with hemolytic disease of the newborn. What is the baby's most probable Rh genotype?

a *r'r'*
b *r'r*
c R_1R_1
d R_1r

75. In an emergency situation, Rh-negative red cells are transfused into an Rh-positive person of the genotype CDe/CDe. The first antibody **most** likely to develop is:

a anti-c
b anti-d
c anti-e
d anti-E

76. A patient's red cells type as shown in this table:

anti-C	anti-D	anti-E	anti-c	anti-e
+	+	+	+	+

Which is the most probable genotype?

a R_1R_1
b R_1r'
c R_0r''
d R_1R_2

77. A patient's red cells type as shown in this table:

anti-D	anti-C	anti-E
4+	0	0

Which of the following genotype would be consistent with these results?

a R_0R_0
b R_1r
c R_1R_2
d R_2r

78. The red cells of a nonsecretor (se/se) will most likely type as:

a Le(a–b–)
b Le(a+b+)
c Le(a+b–)
d Le(a–b+)

79. Which of the following phenotypes will react with anti-f?

a *rr*
b R_1R_1
c R_2R_2
d R_1R_2

80. A patient's red blood cells gave the reactions shown in this table:

anti-D	anti-C	anti-E	anti-c	anti-e	anti-f
+	+	+	+	+	0

The most probable genotype of this patient is:

a R_1R_2
b R_2r''
c R_zr
d R_zR_z

81. A woman types as Rh-positive. She has an anti-c titer of 32 at AHG. Her baby has a negative direct antiglobulin test (DAT) and is not affected by hemolytic disease of the newborn. What is the father's most likely Rh phenotype?

a *rr*
b $r''r$
c R_1r
d R_2r

82. Which of the following red cell typings are most commonly found in the African American donor population?

a Lu(a–b–)
b Jk(a–b–)
c Fy(a–b–)
d K–k–

83. A donor is tested with Rh antisera with the results shown in this table:

anti-D	anti-C	anti-E	anti-c	anti-e	Rh control
+	+	0	+	+	0

What is his most probable Rh genotype?

a R_1R_1
b R_1r
c R_0r
d R_2r

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84. MLS ONLY A family has been typed for HLA because 1 of the children needs a stem cell donor. Typing results are listed in this table:

patient	type
father	A1,3;B8,35
mother	A2,23;B12,18
child #1	A1,2;B8,12
child #2	A1,23;B8,18
child #3	A3,23;B18,?

What is the expected B antigen in child #3?

a A1
b A2
c B12
d B35

85. Which of the following is the immunodominant sugar responsible for the A antigen?

a fucose
b N-acetylgalactosamine
c galactose
d N-acetylglucosamine

86. Which of the following is considered to be a high-prevalence antigen?

a Vel
b Js^a
c s
d K

87. The reason that group O individuals have the most amount of H antigen on their red cells compared to other ABO phenotypes is:

a group O individuals produce more precursor type I chain
b group A, B and AB individuals are heterozygous for the *H* gene
c the *O* gene produces more transferase enzyme, which produces more H antigen
d H antigen is left unchanged by the absence of A and/or B transferase enzymes

88. A patient is typed with the results shown in this table:

patient's cells with		patient's serum with	
anti-A	0	A_1 red cells	2+
anti-B	0	B red cells	4+
anti-A,B	2+	Ab screen	0

The most probable reason for these findings is that the patient is group:

a O; confusion due to faulty group O antiserum
b O; with an anti-A1
c A_x; with an anti-A1
d A_1; with an anti-A

89. Given the serologic reactions shown in this table, what is the most likely A subgroup?

anti-A	anti-B	anti-A,B	A_1 cells	B cells	O cells
2+mf	0	2+mf	1+	4+	0

mf=mixed field agglutination

a A_1
b A_2
c A_3
d A_x

90. The enzyme responsible for conferring H activity on the red cell membrane is alpha-:

a galactosyl transferase
b N-acetylgalactosaminyl transferase
c L-fucosyl transferase
d N-acetylglucosaminyl transferase

91. Even in the absence of prior transfusion or pregnancy, individuals with the Bombay phenotype (O_h) will always have naturally occurring:

a anti-Rh
b anti-K_o
c anti-U
d anti-H

92. Which of the following antibodies in the LU (Lutheran) system is most likely to be IgM and detected as a direct agglutinin?

a anti-Lu^a
b anti-Lu^b
c anti-Lu3
d anti-Au^a

93. Which of the following antibodies is neutralizable by pooled human plasma?
IILS ONLY

a anti-Kn^a
b anti-Ch
c anti-Yk^a
d anti-Cs^a

94. Antibodies from which of the following blood group systems are notorious for causing delayed hemolytic transfusion reactions?

a Rh
b KEL
c FY
d JK

95. HLA antibodies are:
IILS ONLY

a naturally occurring
b induced by multiple transfusions
c directed against granulocyte antigens only
d frequently cause hemolytic transfusion reactions

96. Genes of the major histocompatibility complex (MHC):
IILS ONLY

a code for HLA-A, HLA-B, and HLA-C antigens only
b are linked to genes in the ABO system
c are the primary genetic sex-determinants
d contribute to the coordination of cellular and humoral immunity

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97. Isoimmunization to platelet antigen HPA-1a and the placental transfer of maternal antibodies would be expected to cause newborn:

a erythroblastosis
b leukocytosis
c leukopenia
d thrombocytopenia

98. What antigens would be found in the saliva of an individual with the genotype *Sese Lele AO HH*?

a A, H
b Le^b, A, H
c Le^a, Le^b, A, H
d Le^a

99. (ULS ONLY) Which of the following genes is not in the MHC class I region?

a *HLA-A*
b *HLA-B*
c *HLA-C*
d *HLA-DR*

100. (MLS ONLY) Anti-D and anti-C are identified in the serum of a pregnant woman, gravida 2, para 1. Ten months previously she received Rh immune globulin (RhIG) at 28 weeks' gestation. Test results of the patient, her husband, and the child are shown in this table:

	anti-D	anti-C	anti-E	anti-c	anti-e
patient	0	0	0	+	+
father	0	+	0	+	+
child	0	+	0	+	+

The most likely explanation for the presence of anti-D is that this antibody is:

a actually anti-C^w
b from the RhIG dose
c actually anti-G
d naturally occurring

101. The phenomenon of an Rh-positive person whose serum contains anti-D is best explained by:

a gene deletion
b missing antigen epitopes
c trans position effect
d gene inhibition

102. When the red cells of an individual fail to react with anti-U, they usually fail to react with:

a anti-M
b anti-Le^b
c anti-S
d anti-P1

103. Which of the following red cell antigens are found on glycophorin-A?

a M, N
b Le^a, Le^b
c S, s
d P, P1, P^k

104. Paroxysmal cold hemoglobinuria (PCH) is associated with antibody specificity toward which of the following?

a KEL system antigens
b FY system antigens
c P antigen
d I antigen

105. Which of the following is a characteristic of anti-i?

a associated with warm autoimmune hemolytic anemia
b found in the serum of patients with infectious mononucleosis
c detected at lower temperatures in the serum of normal individuals
d found only in the serum of group O individuals

106. In a case of cold agglutinin disease, the patient's serum would most likely react 4+ at immediate spin with:

a group A cells, B cells and O cells, but not his own cells
b cord cells but not his own or other adult cells
c all cells of a group O cell panel and his own cells
d only penicillin-treated panel cells, not his own cells

107. Cold agglutinin disease is associated with an antibody specificity toward which of the following?

a Fy:3
b P
c I
d Rh:1

108. Which of the following is a characteristic of anti-i?

a often associated with hemolytic disease of the newborn
b reacts best at room temperature or 4°C
c reacts best at 37°C
d is usually IgG

109. MLS ONLY In chronic granulomatous disease (CGD), granulocyte function is impaired. An association exists between this clinical condition and a depression of which of the following antigens?

a Rh
b P
c KEL
d FY

110. The antibodies of the JK blood group system:

a react best by the indirect antiglobulin test
b are predominantly IgM
c often cause allergic transfusion reactions
d do not generally react with antigen-positive, enzyme-treated RBCs

111. Proteolytic enzyme treatment of red cells usually destroys which antigen?

a Jk^a
b E
c Fy^a
d k

112. Anti-Fy^a is:

a usually a cold-reactive agglutinin
b more reactive when tested with enzyme-treated red blood cells
c capable of causing hemolytic transfusion reactions
d often an autoagglutinin

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113. Resistance to malaria is best associated with which of the following blood groups?

a Rh
b I/i
c P
d FY

114. What is linkage disequilibrium in reference to HLA haplotypes?

MLS ONLY

a occurrence of HLA genes in the same haplotype more often than would be expected based on the gene frequencies
b displacement of HLA genes on different chromosomes
c occurrence of HLA genes in the same haplotype less often than would be expected based on the gene frequencies
d recombination of HLA genes during meiosis

115. A 25-year-old Caucasian woman, gravida 3, para 2, required 2 units of Red Blood Cells. The antibody screen is positive, and the results of the antibody panel are shown in this table:

cell	D	C	c	E	e	K	Jk^a	Jk^b	Le^a	Le^b	M	N	P1	IAT
1	+	+	0	0	+	+	+	+	0	+	+	+	+	0
2	+	+	0	0	+	0	+	0	0	+	+	0	0	0
3	+	0	+	+	0	0	+	+	0	+	+	+	+	1+
4	+	+	+	0	+	0	0	+	0	+	+	0	+	1+
5	0	0	+	0	+	0	+	+	0	+	+	0	0	1+
6	0	0	+	+	+	0	+	0	+	0	+	+	0	1+
7	0	0	+	0	+	+	+	+	+	0	+	+	+	1+
8	0	0	+	0	+	0	0	+	0	+	0	+	+	1+
IAT = indirect antiglobulin test													auto	0

What is the most probable genotype of this patient?

a *rr*
b *r′r′*
c R_0r
d R_1R_1

116. Antibodies produced to blood group antigens that are carbohydrate structures are usually:

a IgG and indirect agglutinins
b IgG and react best at 37°C
c IgM and direct agglutinins
d IgM and react best at 37°C

117. Which of the following red blood cells would lack the Lu3 antigen?

a Lu(a+b+)
b Lu(a+b–)
c Lu(a–b+)
d Lu(a–b–)

118. Which of the following is considered a low-prevalence antigen?

a Yt^a
b Jk^b
c Wr^a
d Di^b

119. Which of the following statement is correct?

a antibodies to high-prevalence antigens are most often IgM and room temperature reactive
b antibodies to high-prevalence antigens are usually clinically significant
c antibodies to high-prevalence antigens occur in >98% of the population
d antibodies to high-prevalence antigens include examples of anti-K

120. For which of the following is HLA phenotyping not useful? MLS ONLY

a selection of bone marrow donor for transplantation
b investigation of a hemolytic transfusion reaction
c selection of solid organ donor for transplantation
d investigation of platelet refractoriness in transfusion recipient

121. Fetal and neonatal alloimmune thrombocytopenia (FNAIT) is:

a frequently caused by maternal HLA antibodies
b treated with Cryoprecipitated AHF
c frequently caused by maternal anti-HPA-1a antibodies
d treated with IVIG and washed paternal platelets

122. Which of the following statement regarding the LW blood group system is true?

a D-negative red cells express more LW antigen than D-positive red cells
b Rh_{null} red cells are also LW(a–b–)
c LW antibodies generally react at the room temperature (RT) phase of testing
d the LW system has 3 antigens: low-prevalence LW^a and high-prevalence LW^b and LW^{ab}

123. In which of the following conditions are HNA antibodies not implicated?

a autoimmune neutropenia (AIN)
b transfusion-related acute lung injury (TRALI)
c neonatal alloimmune neutropenia (NAN)
d transfusion-associated circulatory overload (TACO)

C. Role of Blood Groups in Transfusion

124. Anti-K is identified in a patient's serum. If random crossmatches are performed on 10 donor units, approximately how many would be expected to be compatible?

a 1
b 3
c 7
d 9

125. Four units of blood are needed for elective surgery. The patient's serum contains anti-C, anti-e, anti-Fy^a and anti-Jk^b. Which of the following would be the best source of donor blood?

a test all units in current stock
b test 100 group O, Rh-negative donors
c test 100 group-compatible donors
d rare donor registry

126. A patient is group O, Rh-negative with anti-D and anti-K in her serum. What percentage of the general Caucasian donor population would be compatible with this patient?

a 0.5
b 2.0
c 3.0
d 6.0

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127. Which of the following Rh antigens has the highest frequency in Caucasians?

a D
b E
c c
d e

128. The K (KEL1) antigen is:

a absent from the red cells of neonates
b strongly immunogenic
c destroyed by enzymes
d has a frequency of 50% in the random population

129. MLS ONLY What percent of group O donors would be compatible with a serum sample that contained anti-X and anti-Y if X antigen is present on red cells of 5 of 20 donors, and Y antigen is present on red cells of 1 of 10 donors?

a 2.5
b 6.8
c 25.0
d 68.0

130. What is the approximate probability of finding compatible blood among random Rh-positive units for a patient who has anti-c and anti-K? (Consider that 20% of Rh-positive donors lack c and 90% lack K)

a 1%
b 10%
c 18%
d 45%

III. Blood Group Immunology

A. Immune Response

131. Which of the following is responsible for the production of blood group antibodies?

a B cells
b T cells
c NK cells
d dendritic cells

132. In a primary immune response, which immunoglobulin class appears first?

a IgG
b IgM
c IgA
d IgE

133. What is the most common clinical incident that results in alloantibody production?

a viral infection
b solid tumor
c red cell transfusion
d autoimmune disease

134. Macrophages and monocytes have Fc receptors for which of the following immunoglobulins?

a IgG
b IgM
c IgA
d IgD

135. Which of the following blood group antigens is the most immunogenic, or has the greatest ability to initiate antibody production in an individual who lacks the antigen?

a Fy^a
b s
c Jk^b
d D

136. IgG antibody and/or C3b sensitized red blood cells can be phagocytized by cells in the reticuloendothelial system (RES). This phenomenon is clinically recognized as:

a immune tolerance
b allergic sequalae
c extravascular hemolysis
d intravascular hemolysis

B. Immunoglobulins

137. Which immunoglobulin class can cross the placenta?

a IgG
b IgM
c IgA
d IgE

138. Which of the following immunoglobulins is most efficient at activating complement via the classical pathway?

a IgG2
b IgG4
c IgM
d IgA

139. Which of the following immunoglobulins is most efficient at causing direct hemagglutination?

a IgG
b IgM
c IgA
d IgE

140. Which of the following structures of an immunoglobulin molecule is involved in the activation of complement via the classical pathway?

a Fc region
b Fab region
c Hinge region
d J chain

141. Immunoglobulin classes are differentiated according to the molecular structure of:

a light chains
b heavy chains
c Fab fragment
d Fc fragment

142. Which of the following pairs of immunoglobulins is most efficient at activating complement via the classical pathway?

a IgG1 and IgG3
b IgG1 and IgG4
c IgG2 and IgG3
d IgG2 and IgG4

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143. The results in this panel are obtained with the serum of a patient from the prenatal clinic.

cell	D	C	c	E	e	K	Jk^a	Jk^b	Fy^a	Fy^b	M	N	P1	IS	LISS 37°C	IAT
1	+	+	0	0	+	+	+	+	0	+	0	+	+	0	0	0
2	+	+	0	0	+	0	+	0	0	+	+	0	0	3+	2+	2+
3	+	0	+	+	0	0	+	+	0	+	0	+	+	0	0	0
4	+	+	+	0	+	0	0	+	0	+	+	+	+	3+	2+	2+
5	0	0	+	0	+	0	+	+	0	+	+	0	0	3+	2+	2+
6	0	0	+	+	+	0	+	0	+	0	0	+	0	0	0	0
7	0	0	+	0	+	+	+	+	+	0	+	+	+	3+	2+	2+
8	0	0	+	0	+	0	0	+	0	+	0	+	+	0	0	0
IAT = indirect antiglobulin test; LISS = low ionic strength saline													auto	0	0	0

To evaluate potential risk to the fetus, what additional studies should be performed?

a test additional cells to rule-out anti-c
b treat serum with dithiothreitol and repeat panel
c Rh phenotype the patient
d perform D testing on father's blood

144. The immune response to red cell antigens with numerous epitopes results in a heterogeneous population of antibodies referred to as:

a bivalent
b epitope-specific
c monoclonal
d polyclonal

145. A blood sample is taken from a 90-year-old male, who is admitted to the ER for possible GI bleeding. ABO results are shown in this table:

cells tested with		serum tested with	
anti-A	0	A_1 cells	0
anti-B	0	B cells	1+

What might be a likely explanation for these results?

a wrong sample collected
b patient's disease
c patient's age
d subgroup of B

C. Antigen-Antibody Interactions

146. Appropriate antigen-antibody ratios are important to avoid an excess of unbound antibody, which is known as:

a dosage effect
b pH effect
c postzone effect
d prozone effect

147. Many enhancement media used in the blood bank promote hemagglutination in the presence of IgG antibodies by reducing which of the following?

a hydrophilic forces
b low ionic potential
c van der Waals forces
d zeta potential

148. Blood group antigen and antibody hemagglutination reactions are influenced by which of the following?

a temperature
b Ca^{2+} ions
c antigen presenting cells
d memory cells

149. Which of the following blood group antibodies will no longer react with its respective antigens once those antigens are treated with proteolytic enzymes?

a anti-C
b anti-K
c anti-Fy^a
d anti-Jk^a

150. Some blood group antibodies may react stronger with the red cells of individuals who have inherited 2 identical alleles for the antigen to which the antibody is directed. This is known as:

a postzone effect
b dosage effect
c prozone effect
d equivalence effect

151. Which of the following reagents is used to facilitate hemagglutination following the sensitization of red cells with an IgG alloantibody?

a anti-human globulin serum
b low ionic strength saline
c polyethylene glycol
d 22% bovine albumin

152. The addition of antibody-sensitized red cells (Check Cells) to all negative anti-human globulin (AHG) tests ensures that:

a the test was interpreted correctly
b the test was incubated at the correct temperature
c AHG reagent was added to each test
d patient serum was added to each test

153. In which of the following clinical situations will the direct antiglobulin test be positive?

a sickle cell disease
b hemolytic disease of the fetus and newborn
c posttransfusion purpura
d multiple myeloma

154. In which of the following is the indirect antiglobulin test utilized?

a reverse ABO testing
b immediate spin crossmatch
c C antigen testing
d antibody detection (screening) test

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155. Polyspecific AHG reagents contain:

a anti-IgG
b anti-C3d
c anti-IgG and anti-IgM
d anti-IgG and anti-C3d

156. A false-negative direct antiglobulin test can be the result of:

a neutralized AHG reagent
b sample from gel-separator tubes
c overcentrifugation
d dirty glassware

157. A false-positive indirect antiglobulin test can be the result of:

a insufficient saline washing of red cells
b inadequate incubation time
c overcentrifugation
d dissociation of cell bound IgG

158. A negative result using solid phase adherence assays will demonstrate indicator red cells as:

a a red blood cell pellet in the bottom of the well
b a diffuse pattern of red blood cells throughout the well
c red blood cell clumps symmetrically located throughout the well
d a red supernatant, indicating lysis

159. A 4+ positive reactions using gel technology will appear as red blood cells:

a in a pellet at the bottom of the microtubule
b dispersed throughout the gel media
c in a layer at the top of the gel media
d suspended at the mid-point of the gel media

160. One of the advantages of performing antibody screening (detection) studies using gel technology is:

a saline washing is not required
b centrifugation is not required
c special equipment is not required
d precise volumes of serum are not required

161. Monoclonal blood banking reagents have which of the following as a disadvantage?

a little to no batch variation
b cost effectiveness
c high antigen efficiency
d overspecificity

162. The direct antiglobulin test in a suspected case of warm autoimmune hemolytic anemia is positive. Which of the following monospecific reagents would be used in further direct antiglobulin testing?

a anti-C3b
b anti-C3d
c anti-C4
d anti-IgM

163. Low ionic strength saline (LISS) acts as an enhancement medium and facilitates antibody uptake by:

a activating complement
b increasing flexibility in hinge region
c removing water molecules
d reducing zeta potential

D. Complement

164. The membrane attack complex (MAC) formed during the activation of complement via the classical pathway consists of:

a C1q, C1r, C1s
b C3a, C3b, C3d
c C5a, C5b
d C5b through C9

165. In ABO hemolytic transfusion reactions, complement is activated via which of the following pathways?

a alternative
b classical
c lectin
d polyclonal

166. IgG-sensitized red cells that activate complement may not complete complement activation to cell lysis. In addition to IgG on the surface of these red cells, what complement component is present?

a C1q
b C3a
c C3b
d C5b

167. While performing antibody detection studies and compatibility testing, hemolysis is noted following incubation at 37°C. Which of the following alloantibodies might be present?

a anti-D
b anti-PP1P^k
c anti-K
d anti-c

IV. Physiology and Pathophysiology

A. Physiology of Blood

168. Which of the following statements is true about hematopoietic progenitor cells (HPCs)?

a progenitor cells are capable of self-renewal and limitless proliferation
b stem cells make up 95% of total HPCs
c stem cells are unable to undergo self-renewal
d progenitor cells and stem cells are morphologically unrecognizable

169. Key cytokines involved in the differentiation and proliferation of erythrocytes include:

a erythropoietin (EPO) and granulocyte macrophage-colony stimulating factor (GM-CSF)
b interleukin 3 (IL-3) and flt3 ligand (FL)
c granulocyte macrophage-colony stimulating factor (GM-CSF) and flt3 ligand (FL)
d erythropoietin (EPO) and interleukin 3 (IL-3)

170. What hematopoiesis site in adults produces erythrocytes and all of the leukocytes?

a spleen
b bone marrow
c liver
d thymus

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171. What differentiates erythrocytes from leukocytes as they develop into mature cells?

a nuclear chromatin is clumped and condensed
b nucleoli disappear
c size of the cell decreases
d nucleus is removed from the cell

172. What role does the spleen play in an extravascular transfusion reaction?

a produces opsonizing proteins that sensitize incompatible donor red cells
b shears antibody sensitized incompatible donor red cells causing the formation of spherocytes
c stores one-third of the body's granulocytes and platelets
d plays a role in the development of T lymphocytes

173. RBC integral membrane proteins are:

a located on the cytoplasmic side of the membrane
b responsible for RBC deformability
c expand through the membrane and include 30 RBC antigens
d spectrin and actin, which are responsible for cell movement

174. 2,3-DPG's role in RBC metabolism is to:

a protect RBCs from hexose monophosphate shunt degradation
b generate cell energy anaerobically in the form of ATP
c maintain hemoglobin Fe^{2+} to bind O_2 to promote adsorption
d increase the release of O_2 from oxyhemoglobin to the tissues

175. Which of the following cells plays a role as mediators in allergic hypersensitivity reactions?

a eosinophils
b neutrophils
c monocytes
d NK lymphocytes

B. Hemostasis and Coagulation

176. Transfusion of which of the following is needed to help correct hypofibrinogenemia due to DIC?

a Whole Blood
b Fresh Frozen Plasma
c Cryoprecipitated AHF
d Platelets

177. Which of the following is used to treat hemophilia B?

a Factor IX concentrate
b Factor VIII concentrate
c Cryoprecipitated AHF
d DDAVP

178. A unit of Fresh Frozen Plasma is inadvertently thawed and then immediately refrigerated at 4°C on Monday morning. On Tuesday evening, this unit may still be transfused as Thawed Plasma, but will have decreased levels of:

a Factor I
b Factor V
c Factor IX
d Factor XII

179. A newborn demonstrates petechiae, ecchymosis and mucosal bleeding. The preferred blood component for this infant would be:

a Red Blood Cells
b Fresh Frozen Plasma
c Platelets
d Cryoprecipitated AHF

180. Which of the following would be the best source of Platelets for transfusion in a case of fetal and neonatal alloimmune thrombocytopenia (FNAIT)?

a paternal platelets
b maternal platelets
c random donor platelets
d paternal grandfather platelets

181. Fetal and neonatal alloimmune thrombocytopenia (FNAIT) is caused by which of the following maternal antibodies?

a IgM alloantibodies against ABO antigens
b IgG alloantibodies against HLA antigens
c IgM alloantibodies against glycoprotein V (GPV) antigens
d IgG alloantibodies against HPA antigens

182. Which of the following is used to treat posttransfusion purpura?

a DDAVP
b IVIG
c Apheresis Platelets
d Fresh Frozen Plasma

183. In which of the following would platelet transfusions be contraindicated?

a bone marrow transplant patient with a platelet count of 9,000/μL
b actively bleeding patient with a platelet count of 55,000/μL
c thrombocytopenic patient due to functionally abnormal platelets
d patient diagnosed with thrombotic thrombocytopenic purpura (TTP)

C. Hemolytic Disease of the Fetus and Newborn

184. MLS ONLY Blood typing after a normal labor and delivery showed that the mother is group A, D-negative and demonstrates anti-D in her serum. Her slightly jaundiced newborn is anemic and types as group O, D-negative with a 4+ direct antiglobulin test (DAT). Previous lab work shows that the father is group O, D-positive. From the information given, which test result is questionable?

a paternal D type
b newborn D type
c maternal ABO type
d newborn DAT

185. An obstetrical patient has had 3 previous pregnancies. Her first baby was healthy, the second was jaundiced at birth and required an exchange transfusion, while the third was stillborn. Which of the following is the most likely cause?

a ABO incompatibility
b immune deficiency disease
c congenital spherocytic anemia
d Rh incompatibility

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186. MLS ONLY A specimen of cord blood is submitted to the transfusion service for routine testing. The results are shown in this table:

anti-A	anti-B	anti-D	Rh-control	direct antiglobulin test
4+	negative	3+	negative	2+

It is known that the father is group B, with the genotype of *cde/cde*. Of these 4 antibodies, which 1 is the most likely cause of the positive direct antiglobulin test?

a anti-A
b anti-D
c anti-c
d anti-C

187. ABO-hemolytic disease of the newborn:

a usually requires an exchange transfusion
b most often occurs in first born children
c frequently results in stillbirth
d is usually seen only in the newborn of group O mothers

188. Which of the following antigens is most likely to be involved in hemolytic disease of the fetus and newborn?

a Le^a
b P1
c M
d K

189. ABO hemolytic disease of the fetus and newborn (HDFN) differs from Rh HDFN in that:

a Rh HDFN is clinically more severe than ABO HDFN
b the direct antiglobulin test is weaker in Rh HDFN than ABO HDFN
c Rh HDFN occurs in the first pregnancy
d the mother's antibody screen is positive in ABO HDN

190. MLS ONLY Which of the following is the most probable explanation for the results shown in this table?

sample	anti-A	anti-B	anti-D	weak D	DAT	Ab screen
infant	0	0	0	NT	4+	NT
mother	4+	0	0	0	NT	anti-D

DAT = direct antiglobulin test; NT = not tested

a ABO hemolytic disease of the fetus and newborn
b Rh hemolytic disease of the fetus and newborn; infant has received intrauterine transfusions
c Rh hemolytic disease of the fetus and newborn, infant has a false-negative Rh typing
d large fetomaternal hemorrhage

191. A group A, Rh-positive infant of a group O, Rh-positive mother has a weakly positive direct antiglobulin test and a moderately elevated bilirubin 12 hours after birth. The most likely cause is:

a ABO incompatibility
b Rh incompatibility
c blood group incompatibility due to an antibody to a low frequency antigen
d neonatal jaundice **not** associated with blood group

192. In suspected cases of hemolytic disease of the fetus and newborn, what significant information can be obtained from the baby's blood smear?

a estimation of WBC, RBC, and platelet counts
b marked increase in immature neutrophils (shift to the left)
c differential to estimate the absolute number of lymphocytes present
d determination of the presence of spherocytes

193. In treating hyperbilirubinemia in hemolytic disease of the fetus and newborn (HDFN), light phototherapy is used to:

a breakdown conjugated into unconjugated bilirubin
b prevent further hemolysis of fetal red cells
c breakdown unconjugated bilirubin to a nontoxic isomer
d stimulate production of fetal liver enzymes to breakdown unconjugated bilirubin

194. The purpose of performing antibody titers on serum from an immunized, pregnant woman is to:

a determine the identity of the antibody
b identify candidates requiring additional fetal monitoring
c decide if the baby needs an intrauterine transfusion
d determine if early induction of labor is indicated

195. Which unit should be selected for exchange transfusion if the newborn is group A, Rh-positive and the mother is group A, Rh-positive with anti-c?

a A, CDe/CDe
b A, cDE/cDE
c O, cde/cde
d A, cde/cde

196. A mother is group A, with anti-D in her serum. What would be the preferred blood product if an intrauterine transfusion is indicated?

a O, Rh-negative Red Blood Cells, Irradiated, CMV safe
b O, Rh-negative Red Blood Cells, Irradiated, CMV safe, HbS-negative
c A, Rh-negative Red Blood Cells, Irradiated, CMV safe
d A, Rh-negative Red Blood Cells, Irradiated, CMV safe, HbS-negative

197. Laboratory studies of maternal and cord blood yield the results shown in this table:

maternal blood	cord blood
O, Rh-negative	B, Rh-positive
anti-E in serum	direct antiglobulin test (DAT) = 2+
	anti-E in eluate

If an exchange transfusion for the infant is necessary, group O Whole Blood with which of the following Rh phenotypes is the most acceptable to select?

a R_1r
b R_2r
c rr
d r''r

198. A blood specimen from a pregnant woman is found to be group B, Rh-negative and the serum contains anti-D with a titer of 512. What would be the most appropriate type of blood to have available for a possible exchange transfusion for her infant?

a O, Rh-negative
b O, Rh-positive
c B, Rh-negative
d B, Rh-positive

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199. Blood selected for exchange transfusion must:

a lack red blood cell antigens corresponding to maternal antibodies
b be <3 days old
c be the same Rh type as the baby
d be ABO compatible with the father

200. When the main objective of an exchange transfusion is to remove the infant's antibody-sensitized red blood cells and to control hyperbilirubinemia, the blood product of choice is ABO compatible:

a Fresh Whole Blood
b Red Blood Cells (RBC) washed
c RBC suspended in Fresh Frozen Plasma
d heparinized Red Blood Cells

201. To prevent graft-versus-host disease, Red Blood Cells prepared for infants who have received intrauterine transfusions should be:

a saline-washed
b irradiated
c frozen and deglycerolized
d group- and Rh-compatible with the mother

202. Which of the following is the preferred specimen for the initial compatibility testing in exchange transfusion therapy?

a maternal serum
b eluate prepared from infant's red blood cells
c paternal serum
d infant's postexchange serum

203. The results shown in this table are seen on a maternal postpartum sample:

	anti-D	D control	weak D	weak D control
mother's postpartum sample	0	0	$1+^{mf}$	0

^{mf} = mixed field

The most appropriate course of action is to:

a report the mother as Rh-negative
b report the mother as Rh-positive
c perform an elution on mother's RBCs
d investigate for a fetomaternal hemorrhage

204. In cases where antepartum RhIG has been administered and anti-D is detected in the mother's serum at delivery, what laboratory studies may help determine if the anti-D is from the antepartum RhIG, or due to maternal alloimmunization?

a determine antibody titer of maternal anti-D
b repeat antibody identification studies with room temperature incubation
c perform inhibition studies on anti-D with maternal saliva
d repeat antibody identification studies using AET-treated serum

205. What is the best interpretation for the laboratory data in this table?

postpartum	anti-D	Rh control	weak D	weak D control	rosette fetal screen using untreated D+ indicator cells
mother	0	0	0	0	8 rosettes/5 fields
newborn	4+	0	NT	NT	NT

NT = not tested

a newborn needs weak D testing
b mother needs further weak D testing
c mother has a larger than normal FMH
d mother has a negative rosette test

206. The results of a Kleihauer-Betke stain indicate a fetomaternal hemorrhage of 35 mL of whole blood. How many vials of Rh immune globulin would be required?

a 1
b 2
c 3
d 4

207. A fetomaternal hemorrhage of 48 mL of fetal Rh-positive whole blood has been detected in an Rh-negative woman. How many vials of Rh immune globulin should be given?

a 0
b 1
c 2
d 3

208. Criteria determining Rh immune globulin eligibility include:

a mother is Rh-positive
b infant is Rh-negative
c mother has not been previously immunized to the D antigen
d infant has a positive direct antiglobulin test

209. While performing routine postpartum testing for an Rh immune globulin (RhIG) candidate, a weakly-positive antibody screening test is found. Anti-D is identified. This antibody is most likely the result of:

a massive fetomaternal hemorrhage occurring at the time of this delivery
b antenatal administration of Rh immune globulin at 28 weeks' gestation
c contamination of the blood sample with Wharton jelly
d mother having a positive direct antiglobulin test

210. Rh immune globulin administration would not be indicated in an Rh-negative woman who has a(n):

a first trimester abortion
b husband who is Rh-positive
c anti-D titer of 1:4096
d positive direct antiglobulin test

211. A Kleihauer-Betke stain of a postpartum blood film revealed 0.3% fetal cells. What is the estimated volume (mL) of the fetomaternal hemorrhage expressed as whole blood?

a 5
b 15
c 25
d 35

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212. An acid elution stain is made using a 1-hour post-delivery maternal blood sample. Out of 2000 cells that are counted, 30 of them appear to contain fetal hemoglobin. It is the policy of the medical center to add 1 vial of Rh immune globulin to the calculated dose when the estimated volume of the hemorrhage exceeds 20 mL of whole blood. Calculate the number of vials of Rh immune globulin that would be indicated under these circumstances.

a 2
b 3
c 4
d 5

213. The rosette test will detect a fetomaternal hemorrhage (FMH) as small as:

a 10 mL
b 15 mL
c 20 mL
d 30 mL

D. Anemias

214. A 40-year-old man with autoimmune hemolytic anemia due to anti-E has a hemoglobin level of 10.8 g/dL (108 g/L). This patient will most likely be treated with:

a Whole Blood
b Red Blood Cells
c Fresh Frozen Plasma
d no transfusion

215. Pathologic cold autoantibodies differ from benign cold autoantibodies in:

a antibody specificity
b immunoglobulin class
c antibody titer
d ability to bind complement

216. The direct antiglobulin test (DAT) in a patient with warm autoimmune hemolytic anemia (WAIHA) is most often positive for:

a IgG only
b C3 only
c IgM only
d IgG and C3

217. What is the most important consideration for a patient requiring red cell transfusion due to severe anemia when their serum contains a warm autoantibody?

a determine specificity of autoantibody
b determine the immunoglobulin class of the autoantibody
c determine the presence of underlying alloantibody(ies)
d avoid transfusion therapy

218. A drug-induced immune hemolytic anemia caused by a drug-independent antibody would have which of the following results?

a positive direct antiglobulin test (DAT) with IgG; positive antibody screen
b positive DAT with IgG; negative eluate
c positive DAT with C3d; negative antibody screen
d negative DAT; positive antibody screen

219. The autoantibody most often implicated in Paroxysmal Cold Hemoglobinuria (PCH) is:

a cold-reactive, IgG, anti-P
b cold-reactive, IgM, anti-P
c cold-reactive, IgG, anti-I
d cold-reactive, IgM, anti-I

220. Intravascular destruction of red blood cells results when:

a IgG sensitized red cells are destroyed by phagocytes
b IgM antibodies activate complement to completion
c complement-sensitized red cells are destroyed by phagocytes
d IgG antibodies activate complement to C3b

221. Which of the following is likely to result in extravascular red cell destruction?

a transfusion of group A Red Blood Cells to a group O recipient
b transfusion of group A Fresh Frozen Plasma to a group O recipient
c transfusion associated with a delayed transfusion reaction caused by anti-Jk^a
d transfusion of D-positive Red Blood Cells to unsensitized D-negative recipient

E. Transplantation

222. Indications for an autologous hematopoietic progenitor cell (HPC) transplant include patients who have:

a Hodgkin lymphoma with high-dose chemotherapy
b congenital hemoglobinopathies
c congenital immunodeficiency disorders
d inborn errors of metabolism

223. Which of the following statements is correct about hematopoietic progenitor cell (HPC) transplantation?

a successful allogeneic transplantation requires that at least 5 out of 10 HLA loci match
b sources for HPC collection include bone marrow, umbilical cord blood (UCB), and mobilized peripheral blood
c bone marrow collection of HPC is preferred to apheresis collection because of fewer side effects
d HPC products are infused into the patient over a period of at least 6 hours

224. A patient in the immediate post bone marrow transplant period has a hematocrit of 21%. The red cell product of choice for this patient would be:

a packed
b saline washed
c microaggregate filtered
d irradiated

225. Which of the following systems plays an important role in Transfusion-Related Acute Lung Injury (TRALI), transfusion-associated graft-versus-host disease (TA-GVHD), platelet refractoriness, and Febrile Nonhemolytic Transfusion Reactions (FNHTR) as well as in hematopoietic stem and organ transplantation rejection?

a Rh
b HLA
c LE
d DI

226. Which of the following statements is true about Class II HLA antigens?

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a they are found on the surface of most nucleated cells
b Bg antigens are part of HLA Class II
c HLA-DR, HLA-DQ and HLA-DP are all Class II
d they are only located on neurons and platelets

227. The most widely accepted QC test to measure probable Hematopoietic Progenitor Cell (HPC) engraftment is:

a clonogenic assay
b cell viability
c CD34+ cell enumeration
d manual differential

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228. To prevent transfusion-associated graft-versus-host disease (TA-GVHD) in a patient undergoing HPC transplantation, blood components selected for transfusion should be:

a irradiated
b washed
c CMV-seronegative
d leukocyte-reduced

229. Which of the following HPC transplants would be considered an ABO major incompatibility?

a group O donor to a group A recipient
b group A donor to a group AB recipient
c group B donor to a group O recipient
d group O donor to a group B recipient

230. MLS ONLY When selecting blood products for transfusion to a patient who is day 1 post ABO major incompatible HPC transplant, which of the following apply?

a red cell components selected are based on ABO antibodies circulating in the recipient
b platelet and plasma components selected should be ABO identical to the recipient
c red cell components selected should be ABO identical to the donor
d red cell components selected should be group AB only

V. Serologic and Molecular Testing

A. Routine Tests

231. The blood typing results shown in this table are noted on a patient's sample:

anti-A	anti-B	anti-D	A_1 Cells	B Cells	Rh typing results
0	4+	4+	4+	0	C–E–c+e+

What is the patient's likely ethnicity?

a Asian
b Hispanic
c Black
d White

232. Samples from the same patient are received on 2 consecutive days. Test results are summarized in this table:

	day #1	day #2
anti-A	4+	0
anti-B	0	4+
anti-D	3+	3+
A_1 cells	0	4+
B cells	4+	0
Ab screen	0	0

How should the request for crossmatch be handled?

a crossmatch A, Rh-positive units with sample from day 1
b crossmatch B, Rh-positive units with sample from day 2
c crossmatch AB, Rh-positive units with both samples
d collect a new sample and repeat the tests

233. Test results are shown in this table for a unit of blood labeled group A, Rh-negative:

cells tested with		
anti-A	**anti-B**	**anti-D**
4+	0	3+

What should be done next?

a transfuse as a group A, Rh-negative
b transfuse as a group A, Rh-positive
c notify the collecting facility
d discard the unit

234. In what patient population may we observe the results shown in this table?

anti-A	anti-B	anti-D	A_1 cells	B cells
4+	0	4+	0	1+

a labor and delivery patient
b 30-year-old GI bleed patient
c 2-year-old pre-surgical patient
d 16-year-old ACL repair surgery patient

235. A patient is a subgroup of A (A_{sub}), Rh-positive with Anti-A1 in their serum. How many units would you have to screen to find 1 unit that is compatible with the patient's Anti-A1?

a 5
b 10
c 15
d 20

236. A patient is group A, Rh-positive but is receiving a Group O, bone marrow transplant (BMT) on Friday. After the transplant, what hemagglutination pattern will the patient demonstrate when the patient's red cells are tested with anti-A?

a rouleaux
b aggregation
c polyagglutination
d mixed-field

237. A group B, Rh-negative patient has a positive direct antiglobulin test (DAT). Which of the following situations would occur?

a all major crossmatches would be incompatible
b the weak D test and control would be positive
c the antibody screening test would be positive
d the forward and reverse ABO groupings would not agree

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238. The reaction results shown in this table are obtained:

cells tested with			serum tested with	
anti-A	anti-B	anti-A,B	A_1 cells	B cells
4+	3+	4+	2+	4+

The technologist washes the patient's cells with saline, and repeats the forward typing. A saline replacement technique is used with the reverse typing, and obtains the results shown in this second table:

cells tested with			serum tested with	
anti-A	anti-B	anti-A,B	A_1 cells	B cells
4+	0	4+	0	4+

Based on these results, a likely diagnosis for the patient may be:

a acquired immunodeficiency disease
b Bruton agammaglobulinemia
c multiple myeloma
d acquired "B" antigen

239. MLS ONLY What ABO type is found in group A_1 individuals following deacetylation of their A antigens?

a acquired B
b B(A)
c A_{mod}
d A_{int}

240. The Rh-negative phenotype results from the complete deletion of what gene(s)?

a *RHD* and *RHce*
b *RHCE*
c *RHD*
d *RHD* and *RHCE*

241. The results shown in this table are obtained on a patient's blood sample during routine ABO and Rh testing:

cells tested with		serum tested with	
anti-A	0	A_1 cells	4+
anti-B	4+	B cells	2+
anti-D	0		
autocontrol	0		

Select the course of action to resolve this problem:

a enzyme treat the patient's red cells and repeat the forward blood typing
b test the patient's serum with A_2 cells and the patient's red cells with Anti-A1 lectin
c repeat the ABO antigen grouping using 3x washed saline-suspended cells
d perform antibody screening procedure at immediate spin and AHG using group O cells

242. The test for weak D is performed by incubating patient's red cells with:

a different dilutions of anti-D
b anti-D antiserum
c anti-D^u antiserum
d antiglobulin antiserum

243. The results shown in this table are obtained when testing a sample from a 20-year-old, first-time blood donor:

forward group		reverse group	
anti-A	anti-B	A_1 cells	B cells
0	0	0	3+

What is the most likely cause of this ABO discrepancy?

a rouleaux
b acquired B
c phenotype O_h "Bombay"
d weak subgroup of A

244. A mother is Rh-negative and the father Rh-positive. Their baby is Rh-negative. It may be concluded that:

a the father is homozygous for D
b the mother is heterozygous for D
c the father is heterozygous for D
d at least 1 of the 3 Rh typings must be incorrect

245. Some blood group antibodies characteristically hemolyze appropriate antigen-positive red cells in the presence of:

a complement
b anticoagulants
c preservatives
d penicillin

246. Review this set of instructions:

PATIENT SERUM + REAGENT GROUP "O" CELLS
INCUBATE → READ FOR AGGLUTINATION
WASH → ADD AHG → AGGLUTINATION OBSERVED

The next step would be to:

a add "check cells" as a confirmatory measure
b identify the cause of the agglutination
c perform an elution technique
d perform a direct antiglobulin test

247. The most probable cause of the pretransfusion testing results shown in this table is::

	37°C	IAT
screening cell I	0	3+
screening cell II	0	3+
autocontrol	0	3+

IAT = indirect antiglobulin test

a rouleaux
b a warm autoantibody
c a cold autoantibody
d multiple alloantibodies

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248. A patient has an order for a type and screen and 6 units of Red Blood Cells are requested for surgery. At the indirect antiglobulin (IAT) phase of testing, both antibody detection cells and 2 of 6 crossmatched units are incompatible. What is the most likely cause of the incompatibility?

a recipient alloantibody
b recipient autoantibody
c donors have positive direct antiglobulin tests (DATs)
d recipient has a positive DAT

249. Which clinical condition is consistent with the lab results shown in this table?

test	result
hemoglobin	7.4 g/dL (74 g/L)
reticulocyte count	22%

DAT		Ab screen – IAT	
polyspecific	3+	SC I	3+
IgG	3+	SC II	3+
C3	0	auto	3+

DAT = direct antiglobulin test; IAT = indirect antiglobulin test

a cold hemagglutinin disease
b warm autoimmune hemolytic anemia
c penicillin-induced hemolytic anemia
d delayed hemolytic transfusion reaction

250. Based on the results in this table, what is the next step in determination of the patient's ABO/Rh type?

anti-A	anti-B	anti-D	A_1 cells	B cells
4+	4+	4+	0	0

a interpret at AB, D-positive
b interpret at $A_{sub}B$, D-positive
c repeat ABO red cell typing and include a control
d add room temperature incubation with ABO red cell typing

251. The major crossmatch will detect a(n):

a group A patient mistyped as group O
b unexpected red cell antibody in the donor unit
c Rh-negative donor unit mislabeled as Rh-positive
d recipient antibody directed against antigens on the donor red cells

252. Which of the following would most likely be responsible for an incompatible antiglobulin crossmatch?

a recipient's red cells possess a low frequency antigen
b anti-K antibody in donor serum
c recipient's red cells are polyagglutinable
d donor red cells have a positive direct antiglobulin test

253. In the process of identifying an antibody, the technologist observes 2+ reactions with 3 of 10 cells at immediate spin (IS) and room temperature (RT). There are no reactions at 37°C or with antihuman globulin (AHG). What is the most likely antibody?

a anti-Jk^b
b anti-Le^a
c anti-C
d anti-Fy^a

254. DNA arrays evaluate the encoding genes for common blood groups. These technologies detect small differences in the membrane proteins after amplification of what amino acid residues of genomic DNA?

a stop codon
b mRNA
c SNP
d cDNA

255. A 29-year-old male is hemorrhaging severely. He is AB, Rh-negative. Six units of blood are required STAT. Of the following types available in the blood bank, which would be most preferable for crossmatch?

a AB, Rh-positive
b A, Rh-negative
c A, Rh-positive
d O, Rh-negative

256. A patient is group A_2B, Rh-positive and has an antiglobulin-reacting anti-A1 in his serum. He is in the operating room bleeding profusely and group A_2B Red Blood Cells are **not** available. Which of the following blood types is first choice for crossmatching?

a B, Rh-positive
b B, Rh-negative
c A_1B, Rh-positive
d O, Rh-negative

257. A 10% red cell suspension in saline is used in a compatibility test. Which of the following would most likely occur?

a false-positive result due to antigen excess
b false-positive result due to the prozone phenomenon
c false-negative result due to the prozone phenomenon
d false-negative result due to antigen excess

258. A patient received 4 units of blood 2 years previously and now has multiple antibodies. He has not been transfused since that time but is now in need of additional blood transfusions. While determining the specificities for these antibodies, it would also be helpful to:

a phenotype his cells to determine which additional alloantibodies may be produced
b recommend the use of directed donors, which are more likely to be compatible
c use proteolytic enzymes to treat the patient's red cells prior to compatibility testing
d use the patient's serum for antigen typing of compatible units

259. Autoantibodies demonstrating blood group specificity in warm autoimmune hemolytic anemia are associated more often with which blood group system?

a Rh
b I
c P
d FY

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260. An antibody that causes *in vitro* hemolysis and reacts with the red cells of 3 out of 10 AHG-crossmatched donor units is most likely:

a anti-Le^a
b anti-s
c anti-k
d anti-E

261. An antibody identification study is performed with the 5-cell panel shown in this table:

		antigens					
		1	2	3	4	5	test results
Panel cells	I	+	0	0	+	+	+
	II	0	0	+	0	+	0
	III	0	+	+	+	0	0
	IV	0	+	+	0	+	+
	V	+	+	+	0	0	+
						auto	0

An antibody against which of the following antigens could **not** be excluded?

a 1
b 2
c 3
d 4

262. A 25-year-old Caucasian woman, gravida 3, para 2, requires 2 units of Red Blood Cells. The antibody screen is positive and the results of the antibody panel are shown in this table:

cell	D	C	c	E	e	K	Jk^a	Jk^b	Le^a	Le^b	M	N	P1	IAT
1	+	+	0	0	+	+	+	+	0	+	+	+	+	0
2	+	+	0	0	+	0	+	0	0	+	+	0	0	0
3	+	0	+	+	0	0	+	+	0	+	+	+	+	1+
4	+	+	+	0	+	0	0	+	0	+	+	0	+	1+
5	0	0	+	0	+	0	+	+	0	+	+	0	0	1+
6	0	0	+	+	+	0	+	0	+	0	+	+	0	1+
7	0	0	+	0	+	+	+	+	+	0	+	+	+	1+
8	0	0	+	0	+	0	0	+	0	+	0	+	+	1+
IAT = indirect antiglobulin test													auto	0

Which of the following antibodies may be the cause of the positive antibody screen?

a anti-M and anti-K
b anti-c and anti-E
c anti-Jk^a and anti-c
d anti-P1 and anti-c

263. The sample from a 45-year-old Asian woman, gravida 2, para 2 demonstrates a positive antibody screen. The results of the antibody panel are shown in the panel:

cell	D	C	c	E	e	K	Jk^a	Jk^b	Le^a	Le^b	M	N	P1	IAT
1	+	+	0	0	+	+	+	+	0	+	+	+	+	0
2	+	+	0	0	+	0	+	0	0	+	+	0	0	0
3	+	0	+	+	0	0	+	+	0	+	+	+	+	1+
4	+	+	+	0	+	0	0	+	0	+	+	0	+	1+
5	0	0	+	0	+	0	+	+	0	+	+	0	0	1+
6	0	0	+	+	+	0	+	0	+	0	+	+	0	1+
7	0	0	+	0	+	+	+	+	+	0	+	+	+	1+
8	0	0	+	0	+	0	0	+	0	+	0	+	+	1+
IAT = indirect antiglobulin test													auto	0

Which common antibody has not been ruled out by the panel?

a anti-C
b anti-Le^b
c anti-Jk^a
d anti-E

264. MLS ONLY A 5-year-old with chronic upper respiratory infections arrives in the Emergency Room with chronic anemia. The antibody screen and antibody identification panel are all strongly reactive when tested by solid phase automation. The direct antiglobulin test (DAT) is negative, and the patient's phenotype is shown in the panel. With these initial findings, what is suspected?

C	E	c	e	K	k	Fy^a	Fy^b	Jk^a	Jk^b
4+	4+	4+	4+	0	1+	0	0	4+	4+

a paroxysmal cold hemoglobinuria (PCH)
b McLeod syndrome
c warm autoimmune hemolytic anemia (WAIHA)
d cold agglutinin disease (CAD)

265. A 16-year-old female is admitted to the emergency room with a gunshot wound. A massive transfusion protocol has been activated before a blood type can be completed. In the patient's record, the patient has a history of being group A, D-negative. The technologist should:

a release group A, D-negative Red Blood Cells
b release group O, D-negative Red Blood Cells
c release group O, D-positive Red Blood Cells
d refuse to release any blood until a blood type is determined

266. Transfusion of Ch+ (Chido positive) red cells to a patient with anti-Ch has been reported to cause:

a no clinically significant red cell destruction
b clinically significant immune red cell destruction
c decreased ^{51}Cr red cell survivals
d febrile transfusion reactions

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267. Based on the results of the panel, the most likely antibodies are:

MLS ONLY

cell	D	C	c	E	e	K	Jk^a	Jk^b	Le^a	Le^b	M	N	P1	IS	LISS 37°C	IAT
1	+	+	0	0	+	+	+	+	0	+	+	+	+	0	0	2+
2	+	+	0	0	+	0	+	0	0	+	+	0	0	0	0	3+
3	+	0	+	+	0	0	+	+	0	+	+	+	+	1+	1+	3+
4	+	+	+	0	+	0	0	+	0	+	+	0	+	0	0	0
5	0	0	+	0	+	0	+	+	0	+	+	0	0	0	0	2+
6	0	0	+	+	+	0	+	0	+	0	+	+	0	1+	1+	3+
7	0	0	+	0	+	+	+	+	+	0	+	+	+	0	0	2+
8	0	0	+	0	+	0	0	+	0	+	0	+	+	0	0	0
IAT = indirect antiglobulin test; LISS = low ionic strength saline													auto	0	0	0

a anti-M and anti-K
b anti-E, anti-Jk^a and anti-K
c anti-Jk^a and anti-M
d anti-E and anti-Le^b

268. Which characteristics are true of **all 3** of the following antibodies: anti-Fy^a, anti-Jk^a, and anti-K?

a detected at IAT phase and may cause hemolytic disease of the fetus and newborn (HDFN) and transfusion reactions
b not detected with enzyme-treated cells; may cause delayed transfusion reactions
c requires the IAT technique for detection; usually not responsible for causing HDFN
d detected at the room temperature phase of testing; may cause severe hemolytic transfusion reactions

269. Based on the results in the panel, which of the following antibodies is most likely present?

cell	D	C	c	E	e	K	Jk^a	Jk^b	Le^a	Le^b	M	N	P1	IAT	enzyme IAT
1	+	+	0	0	+	+	+	+	0	+	+	+	+	3+	4+
2	+	+	0	0	+	0	+	0	0	+	+	0	0	3+	4+
3	+	0	+	+	0	0	+	+	0	+	+	+	+	0	0
4	+	+	+	0	+	0	0	+	0	+	+	0	+	2+	3+
5	0	0	+	0	+	0	+	+	0	+	+	0	0	0	0
6	0	0	+	+	+	0	+	0	+	0	+	+	0	0	0
7	0	0	+	0	+	+	+	+	+	0	+	+	+	0	0
8	0	0	+	0	+	0	0	+	0	+	0	+	+	0	0
IAT = indirect antiglobulin test													auto	0	0

a anti-C
b anti-E
c anti-D
d anti-K

270. MLS ONLY A pregnant woman has a positive antibody detection test and the antibody identification results are given in this panel:

cell	D	C	c	E	e	K	Jk^a	Jk^b	Fy^a	Fy^b	Le^a	Le^b	M	N	P1	IS	LISS 37°C	IAT	enzyme IAT
1	+	+	0	0	+	+	+	+	0	+	0	+	+	+	+	0	0	0	0
2	+	+	0	0	+	0	+	0	+	0	0	+	+	0	0	1+	1+	2+	0
3	+	0	+	+	0	0	+	+	+	+	0	+	+	+	+	0	0	1+	0
4	+	+	+	0	+	0	0	+	0	+	0	+	+	0	+	0	0	0	0
5	0	0	+	0	+	0	+	+	+	+	0	+	+	0	0	0	0	1+	0
6	0	0	+	+	+	0	+	0	0	0	+	0	+	+	0	0	0	0	0
7	0	0	+	0	+	+	+	+	0	+	+	0	+	+	+	0	0	0	0
8	0	0	+	0	+	0	0	+	+	0	0	+	0	+	+	1+	1+	2+	0
IAT = indirect antiglobulin test; LISS = low ionic strength saline															auto	0	0	0	0

What is the association of this antibody(ies) with hemolytic disease of the fetus and newborn (HDFN)?

a usually fatal HDFN
b may cause HDFN
c is not associated with HDFN
d cannot determine antibody specificity

271. Which of the following tests is most commonly used to detect antibodies attached to a patient's red blood cells *in vivo*?

a direct antiglobulin
b complement fixation
c indirect antiglobulin
d immunofluorescence

272. Anti-I in cold agglutinin disease may cause a positive direct antiglobulin test (DAT) because of:

a anti-I agglutinating the cells
b C3d bound to the red cells
c T-activation
d C3c remaining on the red cells after cleavage of C3b

273. MLS ONLY Which direct antiglobulin test results may be associated with a delayed serologic transfusion reaction in a recently transfused patient?

test result	polyspecific	IgG	C3	control
result A	$+^{mf}$	$+^{mf}$	0	0
result B	1+	0	1+	0
result C	2+	2+	0	0
result D	4+	4+	4+	0

mf=mixed field

a result A
b result B
c result C
d result D

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274. A patient's antibody identification panel demonstrates anti-M. The antibody is most reactive with homozygous M+ cells compared to heterozygous M+ cells. Which of the following cells would demonstrate the strongest reaction?

a M–N+S–s+
b M+N+S+s+
c M+N–S–s+
d M+N+S–s–

275. In the direct antiglobulin test, the antiglobulin reagent is used to:

a mediate hemolysis of indicator red blood cells by providing complement
b precipitate anti-erythrocyte antibodies
c measure antibodies in a test serum by fixing complement
d detect preexisting antibodies on erythrocytes

276. The mechanism that best explains hemolytic anemia due to penicillin is:

a drug-dependent antibodies reacting with drug-coated RBCs
b drug-dependent antibodies reacting in the presence of drug
c drug-independent with autoantibody production
d nonimmunologic protein adsorption with positive DAT

277. Use of EDTA plasma prevents activation of the classical complement pathway by:

a causing rapid decay of complement components
b chelating Mg^{++} ions, which prevents the assembly of C6
c chelating Ca^{++} ions, which prevents assembly of C1
d preventing chemotaxis

278. The drug cephalosporin can cause a positive direct antiglobulin test due to modification of the RBC membrane by the drug, which is independent of antibody production. This mechanism related to the drug cephalosporin is best described as:

a drug-dependent
b complement related drug-dependent
c drug-autoantibody
d nonimmunologic protein adsorption

279. During prenatal studies, a woman is noted to have a positive antibody screen and anti-Kp^a is identified. What percentage of units will be compatible for this patient if transfusion is necessary?

a <2%
b 50%
c 85%
d >98%

280. The purpose of testing with anti-A,B is to detect:

a anti-A1
b anti-A2
c subgroups of A
d subgroups of O

281. A group O, Rh-negative pregnant female has anti-Vel in her serum. If needed, how might blood be provided for her infant?

a maternal donation
b paternal donation
c random ABO-identical unit
d random group O, Rh-negative unit

282. In a prenatal workup, the results shown in the table are obtained:

forward group		reverse group			
anti-A	anti-B	anti-D	Rh control	A_1 cells	B cells
4+	2+	4+	0	0	3+
DAT	negative				
antibody detection test	negative				

This ABO discrepancy is thought to be due to an antibody directed against a component of the ABO typing sera. Which test would resolve this discrepancy?

a test patient cells with anti-A1 lectin
b saline wash patient's cells and repeat forward typing
c test patient cells with anti-A,B
d repeat reverse typing using A_2 cells

283. While performing an antibody screen, a test reaction is suspected to be rouleaux. A saline replacement test is performed and the reaction remains. What is the best interpretation?

a original reaction of rouleaux is confirmed
b replacement test is invalid and should be repeated
c original reaction was due to true agglutination
d antibody screen is negative

284. A 10-year-old girl is hospitalized because her urine has a distinct red color. The patient has recently recovered from an upper respiratory infection and appears very pale and lethargic. Tests are performed with the results show in this table:

test	result
hemoglobin	5 g/dL (50 g/L)
reticulocyte count	15%
DAT	weak reactivity with poly-specific and anti-C3d; anti-IgG is negative
antibody detection test	negative
Donath-Landsteiner test	positive; P– cells showed no hemolysis

The patient probably has:

a paroxysmal cold hemoglobinuria (PCH)
b paroxysmal nocturnal hemoglobinuria (PNH)
c warm autoimmune hemolytic anemia (WAIHA)
d hereditary erythroblastic multinuclearity with a positive acidified serum test (HEMPAS)

285. A transfusion request must show 2 recipient identifiers, the specific type and amount of blood component with any special processing requirements, and what additional information?

a ABO and Rh type of component requested for transfusion
b name of ordering physician or authorized health professional
c date blood typing and antibody detection testing were performed
d name of nursing team to start the blood infusion

286. A 15-year-old bone marrow transplant recipient with a historical blood type of group A, D-positive has a transfusion request for 1 unit of Apheresis Platelets. The transfusion request received in the transfusion service shows that the patient's name and medical record number are cut off and are illegible, although the patient's date of birth is legible. What action should the blood bank technologist do to provide platelets for this patient?

a accept the request since the patient has a historical record for ABO and Rh
b accept the request since the patient is a bone marrow transplant patient
c reject the request due to the incomplete and illegible order
d reject the request since the order is for Apheresis Platelets

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287. When completing recipient pretransfusion ABO testing, the sample must be tested with which of the following reagents?

a anti-A,B; A_1 cells; B cells; O cells
b anti-A; anti-B; A_1 cells; A_2 cells; B cells
c anti-A; anti-B; A_1 cells; B cells
d anti-A; anti-B; anti-A,B; A_1 cells; A_2 cells; B cells

288. A pretransfusion blood typing and antibody detection test is completed on Tuesday at 4:00pm from a patient with an unknown transfusion history. This sample is valid to use for crossmatching for transfusion until what day and time?

a Wednesday at 4:00pm
b Thursday at 11:59pm
c Friday at 11:59pm
d Saturday at 4:00pm

289. What is the purpose of the immediate spin (IS) crossmatch?

a detect clinically significant alloantibodies in recipient
b detect ABO incompatibility between donor and recipient
c verify the correct blood sample from recipient was collected
d verify the presence of IgM alloantibodies in the recipient

290. Use of the computer to detect ABO incompatibility (computer crossmatch) can only be used when:

a the patient has a historical ABO and Rh in the computer system
b the computer system manual states such application is acceptable
c the patient fails to demonstrate clinically significant alloantibodies
d the immediate spin (IS) crossmatch is also negative

291. A unit of Red Blood Cells, Leukocytes Reduced is released from the transfusion service for a patient in ICU. After release, the patient's clinical course changes, and the blood transfusion is no longer needed. The unit is returned to the transfusion service within the hour. The temperature of the unit at the time it is returned is 12°C. The unit has not been entered, is visually acceptable, and multiple segments are still attached to the unit. What action should the transfusion service take?

a discard the unit since it was already released from the transfusion service
b discard the unit as appropriate temperature has not been maintained
c accept unit into inventory since the unit was not entered
d accept unit into inventory since it is visually acceptable with segments attached

292. What are the two tasks a phlebotomist must complete when collecting a pretransfusion blood sample?

a identify the patient and check the patient's pre-phlebotomy vital signs
b check the patient's vital signs and label the tube immediately after phlebotomy
c identify the patient before phlebotomy and label the tube immediately after phlebotomy
d verify the patient's reason for transfusion and label the tube as soon as possible

293. The tie-tag or label attached to each blood component issued for transfusion must include which of the following?

a recipient's 2 independent identifiers
b technologist who performed testing
c date of collection for pretransfusion sample
d date and time of issue from the blood bank

B. Reagents

294. Mixed field agglutination encountered in ABO forward typing with no history of transfusion would most likely be due to:

a Bombay phenotype (O_h)
b T activation
c A_3 red cells
d positive indirect antiglobulin test

295. A patient demonstrates 4+ reactivity with all red cells tested and the autocontrol is nonreactive. This high incidence antibody is suspected to be related to the P1PK blood group system as the patient is the rare p phenotype. What antibody specificity should be suspected?

a anti-IP1
b anti-P2
c anti-PP1P^k
d anti-P1

296. Using the antigen typing results in this table, what is the patient's likely phenotype?

D	C	E	c	e	f	G
+	0	+	+	+	+	+

a R_1R_1
b R_2R_2
c R_2r
d R_1R_2

297. A patient receives 2 units of Red Blood Cells and has a delayed transfusion reaction. Pretransfusion antibody screening records indicate no agglutination except after the addition of IgG-sensitized cells. Repeat testing of the pretransfusion specimen detects an antibody at the antiglobulin phase. What is the most likely explanation for the original results?

a red cells are overwashed
b centrifugation time is prolonged
c patient's serum is omitted from the original testing
d antiglobulin reagent is neutralized

298. Results of a serum sample tested against a panel of reagent red cells gives presumptive evidence of an alloantibody directed against a high incidence antigen. Further investigation to confirm the specificity should include which of the following?

a serum testing against red cells from random donors
b serum testing against red cells known to lack high incidence antigens
c serum testing against enzyme-treated autologous red cells
d testing of an eluate prepared from the patient's red cells

299. Polyspecific reagents used in the direct antiglobulin test should have specificity for:

a IgG and IgA
b IgG and C3d
c IgM and IgA
d IgM and C3d

300. A 56-year-old female with cold agglutinin disease has a positive direct antiglobulin test (DAT). When the DAT is repeated using monospecific antiglobulin sera, which of the following is most likely to be detected?

a IgM
b IgG
c C3d
d C4a

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301. MLS ONLY Inheritance of the rare M^k gene results in the deletion of both *GYPA* and *GYPB*. Which of the following blood group antigens is not expressed on red cells in the presence of a M^k gene?

a K_x
b En^a
c f
d G

302. Which of the following might cause a false-negative indirect antiglobulin test (IAT)?

a over-reading
b IgG-coated screening cells
c addition of an extra drop of serum
d too heavy a cell suspension

303. A patient's serum sample is reactive with all cells except the autocontrol when tested by polyethylene glycol-antihuman globulin (PEG-AHG). The patient's phenotype is confirmed as C–E+c+e+; K–k+, Kp(a–b+), Js(a–b+); Fy(a–b+); Jk(a–b–); M+N+S+s+. Phenotypically similar cells are tested and found to be nonreactive. In what population of donors are we most likely to find a compatible donor for this patient?

a African
b Middle Eastern
c South American
d Polynesian

304. Reagent antibody screening cells may not detect all antibodies. Which of the following antibodies is most likely to go undetected?

a anti-Co^a
b anti-S
c anti-C^W
d anti-Xg^a

305. A 25-year-old pregnant female demonstrated anti-K in her serum. At delivery, the baby's cord blood sample demonstrate a 3+ direct antiglobulin test (DAT) with IgG antibody. What serologic testing should be done to verify that the positive DAT is related to the maternal anti-K?

a test cord serum by polyethylene glycol-indirect antiglobulin testing (PEG-IAT)
b perform and test acid elution from cord blood
c test cord serum by ficin-antihuman globulin (AHG)
d perform and test freeze-thaw elution from cord blood

306. A 32-year-old male types as group O, D-negative with a positive antibody detection test. Antibody identification studies demonstrate anti-D, anti-Jk^a, and anti-K. Patient records confirm that he received 12-units of group O, D-positive Red Blood Cells 2 months earlier following a motor vehicle accident. What testing should be performed to verify the patient's red cell phenotype?

a utilization of monoclonal antisera to test the patient's red cells
b ZZAP treatment of patient's red cells to destroy any remaining transfused cells
c microhematocrit cell separation and testing of autologous reticulocytes with monoclonal antisera
d hypertonic saline wash and testing of recovered autologous cells with polyclonal antisera

C. Application of Special Tests and Reagents

307. In a Group O individual with *Le* and *Se* genes, what ABH and LE antigens are present in their secretions?

a Le^a, Le^b
b Le^a, Le^b, H
c Le^a, H
d Le^b, H

308. Consider the ABO typing results shown in this table:

patient's cells vs		patient's serum vs	
anti-A	anti-B	A_1 cells	B cells
4+	0	1+	4+

Additional testing is performed using patient serum:

sample	IS	RT
screening cell I	1+	2+
screening cell II	1+	2+
autocontrol	1+	2+

IS = immediate spin; RT = room temperature

What is the most likely cause of this discrepancy?

a A_2 with Anti-A1
b cold alloantibody
c cold autoantibody
d acquired-A phenomenon

309. What method may be used to resolve the patient's ABO serum typing?

anti-A	anti-B	anti-D	A_1 cells	B cells
4+	0	4+	2+	4+

antibody detection	immediate spin	RT	LISS 37°C	LISS-IAT
screen cell 1	2+	3+	1+	1+
screen cell 2	2+	3+	1+	1+
auto control	2+	3+	2+	2+

IAT = indirect antiglobulin test; LISS = low ionic strength saline; RT = room temperature

a treat patient's red cells with dithiothreitol (DTT)
b warm saline wash patient's red cells
c enzyme treat patient's red cells
d cold-autoadsorption

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310. The patient's results are listed in this table. What is the most likely specificity based on the reaction pattern?

test	result
antibody detection	reactive with all cells 3+ using solid phase
direct antiglobulin test	anti-IgG positive, anti-C3d negative
antibody identification studies	reactive with all cells 3+ using solid phase
eluate testing	reactive with all cells 3+ using solid phase

a high incidence antibody
b autoantibody
c multiple alloantibodies
d cold reactive antibody

311. Based on these reactions, what is the patient's ABO type?

anti-A	anti-B	*Ulex europaeus*	*Dolichos biflorus*
4+	0	0	4+

a A_1
b A_2
c A_{mod}
d A_x

312. MLS ONLY A patient has a variable reacting anti-P1 pattern in antibody identification studies. What test can be used to verify the specificity of anti-P1 and rule out other common alloantibodies?

a P1 neutralization
b AET-AHG
c chloroquine-AHG
d DTT-AHG

313. A patient serum reacts with 2 of the 3 antibody screening cells at the AHG phase, and 8 of the 10 units crossmatched are incompatible at the AHG phase. All reactions are markedly enhanced by enzymes. These results are most consistent with:

a anti-M
b anti-E
c anti-c
d anti-Fy^a

314. MLS ONLY A patient's serum reacted weakly positive (1+^{w}) with 16 of 16 group O panel cells at the AHG test phase. The autocontrol is negative. Tests with ficin-treated panel cells demonstrated no reactivity at the AHG phase. Which antibody is most likely responsible for these results?

a anti-Ch
b anti-k
c anti-e
d anti-Js^b

315. A male patient's sample demonstrates a pattern most consistent with anti-D. The patient is Rh-negative, and was transfused with Rh-positive blood emergently after a motor vehicle accident 2 years previously. The anti-D shows variable reactivity when tested with D-positive cells. What test would be appropriate to enhance the anti-D reactivity and verify specificity?

a ficin- AHG
b DTT-AHG
c trypsin-AHG
d albumin-AHG

316. Which of the following genes on chromosome 1 encodes for the 4 common antigen combinations ce, cE, Ce and CE.

a *RHD*
b *RHCE*
c *RHD* and *RHCE*
d *RHd* and *RHce*

317. MLS ONLY Which of the following blood bank chemicals produce K_0 red cells (Kell null cells)?

a dithiothreitol (DTT)
b ficin
c formaldehyde
d chloroquine diphosphate

318. Which of the following antigens gives enhanced reactions with its corresponding antibody following treatment of the red cells with proteolytic enzymes?

a Fy^a
b E
c S
d M

319. Based on the results of the panel, which technique would be most helpful in determining antibody specificity?

cell	D	C	c	E	e	K	Jk^a	Jk^b	Fy^a	Fy^b	IS	LISS 37°C	IAT
1	+	+	0	0	+	+	+	+	+	+	0	0	2+
2	+	+	0	0	+	0	+	0	+	+	0	0	2+
3	+	0	+	+	0	0	0	+	+	+	0	1+	3+
4	+	+	0	0	+	0	0	+	0	+	0	0	0
5	0	0	+	0	+	0	+	+	+	+	0	0	2+
6	0	0	+	+	+	0	+	0	+	0	0	1+	3+
7	0	0	+	0	+	+	0	+	+	0	0	0	2+
8	0	0	+	0	+	0	0	+	0	+	0	0	0
										auto	0	0	0

IAT = indirect antiglobulin test; LISS = low ionic strength saline

a proteolytic enzyme treatment
b urine neutralization
c autoadsorption
d saliva inhibition

320. MLS ONLY To confirm a serum antibody specificity identified as anti-P1, a neutralization study is performed and the results obtained are shown in this table:

sample	P1+ RBCs
serum + P1 substance	negative
serum + saline	negative

What conclusion can be made from these results?

a anti-P1 is confirmed
b anti-P1 is ruled out
c a second antibody is suspected due to the results of the saline control
d anti-P1 cannot be confirmed due to the results of the saline control

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321. Plasma neutralization is best used to verify which of the following antibodies?
MLS ONLY

a anti-Lu^b
b anti-M
c anti-Ch/Rg
d anti-V

322. To confirm the specificity of anti-Le^b, an inhibition study using LE substance is performed with the results shown in this table:
MLS ONLY

sample	result with Le(b+) cells
tubes with patient serum + LE substance	0
tubes with patient serum + saline control	+

What conclusion can be made from these results?

a second antibody is suspected due to the positive control
b anti-Le^b is confirmed because the tubes with LE substance are negative
c anti-Le^b is not confirmed because the tubes with LE substance are negative
d anti-Le^b cannot be confirmed because the saline control is positive

323. Which is the correct interpretation of this saliva neutralization testing?
MLS ONLY

	indicator cells		
sample	A	B	O
saliva plus anti-A	+	0	0
saliva plus anti-B	0	+	0
saliva plus anti-H	0	0	0

a group A secretor
b group B secretor
c group AB secretor
d group O secretor

324. Antibody identification results performed using solid phase technology resulted in undetermined results as all common alloantibodies are excluded. What additional testing could determine if a cold reactive antibody is responsible for the questionable results?
MLS ONLY

a DTT-treat patient serum and repeat testing
b test at IS, room temperature (RT) and 37°C with manual tube testing
c repeat all testing using column agglutination technology
d repeat all testing and add albumin as an enhancement media

325. An antibody screen performed using solid phase technology revealed a diffuse layer of red blood cells across the entire well. These results indicate:

a a positive reaction
b a negative reaction
c serum was not added
d red cells have a positive direct antiglobulin test

326. Which of the following genes on chromosome 1 is made up of 2 exons, leading to the expression of the FY glycoprotein and its antigens?

a *DAF*
b *FYAB*
c *FY*
d *ACKR1*

327. Which of the following is useful for removing IgG from red blood cells with a positive direct aniglobuluin test (DAT) to perform a phenotype?

a bromelin
b chloroquine
c low ionic strength saline (LISS)
d dithiothreitol (DTT)

328. A patient's serum contains a mixture of antibodies. One of the antibodies is identified as anti-D. Anti-Jk^a, anti-Fy^a and possibly another antibody are present. What technique(s) may be helpful to identify the other antibody(ies)?

a enzyme panel; select cell panel
b thiol reagents
c lowering the pH and increasing the incubation time
d using albumin as an enhancement media in combination with selective adsorption

329. A sample gives the results shown in this table:

cells with		serum with	
anti-A	3+	A_1 cells	2+
anti-B	4+	B cells	0

Which lectin should be used first to resolve this discrepancy?

a *Ulex europaeus*
b *Arachis hypogaea*
c *Dolichos biflorus*
d *Vicia graminea*

330. A 26-year-old female is admitted with anemia of undetermined origin. Blood samples are received with a crossmatch request for 1 unit of Red Blood Cells. The patient is group A, Rh-negative and has no history of transfusion or pregnancy. The results shown in the table are obtained in pretransfusion testing:

sample	IS	37°C	IAT
screening cell I	0	0	3+
screening cell II	0	0	3+
autocontrol	0	0	3+
donor unit	0	0	3+

IAT = indirect antiglobulin test

The next step to continue this investigation would be:

a do an antibody identification panel
b use the saline replacement technique
c use the pre-warm technique
d perform a warm autoadsorption

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331. MLS ONLY A patient's serum is reactive 2+ in the antiglobulin phase of testing with all cells on a routine panel including their own. There had been 2 units of Red Blood Cells transfused 6-months previously. The optimal adsorption method to remove the autoantibody is:

a autoadsorption using the patient's dithiothreitol and cysteine-activated papain (ZZAP)-treated red cells
b autoadsorption using the patient's low ionic strength saline (LISS)-treated red cells
c adsorption using enzyme-treated red cells from a normal donor
d adsorption using methyldopa-treated red cells

332. MLS ONLY In a cold autoadsorption procedure, pretreatment of the patient's red cells with which of the following reagents is helpful?

a ficin
b phosphate-buffered saline at pH 9.0
c low ionic strength saline (LISS)
d albumin

333. The process of separation of antibody from its antigen is known as:

a diffusion
b adsorption
c neutralization
d elution

334. Which of the following is most helpful to confirm a weak ABO subgroup?

a adsorption-elution
b neutralization
c testing with A1 lectin
d use of anti-A,B

335. Preparation of a two-fold serial dilution of an antibody may be used to characterize the relative amount of antibody present and also determine:

a carbohydrate expression the red cells
b relative antigen expression on the red cells
c sialic acid reduction on the red cells
d antibody specificity

336. Which of the following methods may be useful in determining an accurate phenotype in a transfused patient?

a hypotonic wash
b thiol reagents
c density separation
d titration studies

337. Which of the following genes is analyzed with molecular assays to distinguish weak D from partial D?

a *RHD*
b *RHCE*
c *LW*
d *RHAG*

338. In which of the following clinical situations would DNA-based testing be useful?

a identification of maternal alloantibody that has caused hemolytic disease of the fetus and newborn (HDFN)
b resolution of suspected transfusion-related acute lung injury (TRALI)
c evaluation after extracorporeal photopheresis to treat cutaneous T-cell lymphoma
d evaluation of a patient with warm autoimmune hemolytic anemia and a positive direct antiglobulin test (DAT)

D. Leukocyte/Platelet Testing

339. In solid phase red cell adherence assays (SPRCA) to detect platelet specific antibodies, the wells of the microtiter plates are coated with immobilized:

a red blood cells
b granulocytes
c platelets
d patient serum

340. Testing blood donors for the presence of antibodies to human neutrophil antigens (HNAs) can help prevent which of the following adverse events associated with transfusion?

a transfusion-associated circulatory overload (TACO)
b transfusion-related acute lung injury (TRALI)
c anaphylaxis
d citrate toxicity

341. MLS ONLY The complement-dependent cytotoxicity test combines lymphocytes, patient serum and complement and has been used to detect HLA antibodies. After incubation, visualization of damaged or undamaged cells is completed microscopically after addition of what stain or dye?

a methylene blue
b fluorescent vital
c Congo red
d crystal violet

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E. Quality Assurance

342. Anti-E is identified in a panel at the antiglobulin phase. When check cells are added to the nonreactive tubes and spun, no agglutination is seen. The most appropriate course of action would be to:

a quality control the AHG reagent and check cells and repeat the panel
b open a new vial of check cells for subsequent testing that day
c open a new vial of AHG for subsequent testing that day
d record the check cell reactions and report the antibody panel result

343. A serological calibration is completed for a new centrifuge received in the blood bank.

time in seconds	15	20	25	30
is button delineated?	yes	yes	yes	yes
is supernatant clear?	no	yes	yes	yes
button easy to resuspend?	yes	yes	yes	no
strength of reaction?	+m	1+	1+	1+

Given the data, the centrifuge time for saline tests for this machine should be:

a 15 seconds
b 20 seconds
c 25 seconds
d 30 seconds

344. Which of the following represents an acceptably identified patient for sample collection and transfusion?

a handwritten band with patient's name and hospital identification number is affixed to the patient's leg
b the addressographed hospital band is taped to the patient's bed
c an unbanded patient responds positively when his name is called
d the chart transported with the patient contains his armband not yet attached

345. AHG control cells (Check Cells):

a can be used as a positive control for anti-C3 reagents
b can be used only for the indirect antiglobulin test
c are coated only with IgG antibody
d must be used to confirm all positive antiglobulin reactions

346. Crossmatch results at the antiglobulin phase are negative. When 1 drop of check cells is added, no agglutination is seen. The most likely explanation is that the:

a red cells are overwashed
b centrifuge speed is set too high
c residual patient serum inactivated the AHG reagent
d laboratorian did not add enough check cells

347. A pretransfusion sample is received in the transfusion service. When comparing the label on the blood sample with the transfusion request, the following is noted. The transfusion request documents Smith, Sam L. MR# 7070111, DOB 5.22.70 but the blood sample is labeled Smith, Sam J. MR# 7070111, DOB 6.23.71. What, if any, action should be taken?

a a new blood sample and new transfusion request should be obtained
b the phlebotomist should relabel the sample to match the transfusion request
c the physician should resubmit the transfusion request to match the blood sample
d no action is required since the medical record numbers (MR#) are identical

348. A new lot of anti-B needs quality control (QC) testing. When testing is completed, it is noted that the reagent fails to react with red cells known to be positive for the B antigen and red cells known to be negative for the B antigen. What action should be taken?

a document the results, who completed the testing and place into use
b mark the lot as "not for use" and investigate and resolve the results
c place the lot into use as long as repeat testing yielded a different result
d ask a different technologist to QC this lot and accept those results

349. When preparing an eluate, what is the desired quality control result when testing the last wash?

a reactive with all cells tested
b nonreactive with all cells tested
c reactivity equal to the direct antiglobulin test (DAT) result
d stronger reactivity than the DAT result

350. A sterile connection device requires quality control of the weld at what frequency?

a with each weld
b day of use
c weekly
d quarterly

351. A blood bank serologic centrifuge lid and wiring had to be repaired and replaced. What must be completed before the centrifuge is placed back into service?

a reregister the centrifuge with the Food and Drug Administration (FDA)
b requalify the centrifuge with local board of health
c recalibrate for speed and functionality
d nothing, as the instrument can be used immediately

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VI. Transfusion Practice

A. Indications for Transfusion

352. Deglycerolized Red Blood Cells are used to transfuse patients with:

- **a** an antibody to a high-incidence red cell antigen
- **b** anti-c and anti-K alloantibodies
- **c** chronic anemia
- **d** increased risk of CMV infection

353. The primary indication for granulocyte transfusion is:

- **a** prophylactic treatment for infection
- **b** additional supportive therapy in those patients who are responsive to antibiotic therapy
- **c** clinical situations where bone marrow recovery is not anticipated
- **d** severe neutropenia with an infection that is nonresponsive to antibiotic therapy

354. A 42-year-old male of average body mass has a history of chronic anemia requiring transfusion support. Two units of Red Blood Cells are transfused. If the pretransfusion hemoglobin is 7.0 g/dL (70 g/L), the expected posttransfusion hemoglobin concentration should be:

- **a** 8.0 g/dL (80 g/L)
- **b** 9.0 g/dL (90 g/L)
- **c** 10.0 g/dL (100 g/L)
- **d** 11.0 g/dL (110 g/L)

355. How many units of Red Blood Cells are required to raise the hematocrit of a 70 kg nonbleeding man from 24% to 30%?

- **a** 1
- **b** 2
- **c** 3
- **d** 4

356. For which of the following transfusion candidates would CMV-safe blood be **most** likely indicated?

- **a** renal dialysis patients
- **b** sickle cell patient
- **c** bone marrow and hematopoietic cell transplant recipients
- **d** CMV-seropositive patients

357. Washed Red Blood Cells are indicated in which of the following situations?

- **a** an IgA-deficient patient with a history of transfusion-associated anaphylaxis
- **b** a pregnant woman with a history of hemolytic disease of the newborn
- **c** a patient with a positive direct antiglobulin test (DAT) and red cell autoantibody
- **d** a newborn with a hematocrit of <30%

358. What minimum increment of platelets is expected from each unit of Platelets (Whole Blood-derived) when transfused to a non-HLA-sensitized recipient weighing approximately 70 Kg?

- **a** 3,000/µL
- **b** 5,000/µL
- **c** 20,000/µL
- **d** 25,000/µL

359. Platelet transfusions are of value in treating:

- **a** hemolytic transfusion reaction
- **b** posttransfusion purpura
- **c** functional platelet abnormalities
- **d** immune thrombocytopenia

360. Guidelines for emergency release of blood require that:

a ABO group and Rh typing of recipient be performed before the unit is released
b only group O Whole Blood be issued
c only Rh-negative blood is used for Rh-negative patients
d the physician signs a document authorizing the emergency release

361. In an emergency situation, what type of blood should be given to a female patient of child-bearing age if the ABO group and Rh type are unknown?

a Group O, Whole Blood
b Group O, Rh-negative Red Blood Cells
c Group O, Rh-positive Red Blood Cells
d Group O, Washed Red Blood Cells

362. Which of the following statements regarding red cell transfusion to infants less than 4 months old is correct?

a only phenotype identical units should be issued
b fresher units (generally <7 days) should be issued
c irradiated blood components are contraindicated
d crossmatching with a current sample is required for each transfusion

363. During a massive transfusion event, what other blood components are routinely administered in addition to Red Blood Cells or low-titer group O Whole Blood?

a Cryoprecipitated AHF and Apheresis Platelets, Irradiated
b Fresh Frozen Plasma and Apheresis Platelets
c Deglycerolized Red Blood Cells and Fresh Frozen Plasma
d Granulocytes and Cryoprecipitated AHF

364. When preparing units of blood for emergency release, what information must be written on the label or tie-tag?

a the name of the individual releasing the units
b statement that compatibility testing has not been completed
c statement that the patient's ABO and Rh type are unknown
d physician's statement and signature requesting emergency release

365. A massive transfusion event can result in what type of complication(s)?

a hypercalcemia and hypothermia
b increase in coagulation factors
c citrate toxicity and hypocalcemia
d hyperthermia and platelet dilution

366. Preliminary blood bank testing for neonates <4 months of age must include:

a ABO forward typing and ABO reverse typing
b ABO reverse typing and testing for K antigen
c ABO reverse typing and antibody detection testing
d ABO forward typing and Rh testing for D antigen

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B. Component Therapy

367. Cryoprecipitated AHF transfusion is recommended as a treatment for patients with:

a type 1 vWF disorder
b Factor V coagulation deficiency
c Factor VIII coagulation deficiency
d DIC with hypofibrinogenemia

368. Although ABO compatibility is preferred, ABO incompatible product may be administered when transfusing:

a Single-Donor Plasma
b Cryoprecipitated AHF
c Fresh Frozen Plasma
d Granulocytes

369. Transfusion of Apheresis Platelets Leukocytes Reduced from HLA-compatible donors is the preferred treatment for:

a recently diagnosed cases of TTP with severe thrombocytopenia
b acute leukemia in relapse with neutropenia, thrombocytopenia and sepsis
c immune thrombocytopenic purpura unresponsive to IVIG therapy
d severely thrombocytopenic patients, known to be refractory to random donor platelets

370. Which of the following is consistent with standard blood bank procedure governing the infusion of fresh frozen plasma?

a only blood group specific plasma may be administered
b group O may be administered to recipients of all blood groups
c group AB may be administered to AB recipients only
d group A may be administered to both A and O recipients

371. A patient who is group AB, Rh-negative needs 2 units of Fresh Frozen Plasma. Which of the following units of plasma would be **most** acceptable for transfusion?

a group O, Rh-negative
b group A, Rh-negative
c group B, Rh-positive
d group AB, Rh-positive

372. Fresh Frozen Plasma from a group A, Rh-positive donor may be safely transfused to a patient who is group:

a A, Rh-negative
b B, Rh-negative
c AB, Rh-positive
d AB, Rh-negative

373. A patient admitted to the trauma unit requires emergency release of Fresh Frozen Plasma (FFP). Which of the following blood groups of FFP should be issued?

a A
b B
c AB
d O

374. Fresh Frozen Plasma:

a contains all labile coagulative factors except Factor VIII
b has a higher risk of transmitting hepatitis than does Whole Blood
c should be transfused within 24 hours of thawing
d need not be ABO-compatible

375. The purpose of irradiation of blood components is to:

a prevent posttransfusion purpura
b prevent graft-vesus-host (GVH) disease
c sterilize components
d prevent noncardiogenic pulmonary edema

376. A 70-kg patient is transfused with 1 unit of Apheresis Platelets. What is the expected increase in the patient's platelet count?

a 5,000-10,000/µL
b 10,000-20,000/µL
c 20,000-25,000/µL
d 30,000-50,000/µL

377. Plasma Frozen Within 24 Hours (PF24) is given to patients who have:

a coagulation factor deficiencies
b von Willebrand disease
c factor VIII deficiency
d macroglobulinemia

378. Pathogen reduction technology for blood components:

a promotes antibody formation to microbial antigens
b prevents infections by inhibiting microbial nucleic acid replication
c allows patients to receive blood products without infectious disease testing
d prevents infections by using dithiothreitol to rupture microbial membranes

C. Adverse Effects of Transfusion

379. A common strategy to reduce alloimmunization in chronically transfused patients with Sickle Cell Disease (SCD) is to provide red cell units matched for which of the following antigens?

a Fy^a, Fy^b, and Di^a
b M, N, and S
c Jk^a, Jk^b and s
d C, E, and K

380. MLS ONLY Four units of group A Platelets (Whole Blood-derived) are transfused to a group AB patient because group AB platelets are not availasble. The patient's pretransfusion platelet count is 10,000/µL and the posttransfusion count is 16,000/µL. From this information, the laboratorian would most likely conclude that the patient:

a needs group AB platelets to be more effective
b clinical data does not suggest a need for platelet transfusion
c has developed antibodies to the transfused platelets
d should receive irradiated platelets

381. Hypotension, nausea, flushing, fever and chills are symptoms of which of the following transfusion reactions?

a allergic
b circulatory overload
c hemolytic
d anaphylactic

382. MLS ONLY A patient has become refractory to platelet transfusion. Which of the following are probable causes?

a transfusion of Rh-incompatible platelets
b decreased pH of the platelets
c development of an alloantibody with anti-D specificity
d development of antibodies to HLA antigen

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383. A patient has symptoms indicating a possible hemolytic transfusion reaction. What should be done immediately?

a stop the transfusion and discard the unit
b contact the patient's doctor to ask if the transfusion should be stopped
c stop the transfusion and call the patient's doctor to report the reaction
d have patient blood samples sent to the lab to investigate the reaction

384. Posttransfusion purpura is usually caused by:

a anti-A
b white cell antibodies
c anti-HPA-1a
d platelet wash-out

385. An unexplained fall in hemoglobin and mild jaundice in a patient transfused with Red Blood Cells 1 week previously would most likely indicate:

a paroxysmal nocturnal hemoglobinuria
b posttransfusion hepatitis infection
c presence of HLA antibodies
d delayed hemolytic transfusion reaction

386. In a delayed transfusion reaction, the causative antibody is generally too weak to be detected in routine compatibility testing and antibody screening tests, but is typically detectable at what point after transfusion?

a 3-6 hours
b 2 days-2 weeks
c 60-90 days
d after 120 days

387. Which of the following is a potential complication of massive transfusions?

a citrate toxicity with hypercalcemia
b heparin-induced thrombocytopenia
c hypothermia due to 1-6°C storage temperature of red cells
d iron overload from donor red cells leaking intracellular iron

388. Severe intravascular hemolysis is most likely caused by antibodies of which blood group system?

a ABO
b Rh
c KEL
d FY

389. Which of the following blood group systems is most commonly associated with delayed hemolytic transfusion reactions?

a LE
b JK
c LU
d I

390. After receiving a unit of Red Blood Cells, a patient immediately develops flushing, nervousness, fever spike of 38.9°C, shaking, chills and back pain. The plasma hemoglobin is elevated and there is hemoglobinuria. Laboratory investigation of this adverse reaction would most likely show:

a an error in ABO grouping
b an error in Rh typing
c presence of anti-Fy^a antibody in patient's serum
d presence of gram-negative bacteria in blood bag

391. A trauma patient who has just received 10 units of blood may develop:

a anemia
b polycythemia
c leukocytosis
d thrombocytopenia

392. MLS ONLY Five days after transfusion, a patient becomes mildly jaundiced and experiences a drop in hemoglobin and hematocrit with no apparent hemorrhage. The results of the transfusion reaction workup are in this table:

	anti-A	anti-B	anti-D	A_1 cells	B cells	Ab screen	DAT
patient pretransfusion	neg	4+	3+	4+	neg	neg	neg
patient posttransfusion	neg	4+	3+	4+	neg	1+	1+
donor #1	neg	neg	3+	4+	4+	neg	
donor #2	neg	4+	3+	4+	neg	neg	

DAT = direct antiglobulin test

In order to reach a conclusion, the MLS should:

a retype the pre- and posttransfusion patient samples and donor #1
b request an EDTA tube be drawn on the patient and repeat the DAT
c repeat the pretransfusion antibody screen on the patient's sample
d perform an elution on the posttransfusion sample and identify the antibody in the eluate and the serum

393. The most appropriate laboratory test for early detection of acute posttransfusion hemolysis is:

a visual inspection for free plasma hemoglobin
b plasma haptoglobin concentration
c examination for hematuria
d serum bilirubin concentration

394. During initial investigation of a suspected hemolytic transfusion reaction, it is observed that Blood Bank paperwork and patient sample and blood component labels are correct, the posttransfusion reaction plasma is yellow as is the pretransfusion sample, and the direct antiglobulin test is negative. Repeat ABO typing on the posttransfusion sample confirms the pretransfusion results. What is the next step in this investigation?

a repeat compatibility testing on the suspected unit(s)
b perform plasma hemoglobin and haptoglobin determinations
c use enhancement media to repeat the antibody screen
d no further serological testing is necessary

395. Which of the following transfusion reactions is characterized by high fever, shock, hemoglobinuria, DIC and renal failure?

a bacterial contamination
b circulatory overload
c febrile
d anaphylactic

396. Hemoglobinuria, hypotension and generalized bleeding are symptoms of which of the following transfusion reactions?

a allergic
b circulatory overload
c hemolytic
d anaphylactic

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397. Patients who are chronically transfused with red cell components can develop:

a iron overload
b low ferritin
c hypokalemia
d hypertensive reaction

398. A patient's record shows a previous anti-Jk^b, but the current antibody screen is negative. What further testing should be done before transfusion?

a phenotype the patient's red cells for the Jk^b antigen
b perform a cell panel on the patient's serum
c crossmatch ABO type specific units and release compatible units for transfusion
d phenotype donor units and select Jk(b–) units for compatibility testing

399. Which of the following is associated with a risk of developing transfusion-associated graft-versus-host disease (TA-GVHD)?

a patients receiving leukocyte reduced blood components
b directed donation from first-degree family member
c patients transfused with irradiated blood components
d autologous blood donation prior to surgery

400. A patient is readmitted to the hospital with a hemoglobin level of 7 g/dL (70 g/L) 3 weeks after receiving 2 units of red cells. The initial serological tests are:

test	result
ABO/Rh	A+
antibody screen	negative
DAT	1+ mixed field

Which test should be performed next?

a antibody identification panel on the patient's serum
b repeat the ABO type on the donor units
c perform an elution and identify the antibody in the eluate
d crossmatch the post reaction serum with the 2 donor units

401. In a delayed hemolytic transfusion reaction, the direct antiglobulin test on the posttransfusion sample can be:

a negative
b mixed-field positive
c positive due to complement
d negative when the antibody screen is negative

402. To prevent donor lymphocytes from engraftment in the bone marrow of an immunosuppressed patient, all transfusion products must be:

a washed
b leukocyte-reduced
c treated with UV light
d irradiated

403. For a patient who has suffered an acute hemolytic transfusion reaction, the primary treatment goal should be to:

a prevent alloimmunization
b diminish chills and fever
c prevent hemoglobinemia
d reverse hypotension and minimize renal damage

404. Nine days after being transfused with an HLA-matched platelet transfusion, a patient develops a fever, watery diarrhea, skin rash and demonstrates increased liver enzymes. This patient may have transfusion-associated:

a allergic urticaria reaction
b hepatitis C
c graft-versus-host disease
d septicemia with endocarditis

405. Which of the following transfusion reactions occurs after infusion of only a few milliliters of blood and gives no history of fever?

a febrile
b circulatory overload
c anaphylactic
d hemolytic

406. Fever and chills are symptoms of which of the following transfusion reactions?

a citrate toxicity
b circulatory overload
c allergic
d febrile

407. Hives and itching are symptoms of which of the following transfusion reactions?

a febrile
b allergic
c circulatory overload
d bacterial

408. A temperature rise of 1°C or more occurring in association with a transfusion, with no abnormal results in the transfusion reaction investigation, usually indicates which of the following reactions?

a febrile
b circulatory overload
c hemolytic
d anaphylactic

409. A 65-year-old woman experienced shaking, chills, and a fever of 38.9°C approximately 40 minutes following the transfusion of a second unit of Red Blood Cells. The most likely explanation for the patient's symptoms is:

a transfusion of bacterially contaminated blood
b congestive heart failure
c anaphylactic transfusion reaction
d febrile transfusion reaction

410. Use of only male donors as a source of plasma intended for transfusion is advocated to reduce which type of reaction?

a allergic
b TRALI
c hemolytic
d TACO

411. Symptoms of dyspnea, hypoxemia, and pulmonary edema within 6 hours of transfusion is most likely which type of reaction?

a anaphylactic
b hemolytic
c febrile
d TRALI

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412. Coughing, hypoxemia and difficult breathing are symptoms of which of the following transfusion reactions?

a febrile
b allergic
c TACO
d hemolytic

413. Congestive heart failure, severe headache and/or peripheral edema occurring soon after transfusion is indicative of which type of transfusion reaction?

a hemolytic
b febrile
c anaphylactic
d TACO

414. A patient becomes hypotensive and goes into shock after receiving 50 mL of a unit of Red Blood Cells. She has a shaking chill and her temperature rises to 40.4°C. A transfusion reaction investigation is initiated but no abnormal results are seen. What additional testing should be performed?

a Gram stain and culture of the donor unit
b lymphocytotoxicity tests for leukoagglutinins
c plasma IgA level
d elution and antibody identification

415. The most frequent transfusion-associated disease complication of blood transfusions is:

a cytomegalovirus (CMV)
b syphilis
c hepatitis
d HIV-1/2

416. Which of the following patient groups is at risk of developing graft-versus-host disease?

a full-term infants
b patients with history of febrile transfusion reactions
c patients with a positive direct antiglobulin test
d recipients of blood donated by immediate family members

417. Transfusion-associated HTLV I/II incidence is low due to the following laboratory testing:

a nucleic acid testing for HTLV I/II
b HTLV I core antibody testing by ELISA
c HTLV II surface antigen testing by chemiluminescent immunoassay (ChLIA)
d IgG antibody testing for HTLV I/II by ELISA or ChLIA

418. Which viral diseases have a lower incidence of transfusion-associated infections due to nucleic acid testing (NAT)?

a HIV-1, HCV, WNV
b HIV-2, HBV, CMV
c HTLV I/II, HCV, WNV
d CMV, WNV, HCV

419. Two hours after receiving a blood transfusion, the recipient experienced fever, chills, back pain and hypotension. Patient blood samples are collected along with a urine sample and sent to the laboratory. Which of the following results could indicate that an acute hemolytic transfusion reaction is taking place?

a decrease in the patient's WBC count
b hemolysis in posttransfusion sample only
c absence of hemoglobin in the urine
d hemolysis in pre- and posttransfusion samples

420. What can cause hypothermia in a blood recipient?

a infusion of citrate from multiple transfusion
b bacterial contamination in donor unit
c air emboli in the IV line
d rapid infusion of cold blood

D. Apheresis and Extracorporeal Circulation

421. The preferred replacement fluid to maintain normal oncotic pressure and intravascular fluid levels for patients who have therapeutic plasma exchange (TPE) for thrombotic thrombocytopenic purpura is:

a IVIG
b plasma
c crystalloid solutions
d isotonic saline

422. Which of the following disease states is treated with therapeutic plasma exchange (TPE)?
MLS ONLY

a gout
b myasthenia gravis
c intrahepatic cholestasis
d Crohn disease

423. A 50-year-old patient with acute lymphocytic leukemia has symptoms of dyspnea, visual abnormalities, and headache. The blast count is greater than 100,000/μL. What type of apheresis is indicated to treat this patient?
MLS ONLY

a extracorporeal photopheresis (ECP) to remove immature lymphocytes
b selective absorption apheresis to remove immature lymphocytes
c cytapheresis to remove immature lymphocytes
d therapeutic plasma exchange (TPE) to remove antibodies

424. Therapeutic plasmapheresis is performed in order to:

a harvest granulocytes
b harvest platelets
c treat patients with polycythemia
d treat patients with plasma abnormalities

425. Plasma exchange is recommended in the treatment of patients with macroglobulinemia in order to remove:

a antigen
b excess IgM
c excess IgG
d abnormal platelets

426. Extracorporeal photopheresis can be used to treat which of the following disorders?
MLS ONLY

a solid organ transplant rejection
b multiple myeloma
c familial hypercholesterolemia
d macroglobulinemia

427. For patients with familial hypercholesterolemia, what type of therapeutic apheresis will remove LDL cholesterol?
MLS ONLY

a plasma exchange
b selective adsorption
c cytapheresis
d extracorporeal photopheresis

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428. Sickle cell anemia may be treated with:

a leukocytapheresis to remove patient WBCs attached to sickled RBCs
b extracorporeal photopheresis to destroy sickled RBCs
c thrombocytapheresis to remove platelets to prevent clotting
d erythrocytapheresis to remove sickled RBCs

E. Blood Administration and Patient Blood Management

429. Prior to initiating a blood transfusion, the transfusionist and another qualified individual must:

a match the blood component to the recipient using 2 independent identifiers
b verify that the recipient's vital signs are within normal limits
c review the patient's medical record to verify the need for a transfusion
d order a baseline hemoglobin and hematocrit before the transfusion is started

430. Which of the following must be verified in the transfusion service prior to the issue of blood products?

a antibody detection test result
b date and time when the patient's blood sample was drawn
c name of the transfusionist administering the blood product
d expiration date and, if applicable, expiration time of the blood product

431. During the issue of an autologous unit of Whole Blood, the supernatant plasma is observed to be dark red. What would be the best course of action?

a the unit may be issued only for autologous use
b remove the plasma and issue the unit as Red Blood Cells
c issue the unit only as washed Red Blood Cells
d quarantine the unit for further testing

432. Which of the following must be performed on a patient before, during, and after receiving a blood transfusion?

a blood pressure, pulse, respiration rate and temperature
b blood pressure, pulse, temperature and urine occult blood
c blood pressure, temperature, respiration rate and hematocrit
d blood pressure, temperature, pulse and hematocrit

433. An intraoperative strategy for patient blood management is:

a anemia assessment and treatment
b autologous blood donation
c acute normovolemic hemodilution
d limiting phlebotomy-related blood loss

434. The most important step in the safe administration of blood is to:

a perform compatibility testing accurately
b get an accurate patient history
c exclude disqualified donors
d accurately identify the donor unit and recipient

435. What information related to receiving a blood transfusion needs to be documented in a patient's medical record?

a result of patient direct antiglobulin testing (DAT)
b identification of personnel performing crossmatch
c signed patient consent for transfusion
d crossmatch test results and interpretation

436. In addition to verifying the blood recipient's 2 independent identifiers, ABO group and Rh type, what else needs to be verified immediately before starting a transfusion?

a name of the donation facility where blood was collected
b date and time the crossmatch was completed
c patient CBC results and date of last transfusion
d donation identification number on unit and donor's ABO and Rh

437. Preoperative autologous blood donation can be used for patients who:

a have a history of adverse transfusion reactions
b have an alloantibody to a high-incidence antigen
c need to mitigate iatrogenic blood loss
d want to be discharged the same day as surgery

438. Intraoperative strategies for blood management include:

a decreased amounts of blood drawn for lab tests to reduce iatrogenic blood loss
b more orders to transfusion a single unit of Red Blood Cells vs 2 units of Red Blood Cells
c reinfusing shed blood recovered during surgery that has been washed with normal saline
d more reliance on a transfusion trigger to determine need for additional transfusions

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I. Blood Products

A. Donors

1. **c** The hemoglobin for women shall be ≥12.5 g/dL (hematocrit ≥38%) and the hemoglobin for men shall be ≥13.0 g/dL (hematocrit ≥39%).
[QCMLS 2021: 1.1.1.1] [AABB BB/TS Standards 2022, Reference Standard 5.4.1A, p67]

2. **c** Present or past clinical or laboratory evidence of infection with *Trypanosoma cruzi* (Chagas disease) is cause for an indefinite deferral from blood donation. A reactive test for *Babesia* species is a 2-year deferral, residence in a malarial endemic area for 5 years is a 3-year deferral after departure, and chicken pox vaccination is a 4-week deferral.
[QCMLS 2021: 1.1.1.1] [AABB BB/TS Standards 2022, Reference Standard 5.4.1A, p70-73]

3. **d** As there is no upper age limit for blood donation and this donor's hemoglobin is ≥13.0 g/dL, this individual is eligible to donate. All other potential donors have physical examination requirements outside of acceptable values. Systolic blood pressure must range from 90-180 mm Hg and the diastolic blood pressure must range between 50-100 mm Hg. Donors must weigh at least 110 lb. Pulse must be 50-100 bpm.
[QCMLS 2021: 1.1.1.1] [AABB BB/TS Standards 2022, Reference Standard 5.4.1A, p67]

4. **d** A positive test for HBsAg at any time is a permanent deferral.
[QCMLS 2021: 1.1.1.1] [AABB BB/TS Standards 2022, Reference Standard 5.4.1A, p70]

5. **b** An individual taking HIV prevention drugs (PrEP & PEP) is eligible to donate 3 months after the last dose. Hepatitis B immune globulin administration is a 12-month deferral. A potential donor is deferred for 3-months after receiving a blood transfusion. Travel to a malarial-endemic area is a 3-month deferral after departure.
[QCMLS 2021: 1.1.1.1] [AABB BB/TS Standards 2022, Reference Standard 5.4.1A, p68-69, 73]

6. **a** The minimum platelet count required for plateletpheresis donors is 150,000/µL. The results of platelet counts performed before or after a procedure may be used to qualify the donor for the next procedure.
[QCMLS 2021: 1.1.1.4] [AABB BB/TS Standards 2022, p22-23]

7. **d** The arm scrub must use povidone-iodine. Donors who are sensitive to iodine can have the area cleaned with a preparation of 2% chlorhexidine and 70% isopropyl alcohol. Green soap must not be used.
[QCMLS 2021: 1.1.1.4] [AABB Tech Manual 2020, p203-205, Method 6-2]

8. **c** Donation of 2 units of Apheresis Red Blood Cells will result in a 16-week deferral from blood donation.
[QCMLS 2021: 1.1.1.1, 1.1.1.4] [AABB BB/TS Standards 2022, p23]

9. **b** There are no current tests to detect the presence of *Plasmodium* parasites in blood donors. Protection from transfusion-transmitted malaria relies solely on ascertaining the travel history of prospective donors.
[QCMLS 2021: 1.1.1.1] [AABB Tech Manual 2020, p212; AABB BB/TS Standards 2022, Reference Standard 5.4.1A, p72-73]

10. **b** Diversion of the first 10-40 mL of donor blood that may contain contaminated skin into a separate pouch helps prevent bacterial contamination in blood products. This is particularly true of platelet products as their storage at 20-24°C can facilitate bacterial growth.
[QCMLS 2021: 1.1.1.5] [AABB BB/TS Standards 2022, p12-13, 23; AABB Tech Manual 2020, p142, 204]

11. **c** If the collecting and transfusing facility are the same, infectious disease screening is not required. ABO and Rh testing must be done for each autologous donation. A physician's order and evaluation for potential infection in the autologous donor are also required.
[QCMLS 2021: 1.1.1.2] [AABB BB/TS Standards 2022, p20, 34, 36] [Harmening 2019, p293]

12. **a** Autologous blood donors must have a hemoglobin of ≥11 g/dL.
[QCMLS 2021: 1.1.1.2] [AABB BB/TS Standards 2022, p20]

B. Processing

13. **c** The label on any blood product must include the unique facility identifier, such as FDA registration number and facility name.
[QCMLS 2021: 1.1.2.4] [AABB Tech Manual 2020, p164-165; AABB BB/TS Standards 2022, p13-14]

14. **d** Testing for syphilis was the first mandated donor screening test for infectious disease and is still part of infectious disease testing.
[QCMLS 2021: 1.1.2.1] [AABB BB/TS Standards 2022, p36]

15. **a** The use of green soap to disinfect the venipuncture site is unacceptable. Products containing povidone iodine or isopropyl alcohol and iodine tincture are suitable to prepare the venipuncture site. Diversion of the first 10-40 mL of blood at the start of phlebotomy, culture-based testing strategies, and pathogen reduction are all acceptable strategies to avoid bacterial contamination in platelet components.
[QCMLS 2021: 1.1.2.1] [AABB Tech Manual 2020, p142, 203-205, 215]

16. **a** One of the mitigation strategies for Transfusion Related Acute Lung Injury (TRALI) is to use plasma components, whole blood, and platelets from male donors as they are unlikely to demonstrate HLA antibodies. Never-pregnant female donors, or female donors who have been found to be negative for HLA antibodies (and in some donors, negative for HNA antibodies) since their last pregnancy, are also acceptable as donors for plasma components. Although these measures reduce the risk of TRALI, they do not eliminate TRALI completely.
[QCMLS 2021: 1.6.3.3] [AABB Tech Manual 2020, p640-643]

17. **a** Nucleic acid testing (NAT) for HBV DNA is required for each blood donation.
[QCMLS 2021: 1.1.2.1] [AABB BB/TS Standards 2022, p36]

18. **b** The primary cause of residual disease transmission in the blood supply is from donors in the window period of early infection before test results become positive.
[QCMLS 2021: 1.6.3.8] [AABB Tech Manual 2020, p196-198]

19. **b** If Apheresis Platelets undergo pathogen reduction technology, a culture-based test to detect bacteria in the product is not required. Pathogen reduction technology is an acceptable bacterial risk control strategy.
[QCMLS 2021: 1.1.2.1] [AABB BB/TS Standards 2022, p12]

20. **c** Before labeling, weak D testing must be performed on all samples that initially test as D-negative. If weak D testing and a control are negative, blood components are labeled as Rh-negative. If weak D testing is positive (and the control is negative), blood components are labeled as Rh-positive.
[QCMLS 2021: 1.1.2.1] [AABB BB/TS Standards 2022, p34]

21. **c** Platelet components undergoing culture-based testing for the detection of bacteria are usually released to inventory as incubation of those cultures continues. If a culture becomes positive, the first action is to retrieve products issued from inventory as they may not have been transfused. This retrieval can prevent an adverse reaction in a blood recipient. Determination of the identity of the organism also needs to be done, but this can occur after retrieval of the issued component.
[QCMLS 2021: 1.1.2.1] [AABB BB/TS Standards 2022, p12-13, 93]

22. **a** Each blood donation is tested for HIV-1/2, HTLV-I/II, WNV, HCV, HBV, and syphilis and can include multiple assays for each infectious agent. Depending on geographic location, each donation may also be tested for *Babesia* species. Testing for *Trypanosoma cruzi* is a 1-time test. In May 2021, the FDA established a policy that Zika virus is no longer a relevant transfusion-transmitted infection and testing is no longer required.
[QCMLS 2021: 1.1.2.1] [AABB BB/TS Standards 2022, p36]

C. Storage

23. **b** The transport temperature for all red cell blood products is 1-10°C.
[QCMLS 2021: 1.1.3] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p57-58]

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24. **d** The transport temperature for all platelet products is 20-24°C.
[QCMLS 2021: 1.1.3] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p59-61]

25. **d** Sterile connecting devices allow entry into donor units without affecting the expiration date of the product.
[QCMLS 2021: 1.1.2.5] [AABB BB/TS Standards 2022, p26]

26. **c** If storage devices do not have automated temperature recording, ambient temperature must be monitored and recorded every 4 hours.
[QCMLS 2021: 1.1.3] [AABB BB/TS Standards 2022, p16]

27. **b** Fresh Frozen Plasma is stored at ≤–18°C for 12 months.
[QCMLS 2021: 1.1.3] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p53]

28. **b** Blood may be returned to the blood bank after issue provided that 1) the container closure has not been disturbed, 2) at least 1 sealed segment is attached to the container, 3) visual inspection of the unit is satisfactory and documented, and 4) the unit has been maintained at the appropriate storage or transport temperature.
[QCMLS 2021: 1.5.1.8.1] [AABB BB/TS Standards 2022, p48]

29. **a** Red Blood Cells Frozen with 40% glycerol are stored at ≤–65°C.
[QCMLS 2021: 1.1.3] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p58]

30. **a** Red Blood Cells, Leukocytes Reduced are stored at 1-6°C.
[QCMLS 2021: 1.1.3] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p58]

31. **b** If the seal is broken or entered during processing or storage, components are considered to be in an open system, rather than a closed system. The expiration time for Red Blood Cells in an open system is 24 hours.
[QCMLS 2021: 1.1.4] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p57]

32. **b** Cryoprecipitated AHF is stored at ≤–18°C.
[QCMLS 2021: 1.1.3] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p62]

33. **c** Plasma Frozen Within 24 Hours After Phlebotomy and thawed has an expiration of 24 hours.
[QCMLS 2021: 1.1.4] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p63]

34. **d** Apheresis Platelets are stored at 20-24°C.
[QCMLS 2021: 1.1.3] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p60]

35. **c** Cryoprecipitated AHF has a shelf life of 12 months if stored at ≤–18°C in the frozen state.
[QCMLS 2021: 1.1.4] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p62]

36. **b** Thawed Plasma is stored at 1-6°C.
[QCMLS 2021: 1.1.3] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p64]

37. **c** 2,3-DPG declines during storage of Red Blood Cells, causing a "shift-to-the-left" in the oxygen dissociation curve and an impaired ability to deliver oxygen to the tissues.
[QCMLS 2021: 1.1.2.2] [Harmening 2019, p8-9]

38. **d** Clots in the unit may indicate bacterial contamination. Other visible signs of possible contamination could include particulate matter, discoloration of the product, and bubbles/ frothiness.
[QCMLS 2021: 1.5.1.1, 1.6.3.7] [AABB Tech Manual 2020, p522]

39. **b** Thawed Plasma has an expiration date of 5 days from the date the product is thawed. This product was already thawed for 24 hours, leaving 4 days before expiration.
[QCMLS 2021: 1.1.4] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p65]

40. **b** Since the original weld for this unit was incomplete, this product was prepared in an open system. The expiration date will be 24 hours.
[QCMLS 2021: 1.1.4] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p58]

D. Blood Components

41. **c** Each Apheresis Platelets, Leukocytes Reduced will increase the platelet count by approximately 30,000-50,000/µL in a 70-kg adult. Lesser increases suggest refractoriness that may be immune in origin including the presence of HLA-antibodies or clinical refractoriness, such as DIC, bleeding, and sepsis.
[QCMLS 2021: 1.1.4] [AABB Tech Manual 2020, p565]

42. **b** The method used to produce Red Blood Cells Leukocytes Reduced must assure <5.0 × 10^6 residual leukocytes per unit.
[QCMLS 2021: 1.1.5] [AABB BB/TS Standards 2022, p28-29]

43. **b** When red blood cell products are irradiated, the expiration date will be the original date or 28 days from the date of irradiation, whichever is sooner.
[QCMLS 2021: 1.1.4.1] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p58]

44. **a** Cryoprecipitated AHF must be transfused within 4 hours after thawing and pooling in an open system. If thawed and then pooled using a sterile connection, the expiration time is 6 hours.
[QCMLS 2021: 1.1.4] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p62]

45. **a** Per AABB Standards, thawed Fresh Frozen Plasma should be stored at 1-6°C for no more than 24 hours.
[QCMLS 2021: 1.1.4] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p63]

46. **a** Cryoprecipitated AHF is used primarily for fibrinogen replacement. It is stored at room temperature (20-24°C) after thawing and must be infused within 6 hours as single units or if pooled using a sterile connecting device. If pooled in an open system, it must be infused within 4 hours.
[QCMLS 2021: 1.1.4] [Harmening 2019, p362; AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p62]

47. **c** Blood products from blood relatives containing viable lymphocytes must be irradiated to inhibit the proliferation of T cells and prevent transfusion-associated GVHD.
[QCMLS 2021: 1.6.3.2] [Harmening 2019, p385-387]

48. **b** Irradiation inhibits proliferation of donor T lymphocytes and renders them immunoincompetent and unable to cause transfusion-associated GVHD.
[QCMLS 2021: 1.6.3.2] [Harmening 2019, p385-387]

49. **c** FFP thawed in a water bath should be protected with a plastic overwrap so that entry ports are not contaminated with water. FDA-cleared dry-thawing devices may also be used to thaw FFP and do not require a protective overwrap.
[QCMLS 2021: 1.1.4] [AABB Tech Manual 2020, p156]

50. **b** Fresh Frozen Plasma (FFP) must be separated and frozen within 8 hours of Whole Blood collection.
[QCMLS 2021: 1.1.4] [AABB BB/TS Standards 2022, Reference Standard 5.1.8A, p63]

51. **a** Whole blood-derived Platelets are prepared by a light spin to separate the Red Blood Cells from the platelet-rich plasma (PRP), followed by a heavy spin of the PRP to concentrate the platelets.
[QCMLS 2021: 1.1.2.2] [Harmening 2019, p337-339]

52. **d** Leukoreduction of blood products reduces donor leukocytes to less than 5 × 10^6 leukocytes per unit. and decreases the risk of HLA alloimmunization.
[QCMLS 2021: 1.1.5, 1.6.3.5] [Harmening 2019, p339-340]

E. Blood Component Quality Control

53. **a** Apheresis Granulocytes must yield a minimum of 1.0 × 10^{10} granulocytes in at least 75% of units tested.
[QCMLS 2021: 1.1.5] [AABB BB/TS Standards 2022, p34]

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54. **b** The pH of all platelet products should be maintained at 6.2 or above throughout the storage period.
[QCMLS 2021: 1.1.5] [AABB BB/TS Standards 2022, p31-33]

55. **c** Per AABB Standards, Apheresis Platelets shall demonstrate with 95% confidence that >75% of units contain ≥3.0 × 10^{11} platelets.
[QCMLS 2021: 1.1.5] [AABB BB/TS Standards 2022, p32-33]

56. **a** Per AABB Standards, at least 90% of the Platelets prepared from Whole Blood that are sampled must contain ≥5.5 × 10^{10} platelets.
[QCMLS 2021: 1.1.5] [AABB BB/TS Standards 2022, p32]

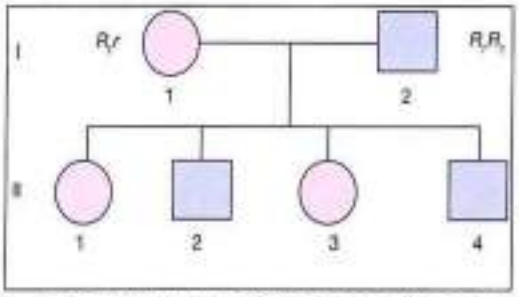
57. **b** Per AABB Standards, Apheresis Platelets shall demonstrate with 95% confidence that >95% of units have a pH ≥6.2 at the time of issue or within 12 hours after expiration.
[QCMLS 2021: 1.1.5] [AABB BB/TS Standards 2022, p32-33]

58. **c** Methods for preparing Red Blood Cells Leukocytes Reduced must retain at least 85% of the original red cells.
[QCMLS 2021: 1.1.5] [AABB BB/TS Standards 2022, p28]

59. **c** AABB Standards require that units be tested for Factor VIII levels and demonstrate 80 IU or higher. If the average value is <80 IU of Factor VIII, corrective action must be taken.
[QCMLS 2021: 1.1.5] [AABB BB/TS Standards 2022, p31]

II. Blood Group Systems

A. Genetics



60. **b** The mother has a 50% chance of passing on R_1 and 50% chance of passing on *r*. The father will always pass on R_1. Statistically, 50% of the children will be R_1r and 50% will be R_1R_1.
[QCMLS 2021: 1.2.1.1] [AABB Tech Manual 2020, p266-270]

61. **d** The HLA genes located on each copy of chromosome 6 are inherited together and called a haplotype.
[QCMLS 2021: 8.1.1.3] [Harmening 2019, p500-501]

62. **d** Most blood group genes are autosomal; the XG blood group is found on the X chromosome. Whenever a blood group gene is inherited, the antigen (the trait) is expressed on the red blood cells, and is known as codominant inheritance.
[QCMLS 2021: 1.2.1.1] [AABB Tech Manual 2020, p266-270]

63. **c** The XG blood group system is unique in that the gene encoding Xg^a is located on the X chromosome. An Xg(a–) mother would not have Xg^a to pass on. An Xg(a+) father would, however, transmit Xg^a to all his daughters.
[QCMLS 2021: 1.2.1.1] [Harmening 2019, p214-215]

64. **c** Use the Hardy-Weinberg equation: $p^2 + 2pq + q^2 = 1.0$. In this example, the gene frequency of *Jk*^{a} is represented by p and the gene frequency of *Jk*^{b} is represented by q and is expressed in the formula $p + q = 1.0$. p^2 is the homozygous population of *Jk*^{a}, or those who are Jk(a+b–). The square root of $p^2 = p$, which is the *gene frequency* of *Jk*^{a} in this population. Out of 400 people, 122, or 30% are homozygous. The square root of 0.30 = is 0.55.
[QCMLS 2021: 1.2.1.1] [AABB Tech Manual 2020, p276-278]

65. **b** The Hardy-Weinberg equation states p + q= 1.0. When the equation is expanded, it is $p^2 + 2pq + q^2 = 1.0$. p^2 equals those that are homozygous for D, 2pq equals those that are heterozygous for D and q^2 represents those that are D-negative (*rr*). Homozygous population that is D-negative equals 0.16 (*rr*).The square root of 0.16 is 0.4 (q) and p can be determined by subtracting q from 1 where p equals 0.6 (0.4+0.6=1.0). Use this in the expanded equation to determine the D-positive population that is heterozygous for D, or 2pq or 2(0.6)(0.4) = 0.48 or 48%.
[QCMLS 2021: 1.2.1.2] [AABB Tech Manual 2020, p276-278]

66. **d** This is an example of autosomal codominant inheritance. The female will pass either *Jk*^{a} or *Jk*^{b}. The male is *Jk*^{b} *Jk*^{b} or *Jk Jk*^{b}. The resulting offspring cannot be negative for both Jk^a and Jk^b antigens.
[QCMLS 2021: 1.2.1.1] [AABB Tech Manual 2020, p266-270]

67. **c** Homozygous is the term used when identical alleles at a given locus are inherited. The Fy(a–b+) phenotype may be the result of *Fy*^{b} *Fy*^{b} genotype. M+N+ and Jk(a+b+) would result from inheriting non-identical alleles at a given locus, which is termed heterozygous. LE antigens are not the result of codominant alleles, but rather the interactions of 2 genes, *Le* and *Se*, at different loci.
[QCMLS 2021: 1.2.1.1] [AABB Tech Manual 2020, p266-268, 314]

68. **d** These are all examples of blood group phenotypes that arise from the inheritance of 2 autosomal recessive genes. The O_h phenotype results from the inheritance of 2 "h" genes (nonfunctional *FUT1* and nonfunctional *FUT2* genes). The group O phenotype results from the inheritance of 2 recessive *O* genes and the Lu(a–b–) phenotype from the inheritance of 2 recessive *LU* genes.
[QCMLS 2021:1.2.1.1] [Harmening 2019, p28, 134]

B. Chemistry, Antigens

69. **b** Anti-P1 is usually IgM and clinically insignificant. Anti-P and anti-P^k are clinically significant antibodies found in individuals of the rare p phenotype. Anti-p is not an antibody.
[QCMLS 2021: 1.2.5] [AABB Tech Manual 2020, p318-322]

70. **d** The patient lacks E. Since C and c are antithetical antigens, C is inherited from one parent and c from the other. Since the person is homozygous for e, one of the genes needs to code for ce (*RHce*) and the other Ce (*RHCe*). The *RHD* gene is more likely inherited with *Ce* than *ce*, so the person's most probable genotype is *DCe/dce*.
[QCMLS 2021: 1.2.6.2] [Harmening 2019, p155-157]

71. **c** The blood donor lacks the *H* gene so there is no H precursor substance. In the absence of H antigen, no A or B antigens are expressed. These red cells appear to type as group O as they do not react with anti-A or anti-B.
[QCMLS 2021: 1.2.2.6] [AABB Tech Manual 2020, p310-312]

72. **a** This individual cannot have the *k* antigen on their cells and therefore, *KK* is the most probable genotype. K_oK_o is rare and no KEL system antigens are detected on the red blood cells. Those individuals usually produce antibodies that are reactive with all normal cells.
[QCMLS 2021: 1.2.8.1] [Harmening 2019, p193]

73. **c** The Fy3 antigen is present on cells that express the Fy^a and/or Fy^b antigens. Fy3 antigen is not found on Fy(a–b–) red cells. The Fy3 antigen is resistant to enzyme treatment.
[QCMLS 2021: 1.2.9.1] [AABB Tech Manual 2020, p369]

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74. **a** The baby is D-negative and also c-negative as there is no evidence of HDFN. The gene responsible for a lack of both the D and c antigens is designated as *r' (Ce or RHCe)*. The baby must have inherited this gene from both parents and is homozygous *r'r'*.
[QCMLS 2021: 1.2.6.2] [Harmening 2019, p155-157]

75. **a** The most common genotype in Rh negative individuals is *rr (ce/ce)*. Anti-e would not be formed because the recipient's red cells express the e antigen but lack the c antigen. The first antibody most likely to develop would be anti-c.
[QCMLS 2021: 1.2.6.2] [Harmening 2019, p155-157]

76. **d** All common Rh antigens are present on the red blood cells. Based on gene frequencies, the most probable genotype is R_1R_2 (*DCe/DcE*).
[QCMLS 2021: 1.2.6.2] [Harmening 2019, p155-157]

77. **a** R_0R_0 is the only correct choice. R_0 = D+C–E–c+e+. All other choices are positive for C and/or E antigen.
[QCMLS 2021: 1.2.6.2] [Harmening 2019, p155-157]

78. **c** The LE antigens are developed by gene interaction. Both the *LE* and *SE* gene are required for red cells to type as Le(a–b+). If a person has a *LE* gene, but not a *SE* gene, then the cells type as Le(a+b–). The Le(a–b–) phenotype is derived when the *LE* gene is absent and the *SE* gene may or may not be present. The Le(a+b–) phenotype occurs in 22% of the population, and Le(a–b–) occurs in 6%, so the most likely phenotype of a nonsecretor (se/se) is Le(a+b–).
[QCMLS 2021: 1.2.3] [Harmening 2019, p178-181]

79. **a** Anti-f will react with cells that carry c and e on the same Rh polypeptide. No other listed genotype produces an Rh polypeptide that carries both c and e.
[QCMLS 2021: 1.2.6.5] [Harmening 2019, p165]

80. **a** Nonreactivity with anti-f indicates the cells do not have an Rh polypeptide that possesses both c and e, which is necessary to type as f+. R_1R_2 is the most likely genotype.
[QCMLS 2021: 1.2.6.5] [Harmening 2019, p165]

81. **c** The baby appears to lack c since no HDFN is evident. The mom is most likely R_1R_1, so had to pass R_1 onto the baby. The father must have passed on an Rh gene that also did not produce c. Given the choices, the father has to be R_1r.
[QCMLS 2021: 1.2.6.2] [Harmening 2019, p155-157]

82. **c** The Fy(a–b–) phenotype occurs in 68% of the population of African descent, but is extremely rare in the other ethnic backgrounds. Lu(a–b–), Jk(a–b–) and K–k– are very rare in all ethnic backgrounds.
[QCMLS 2021: 1.2.9.1] [Harmening 2019, p198]

83. **b** The most likely genotype is R_1r *(DCe/dce)*.
[QCMLS 2021: 1.2.6.2] [Harmening 2019, p155-157]

84. **d** From the first 2 children it can be determined the mom has the haplotypes A2B12 and A23B18. The dad has the haplotypes A1B8 and A3B35. The expected B antigen in child #3 is B35.
[QCMLS 2021: 8.1.1.3] [Harmening 2019, p500-501]

85. **b** The immunodominant sugar for the A antigen is N-acetylgalactosamine. The immunodominant sugar for the H antigen is fucose and the immunodominant sugar for the B antigen is galactose.
[QCMLS 2021: 1.2.2.2] [Harmening 2019, p124-125]

86. **a** Vel is a high-prevalence antigen, occurring in >98% of the random population.
[QCMLS 2021: 1.2.13] [Harmening 2019, p220-221]

87. **d** H antigen is the precursor for the A and B antigens. In group O individuals, H antigen remains unmodified as the O gene is an amorph and produces no detectable transferase.
[QCMLS 2021: 1.2.2.2] [Harmening 2019, p124-125]

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68. **c** A_x cells are more strongly reactive with anti-A,B than with anti-A, and the plasma frequently has anti-A1 present.
[QCMLS 2021: 1.2.2.4] [Harmening 2019, 131-132]

89. **c** Mixed-field reactivity with anti-A and anti-A,B is a typical finding for A_3 subgroups.
[QCMLS 2021: 1.2.2.4] [Harmening 2019, p131-132]

90. **c** The genetic product of the *H* gene is alpha-L-fucosyltransferase. This enzyme confers H antigen specificity by adding fucose to a terminal galactose residue.
[QCMLS 2021: 1.2.2.2] [Harmening 2019, p124-125]

91. **d** Bombay phenotypes (O_h) lack H antigen on their red cells and produce naturally occurring anti-H in their serum. The anti-H is clinically significant.
[QCMLS 2021: 1.2.2.6] [Harmening 2019, p134-135]

92. **a** Most examples of anti-Lu^a agglutinate saline suspended cells. Most examples of other LU antibodies are IgG and react at 37°C by IAT.
[QCMLS 2021: 1.2.11.1] [AABB Tech Manual 2020, p364; Harmening 2019, p202]

93. **b** Anti-Ch and anti-Rg react at IAT with trace amounts of C4 (a component of complement) found as a soluble substance in plasma and capable of attaching to the surface of red blood cells. Neutralization studies with pooled plasma containing CH/RG substance can help confirm the antibody reactivity in a patient's sample. If test procedures are performed to coat cells with C4, a patient with anti-Ch or anti-Rg may agglutinate the cells directly.
[QCMLS 2021: 1.3.7] [AABB Tech Manual 2020, p377]

94. **d** Anti-Jk^a and anti-Jk^b are often weakly reactive and often drop to undetectable levels in the serum. Failure to detect these antibodies in pretransfusion testing may lead to a delayed transfusion reaction.
[QCMLS 2021: 1.2.10.2] [AABB Tech Manual 2020, p371]

95. **b** HLA antibodies are formed in response to pregnancy, transfusion or transplantation and are therefore not naturally occurring. They are associated with febrile nonhemolytic transfusion reactions, refractoriness to platelet transfusions, and TRALI. They are directed against antigens found on platelets and most other nucleated cells in the body.
[QCMLS 2021: 8.3.2] [AABB Tech Manual 2020, p491-492]

96. **d** MHC consists of both class I and class II HLA antigens. Discrimination of self from non-self, immune response to antigenic stimuli, and the coordination of cellular and humoral immunity are the primary functions of the HLA system.
[QCMLS 2021: 8.1.1.3-8.1.1.4] [AABB Tech Manual 2020, p479-484]

97. **d** HPA-1a is a platelet specific antigen, and maternal antibody to this antigen is the most common cause of fetal and neonatal alloimmune thrombocytopenia (FNAIT).
[QCMLS 2021: 1.2.15.2] [AABB Tech Manual 2020, p666-667]

98. **c** Due to the inheritance of the *Le (FUT3)* and *Se (FUT2)* genes, Le^a and Le^b antigens will be present in the saliva. Due to the inheritance of the *Se*, *A*, and *H* genes, A and H antigens will also be present.
[QCMLS 2021: 1.2.3.2] [Harmening 2019, p178-181]

99. **d** Class I region contains HLA-A, HLA-B and HLA-C. Class II region contains HLA-DR, -DP and -DQ.
[QCMLS 2021: 1.2.15.1 and 8.1.1.3] [Harmening 2019, p500-501]

100. **c** The G antigen is present on red cells possessing either C or D. Anti-G reacts with panel cells that are D positive and/or C-positive and appears to be anti-C and anti-D. The mother's antibody was likely stimulated by the G antigen expressed on the child's C-positive red blood cells. After 10 months, passively-acquired anti-D from the administration of RhIG would no longer be present.
[Tech Manual 2020, p343-344; Harmening 2019, p165]

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101. **b** Individuals who are partial D are missing epitopes of the D antigen and can develop alloantibodies toward the epitopes they lack. Since all normal D antigens have all epitopes, the specificity of the person's antibody is anti-D. This antibody will not react with the red cells of other genetically identical partial D individuals.
[QCMLS 2021: 1.2.6.4] [Harmening 2019, p160]

102. **c** The U antigen is a high incidence antigen found on the RBCs of all individuals except 1% of African-Americans, who lack glycophorin B and type S–s–U–.
[QCMLS 2021: 1.2.7.1] [Harmening 2019, p191-192]

103. **a** The M and N antigens are found on glycophorin A. The S and s antigens are found on glycophorin B.
[QCMLS 2021: 1.2.7.1] [Harmening 2019, p189]

104. **c** Autoanti-P, a cold-reactive IgG autoantibody described as a biphasic hemolysin, is associated with paroxysmal cold hemoglobinuria.
[QCMLS 2021: 1.4.2.4.1] [Harmening 2019, p184]

105. **b** Patients with infectious mononucleosis often demonstrate potent examples of anti-i that are transient in nature.
[QCMLS 2021: 1.4.2.3] [Harmening 2019, p186]

106. **c** Anti-I is an autoantibody commonly found in all individuals, but when it causes hemolysis, the titer may be high and have a broad thermal range with reactivity at 30°C or higher. Cold agglutinin disease can be associated with *M. pneumoniae* infection and in lymphoproliferative diseases.
[QCMLS 2021: 1.4.2.3.1] [Harmening 2019, p447-451]

107. **c** Anti-I is associated with cold agglutinin disease.
[QCMLS 2021: 1.4.2.3.1] [Harmening 2019, p447-451]

108. **b** Anti-i is an IgM antibody that reacts with cord cells and i adult cells. It is not associated with hemolytic disease of the newborn since IgM antibodies do not cross the placenta.
[QCMLS 2021: 1.2.4.1] [Harmening 2019, p447-451]

109. **c** Red blood cells of individuals with the McLeod phenotype lack K_x and K_m and have significant depression of other KEL antigens. The McLeod phenotype has been found in patients with chronic granulomatous disease (CGD).
[QCMLS 2021: 1.2.8.1] [Harmening 2019, p196-197]

110. **a** Antibodies in the JK blood group system are IgG and react best at the antiglobulin phase. These antibodies are associated with delayed hemolytic transfusion reactions and reactivity can be enhanced by testing with enzyme-pretreated cells.
[QCMLS 2021: 1.2.10.1] [Harmening 2019, p200]

111. **c** The Fy^a and Fy^b antigens are sensitive to denaturation by proteolytic enzymes. Serum containing anti-Fy^a reacts with untreated Fy(a+) cells, but not with enzyme-treated Fy(a+) cells. The Jk^a, E and k antigens are not destroyed by proteolytic enzymes.
[QCMLS 2021: 1.2.9.1] [Harmening 2019, p198]

112. **c** Anti-Fy^a is an IgG antibody that reacts best at the AHG phase, does not react with enzyme-treated red cells, is capable of causing hemolytic disease of the newborn, and is not known to be an autoagglutinin.
[QCMLS 2021: 1.2.9.2] [Harmening 2019, p198]

113. **d** The FY glycoprotein on red cells is a receptor for the malarial parasite *Plasmodium vivax*. Red cells with the phenotype Fy(a–b–) are resistant to invasion by *P. vivax*.
[QCMLS 2021: 1.2.9.1] [Harmening 2019, p196]

114. **a** Linkage disequilibrium results when genes occur more frequently in a haplotype than is expected based on chance alone. This is commonly seen in the HLA system and the MNS blood group system.
[QCMLS 2021: 8.1.1.3] [Harmening 2019, p187, 501]

115. **d** After performing rule outs, the most likely antibody is anti-c. To form anti-c, the patient would need to inherit a gene from both parents that does not produce the c antigen. Individuals with the R_1R_1 or *r'r'* genotype would lack the c antigen but the *r'r'* genotype is very rare.
[QCMLS 2021: 1.2.6.2] [Harmening 2019, p156]

116. **c** Blood groups with carbohydrate antigens include ABO, H, LE, I, P1PK and GLOB. Antibodies to these antigens are usually naturally occurring, IgM, direct agglutinins and react best at room temperature.
[QCMLS 2021: 1.2.1] [Harmening 2019, p62]

117. **d** Lu3 is a high-prevalence antigen found on all red cells except those of the rare Lu(a–b–) phenotype.
[QCMLS 2021: 1.2.11] [Harmening 2019, p204]

118. **c** Wr^a is a low-prevalence antigen. Low-prevalence antigens are those that occur in <10% of the population. Yt^a and Di^b are both high-prevalence antigens, occurring in >98% of the population. Jk^b is neither a high-prevalence nor a low-prevalence antigen, occurring in approximately 74% of the white population and 48% of the black population.
[QCMLS 2021: 1.2.13, 1.2.10]

119. **b** Antibodies to high-prevalence antigens are usually immune in origin, clinically significant, IgG, bind to their respective antigens at 37°C and are reactive at the antiglobulin phase of testing. High-prevalence antigens occur in >98% of the population, but antibodies to high-prevalence antigens are uncommon. Anti-K is a commonly encountered alloantibody and the K antigen is not a high-prevalence antigen, as it is present on the red cells of approximately 9% of the population.
[QCMLS 2021: 1.2.13]

120. **b** Platelets express HLA Class I antigens. Antibodies to these antigens may be the cause of platelet refractoriness. Both bone marrow and solid organ transplantation require donor and recipient matching for select HLA antigens. A hemolytic transfusion reaction is the result of recipient antibodies directed against donor red cell antigens and is unrelated to HLA antigens.
[QCMLS 2021: 1.2.15] [AABB Tech Manual 2020, p491, 493-495]

121. **c** Fetal and neonatal alloimmune thrombocytopenia (FNAIT) results from platelet destruction due to anti-HPA antibodies. The most implicated specificity is anti-HPA-1a. IVIG is a treatment for FNAIT and could be paired with maternal washed platelets transfused to the affected fetus or neonate. The father can be assumed to be positive for the HPA-1a or other HPA antigen responsible for FNAIT and would not be an appropriate source for transfused platelets to the affected fetus or neonate.
[QCMLS 2021: 1.2.15.2] [AABB Tech Manual 2020, p464]

122. **b** Rh_{null} red cells are also LW(a–b–). D-positive cells express more LW antigen than D-negative cells. LW antibodies usually react at the indirect antiglobulin phase of testing. LW^b is the low-prevalence antigen and LW^a and LW^{ab} are high-prevalence antigens.
[QCMLS 2021: 1.2.12] [AABB Tech Manual 2020, p275-277]

123. **d** TACO (transfusion-associated circulatory overload) is a complication of transfusion resulting from volume overload and is not immunologic in origin. AIN can be seen in infants or adults. In infants, the autoantibody specificity is usually anti-HNA-1a. TRALI is a transfusion adverse event that can be initiated by HLA and HNA antibodies that are present in the donor's blood. NAN is caused by maternal IgG HNA antibodies.
[QCMLS 2021: 1.2.15.3] [AABB Tech Manual 2020, p425-428 and 582-585]

C. Role of Blood Groups in Transfusion

124. **d** The K antigen is absent in 91% of the White population and 98% of the Black population.
[QCMLS 2021: 1.2.8.1] [Harmening 2019, p194-195]

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125. **d** The frequency of compatible donors for this patient can be calculated by multiplying the percentage of the population that is e– × C– × Fy(a–) × Jk(b–). In the random White population, 2% are e–, 32% are C–, 32% are Fy(a–) and 26% are Jk(b–). Multiplication shows that finding compatible units will be rare (.05 units per 100 units) and random screening for compatible units would be a challenge. The blood supplier's immunohematology reference laboratory (IRL) may have units in stock or can request blood from other IRLs through the rare donor registry.
[QCMLS 2021: 1.2.14.2] [Harmening 2019, p248]

126. **d** Determination of compatibility can be determined by multiplying the percentage of compatibility of each antigen. 45% of the population is group O, 15% are D-negative, and 91% are K-negative. 0.45 × 0.15 × 0.91 = 0.06, or 6%.
[QCMLS 2021: 1.2.14.2] [Harmening 2019, p248]

127. **d** The overall incidence of the e antigen is 98%. The overall incidence of c is 80%, D is 85% and E is 30%.
[QCMLS 2021: 1.2.6.1] [Harmening 2019, p158]

128. **b** The K antigen is highly immunogenic. It is present on the red cells of up to 9% of adults and neonates and is not affected by enzymes.
[QCMLS 2021: 1.2.8.2] [Harmening 2019, p194-195]

129. **d** 75% of donors would be compatible with anti-X and 90% with anti-Y. The frequency of compatibility for both antigens is determined by multiplying the 2 compatibility percentages: 0.75 × 0.90 = 0.675 or 68%.
[QCMLS 2021:1.2.14.2] [Harmening 2019, p248]

130. **c** Multiplication of the individual compatibility frequencies results in the percentage of compatible donors that would lack both antigens. 0.20 × 0.90 = 0.18, or 18%.
[QCMLS 2021: 1.2.14.2] [Harmening 2019, 248]

III. Blood Group Immunology

A. Immune Response

131. **a** B lymphocytes (B cells) mature in the bone marrow and evolve into plasma cells that secrete antibody when stimulated by antigen. Plasma cells and resulting memory B cells are an integral part of humoral immunity.
[QCMLS 2021: 1.3.1] [Harmening 2019, p50-51]

132. **b** During a primary response, IgM antibodies are the first to be produced, followed by the production of IgG antibodies.
[QCMLS 2021: 1.3.1] [Harmening 2019, p52]

133. **c** Alloimmunization resulting in the production of blood group antibodies is associated with red cell transfusions. Exposure to foreign red blood cell antigens from pregnancy or transplantation may also result in alloantibody production.
[QCMLS 2021: 1.3.1] [Harmening 2019, p62-63]

134. **a** Macrophages and monocytes have receptors for the Fc portion of IgG molecules. This is one of the ways that IgG sensitized red cells are removed from the circulatory system and phagocytized.
[QCMLS 2021: 1.4.25.1.2] [Harmening 2019, p59; AABB Tech Manual 2020, p246-247]

135. **d** The most immunogenic blood group antigen is D (Rh_0). If D-negative individuals are exposed to even small amounts of D-positive red blood cells, anti-D will be produced in a significant number of individuals.
[QCMLS 2021: 1.2.6] [Harmening 2019, p 62, 158]

136. **c** Removal of IgG and/or C3b-sensitized red blood cells by the RES system can be noted as extravascular hemolysis and a delayed transfusion reaction.
[QCMLS 2021: 1.4.2.1.2] [Harmening 2019, p61; AABB Tech Manual 2020, p247]

B. Immunoglobulins

137. **a** Only IgG antibodies can cross the placenta.
[QCMLS 2021: 1.3.2] [Harmening 2019, p55, 58]

138. **c** IgM is a large molecule with 5 Fc monomers in close proximity, facilitating the activation of complement via the classical pathway.
[QCMLS 2021: 1.3.2] [Harmening 2019, p57, 59-61]

139. **b** Each IgM molecule has 10 Fab sites, or antigen binding sites. This allows this molecule to attach to multiple antigen sites on adjacent red blood cells, facilitating direct hemagglutination.
[QCMLS 2021: 1.3.2] [Harmening 2019, p55, 57, 67]

140. **a** Binding of complement C1 protein to the Fc fragment of an immunoglobulin molecule initiates the activation of complement via the classical pathway.
[QCMLS 2021: 8.1.4.1] [Harmening 2019, p59]

141. **b** Ig classes are differentiated based on their amino acid sequences of the constant region of the heavy chains.
[QCMLS 2021: 1.3.2] [Harmening 2019, p55-56; AABB Tech Manual 2020, p243-246]

142. **a** The IgG1 and IgG3 subclasses of IgG are the most efficient at activating complement via the classical pathway.
[QCMLS 2021: 1.3.2] [Harmening 2019, 58-59]

143. **b** This red cell antibody identification panel depicts anti-M. While usually IgM, this antibody may also be IgG and the reactions observed suggest that the antibody may have an IgG component. Treating the serum with sulfhydryl reagents such as dithiothreitol (DTT) will cleave IgM molecules, rendering them nonreactive. DTT has no effect on IgG molecules. As only IgG antibodies are capable of placental transfer, it is important to determine if this antibody is IgM, IgG, or a mixture of both.
[QCMLS 2021: 1.4.1.2.1] [Harmening 2019, p57, 432]

144. **d** B cells, when stimulated, produce antibodies to multiple epitopes on an antigen. The different epitopes on a single antigen induce proliferation of different B cells clones that result in a heterogeneous population of antibodies. These antibodies are referred to as polyclonal.
[QCMLS 2021: 8.1.2.5] [Harmening 2019, p61-62]

145. **c** Various host factors play a role in an individual's immune response and one of those factors is age. A decrease in antibody levels may be seen in older individuals and may result in a false-negative reaction as seen in reverse ABO typing.
[QCMLS 2021: 1.2.2.5] [Harmening 2019, p64]

C. Antigen-Antibody Interactions

146. **d** Excess antibody, known as prozone, may lead to false-negative agglutination results.
[QCMLS 2021: 1.3.4] [Harmening 2019, p66]

147. **d** Many enhancement media promote hemagglutination by reducing the zeta potential. Red blood cells naturally repel each other due to the net negative charge that surrounds each cell. A potential is created because of the cations (positively charged ions) that are attracted to the zone of negative charges. This potential is the zeta potential. Reducing the zeta potential allows the more positively charged antibodies to get closer to the negatively charged red cells, facilitating agglutination.
[QCMLS 2021: 1.3.3] [Harmening 2019, p67-68]

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148. **a** Several factors influence hemagglutination reactions including antigen-antibody ratio, pH, temperature, enhancement media and the use of anti-human globulin reagents.
[QCMLS 2021: 1.3.4] [Harmening 2019, p66-67]

149. **c** The Fy^a, Fy^b, M, and N antigens are destroyed when red cells are treated with proteolytic enzymes. The S antigen may be destroyed or weakened when red cells are treated with proteolytic enzymes.
[QCMLS 2021: 1.5.3] [Harmening 2019, p70]

150. **b** Dosage effect reflects different amounts of antigen expressed on the red cell surface. For example, an individual that inherits 2 identical *MM* genes will express more M antigen than an individual of the genotype *MN*. Some examples of anti-M will react stronger with red cells that are MM vs red cells that are MN.
[QCMLS 2021: 1.5.1.9] [Harmening 2019, p66]

151. **a** Anti-human globulin serum will facilitate hemagglutination of red blood cells sensitized with IgG antibody.
[QCMLS 2021: 1.3.5] [Harmening 2019, p70, 104]

152. **c** All negative antiglobulin tests must be verified with the addition of IgG-sensitized RBCs (Check Cells). Failure to demonstrate hemagglutination following the addition of IgG-sensitized RBCs invalidates the test and suggests the possibility of inadequate washing to remove unbound globulin, failure to add AHG reagent, or faulty/nonreactive AHG.
[QCMLS 2021: 1.3.5] [Harmening 2019, p114]

153. **b** Hemolytic disease of the fetus and newborn will demonstrate a positive direct antiglobulin test.
[QCMLS 2021: 1.4.1.2.2] [Harmening 2019, p109-111]

154. **d** Following incubation at 37°C, antibody detection (screening) tests must include an indirect antiglobulin phase to facilitate detection of clinically significant antibodies.
[QCMLS 2021: 1.3.5] [Harmening 2019, p112-114; AABB BB/TS Standards, 2022, p40]

155. **d** Polyspecific AHG reagents must contain anti-IgG and anti-C3d. Monospecific AHG reagents contain only one antibody specificity and can include anti-IgG, anti-C3d and anti-C3b.
[QCMLS 2021: 1.3.5] [Harmening 2019, p104-108]

156. **a** Neutralized AHG reagent, no longer able to bind to IgG immunoglobulin, can result in a false-negative test result.
[Harmening 2019, p114]

157. **c** Overcentrifugation of the test can result in a false-positive result.
[QCMLS 2021: 1.3.5] [Harmening 2019, p114]

158. **a** Negative reactions using solid-phase technology will appear as a button at the center of the microplate well following centrifugation.
[QCMLS 2021: 1.3.6] [Harmening 2019, p272-276]

159. **c** A 4+ reaction using gel technology will appear as a solid band of agglutinated cells at the top of the gel column.
[QCMLS 2021: 1.3.6] [Harmening 2019, p269-276]

160. **a** One of the advantages of gel technology containing anti-IgG, as used in antibody detection testing, is that saline washing is not required.
[QCMLS 2021: 1.3.6] [Harmening 2019, p271]

161. **d** Overspecificity can be a disadvantage of monoclonal reagents as all antibodies produced by the clone recognize only a single epitope. Blending these reagents with other monoclonal reagents or polyclonal reagents to recognize multiple epitopes has been useful in overcoming this disadvantage.
[QCMLS 2021: 1.3.5] [Harmening 2019, p106]

162. **b** Polyspecific anti-human globulin reagents used to evaluate suspected cases of autoimmune hemolytic anemia must contain anti-C3d and anti-IgG. Once a positive direct antiglobulin is noted using a polyspecific reagent, testing with monospecific anti-C3d and monospecific anti-IgG would be done.
[QCMLS 2021: 1.4.2] [Harmening 2019, p109-110]

163. **d** Low ionic strength saline (LISS) facilitates hemagglutination by decreasing ionic strength and reducing the zeta potential, allowing for antibodies to react with their respective red cell antigens. LISS may also increase antibody uptake and reduce incubation times.
[QCMLS 2021: 1.3.4] [Harmening 2019, p69-70]

D. Complement

164. **d** Once activated, the membrane attack complex (MAC) results in cell lysis and includes C5b through C9.
[QCMLS 2021: 1.4.2.1.1] [Harmening 2019, p60-61]

165. **b** ABO antibodies are very efficient at activating complement via the classical pathway. ABO antibodies can be potent hemolysins and cause intravascular destruction of incompatible red cells and acute hemolytic transfusion reactions that can be fatal.
[QCMLS 2021: 1.2.2.1] [Harmening 2019, p59, 119, 378-379]

166. **c** Some blood group IgG antibodies are only capable of activating the complement cascade to where C3b is bound to the red cell membrane. Cells of the RES system have receptors for C3b and sensitized cells may be cleared from the circulation.
[QCMLS 2021: 1.4.2.1.2] [Harmening 2019, p59-61]

167. **b** Anti-PP1P^k is a clinically significant alloantibody found in individuals with the rare p phenotype. This IgM or IgM and IgG alloantibody is very efficient at complement activation and often demonstrates hemolysis following incubation at 37°C.
[QCMLS 2021: 1.3.3] [Harmening 2019, p184]

IV. Physiology and Pathophysiology

A. Physiology of Blood

168. **d** Both stem and progenitor cells are unrecognizable morphologically. Stem cells make up only 0.5% of HPCs, while progenitor cells are close to 3%. Stem cells can undergo self-renewal while progenitor cells cannot.
[QCMLS 2021: 4.1.1]

169. **a** EPO and GM-CSF are factors that regulate differentiation and proliferation of erythrocytes. Different ILs are involved in regulating production and maturation of HPCs and leukocytes, including granulocytes, eosinophils, basophils, lymphocytes, and monocytes and megakaryocyte development. Flt ligands help regulate HPCs, B and T lymphocytes and stem cells.
[QCMLS 2021: 4.1.1]

170. **b** The bone marrow is the hematopoietic site for erythrocytes and all leukocytes beginning in the 6th month of gestation and continuing through the adult years. The liver functions as the site of hematopoiesis during the 3rd month of gestation and the spleen, during 3-6 months of fetal life. The thymus is involved in the development of T-cells.
[QCMLS 2021: 4.1.2]

171. **d** Erythrocytes and leukocytes have similar maturation characteristics. These cells develop condensed, clumped chromatin; the nucleoli disappear; and the cell size decreases with maturation. The erythrocyte is the only one to lose its nucleus. The leukocyte nucleus is very visible in mature cells.
[QCMLS 2021: 4.1.3, 4.1.4]

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172. **b** In an extravascular transfusion reaction, red cell destruction and removal takes place in the spleen where incompatible donor red cells sensitized with recipient antibodies undergo shearing and become spherocytes. The thymus is the site of T lymphocyte development. The other choices are all functions of the spleen.
[QCMLS 2021: 4.2]

173. **c** Integral proteins are located from the membrane's outer side and through the entire RBC structure to the cytoplasmic side of the membrane. Integral proteins are responsible for a variety of functions including active and passive transport of electrolytes. They also are the location for RBC antigens found on glycophorin and band 3. Peripheral proteins such as actin, spectrin, and ankyrin are located on the cytoplasmic side of the membrane and manage red cell deformability.
[QCMLS 2021: 4.3.1] [Harmening 2019, p4-5]

174. **d** 2,3-DPG is produced in the Luebering-Rapaport pathway and accumulates in the red cell. The concentration of 2,3-DPG affects the hemoglobin-O_2 dissociation curve and facilitates release of O_2 to the tissues. The majority of ATP is generated by the breakdown of glucose during glycolysis. 2,3-DPG is not involved with the hexose monophosphate shunt or any other RBC metabolic pathways.
[QCMLS 2021: 4.3.3] [Harmening 2019, p6-7]

175. **a** Eosinophils are involved in hypersensitivity reactions as mediators and can bind IgE in allergic reactions. These cells become activated, causing their granules to release enzymes, which trigger allergic symptoms. Neutrophils phagocytize bacteria; lymphocytes are involved in humoral and cellular immunity. NK cells are lymphocytes that are involved in cytotoxicity.
[QCMLS 2021: 4.5.1]

B. Hemostasis and Coagulation

176. **c** Cryoprecipitated AHF is used primarily for fibrinogen replacement. Fibrinogen level is decreased in patients with DIC, due to uncontrolled thrombin generation.
[QCMLS 2021: 1.1.4] [Harmening 2019, p358, 362]

177. **a** Recombinant Factor IX concentrates are used to treat hemophilia B. The other treatment modalities listed may be used to treat hemophilia A.
[QCMLS 2021: 5.6.1.4]

178. **b** Fresh Frozen Plasma (FFP) that has been thawed expires in 24 hours. However, expired FFP can be converted to Thawed Plasma, providing a shelf-life of an additional 4 days. Factors V and VIII decline in Thawed Plasma while other factor levels are acceptable to support hemostasis.
[QCMLS 2021: 1.1.4] [AABB Tech Manual 2020, p518-519]

179. **c** These are symptoms of a low platelet count. If the mother's platelet count is normal, the newborn likely has fetal and neonatal alloimmune thrombocytopenia (FNAIT), caused by maternal antibody to the infant's platelet antigens.
[QCMLS 2021: 1.2.15.2] [AABB Tech Manual 2020, p464-465]

180. **b** When platelets are needed in a case of FNAIT, maternal platelets can be transfused. These platelets will lack the HPA antigen to which the anti-HPA antibody is directed. Paternal platelets, random donor platelets and paternal grandfather platelets will most likely express the HPA antigen to which the causative anti-HPA antibody is directed. Maternal platelets used to treat FNAIT are washed prior to transfusion to remove maternal antibody.
[QCMLS 2021: 1.2.15.2] [AABB Tech Manual 2020, p464-465, 566-567]

181. **d** FNAIT is due to maternal IgG HPA alloantibodies crossing the placenta and reacting with fetal platelet HPA antigens, leading to thrombocytopenia.
[QCMLS 2021: 1.2.15.2] [AABB Tech Manual 2020, p464-465, 566-567]

182. **b** Intravenous IVIG is the current treatment of choice for posttransfusion purpura, a rare but profound complication of transfusion. Antibodies to platelet antigens (often anti-HPA-1a) destroy autologous platelets, resulting in profound thrombocytopenia.
[QCMLS 2021: 5.5.1.2.2] [Harmening 2019, p387]

183. **d** Indications for platelet transfusions include those that are prophylactic following bone marrow or stem cell transplant, actively bleeding thrombocytopenic patients, and those with qualitative platelet dysfunctions. Platelet transfusions are contraindicated in patients with TTP, because the treatment of choice is therapeutic plasma exchange.
[QCMLS 2021: 1.6.4.1] [AABB Tech Manual 2020, p567]

C. Hemolytic Disease of the Fetus and Newborn

184. **b** The questionable test result is the newborn's Rh (D) typing. The newborn has a 4+ DAT caused by maternal anti-D. This maternal antibody, bound to fetal D antigen sites, can prevent anti-D typing reagents access to fetal D antigen sites, resulting in a false-negative result, also known as "blocking" phenomenon.
[QCMLS 2021: 1.2.6.4] [AABB Tech Manual 2020, p348]

185. **d** HDFN is caused by maternal antibody crossing the placenta and destroying fetal antigen-positive red cells. Unlike ABO antibodies, which are naturally-occurring and can affect the first pregnancy, Rh antibodies are not produced until the mother has been exposed to Rh-positive red cells. This can be caused by small amounts of fetomaternal hemorrhage (FMH) which can occur throughout a normal pregnancy and may increase in frequency during the 2nd and 3rd trimesters. Larger amounts of fetal cells can enter maternal circulation during procedures such as cord blood sampling and at delivery of the first Rh-positive child. Once immunized, subsequent pregnancies with Rh-positive infants are affected, usually with increasing severity.
[QCMLS 2021: 1.4.1.1] [Harmening 2019, p429]

186. **c** HDFN is caused by maternal antibodies against antigens on fetal red cells inherited from the father. Since the father is homozygous for c, the baby's red cells must be c+, and could react with maternal anti-c if present. The father lacks the A, D, and C antigens and therefore cannot pass these antigens to the child. The A antigen must have come from the mother so anti-A would not be the cause of the positive DAT.
[QCMLS 2021: 1.4.1] [Harmening 2019, p432-433]

187. **d** ABO HDFN is a mild disease, not usually requiring transfusion. It may occur in any pregnancy in which there is ABO incompatibility. High-titer IgG antibodies are more frequently seen in group O mothers than in A or B mothers.
[QCMLS 2021: 1.4.1.1] [Harmening 2019, p428]

188. **d** HDFN is caused by maternal IgG antibodies. Outside the Rh system, the most clinically significant antibody for HDFN is IgG anti-K. Anti-Le^a, anti-P1 and anti-M are usually IgM, and IgM antibodies do not cross the placenta.
[QCMLS 2021: 1.4.1] [Harmening 2019, p429]

189. **a** ABO HDFN is a mild disease that may occur in any ABO-incompatible pregnancy, including the first, since the antibodies are naturally occurring. Rh HDFN does not occur until the mother has become immunized. Once this happens, subsequent pregnancies may be quite severely affected. The DAT is typically weak or even negative in ABO HDFN, and strongly positive in Rh HDFN.
[QCMLS 2021: 1.4.1.1] [Harmening 2019, p428-429]

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190. **c** The mother has anti-D; the baby has a positive DAT; yet the baby appears to be Rh-negative. Rarely, an infant with a strongly positive DAT may have RBCs so heavily sensitized with maternal antibody that the D antigen sites are blocked and cannot react with anti-D reagent, causing a false-negative Rh type. Since the infant is group O, ABO hemolytic disease of the fetus and newborn (HDFN) does not fit this example. If the fetus had received enough D-negative intrauterine transfusions to cause the red cells to type as D-negative, they would not demonstrate a 4+ positive DAT, as shown in this example. There is no indication of a fetomaternal hemorrhage.
[QCMLS 2021: 1.2.6.4] [Harmening 2019, p434]

191. **a** ABO HDFN occurs most commonly in group A (or group B) babies born to group O mothers and usually has a mild course. The DAT is typically weak or negative and jaundice develops 12-48 hours after birth. The mother and baby in this clinical case are both Rh-positive.
[QCMLS 2021: 1.4.1.1] [Harmening 2019, p428, 431]

192. **d** Spherocytosis is characteristic of ABO HDFN but not of Rh HDFN.
[QCMLS 2021: 4.7.1.3.2] [Harmening 2019, p428]

193. **c** Phototherapy lights break down the toxic, unconjugated bilirubin to a nontoxic, water-soluble isomer that can be excreted by the baby.
[QCMLS 2021: 1.4.2.2] [AABB Tech Manual 2020, p662; Harmening 2019, p435]

194. **b** Antibody titers do not themselves predict the severity of HDFN or determine the treatment needed. Instead, titers above a critical level identify candidates requiring additional monitoring to determine necessary intervention.
[QCMLS 2021: 1.4.1.2.1] [AABB Tech Manual 2020, p661; Harmening 2019, p432]

195. **a** Mother and newborn are the same blood type (group A) so group A units may be selected. In addition, the unit for an exchange transfusion must lack the antigen to maternal antibodies that have entered the infant's circulation and are reactive at 37°C or AHG.
[QCMLS 2021: 1.4.2.2] [AABB Tech Manual 2020, p677-678]

196. **b** For intrauterine transfusion, group O red blood cells are selected. The unit must lack the antigen to the antibody the mother has produced. The unit is irradiated to prevent TA-GVHD, is CMV safe (leukoreduced or CMV-negative) and HbS-negative to prevent sickling when O_2 levels are low.
[QCMLS 2021: 1.4.2.2] [Harmening 2019, p434]

197. **a** Blood selected for exchange transfusion is usually crossmatched with the mother's blood and should be ABO-compatible. It should also be negative for the antigen that she has produced antibody against. Unless the HDFN is caused by anti-D, the baby's Rh type is selected. In this case, group O, Rh-positive, E-negative, is the best choice for the exchange transfusion. The R_1r phenotype is D+, C+, E–, c+ and e+. The R_2r phenotype is D+, C–, E+, c+ and e+ and is unacceptable for exchange transfusion. The r''r is D–, C–, E+, c+ and e+ and is also unacceptable. The rr unit (D–, C–, E–, c+, e+) should be conserved for transfusion to Rh-negative recipients.
[QCMLS 2021: 1.4.2.2] [Harmening 2019, p434]

198. **a** Blood selected for exchange transfusion should be ABO-compatible with the mother and baby, and antigen negative for maternal alloantibody(ies). Prenatal antibody titers above 16 or 32 are considered significant, and the condition of the fetus should be monitored. In this case, the fetus ABO type is unknown so group O blood would be selected.
[QCMLS 2021: 1.4.2.2] [Harmening 2019, p434]

199. **a** Blood selected for exchange transfusion should be antigen negative and ABO-compatible with the mother and baby. Red Blood Cells are usually less than 7 days old, CMV-negative (or CMV-safe, including leukocyte-reduced products), hemoglobin S-negative, and irradiated.
[QCMLS 2021: 1.4.2.2] [AABB Tech Manual 2020, p661]

200. **c** For exchange transfusion, antigen negative Red Blood Cells may be suspended in ABO-compatible thawed Fresh Frozen Plasma. This is referred to as Reconstituted Whole Blood.
[QCMLS 2021: 1.4.2.2] [Harmening 2019, p434]

201. **b** Blood selected for intrauterine transfusion and transfusion to premature infants should be irradiated to prevent transfusion-associated graft-versus-host disease as these transfusion recipients are immunocompromised.
[QCMLS 2021: 1.4.2.2] [AABB Tech Manual 2020, p661, 694]

202. **a** If the initial antibody screen using either the mother's or baby's serum is positive, the preferred specimen for compatibility testing is the maternal serum as this is the source of the clinically significant antibody. Antigen negative or AHG-crossmatch-compatible units are selected for transfusion.
[QCMLS 2021: 1.4.2.2] [AABB Tech Manual 2020, 677-678; AABB BB/TS Standards 2022, p44; Harmening 2019, p264]

203. **d** The presence of D-positive fetal RBCs in the maternal circulation due to significant fetomaternal hemorrhage can cause the weak D test to show mixed-field agglutination in the post-partum sample. The suspected fetomaternal hemorrhage would be corroborated with a rosette test and subsequent quantitation of fetomaternal hemorrhage. Prenatal D and weak D typing results should also be verified.
[QCMLS 2021: 1.4.1.4] [Harmening 2019, p435-436; AABB Tech Manual 2020, p343, 663]

204. **a** Antepartum RhIG usually has an anti-D titer ≤4. By contrast, anti-D titers due to maternal alloimmunization are generally >16.
[QCMLS 2021: 1.4.1.3.2] [Harmening 2019, p435]

205. **c** When untreated indicator cells are used in the rosette test, a positive result occurs when there are >6 rosettes/5 fields. In this case, the results indicate that the mother had a larger than normal (≥10 mL) fetomaternal hemorrhage (FMH). Additional testing needs to be performed to calculate the FMH size.
[QCMLS 2021: 1.4.1.4] [AABB Tech Manual 2020, p664-665, Method, 5-2; Harmening 2019, p435-436]

206. **b** One dose of RhIG will protect the mother from a bleed of 30 mL of whole blood or 15 mL of packed red cells. In this case, divide the volume of the fetomaternal hemorrhage (35 mL of whole blood) by 30; round down to 1, then add 1 extra vial = 2 vials of RhIG.
[QCMLS 2021: 1.4.1.4] [AABB Tech Manual 2020, p664-665, Method 5-2; Harmening 2019, p436]

207. **d** One vial of Rh immune globulin protects against a fetomaternal hemorrhage of 30 mL of whole blood or 15 mL of packed red cells. Divide the volume of fetomaternal hemorrhage (48 mL of whole blood) by 30; round up to 2, then add 1 extra vial = 3 vials of RhIG.
[QCMLS 2021: 1.4.1.4] [AABB Tech Manual 2020, p664-665, Method 5-2; Harmening 2019, p436]

208. **c** RhIG should be given to nonimmunized D-negative females who are pregnant or have delivered a D positive infant. In addition, females who have a miscarriage, abortion, terminated ectopic pregnancy and undergo cordocentesis or an intrauterine fetal transfusion may also be eligible for RhIG administration.
[QCMLS 2021: 1.4.1.3.2] [Harmening 2019, p435-436]

209. **b** Approximately 10% of the antepartum dose of RhIG may still be present at delivery so the antibody screen may detect weak anti-D. This should not be interpreted as active immunization. The titer of anti-D is usually ≤4.
[QCMLS 2021: 1.4.1.3.2] [Harmening 2019, p435]

210. **c** RhIG is of no benefit once a person has been actively immunized. A saline-AHG Rh titer ≥1:16 indicates formation of anti-D during the pregnancy.
[QCMLS 2021: 1.4.1.3.2] [AABB Tech Manual 2020, p663; Harmening 2019, p432]

211. **b** The formula to calculate the FMH volume (percentage) assumes the mother's blood volume as 5000 mL. 0.003 × 5000 mL=15 mL. Another calculation method is to take the percentage of fetal cells (0.3%) and multiply by 50. In other words, 0.3 × 50 = 15 mL.
[QCMLS 2021: 1.4.1.4] [AABB Tech Manual 2020, p664-665, Method 5-2]

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212. **c** Use the formula: (fetal cells counted / cells counted) × (maternal blood volume). Assume the mother's blood volume is 5000 mL. In this example, (30 fetal cells / 2000 cells counted) × 5000 mL = 75 mL. RhIG protects against 30 mL. 2.5 vials are needed, and this is rounded up to 3 since the number following the decimal is ≥5. To these 3 vials, 1 additional vial will be added, giving 4 vials of RhIG for administration.
[QCMLS 2021: 1.4.1.4] [AABB Tech Manual 2020, p664-665, Method 5-2]

213. **a** The rosette test is a sensitive method that can measure a FMH ≥10 mL.
[QCMLS 2021: 1.4.1.4] [AABB Tech Manual 2020, p664-665, Method 5-1]

D. Anemias

214. **d** Transfusion should generally be avoided except in cases of life-threatening anemia. Hemoglobin trigger values are usually <7 g/dL for most patients. For some patients with certain cardiac conditions, they are higher (≤8 g/dL). A hemoglobin of 10.8 g/dL (108 g/L) is not life-threatening, especially if the patient is not actively bleeding.
[QCMLS 2021: 1.6.1] [AABB Tech Manual 2020, p553-556; Harmening 2019, p356]

215. **c** A pathologic cold autoantibody will react at a wider thermal range (≥30°C) than a benign cold agglutinin (<20 24°C). The titer of a pathologic cold autoantibody is 1000 or greater whereas a benign cold autoantibody will have a titer of 64 or less. Pathologic and benign cold autoantibodies are IgM and often have anti-I specificity.
[QCMLS 2021: 1.4.2.3.1] [Harmening 2019, p443-444, 449-450]

216. **d** In 67% of WAIHA cases, the DAT is positive due to both IgG and C3. Approximately 20% of cases demonstrate a DAT with only IgG and 13%, only C3.
[QCMLS 2021: 1.4.2] [AABB Tech Manual 2020, p435; Harmening 2019, 453-454]

217. **c** The autoantibody in WAIHA is usually of broad specificity and can mask underlying alloantibodies. That is why autologous or allogeneic adsorption may be necessary to evaluate the potential for underlying alloantibodies. The red blood cells selected for transfusion will be crossmatch incompatible due to the autoantibody. It is important to ensure that these red cells lack antigens to any alloantibodies identified in the sample.
[QCMLS 2021: 1.4.2.2.1] [AABB Tech Manual 2020, p439-441]

218. **a** The antibody detected in drug-independent immune hemolytic anemia is indistinguishable from warm autoantibodies. The DAT is positive with IgG, the eluate and serum react with all cells tested. Drugs that are known to cause this are methyldopa and fludarabine.
[QCMLS 2021: 1.4.2.6.3] [AABB Tech Manual 2020, p447; Harmening 2019, 466-467]

219. **a** The causative antibody in PCH is unique in that it is a cold-reactive, complement-binding, IgG antibody. The most common specificity is anti-P.
[QCMLS 2021: 1.4.2.4.1] [Harmening 2019, p451-452]

220. **b** Intravascular destruction of red blood cells in the circulation occurs when IgM (and sometimes IgG) antibodies activate complement via the classical pathway resulting in the formation of the membrane attack complex (MAC) and lysis of the cells. This is what is seen when ABO incompatible red cells are transfused. Extravascular destruction occurs when IgG and/or complement-sensitized red cells are removed in the reticuloendothelial system.
[QCMLS 2021: 1.4.2.1.1] [AABB Tech Manual 2020, p247-248]

221. **c** A delayed transfusion reaction associated with anti-Jk^a is likely to result in extravascular red cell destruction. This occurs when IgG and/or complement-sensitized red cells are destroyed by macrophages in the reticuloendothelial system (RES). ABO incompatibility as would be noted if a group O recipient receives a transfusion of Group A blood results in intravascular hemolysis. Group O recipients can receive group A Fresh Frozen Plasma. A D-negative recipient that has not been sensitized to the D antigen can receive D-positive blood. The recipient may respond and produce anti-D and this would impact future transfusion decisions.
[QCMLS 2021: 1.4.2.1.2] [AABB Tech Manual 2020, p247-248] [Harmening 2019, p200]

E. Transplantation

222. **a** Autologous HPC transplants are used for the rescue of bone marrow function following high-dose chemotherapy and patients with Hodgkin lymphoma are often ideal candidates. The other clinical conditions are genetic in nature so allogeneic HPC transplants (and not autologous) would be appropriate.
[QCMLS 2021: 1.4.3.2] [AABB Tech Manual 2020, p739-740; Harmening 2019, p418-419]

223. **b** Bone marrow, umbilical cord blood, and mobilized peripheral blood are the most common sources of HPCs. Successful allogenic transplants require 8/10 or 10/10 HLA loci matches. Apheresis HPC collections have fewer adverse effects than bone marrow collection. HPC products are generally administered by central line infusion and as quickly as the patient can tolerate.
[QCMLS 2021: 1.4.3.2.2] [AABB Tech Manual 2020, p741-743, 754-755; Harmening 2019, p419-420]

224. **d** Bone marrow transplant patients are at risk for transfusion-associated graft-versus-host disease (TA-GVHD) and therefore should receive irradiated blood products. Pathogen reduction of blood components can also prevent TA-GVHD.
[QCMLS 2021: 1.2.2.3, 1.4.3.2.4] [AABB Tech Manual 2020, p773; Harmening 2019, p421]

225. **b** Human leukocyte antigens (HLA) are found on most nucleated cells with some exceptions such as neurons. The antigens and antibodies of this system play a significant role in several different transfusion related reactions and in transplantation rejections.
[QCMLS 2021: 1.2.15, 1.4.3, 1.6.3.2, 1.6.3.3] [AABB Tech Manual 2020, p493-495, 741-742; Harmening 2019, p423, 508-510]

226. **c** HLA-DR, HLA-DQ and HLA-DP are Class II antigens, which are only found on monocytes, macrophages, B lymphocytes, dendritic cells, early hematopoietic cells and intestinal epithelial cells.
[QCMLS 2021: 1.2.15.1, 6.1.1.4] [AABB Tech Manual 2020, p479-484; Harmening 2019, p498]

227. **c** CD34+ enumeration is the most widely used and accepted, albeit surrogate, QC test to evaluate the likelihood of HPC engraftment.
[QCMLS 2021: 1.4.3.2.3] [AABB Tech Manual 2020, p752-753; Harmening 2019, p420]

228. **a** Blood components selected to support the recipient through engraftment should be irradiated or pathogen reduced to prevent transfusion-associated graft-versus-host disease. These products should also be leukocyte-reduced to prevent HLA alloimmunization and CMV infection.
[QCMLS 2021: 1.4.3.2.4] [AABB Tech Manual 2020, p772-773]

229. **c** The group O recipient would have ABO antibodies directed to the group B donor ABO antigens and this would constitute a major incompatibility. The other scenarios are all considered minor incompatibilities where the donor ABO antibodies are directed to the recipient ABO antigens.
[QCMLS 2021: 1.4.3.2.5] [Harmening 2019, p421-422]

230. **a** If recipient ABO antibodies are still circulating, as would be the case at day-1 posttransplant, red cells selected for transfusion must be compatible with those antibodies and therefore identical to the recipient when there is a major ABO incompatible HPC transplant. While recipient ABO antibodies continue to circulate, red cells identical to the donor would not be acceptable. Once recipient ABO antibodies are no longer detectable, red cells identical to the donor are acceptable for transfusion. Platelet and plasma transfusions must be ABO identical to the donor. Group AB red cells would be incompatible with the HPC transplant recipient's circulating ABO antibodies at day-1 post-transplant.
[QCMLS 2021: 1.4.3.2.5] [Harmening 2019, p421-422]

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V. Serologic and Molecular Testing

A. Routine Tests

231. c 20% of persons of African ethnicity are Group B compared to only 11% of those of European ethnicity. For the Rh phenotype results, 44% of blacks compared to 4% of whites and 3% of Asians will demonstrate the Dce phenotype.
[QCMLS 2021: 1.2.2.2, 1.2.6.2] [AABB Tech Manual 2020, p299, 333]

232. d Results of ABO and Rh testing on a current specimen must always be compared to that of a previous transfusion record. Errors in typing or patient identification may be detected when discrepancies are found. Collection of a new sample allows determination of which sample was incorrectly collected.
[QCMLS 2021: 1.5.1.4] [AABB Tech Manual 2020, p504]

233. c A serologic test to confirm the ABO on all RBC units and Rh on units labeled as Rh-negative must be performed prior to transfusion. Any errors in labeling must be reported to the collection facility. The component may not be used until the typing discrepancy is resolved.
[QCMLS 2021: 1.5.1.1] [AABB Tech Manual 2020, p523]

234. c ABO antibodies are not present at birth and if detected are of maternal origin. ABO antibodies develop in response to the environment from 3-6 months of age and continue to increase in titer until displaying appropriate agglutination strength at 12 months. ABO antibodies continue to increase over time, achieving adult levels at 5-10 years of age.
[QCMLS 2021: 1.2.2.1] [AABB Tech Manual 2020, p299-300]

235. a 20% of the population is A1 negative. To find one unit that is A1 negative, 5 units would have to be screened (20/100 = 1/x, 20x = 100, x = 100/20, x = 5 units).
[QCMLS 2021: 1.2.2.4] [AABB Tech Manual 2020, p303]

236. d A red cell suspension with 2 cell populations may lead to a mixed-field agglutination pattern. Mixed-field may occur from transfusion, stem cell transplant or rare genetic chimera. A chimera is a person with a dual population of cells derived from more than one zygote.
[QCMLS 2021: 1.4.3.2.5] [AABB Tech Manual 2020, p307]

237. b A positive DAT will interfere with weak D testing causing both the patient and control to demonstrate positive results. Any positive result in the control tube invalidates any results.
[QCMLS 2021: 1.4.2] [AABB Tech Manual 2020, p348-349]

238. c Patients with multiple myeloma may demonstrate rouleaux formation, which can cause the appearance of agglutination. If the cells are washed to remove residual plasma, and tests repeated, an accurate red cell typing is obtained. By performing a saline replacement with the reverse typing, true agglutination will remain when the cell buttons of the reverse cells are resuspended in saline.
[QCMLS 2021: 1.5.3] [AABB Tech Manual 2020, p309-310, 414]

239. a In a group A patient, Acquired B should be suspected when the patient initially types as group AB but with a weaker expression of the B antigen in the ABO red cell testing. To resolve Acquired B one needs to repeat the typing with a second source of antibody or acidified human anti-B.
[QCMLS 2021: 1.2.2.5] [AABB Tech Manual 2020, p305]

240. c The Rh-negative, or D-negative, phenotypes are most often due to the complete deletion of the *RHD* gene. Rh-negative individuals express *RHCE* genes only.
[QCMLS 2021: 1.2.6.1] [AABB Tech Manual 2020, p340-341]

241. d Unexpected reactivity with reverse cells should include a test with antibody detection (screening) cells at immediate spin to determine if alloantibodies are present. Resolution of the ABO discrepancy can be performed with group B cells that lack the corresponding antigen for the identified alloantibody.
[QCMLS 2021: 1.2.2.5] [AABB Tech Manual 2020, p309]

242. **b** Although monoclonal anti-D reacts with most D-positive red blood cells, cells with fewer antigen sites requires testing with an indirect antiglobulin test. This is referred to as weak D testing.
[QCMLS 2021: 1.2.6.4] [AABB Tech Manual 2020, p342]

243. **d** Some subgroups of A will forward ABO type as a group O, but reverse type as a group A. Confirmation of weak A antigen on the red cells can be accomplished by adsorption and elution studies using anti-A. If rouleaux were present, what appears to be agglutination would be observed with the B cells in the reverse ABO typing. O_h red cells would demonstrate anti-B in the serum, in addition to a potent anti-H. There is no serologic evidence of an acquired B antigen being present on the red cells.
[QCMLS 2021: 1.2.2.4] [Harmening 2019, p131-132, 134-135]

244. **c** The mother does not have the *D* gene. The father would have to have inherited one gene that produces D and another gene that does not produce D. The mother and father both passed on genes that do not produce D.
[QCMLS 2021: 1.2.1.1] [AABB Tech Manual 2020, p257-258]

245. **a** Some blood group antibodies, in the presence of their corresponding antigen and complement, activate the complement cascade and demonstrate *in-vitro* hemolysis.
[QCMLS 2021: 1.3.3] [Harmening 2019, p61]

246. **b** Agglutination at AHG phase indicates the presence of clinically significant antibody, indicating the need for antibody identification studies. Since agglutination is observed at the AHG phase of testing, check cells are not indicated. A DAT and preparation of an eluate are also not indicated.
[QCMLS 2021: 1.5.1.4] [Harmening 2019, p112]

247. **b** Presence of agglutination at AHG phase with both screening cells and autocontrol is indicative of warm autoantibody.
[QCMLS 2021: 1.5.1.9] [Harmening 2019, p455]

248. **a** Presence of agglutination at the IAT phase of testing with antibody detection cells and 2 out of 6 donor units indicates alloantibody in the patient's serum to antigen(s) on the antibody detection cells and donor cells. The presence of an autoantibody would most likely react with all cells tested.
[QCMLS 2021: 1.5.1.9] [Harmening 2019, p233]

249. **b** Reaction with anti-IgG in the DAT and reactivity with both screening cells and autocontrol at the AHG phase of testing is indicative of a warm autoantibody. The patient's hemoglobin is low, and the reticulocyte counts is elevated.
[QCMLS 2021: 1.4.2.2.1] [Harmening 2019, p111, 441-443]

250. **c** When results with anti-A, anti-B and anti-D are all positive, there is no way to determine the accuracy of the results. Repeat testing with an inert control, verified as negative, is the only way to confirm the accuracy of these results.
[QCMLS 2021: 1.5.5.3] [Harmening 2019, p162]

251. **d** The major crossmatch tests the recipient's plasma or serum with donor's red cells. This would detect any antibody in the recipient that would react with antigens on the donor's RBCs. If a patient had been mistyped as a group O rather than group A, group O cells would be selected for crossmatching and no incompatibility would be detected.
[QCMLS 2021: 1.5.1.6] [AABB Tech Manual 2020, p508-509]

252. **d** Crossmatching is a test between the patient's plasma or serum and donor's red cells. If the patient antibody detection test is negative for clinically significant antibodies to common antigens, an incompatible crossmatch at the antiglobulin phase of testing is due to either a positive DAT on the donor's red cells or the patient has an antibody to a low-incidence antigen that the donor's red cells possess.
[QCMLS 2021: 1.5.1.6] [AABB Tech Manual 2020, p535]

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253. **b** Reactions at immediate spin and room temperature phases of testing but not at 37°C or AHG phases suggest an antibody that is IgM. JK, Rh, and FY antibodies are usually IgG and react at the AHG phase of testing. Most LE antibodies are IgM and reactive at immediate spin and room temperature phases of testing.
[QCMLS 2021: 1.2.3.1] [AABB Tech Manual 2020, p402, 406]

254. **c** DNA arrays amplify single nucleotide polymorphisms (SNPs) that demonstrate differences in blood group antigen alleles. The array platforms use genomic DNA (gDNA) rather than RNA (mRNA), and following amplification, probes or primers are able to detect specific products that are recognized as a blood group antigen.
[QCMLS 2021:1.5.3.1 and 9.5.5] [AABB Tech Manual 2020, p235]

255. **b** When group-specific units of Red Blood Cells are not available, group-compatible units are selected. Since the patient is AB, group A would be selected to conserve group O units for group O patients. Rh-negative patients should receive Rh-negative units of red blood cells.
[QCMLS 2021: 1.5.1.5] [Harmening 2019, p263-264]

256. **a** This patient has an anti-A1, which eliminates A_1B cells immediately. Rh-negative units should be conserved for Rh-negative patients when Rh-positive units are available. Selection of group B, Rh-positive units provides compatible units quickly.
[QCMLS 2021: 1.5.1.5] [AABB Tech Manual 2020, p306, 507]

257. **d** Agglutination is dependent upon optimal antigen to antibody ratio. Excessive amount of antigen, as can be found in a cell suspension that is too heavy, does not allow maximal uptake of antibody per red cell and therefore agglutination is negatively affected leading to weaker or negative results.
[QCMLS 2021: 1.3.4] [Harmening 2019, p66-67]

258. **a** Determining the patient's phenotype allows focusing identification procedures to antibodies the patient can develop. An alloantibody cannot be produced to a particular antigen expressed on the patient's red cells. Treating the patient's red cells with enzymes will not aid in antibody identification and directed donors are unlikely to be more compatible than randomly selected donors. The patient's serum would be used for crossmatching but should not be used for antigen typing.
[QCMLS 2021: 1.5.1.9] [Harmening 2019, p240]

259. **a** Cold autoantibodies often have a preference for the I, IH or P antigens. Warm autoantibodies most often exhibit Rh specificity although specificities in other blood groups have been rarely reported.
[QCMLS 2021: 1.4.2.2.1] [Harmening 2019, p462-463]

260. **a** Anti- E, anti-k, and anti-s have not been reported to cause *in vitro* hemolysis. LE antibodies may bind complement and fresh serum that contains anti-Le^a may hemolyze Le(a+) red cells *in vitro*. Approximately 22% of the population is Le(a+).
[QCMLS 2021: 1.2.3.1] [Harmening 2019, p180]

261. **a** Antibodies to antigens 2, 3, 4, and 5 can be ruled out with panel cells II and III. There is no reactivity with the patient's serum and panel cells II and III. Cells II and III express antigens 2, 3, 4, and 5. Antibody to antigen 1 cannot be excluded.
[QCMLS 2021: 1.5.1.9] [Harmening 2019, p237-241]

262. **b** Anti-K and anti-P1 can be ruled out on cell 1 since there is no agglutination of cell 1 with the patient's sample. Anti-M and anti-Jk^a can be eliminated on cell 2, which has a double-dose antigen expression of both M and Jk^a.
[QCMLS 2021: 1.5.1.9] [AABB Tech Manual 2020, p397-401; Harmening 2019, p237-241]

263. **d** Antibodies to C, Le^b and Jk^a can be eliminated due to the lack of agglutination with panel cells 1 and 2. Panel cells 1 and 2 possessed the C, Le^b and Jk^a antigens. Only anti-E remains.
[QCMLS 2021: 1.5.1.9] [Harmening 2019, p237-241]

264. **b** Since the direct antiglobulin test (DAT) is negative, warm autoimmune hemolytic anemia (WAIHA) is unlikely (as are cold agglutinin disease [CAD] and paroxysmal cold hemoglobinuria [PCH]) because no reactions at immediate spin or room temperature are noted. The McLeod phenotype demonstrates weakened KEL system antigens and the antigens K_m and K_x are absent. McLeod syndrome is an X-linked disease state associated with acanthocytosis, muscular, neurologic, and psychiatric symptoms. Many also have chronic granulomatous disease and recurrent infections.
[QCMLS 2021: 1.2.8.1] [AABB Tech Manual 2020, p368]

265. **b** Emergent release of blood cannot rely on previous records. The release of blood should not be delayed in an emergency where a massive transfusion protocol has been activated and the physician has signed a statement indicating the clinical situation is sufficiently urgent to release blood prior to completion of all testing. In this case, the release of group O, Rh-negative blood would be the preferred course of action as there is some evidence that the recipient is Rh-negative and of child-bearing potential.
[QCMLS 2021: 1.6.2.2 and 1.6.2.3] [AABB Tech Manual 2020, p525-526]

266. **a** Although Chido antigens are present in >90% of the population, Chido antibodies do not cause red cell destruction and are considered clinically insignificant.
[QCMLS 2021: 1.2.13] [Harmening 2019, p205, 209]

267. **b** Reactivity at 37°C and IAT indicates the presence of an IgG antibody. Anti-M, although usually IgM, may be partly or wholly IgG. Anti-M is ruled out on cell 4. Anti-Le^b is usually IgM and can be ruled out on cells 4 and 8. This leaves anti-E, anti-Jk^a and anti-K.
[QCMLS 2021: 1.5.1.9] [Harmening 2019, p237-241]

268. **a** All 3 antibodies can cause HDFN and delayed transfusion reactions. These are IgG antibodies and react at the antiglobulin phase of testing.
[QCMLS 2021: 1.2.8, 1.2.9, 1.2.10] [Harmening 2019, p193-201]

269. **a** Rh antibodies demonstrate enhanced reactivity with enzyme-pretreated cells. Antibodies in the KEL system do not have enhanced reactivity with enzyme-pretreated cells. Anti-E and anti-D are ruled out on cell 3, and anti-K is ruled out on cell 7.
[QCMLS 2021: 1.2.6.3] [Harmening 2019, p237-241]

270. **b** Anti-Fy^a may cause mild to rarely severe hemolytic disease of the fetus and newborn.
[QCMLS 2021: 1.2.9.2] [Harmening 2019, p198]

271. **a** The direct antiglobulin test (DAT) is used to identify red blood cells that have been sensitized or coated with antibody *in vivo*.
[QCMLS 2021: 1.3.5] [AABB Tech Manual 2020, p430]

272. **b** In cold agglutinin syndrome, anti-I acts as a complement-binding antibody with a high titer and high thermal amplitude. The complement cascade is activated and C3d remains on the red cell membrane of circulating cells.
[QCMLS 2021: 1.4.2.3.1] [Harmening 2019, p185-186]

273. **a** A delayed serologic transfusion reaction occurs within days to weeks following a transfusion. The DAT is positive and usually appears as a mixed-field reaction as only the transfused cells are sensitized with patient alloantibody. Patients with a delayed serologic transfusion reaction are usually asymptomatic, and the serology is often an accidental laboratory finding. Preparation of an eluate may be useful in antibody identification.
[QCMLS 2021: 1.6.3.1.2] [AABB Tech Manual 2020, p648-649]

274. **c** The strength of reactions for some antibodies varies due to dosage. The strongest reactions are obtained on double-dose/homozygous red cells compared to single-dose/heterozygous red cells.
[QCMLS 2021: 1.5.1.9] [AABB Technical Manual 2020, p390]

275. **d** Antiglobulin reagent is used to detect the presence of red cells, coated *in vivo* with IgG and/or C3d. Antiglobulin reagent may be polyspecific (contains an anti-IgG and anti-C3d) or monospecific (anti-IgG or anti-C3d).
[QCMLS 2021: 1.3.5] [Harmening 2019, p110]

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276. **a** Drugs operating through the drug-adsorption mechanism, including penicillin, bind firmly to proteins on the RBC membrane. Antibodies to the drug are then able to react with cell-bound drug, possibly resulting in RBC destruction. Detection of antibodies to penicillin requires treatment of test cells with penicillin and the subsequent testing of the patient's plasma and eluate. Test cells that have not been treated with penicillin do not react.
[QCMLS 2021: 1.4.2.6.1] [Harmening 2019, p463-465]

277. **c** EDTA chelates calcium preventing blood from clotting. This chelation of calcium also will stop the complement cascade. Calcium ions are necessary for C1 to attach to IgG on the red blood cells.
[QCMLS 2021: 7.6.1.2] [AABB Tech Manual 2020, p431, 506]

278. **d** Cephalosporins were originally associated with positive DAT and nonimmunologic protein adsorption (or membrane modification). Other drugs that cause nonimmunologic protein adsorption and a positive DAT include tazobactam, sulbactum, cisplatin, oxaliplatin and diglycoaldehyde.
[QCMLS 2021: 1.4.2.6.4] [AABB Tech Manual 2020, p447; Harmening 2019, p466]

279. **d** Anti-Kp^a is an antibody to a low-incidence antigen. The Kp^a antigen is present in 2% of Caucasian populations and <1% of black populations. Therefore, more than 98% of random donor units should be compatible if transfusion is needed for this patient.
[QCMLS 2021: 1.2.8.1] [AABB Technical Manual 2020, p364-366]

280. **c** Patterns of reactivity with anti-A,B can be helpful for ABO discrepancy investigations and investigation of ABO subgroups. A_x cells react more strongly with anti-A,B (1-2+) than with anti-A, which may be nonreactive or only weakly positive.
[QCMLS 2021: 1.2.2.4] [AABB Tech Manual 2020, p303-304]

281. **a** Anti-Vel has been associated with transfusion reactions and rarely, HDFN. If blood is needed, maternal red cells, washed to remove the anti-Vel, can be a transfusion source for the infant. Paternal and random donor units would likely be Vel-positive since Vel is a high-incidence antigen.
[QCMLS 2021: 1.2.13] [AABB Tech Manual 2020, p381, 661]

282. **b** Patients may have reagent-dependent antibodies to additional components in commercial blood bank antisera. For example, antibodies to acriflavine, the yellow dye used in some commercial anti-B reagents has been responsible for a false-positive result. Washing the patient's cells with saline prior to testing to remove their plasma from the cell suspension will resolve this type of reactivity with anti-B.
[QCMLS 2021: 1.2.2.5] [Harmening 2019, p141-142]

283. **c** Rouleaux will readily disperse in saline, whereas true agglutination will remain after saline replacement.
[QCMLS 2021: 1.2.2.5] [AABB Tech Manual 2020, p414]

284. **a** The Donath-Landsteiner test is diagnostic for paroxysmal cold hemoglobinuria (PCH). The antibody is IgG and is biphasic: hemolysis occurs when the antibody is incubated with cells at cold temperatures and then incubated at 37°C. Often the antibody demonstrates specificity towards the high-incidence antigen P (not to be confused with P1). The antibody detection test is usually negative, and the patient's red cells are sensitized with complement.
[QCMLS 2021: 1.4.2.4] [AABB Tech Manual 2020, p322, 444-445]

285. **b** All requests for blood and blood component transfusions must be legible, accurate and complete. Transfusion requests must have recipient identification that contains 2 independent identifiers such as patient name, date of birth or facility's unique identification number. The request must include the specific type and amount of blood component requested and any special processing requirements such as irradiation or volume-reduction. The request must also include the name of the ordering physician or authorized health professional.
[QCMLS 2021: 1.5.1.2] [AABB Tech Manual 2020, p503]

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286. **c** This order must be rejected and a new request ordered as the original request is both incomplete and illegible. All requests for transfusion must be legible, accurate and complete.
[QCMLS 2021: 1.5.1.2] [AABB Tech Manual 2020, p503]

287. **c** Pretransfusion ABO testing requires anti-A and anti-B antisera for forward ABO typing and A_1 cells and B cells for ABO reverse typing. The use of anti-A,B, A_2 cells or O cells may be useful in resolving ABO discrepancies.
[QCMLS 2021: 1.5.1.4] [AABB BB/TS Standards 2022, p40]

288. **c** When the history of previous transfusion (or pregnancy) is unknown or unavailable, a pretransfusion sample is good for 3 days. The day of collection is day 0. In this particular case, the sample can be used for crossmatching until Friday at 11:59 PM.
[QCMLS 2021: 1.5.1.4] [AABB Tech Manual 2020, p506; AABB BB/TS Standards 2022, p40]

289. **b** If a transfusion recipient has a negative antibody detection test and no history of clinically significant antibodies, the immediate spin (IS) crossmatch is all that is required. The IS serologic crossmatch uses recipient serum and donor red cells and is intended to detect ABO incompatibility between the donor and recipient.
[QCMLS 2021: 1.5.1.6] [AABB Tech Manual 2020, p508; AABB BB/TS Standards 2022, p42]

290. **c** In order to utilize the computer crossmatch to detect ABO incompatibility, the patient must not demonstrate clinically significant alloantibodies nor have a history of such alloantibodies. Regardless of what a manual may state, the computer system must be validated on-site to ensure only ABO compatible products are selected for transfusion. There must also be a method to verify correct data entry and alert the user to ABO discrepancies and incompatibilities.
[QCMLS 2021: 1.5.1.7] [AABB Tech Manual 2020, p508; AABB BB/TS Standards 2022, p43]

291. **b** A transfusion service may accept issued blood components back into inventory for reissue if the appropriate temperature has been maintained. In this case, the unit exceeded the storage temperature of 1-6°C. Other factors for acceptance and reissue include that the unit has not been entered (closure disturbed) and that at least one segment remains attached to the unit. Documentation needs to exist that the unit has been visually inspected and is acceptable for reissue.
[QCMLS 2021: 1.5.1.8.1] [AABB Tech Manual 2020, p529; AABB BB/TS Standards 2022, p48]

292. **c** The pretransfusion blood sample is drawn after verifying the patient's identification and the sample must be labeled at the bedside immediately after phlebotomy. The patient does not need vital signs taken prior to phlebotomy nor does the phlebotomist need to verify the reason for transfusion.
[QCMLS 2021: 1.5.1.3] [AABB BB/TS Standards 2022, p39]

293. **a** The tie-tag or label on each blood component intended for transfusion must indicate the intended recipient's 2 independent identifiers, the donation identification number or pool number, and the interpretation of compatibility testing, if performed.
[QCMLS 2021: 1.5.1.5] [AABB BB/TS Standards 2022, p47]

B. Reagents

294. **c** T-activation, the Bombay phenotype, and a positive IAT result are not associated with a mixed-field pattern of reactivity in an ABO forward typing result. Mixed-field reactivity with anti-A is a characteristic of the A_3 subgroup. A history of transfusion or a possible chimera and presence of 2 distinct cell populations should be considered as these can also present with mixed-field reactivity.
[QCMLS 2021: 1.2.2.4] [AABB Tech Manual 2020, p303-304; Harmening 2019, p131-132]

295. **c** Anti-$PP1P^k$ is the antibody to high incidence antigens made by individuals with the p phenotype. This antibody has an association with early spontaneous abortion and is clinically significant.
[QCMLS 2021: 1.2.5] [Harmening 2019, p184]

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296. **c** These red cells are negative for the C antigen. Therefore, the most likely Rh phenotype is R_2r.
[QCMLS 2021: 1.2.6.2] [AABB Tech Manual 2020, p336-337]

297. **c** The initial result is most likely a false-negative result due to the omission of patient serum. This would explain the initial negative result followed by the subsequent positive result.
[QCMLS 2021: 1.3.5] [AABB Tech Manual 2020, p534]

298. **b** The patient's serum should be tested against a selected antibody identification panel that includes red cells known to lack specific high-incidence antigens. Lack of agglutination of patient serum with red cells that lack a known high-incidence antigen would be the first step to confirm the specificity of the antibody.
[QCMLS 2021: 1.2.13] [AABB Tech Manual 2020, p410]

299. **b** Polyspecific AHG contains anti-IgG and anti-C3d.
[QCMLS 2021: 1.3.5] [AABB Tech Manual 2020, p431]

300. **c** Cold agglutinin disease is associated with cold-reactive antibodies that typically activate complement. Red cells that do not undergo lysis due to complement activation have C3d attached to their surface.
[QCMLS 2021: 1.4.2.3.1] [Harmening 2019, p110, 449-450]

301. **b** The rare M^k gene leads to a deletion of the *GYPA* and *GYPB* genes. These red cells have a normal morphology and phenotype as M–N–S–s–U–; En(a–); Wr(a–b–). M^kM^k is a null phenotype in the MNS system, and is associated with reduced RBC sialic acid and increased glycosylation of Band 4.1 and Band 3.
[QCMLS 2021: 1.2.7.1] [Harmening 2019, p192]

302. **d** Weak antibodies may be missed if there are excess RBC antigens as there may be too few antibodies to bind to red cell antigens, resulting in failure to facilitate hemagglutination.
[QCMLS 2021: 1.3.5] [Harmening 2019, p114]

303. **d** As high as 1 in 400 people of Polynesian ethnicity have a silent gene at the *JK* locus, leading to the Jk(a–b–) phenotype.
[QCMLS 2021: 1.2.10] [Harmening 2019, p201]

304. **c** Anti-C^W is an antibody to a low-incidence antigen in the Rh blood group system. These antigens are present on the red cells of <1% of the population. Not all low incidence antigens are represented on the typical antibody screening cells.
[QCMLS 2021: 1.2.13] [AABB Tech Manual 2020, p330, 413]

305. **b** The positive DAT on the cord blood is highly suggestive that maternal anti-K has sensitized fetal red cells and likely causing HDFN. To confirm the presence of anti-K, an eluate from the cord blood should be prepared. For IgG antibodies such as anti-K, an acid elution or the use of an organic solvent would be the methods of choice. Freeze-thaw elution is most useful in the detection of ABO HDFN.
[QCMLS 2021: 1.5.3] [AABB Tech Manual 2020, p433-434]

306. **c** Due to multiple transfusions within the last two months, direct typing of the current specimen will likely give an inaccurate result for the patient's genetic phenotype. But techniques like microhematocrit density separation can be used to successfully recover patient autologous reticulocytes from any transfused red cells that can then be used to determine a patient's accurate genetic phenotype.
[QCMLS 2021: 1.5.3] [AABB Tech Manual 2020, p415]

C. Application of Special Tests and Reagents

307. **b** When the genes Le and Se are inherited in a group O individual, Le^a, Le^b, and H soluble antigens will be in the secretions. Le^b will be expressed on the red cells.
[QCMLS 2021: 1.2.3.2] [Harmening 2019, p180-181]

308. **c** Presence of agglutination with A_1 cells, group O screening cells and autocontrol at immediate spin (IS) and room temperature (RT) is indicative of a cold autoantibody.
[QCMLS 2021: 1.4.2.3.1] [Harmening 2019, p136-145]

309. **d** Unexpected ABO serum reactions due to a cold reactive autoantibody can be dispersed with testing at 37°C or by performing a cold autoadsorption and using that autoadsorbed serum to repeat the reverse ABO typing and antibody detection testing.
[QCMLS 2021: 1.4.2.3.1] [Harmening 2019, p141]

310. **b** Reactivity of the patient's serum and eluate with all cells tested and the DAT results indicate the presence of warm autoantibody.
[QCMLS 2021: 1.4.2.2.1] [AABB Tech Manual 2020, p435-437]

311. **a** Reaction with anti-A indicates group A. Reaction with anti-A1 (*Dolichos biflorus*) and nonreactivity with anti-H (*Ulex europaeus*) verifies the reaction pattern as A_1.
[QCMLS 2021: 1.2.2.4] [AABB Tech Manual 2020, p303]

312. **a** Anti-P1 can be verified by P1 neutralization with hydatid cyst fluid or P1 substance from pigeon eggs. Enzymes may also be used to enhance the reactions of anti-P1.
[QCMLS 2021: 1.2.5] [Harmening 2019, 183-184]

313. **c** Rh antibodies show enhanced reactivity with enzyme-pretreated cells. The M and Fy^a antigens are cleaved from enzyme-pretreated cells, and therefore there would be no reaction between enzyme-pretreated cells and serum containing anti-M or anti-Fy^a. The incidence of the c antigen is 80% in whites and 96% in blacks. The incidence of the E antigen is 29% in whites and 22% in blacks. Increased reactivity with enzyme-pretreated cells and incompatible results with 8 of 10 donor units is most likely due to anti-c.
[QCMLS 2021: 1.2.6.1, 1.2.6.3] [Harmening 2019, p158, 205]

314. **a** The reactivity of anti-k and anti-Js^b with enzyme-pretreated cells is unchanged and anti-e would show enhanced reactivity with enzyme-treated cells. Chido antigens are sensitive to treatment with most enzymes, and anti-Ch would therefore not react with enzyme-pretreated cells. The Chido antigen is a high-incidence antigen.
[QCMLS 2021: 1.5.3] [Harmening 2019, p216-217, 205]

315. **a** Red cells treated with enzymes like ficin and papain are known to increase reactivity of some blood group antibodies (Rh, P, I, LE and JK).
[QCMLS 2021: 1.2.6.3] [AABB Tech Manual 2020, p416]

316. **b** *RHD* and *RHCE* are closely linked genes on chromosome 1. *RHD* codes for the D antigen. *RHCE* codes for CE antigens in the 4 combinations of ce, CE, cE, and Ce.
[QCMLS 2021: 1.2.6.1] [AABB Tech Manual 2020, p333-337]

317. **a** Sulfhydryl reagents (DTT, AET or 2-ME) may be used to eliminate certain blood group system antigens. These antigens include those in the KEL, DO, YT, LW, and KN systems. The use of DTT would eliminate KEL blood group system antigens from red cells, thus creating K_0 red cells (KEL null cells).
[QCMLS 2021: 1.5.3] [AABB Tech Manual 2020, p405-406, 412]

318. **b** Rh antibodies show enhanced reactivity with enzyme-pretreated cells. Treatment of red cells with enzymes weakens reactivity with antibodies in the MNS and FY systems.
[QCMLS 2021: 1.2.6.3] [Harmening 2019, p161, 205]

319. **a** Enzyme treatment would allow for differentiation of the remaining antibodies after rule out. The Fy^a antigen would be denatured, allowing determination of whether anti-Jk^a and anti-K are present, and to confirm anti-E.
[QCMLS 2021: 1.2.9.1, 1.5.3] [Harmening 2019, p238-243; AABB Tech Manual 2020, p405-406, 412]

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320. **d** For neutralization studies to be valid, the saline dilutional control must be reactive. Since neutralization studies involve adding a substance to the patient's plasma, nonreactivity in test tubes may be due to simple dilution. The saline control acts as the dilutional control and must be reactive. When the saline control is reactive, then if the tube with the substance is nonreactive, the interpretation that neutralization has occurred is made. If it is reactive, neutralization did not occur.
[QCMLS 2021: 1.3.7] [AABB Tech Manual 2020, p417-418]

321. **c** Binding of CH/RG (Chido/Rodgers) antibodies to red cells is readily inhibited by plasma from Ch/Rg-positive individuals.
[QCMLS 2021: 1.3.7] [AABB Tech Manual 2020, p417-418]

322. **b** Anti-Le^b is confirmed because the tubes with LE substance are negative. Nonreactivity of the serum with Le(b+) cells indicates the anti-Le^b in the serum is neutralized by the LE substance. The test is valid since the patient's serum with saline rather than LE substance is still reactive with the Le(b+) cells.
[QCMLS 2021: 1.3.7, 1.5.3] [AABB Tech Manual 2020, p417-418]

323. **d** Reactivity with anti-H is no longer demonstrable, which indicates H substance is present. There is no A or B substance in the saliva as evidenced by the ability of anti-A and anti-B reacting with respective cells. People with H substance and no A or B substance are group O secretors.
[QCMLS 2021: 1.2.3.2, 1.3.7] [AABB Tech Manual 2020, p310-313]

324. **b** Solid phase testing for antibody identification only relies on reactions at the antihuman globulin (AHG) phase of testing. Testing the sample at various temperatures that include immediate spin (IS), room temperature (RT) and 37°C may clarify optimal temperature for reactivity and facilitate antibody identification.
[QCMLS 2021: 1.5.3] [AABB Tech Manual 2020, p420]

325. **a** In solid phase technology, the antibody screening cells are bound to the surface of the well. Antibody specific for antigen on the red blood cells attaches, resulting in a diffuse pattern of red blood cells across the entire well. A negative reaction would have manifested as a pellet of red blood cells in the bottom of the well.
[QCMLS 2021: 1.3.6] [Harmening 2019, p273-274]

326. **d** The *ACKR1* (formerly *DARC*) gene encodes for the FY glycoprotein. *ACKR1* stands for atypical chemokine receptor 1.
[QCMLS 2021: 1.2.9.1] [AABB Tech Manual 2020, p370]

327. **b** Two reagents used for removing IgG from red blood cells are chloroquine diphosphate (CDP) and EDTA glycine acid (EGA). Using either of these procedures is useful to reduce the amount of immunoglobulin on the RBCs of a patient with a positive DAT and potentially facilitating phenotyping of the patient's RBCs with IAT reactive antisera.
[QCMLS 2021: 1.5.3] [AABB Tech Manual 2020, p436; Harmening 2019, p455]

328. **a** Anti-Fy^a would not react with enzyme-pretreated cells and a select cell panel would allow for the possible identification of the remaining 2 antibodies. Thiol reagents would be used to disperse agglutination of IgM antibodies but these antibodies in question are IgG. Lowering the pH and increasing incubation time and the use of albumin with selective adsorption would not be useful in antibody identification.
[QCMLS 2021: 1.2.9.1] [AABB Tech Manual 2020, p405-406, 412]

329. **c** *Dolichos biflorus* plant seed extract forms complexes with N-acetylgalactosamine. When properly diluted, it can distinguish between A_1 donor cells and all other subgroups of A.
[QCMLS 2021: 1.2.2.4] [AABB Tech Manual 2020, p303]

330. **d** Since the auto control is positive after the AHG phase and no reactivity is detected at immediate spin, the serology is most consistent with a warm autoantibody. An adsorption with autologous cells to remove the antibody and subsequent testing of the adsorbed plasma or serum for alloantibody detection is the next step.
[QCMLS 2021: 1.4.2.2.1] [AABB Tech Manual 2020, p435-439; Harmening 2019, p243-245, 455-458]

331. **a** In this case, autoadsorption is acceptable as the patient was transfused approximately 180 days previously. ZZAP is a reagent that will remove IgG from the patient's own cells and allow better adsorption of IgG autoantibody from the patient's plasma onto their cells. The intent of autoadsorption is to remove autoantibody but leave any clinically significant alloantibody(ies) in the autoadsorbed serum. Additional testing with the autoadsorbed serum will identify if alloantibody is present, followed with identification of those alloantibodies.
[QCMLS 2021: 1.4.2.2.1] [AABB Tech Manual 2020, p437]

332. **a** Treating autologous cells with a proteolytic enzyme such as ficin enhances the adsorption of the cold reactive antibody.
[QCMLS 2021: 1.4.2.3.1] [AABB Tech Manual 2020, p443]

333. **d** Elution is the process of removal of antibody from red blood cells. The product of the elution method is an eluate. The eluate contains the removed antibody and can be used in antibody identification studies.
[QCMLS 2021: 1.5.3] [Harmening 2019, p455]

334. **a** Adsorption and elution techniques are used to detect ABO antigens that are not detectable by direct agglutination. The cells are incubated with the antibody (anti-A or anti-B) to the antigen expected on the red blood cells. An elution method is performed and the antibody in the eluate is tested for recovering anti-A (or anti-B).
[QCMLS 2021: 1.2.2.4] [AABB Tech Manual 2020, p309]

335. **b** A two-fold serial dilution can confer the relative strength of antigen expression on red cells. Red cells that express a double-dose of an antigen will usually demonstrate a higher titer with their respective antibody than those cells that express only a single-dose of the antigen. It cannot verify the amount of sialic acid or carbohydrates expressed on red cells. Two-fold serial dilutions may be used to characterize an antibody such as high-titer, low-avidity but does not determine specificity.
[QCMLS 2021: 1.5.3] [AABB Tech Manual 2020, p420]

336. **c** Density separation with centrifugation may be useful in separating younger autologous red cells from older transfused red cells as the autologous cells have a lower density. Once separated, the autologous red cells can be used to determine an accurate phenotype in a transfused individual. Hypotonic wash is used to separate hemoglobin SS cells from hemoglobin AA cells. Thiol reagents are used to differentiate IgM from IgG antibodies and titration is most frequently used to semi-quantify relative amounts of antibody.
[QCMLS 2021: 1.5.3] [AABB Tech Manual 2020, p415]

337. **a** Molecular analysis of the *RHD* gene distinguishes weak D types from partial D. These types of analyses are useful when determining RhIG candidacy. *RHCE* does not encode the D antigen nor does *LW* and *RHAG*.
[QCMLS 2021: 1.5.3.1] [AABB Tech Manual 2020, p287]

338. **d** DNA-based testing can be useful in determining red cell genotype in a patient with a positive DAT as seen in warm autoimmune hemolytic anemia. Knowing the red cell genotype can aid in differentiating alloantibody from autoantibody, particularly if transfusion therapy is needed. RBC genotype information may aid in detecting fetal antigens, which may have stimulated a maternal antibody; however, the identification of maternal antibody specificity does not require molecular testing. DNA-based testing is not utilized in the investigation of transfusion complications such as TRALI or TACO. Evaluation of the effectiveness of therapeutic apheresis is not related to DNA-based testing but rather clinical condition and selected chemistry assays.
[QCMLS 2021: 1.5.3.1] [Harmening 2019, p95-97]

D. Leukocyte/Platelet Testing

339. **c** Platelets are immobilized on the wells of microtiter plates and incubated with patient serum. After washing, red cells coated with antihuman IgG are added, centrifuged and examined visually.
[QCMLS 2021: 1.5.4] [AABB Tech Manual 2020, p468]

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340. **b** One mitigation strategy to help prevent TRALI is to eliminate the transfusion of plasma-containing components that demonstrate both HNA and HLA antibodies. Other mitigation strategies include the use of plasma-containing products from only male donors, never-pregnant females or females tested since their last pregnancy and found to be negative for HLA and HNA antibodies.
[QCMLS 2021: 1.6.3.3] [AABB Tech Manual 2020, p469, 640-643]

341. **b** Methylene blue, Congo red and crystal violet stains are simple stains. For cytotoxicity testing, a fluorescent vital stain that enters damaged cells following an antigen-antibody reaction is utilized.
[QCMLS 2021: 1.5.4] [Harmening 2019, p504-505]

E. Quality Assurance

342. **a** Negative check cells means the results of tubes with the negative reactions are invalid. The reactivity of the check cells should be verified with anti-IgG since anti-E is detected, indicating the anti-IgG is reactive. All tests that are nonreactive with the check cells require repeat test performance.
[QCMLS 2021: 1.3.5, 1.5.5.3] [Harmening 2019, p114]

343. **b** A new centrifuge should be calibrated when received, after any repairs, and periodically. Calibration verifies the optimal time of centrifugation in serologic blood banking. Centrifuge time is determined by the time that is the shortest time required to fulfill the following criteria: (1) the supernatant fluid is clear; (2) the cell button is clearly delineated; (3) the cell button is easily resuspended; (4) agglutination in the positive tubes is as strong as determined in preparing reagents and (5) the negative reactions are clearly determined.
[QCMLS 2021: 1.5.5.4] [AABB Tech Manual 2020, Method 8-7]

344. **a** Samples must be labeled with 2 independent patient identifiers and the date of collection. This information should be identical to that on the patient's identification band and request.
[QCMLS 2021: 1.5.1.3, 1.5.5.1] [AABB Tech Manual 2020, p503-504]

345. **c** AHG control cells are IgG-sensitized cells that react with the anti-IgG in the AHG reagent to demonstrate AHG was added and not neutralized by insufficient washing of the tests prior to its addition.
[QCMLS 2021: 1.3.5] [Harmening 2019, p114]

346. **c** A negative reaction after the addition of check cells indicates AHG serum is not present. Inadequate washing of red cells may leave residual patient serum behind, which can neutralize AHG serum.
[QCMLS 2021: 1.3.5] [Harmening 2019, p114]

347. **a** The pretransfusion blood sample label and the transfusion request must be in agreement for patient identifiers that include name, medical record number and date of birth. If not in agreement, a new blood sample and new transfusion request should be obtained.
[QCMLS 2021: 1.5.1.3 and 1.5.5.1] [AABB BB/TS Standards 2022, p38]

348. **b** Results must be acceptable for critical supplies and if the quality control is invalid, the reagent should be marked as "not for use" until the results are investigated and resolved. Testing with known antigen-positive and known antigen-negative red cells is required each day of use.
[QCMLS 2021: 1.5.5.2] [AABB Tech Manual 2020, p31; AABB BB/TS Standards 2022, p12]

349. **b** Evaluation of the last wash for nonreactivity helps assure that antibody recovered in the eluate is bound to the RBCs and not residual antibody in the saline wash solution.
[QCMLS 2021: 1.5.5.3] [AABB Tech Manual 2020, p419]

350. **a** Due to the criticality of the completeness of the weld in the preparation of aliquots for transfusion, the weld should be checked for its integrity with each use. The sterile connection device has a functional check annually but no other weekly or day-of-use checks.
[QCMLS 2021: 1.5.5.4] [AABB BB/TS Standards 2022, p26]

351. **c** Requalification or recalibration of instrument functionality is required after repair and before the instrument is placed back into service. For serologic centrifuges, this includes assessment of speed and time for agglutination and adequate washing. Blood bank centrifuges do not require any registration of requalification with federal, state or local authorities.
[QCMLS 2021: 1.5.5.4] [AABB BB/TS Standards 2022, p6]

VI. Transfusion Practice

A. Indications for Transfusion

352. **a** Red cell units that lack high-incidence antigens are often frozen and stored as rare donor units. These units can be thawed, washed, and issued as Deglycerolized Red Blood Cells to patients that develop antibodies to high-incidence antigens. Units of Deglycerolized Red Blood Cells can also be used to transfuse patients needing autologous transfusion where units were previously frozen, patients with Paroxysmal Nocturnal Hemoglobinuria (PNH), and for transfusion of IgA-deficient recipients with anti-IgA.
[QCMLS 2021: 1.1.4] [Harmening 2019, p347, 357]

353. **d** Granulocyte transfusions may be indicated for severely neutropenic patients with infections not controlled by antibiotic therapy. They may also be used for severely neutropenic patients on chemotherapy, stem cell transplant patients and those with reversible bone marrow hypoplasia.
[QCMLS 2021: 1.6.1] [Harmening 2019, p361]

354. **b** Each unit of RBCs is expected to increase the hemoglobin level by 1 g/dL (10 g/L).
[QCMLS 2021: 1.1.4] [Harmening 2019, p359]

355. **b** Each unit of RBCs is expected to increase the hematocrit level by 3-5%, so it would take 2 units to raise the level 6%.
[QCMLS 2021: 1.1.4] [Harmening 2019, p359]

356. **c** CMV-seronegative or leukoreduced (CMV-safe) blood products should be administered to immunocompromised patients including fetuses, low-birthweight premature infants born to CMV-seronegative mothers and CMV-seronegative recipients of solid organ and hematopoietic stem cell transplants.
[QCMLS 2021: 1.4.3.2.4] [AABB Tech Manual 2020, p194-195, 694]

357. **a** Patients with IgA deficiency who have had anaphylactic transfusion reactions should receive washed RBCs. Anaphylactic reactions in these patients are typically caused by anti-IgA in the recipient reacting with plasma IgA from the donor. Washing removes plasma IgA from the donor unit.
[QCMLS 2021: 1.6.3.6] [Harmening 2019, p344-345]

358. **b** One unit of Platelets derived from Whole Blood should cause the platelet count to increase by approximately 5,000-10,000/µL in a 70-kg person.
[QCMLS 2021: 1.1.4] [Harmening 2019, p360]

359. **c** Functional abnormalities are found in both congenital and acquired platelet disorders and platelet transfusions may be beneficial. Decreased platelets is not an outcome of a hemolytic transfusion reaction. Posttransfusion purpura is usually self-limiting and is due to an antibody to a specific platelet antigen. Immune thrombocytopenia patients have low platelet counts, but rarely have hemorrhage.
[QCMLS 2021: 1.6.1] [Harmening 2019, p357, 359-360]

360. **d** In a medical emergency where a blood transfusion is urgent, the Blood Bank must have in their records a signed physician statement asking for the immediate release of the blood for emergency use before typing and crossmatch tests are completed.
[QCMLS 2021: 1.6.2.2] [AABB Tech Manual 2020, p525-526; AABB BB/TS Standards 2022, p49]

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361. **b** In an emergency situation, group O red cells are given if the patient's ABO and Rh are unknown. In the case of women with child-bearing potential, it is preferential to provide group O, Rh-negative red cells. This will prevent Rh immunization to the D antigen, which can affect future pregnancies if the fetus is Rh-positive.
[QCMLS 2021: 1.6.2.2] [AABB Tech Manual 2020, p525-526]

362. **b** For infants <4 months old, red cell units should be, in general, <7 days old to have optimal 2,3, DPG levels to facilitate O_2 delivery to the tissues. Fresher units will also minimize the amount of K^+ ions transfused to the infant.
[QCMLS 2021: 1.6.2.4] [AABB Tech Manual 2020, p780-781]

363. **b** When a patient is massively transfused, hemostatic abnormalities can develop as both coagulation factors and platelets are diluted by transfusion of Red Blood Cells or low-titer group O Whole Blood and other IV fluids. Therefore, massive transfusion protocols include the transfusion of Fresh Frozen Plasma to replace coagulations factors and the transfusion of Apheresis Platelets or pooled Platelets (Whole Blood derived).
[QCMLS 2021: 1.6.2.3] [AABB Tech Manual 2020, p526-528, 605]

364. **b** For emergency release of blood components, there must be a statement on the unit label or tie-tag that compatibility testing has not been completed.
[QCMLS 2021: 1.6.2.2] [AABB BB/TS Standards 2022, p48]

365. **c** Massive transfusion can result in several complication including hemostatic abnormalities, hypothermia, hypokalemia, citrate toxicity and hypocalcemia. Stored blood components contain citrate, which is a component of the anticoagulants used in the collection of Whole Blood. As plasma citrate levels increase in the recipient and bind to calcium, hypocalcemia can result. Symptoms of hypocalcemia include perioral tingling and muscle cramps.
[QCMLS 2021: 1.6.2.3] [AABB Tech Manual 2020, p646-648]

366. **d** For neonates <4 months of age, a forward ABO typing and testing for the D antigen are required. Antibody detection testing using maternal or neonatal serum is also required. ABO reverse testing is not required. Testing for the K antigen is not required, but might be performed if the maternal serum contains anti-K.
[QCMLS 2021:1.6.2.4] [AABB BB/TS Standards 2022, p43-44]

B. Component Therapy

367. **d** Cryoprecipitated AHF is generally transfused for fibrinogen replacement. Patients with DIC often have fibrinogen consumption, and Cryoprecipitated AHF transfusions may be recommended.
[QCMLS 2021: 1.6.1] [AABB Tech Manual 2020, p569, 636; Harmening 2019, p357]

368. **b** In selecting units of Cryoprecipitated AHF for transfusion, all ABO groups are acceptable. However, when large volumes of the component are to be transfused, ABO compatibility may be a consideration.
[QCMLS 2021: 1.5.1.5] [AABB Tech Manual 2020, p507]

369. **d** Patients who are repeatedly transfused or pregnant may develop antibodies to HLA or platelet-specific antigens. These antibodies may destroy randomly selected platelet products when transfused and result in alloimmune refractoriness. The selection of Apheresis Platelets Leukocytes Reduced that are matched for class I HLA antigens and/or platelet-specific antigens can result in better posttransfusion platelet counts in these patients.
[QCMLS 2021: 1.2.15.1, 1.2.15.2] [AABB Tech Manual 2020, p462-464; Harmening 2019, p359-360]

370. **d** FFP should be ABO compatible with the recipient's RBCs. Avoid FFP with antibodies to A or B antigens the patient may have. Group A plasma has anti-B and should only be transfused to A or O recipients.
[QCMLS 2021: 1.5.1.5] [Harmening 2019, p357-358]

371. **d** FFP should be ABO compatible with the recipient's RBCs. Avoid FFP with ABO antibodies to A or B antigens the patient may have. Rh phenotype of the units of FFP is not of concern.
[QCMLS 2021: 1.5.1.5] [Harmening 2019, p357-358]

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372. **a** FFP should be ABO compatible with the recipient's RBCs. Avoid FFP with ABO antibodies to A or B antigens the patient may have. Rh phenotype of the units of FFP is not of concern.
[QCMLS 2021: 1.5.1.5] [Harmening 2019, p357-358]

373. **c** FFP should be ABO compatible with the recipient's RBCs. If patient's type has not been determined (currently), plasma lacking anti-A and anti-B should be given.
[QCMLS 2021: 1.5.1.5] [Harmening 2019, p357-358]

374. **c** FFP contains all coagulation factors, including factor VIII. It does not have a higher risk of transmitting hepatitis than Whole Blood. It must be transfused within 24 hours of thawing and must be ABO compatible.
[QCMLS 2021: 1.1.4] [Harmening 2019, p357-358]

375. **b** Irradiation inhibits proliferation of T cells, preventing transfusion-associated GVHD.
[QCMLS 2021: 1.6.3.2] [Harmening 2019, p364-365]

376. **d** When a 70-kg individual is transfused with 1 unit of Apheresis Platelets, the expected increase is approximately 30-50,000 platelets/µL for each unit.
[QCMLS 2021: 1.1.4] [AABB Tech Manual 2020, p565; Harmening 2019, p380]

377. **a** Plasma Frozen Within 24 Hours (PF24) is prepared by freezing at ≤–18°C in the first 24 hours after collection. Compared to Fresh Frozen Plasma (FFP), it has lower concentrations of Factors V and VIII and Protein C, but comparable concentrations of other plasma factors and proteins. It can be used to treat patients with coagulation deficiencies caused by DIC or liver failure.
[QCMLS 2021: 1.1.4] [AABB Tech Manual 2020, p156-157]

378. **b** Pathogen inactivation strategies approved for clinical use and those currently under development target nucleic acids and prevent pathogen replication. Dithiothreitol is a reagent used in blood bank testing.
[QCMLS 2021: 1.1.2.3] [AABB Tech Manual 2020, p162-164, 215-217]

C. Adverse Effects of Transfusion

379. **d** Approximately 25-30% of SCD patients develop alloantibodies due to repeated red cell transfusions. A recommended strategy to reduce the number of patients who develop alloantibodies is to phenotype for C, E, and K antigens and transfuse with phenotypically identical units. If patients do become alloimmunized to these antigens, additional phenotyping for red cell antigens in the JK and FY systems and Ss is recommended, followed by transfusion with phenotypically identical units.
[QCMLS 2021: 1.6.2.1] [AABB Tech Manual 2020, p682-684]

380. **c** A patient count of 10,000/µL is usually considered the trigger value for prophylactic platelet transfusion. Studies have shown that when ABO identical platelets are not available, ABO compatible platelets (ie, group A donor platelets for a group AB patient) are more effective than ABO incompatible (ie, group O platelets in a non-group O recipient). Each platelet unit (Whole Blood derived) should increase the count by approximately 5,000 platelets/µL. Since this patient received 4 units, the platelet count should have increased by approximately 20,000 platelets/µL. The most likely cause of the smaller increase in the patient's platelet count is the formation of antibodies against Class I HLA antigens. Platelet irradiation does not prevent this.
[QCMLS 2021: 1.1.4, 1.2.15.1] [AABB Tech Manual 2020, p565-567; Harmening 2019, p380]

381. **c** Symptoms of hemolytic transfusion reactions are fever, chills, flushing, chest and back pain, hypotension, nausea, dyspnea, shock, renal failure, and DIC. Circulatory overload, allergic, and anaphylactic reactions are not characterized by fever.
[QCMLS 2021: 1.6.3.1.1] [AABB Tech Manual 2020, p628, 634-636]

382. **d** Alloimmunization to HLA antigens can result in refractoriness to random donor platelet transfusions.
[QCMLS 2021: 1.25.15.1] [AABB Tech Manual 2020, p565-567; Harmening 2019, p387]

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383. **c** AABB Standards state that any sign of a possible transfusion reaction requires that the transfusion be stopped, and the patient's doctor contacted immediately.
[QCMLS 2021: 1.6.3.1] [AABB BB/TS Standards 2022, p95; Harmening 2019, p374-375]

384. **c** Posttransfusion purpura (PTP) is caused by platelet-specific alloantibody in a previously immunized recipient. Transfused donor platelets in blood products are destroyed, with concomitant destruction of the recipient's own platelets, through unknown mechanisms. The usual antibody specificity is HPA 1a.
[QCMLS 2021: 1.2.15.2] [AABB Tech Manual 2020, p650-652; Harmening 2019, p387]

385. **d** Previously immunized patients may have an undetectable level of antibody. Transfusion of antigen-positive donor red cells may cause an anamnestic response and result in a delayed hemolytic transfusion reaction. Symptoms may be mild, and present only as jaundice and unexplained anemia.
[QCMLS 2021: 1.6.3.1.2] [AABB Tech Manual 2020, p631, 648-649; Harmening 2019, p200, 379]

386. **b** Delayed hemolytic transfusion reactions are caused by a secondary anamnestic response in a previously alloimmunized recipient. Unlike a primary response, a secondary response is rapid. Antibody may be detectable days to weeks from the time of transfusion.
[QCMLS 2021: 1.6.3.1.2] [AABB Tech Manual 2020, p648-649; Harmening 2019, p379-380]

387. **c** Massive transfusion occurs when a patient receives >10 red cells units over 24 hours. Rapid infusion of large amounts of red cells can result in hypothermia due to blood storage at 1-6°C and cause adverse responses such as apneic events and metabolic changes.
[QCMLS 2021: 1.6.2.3] [AABB Tech Manual 2020, p646-648]

388. **a** ABO antibodies activate complement and may cause intravascular hemolysis. Rh, KEL, and FY antibodies are primarily associated with extravascular hemolysis.
[QCMLS 2021: 1.6.3.1.1] [AABB Tech Manual 2020, p634-636, 648-649; Harmening 2019, p122, 195, 198]

389. **b** Antibodies in the JK system activate complement and may cause intravascular hemolysis. The antibodies often decline *in vivo*, are weak, show dosage, and are difficult to detect *in vitro*, making them prime candidates for causing anamnestic delayed hemolytic transfusion reactions.
[QCMLS 2021: 1.6.3.1.2] [AABB Tech Manual 2020, p371; Harmening 2019, p201]

390. **a** ABO antibodies activate complement and may cause intravascular hemolysis. The symptoms described are typical of what is seen in an ABO incompatible transfusion reaction. ABO antibodies are naturally occurring and directed against the A and B antigens the individual lacks. Rh and FY antibodies may also cause hemolytic transfusion reactions, but the antibodies are the results of alloimmunization and not naturally present in individuals who lack the antigen. The incidence of septic transfusion reactions from bacterial contamination of Red Blood Cells is rare and patients do not have increased plasma hemoglobin or hemoglobinuria.
[QCMLS 2021: 1.6.3.1.1] [AABB Tech Manual 2020, p347, 369, 634; Harmening 2019, p378]

391. **d** Patients receiving >1 blood volume replacement due to massive blood loss often develop thrombocytopenia and require platelet transfusions. Stored Red Blood Cells or Whole Blood units have few, if any, functional platelets. Transfusion of Whole-Blood derived Platelets or Apheresis Platelets are needed to restore platelet functionality in these massively transfused individuals.
[QCMLS 2021: 1.6.2.3] [AABB Tech Manual 2020, p647]

392. **d** A positive DAT in a posttransfusion blood sample usually indicates that the patient is producing alloantibody against an antigen present on the transfused donor red cells. An elution should be performed to remove the antibody from the red cells and identify it. Free antibody may also be present in the serum. In this case, the antibody screen is positive and antibody identification studies can be undertaken with the patient's serum.
[QCMLS 2021: 1.6.3.1.2] [AABB Tech Manual 2020, p432-434]

393. **a** Free hemoglobin released from destruction of transfused donor red cells will impart a distinct pink or red color in the posttransfusion sample plasma.
[QCMLS 2021: 1.6.3.1.1] [AABB Tech Manual 2020, p633; AABB BB/TS Standards 2022, p95; Harmening 2019, p374-376]

394. **d** The immediate steps required to investigate a transfusion reaction include a clerical check of records and labels, visual inspection of postreaction plasma for hemolysis compared to the pretransfusion sample, direct antiglobulin test and repeat ABO typing on the postreaction sample. Since the results of the lab investigation show no clerical mistakes, no plasma hemolysis in the postreaction sample as well as confirmed the DAT and ABO results of the pretransfusion sample, no further laboratory testing is necessary. Findings should be reported to the blood bank supervisor or medical director.
[QCMLS 2021: 1.6.3.1] [AABB Tech Manual 2020, p633-634; AABB BB/TS Standards 2022, p95-96; Harmening 2019, p376-377]

395. **a** In septic transfusion reactions, patients can experience fever >101°F (38.3°C), hypotension and chills with shaking. In severe reactions, patients can have shock, hemoglobinuria, renal failure, and DIC.
[QCMLS 2021: 1.6.3.7] [AABB Tech Manual 2020, p637-638; Harmening 2019, p389-390]

396. **c** Clinical signs of a hemolytic transfusion reaction include fever and chills, and, in severe cases, DIC. Circulatory overload, allergic and anaphylactic reactions are not characterized by fever and DIC.
[QCMLS 2021: 1.6.3.1.1] [AABB Tech Manual 2020, p633-634; Harmening 2019, p374-375]

397. **a** Iron overload can occur in nonbleeding patients who receive multiple transfusions over extended periods of time for diseases such as thalassemia. When red cells break down, most of the iron is stored in the reticuloendothelial system and organs including the heart and cannot be excreted. When the accumulation of iron overwhelms the capacity for safe storage, tissue damage can ensue including liver and heart failure and diabetes.
[QCMLS 2021: 1.6.3] [AABB Tech Manual 2020, p652]

398. **d** Delayed hemolytic transfusion reactions may occur in recipients who are previously immunized but do not have detectable antibody at the time of testing. JK alloantibodies are often associated with this finding. If they receive blood with the corresponding antigen, they will have an anamnestic response days to weeks later, resulting in antibody production causing a transfusion reaction. When there is a history of clinically significant antibodies, donor red cells should be phenotyped and antigen-negative blood selected. A complete antiglobulin crossmatch must also be performed.
[QCMLS 2021: 1.5.1.3] [AABB Tech Manual 2020, p648-649; AABB BB/TS Standards 2022, p42; Harmening 2019, p200]

399. **b** Patients who are immunocompromised, on immunosuppressive drugs, of neonatal age, and those with disorders such as acute leukemia and lymphoma are at risk for TA-GVHD. TA-GVHD can also occur in non-immunocompromised patients if the donor is homozygous for an HLA haplotype and the recipient is heterozygous for that allele, as could be the case in a first-degree relative and a directed donation. This allows the donor lymphocytes to mount an immune response against the recipient, resulting in nearly always fatal TA-GVHD. Blood components for at risk patients should be irradiated to inactivate donor lymphocytes.
[QCMLS 2021: 1.6.3.2] [AABB Tech Manual 2020, p519-520, 649-650; Harmening 2019, p364-365]

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400. **c** Lack of expected rise in hemoglobin after transfusion and a positive DAT that demonstrates a mixed-field reaction may be a sign of a delayed hemolytic transfusion reaction. If the DAT is positive, a repeat DAT using anti-IgG and anti-C3d reagents should be performed to determine if the red cells are coated with IgG, C3d, or both. A positive anti-IgG result suggests that an IgG antibody(ies) has sensitized the transfused donor cells, as would be seen in a delayed transfusion reaction. An elution should be performed to remove and identify the antibody coating the transfused donor red cells. Antibody is not detectable in the patient's serum as the antibody screen is negative. It is likely all alloantibody is bound to the antigen-positive transfused cells, so a routine cell panel using the patient's serum would not be helpful. Since the transfusion occurred 3 weeks previously, it is likely that donor samples are not available for testing as AABB Standards only require recipient and donor samples be stored for a minimum of 7 days following transfusion.
[QCMLS 2021: 1.6.3.1.2] [AABB Tech Manual 2020, p432-434, 648-649; AABB BB/TS Standards 2022, p39; Harmening 2019, p376, 379]

401. **b** Delayed hemolytic transfusion reactions are associated with extravascular hemolysis, rather than intravascular. Alloantibody sensitizes the transfused antigen-positive donor cells in the recipient's circulation, producing a mixed-field positive reaction in the DAT.
[QCMLS 2021: 1.6.3.1.2] [AABB Tech Manual 2020, p409]

402. **d** Donor lymphocytes from transfused blood can cause transfusion associated graft-versus-host-disease (TA-GVHD) in immunosuppressed patients. Irradiation of blood products will inactivate lymphocytes and prevent TA-GVHD. Pathogen reduction technologies can also inactivate donor lymphocytes and prevent TA-GVHD.
[QCMLS 2021: 1.6.3.2] [AABB Tech Manual 2020, p162-163, 519, 649-650; Harmening 2019, p385-387]

403. **d** Treatment of acute hemolytic transfusion reactions focuses on supportive measures and control of DIC, hypotension, shock, and acute renal failure.
[QCMLS 2021: 1.6.3.1.1] [AABB Tech Manual 2020, p636; Harmening 2019, p378]

404. **c** Patients who develop TA-GVHD have these types of symptoms. This complication of transfusion has a mortality rate higher than 90%.
[QCMLS 2021: 1.6.3.2] [AABB Tech Manual 2020, p649-650; Harmening 2019, p385-387]

405. **c** Two distinguishing features of anaphylactic transfusion reactions are that 1) symptoms occur with transfusion of only small amounts of blood, and 2) the patient has no fever.
[QCMLS 2021: 1.6.3, 1.6.3.6] [AABB Tech Manual 2020, p629; Harmening 2019, p385]

406. **d** Febrile nonhemolytic transfusion reactions are defined as fever of 1°C or greater (over baseline temperature) during or after transfusion, with no other reason for the elevation than transfusion, and no evidence of hemolysis in the transfusion reaction investigation. Patients often exhibit chills. Allergic reactions, citrate toxicity, and circulatory overload are not characterized by fever or chills.
[QCMLS 2021: 1.6.3.5] [AABB Tech Manual 2020, p628, 638-639; Harmening 2019, p383-384]

407. **b** Allergic reactions are a type 1 immediate hypersensitivity reaction to an allergen in plasma. Most are mild reactions shown by urticaria (hives, swollen red wheals), which may cause itching.
[QCMLS 2021: 1.6.3.6] [AABB Tech Manual 2020, p639-640; Harmening 2019, p384]

408. **a** Symptoms for febrile nonhemolytic transfusion reactions (FNHTR) include a rise in fever of ≥1°C during or <4 hours after transfusion, chills, mild nausea/vomiting, and headache. Some of these symptoms are also found in more serious transfusion complications. Recipient antibodies to donor WBCs and proinflammatory cytokines released during blood component storage are responsible for these reactions. The use of Red Blood Cells Leukocytes Reduced has greatly decreased the incidence of these reactions. There is also no evidence of hemolysis in the transfusion reaction investigation.
[QCMLS 2021: 1.6.3.5] [AABB Tech Manual 2020, p628, 638-639; Harmening 2019, p383-384]

409. **d** Febrile nonhemolytic transfusion reactions occur in about 1% of transfusions, making it one of the most common types of reaction. Neither transfusion-associated circulatory overload (TACO) or anaphylactic transfusion reactions are characterized by fever, shaking or chills. Bacterially contaminated Red Blood Cells are rare, and rapidly produce severe symptoms upon transfusion.
[QCMLS 2021: 1.6.3.5] [AABB Tech Manual 2020, p628, 638-639; Harmening 2019, p383-384]

410. **b** Transfusion-related acute lung injury (TRALI) is most commonly caused by donor HLA or HNA antibodies that react with recipient antigens, causing damage to the lung basement membrane and bilateral pulmonary edema within 6 hours of transfusion. Multiparous females are more likely than males to have these antibodies. Using male donors as the sole source of plasma products is a strategy for reducing the risk of TRALI.
[QCMLS 2021: 1.6.3.3] [AABB Tech Manual 2020, p640-643; Harmening 2019, p380-381]

411. **d** Noncardiogenic pulmonary edema, dyspnea, hypotension, and hypoxemia occurring within 6 hours of transfusion are clinical symptoms of TRALI.
[QCMLS 2021: 1.6.3.3] [AABB Tech Manual 2020, p640-643; Harmening 2019, p380-381]

412. **c** Transfusion-associated circulatory overload (TACO) is hypervolemia manifested by coughing, hypoxemia, difficulty breathing, cyanosis, and cardiogenic pulmonary edema.
[QCMLS 2021: 1.6.3.4] [AABB Tech Manual 2020, p630, 643; Harmening 2019, p381-383]

413. **d** Transfusion-induced hypervolemia causing edema and congestive heart failure is a feature of transfusion-associated circulatory overload (TACO). Hypervolemia is not a complication of a hemolytic, febrile, or anaphylactic transfusion reaction.
[QCMLS 2021: 1.6.3.4] [AABB Tech Manual 2020, p643-645; Harmening 2019, p381-383]

414. **a** Septic transfusion reactions due to contaminated blood products are manifested by high fever, chills, hypotension, shock, renal failure, and DIC. Symptoms usually appear rapidly. Transfusion reaction investigation shows no evidence of unexpected blood group antibodies. A Gram stain and blood culture of the donor unit may detect the presence of aerobic or anaerobic organisms.
[QCMLS 2021: 1.6.3.7] [AABB Tech Manual 2020, p637-638; Harmening 2019, p389-390]

415. **c** Hepatitis transmission is unlikely but has a higher risk of transmission through blood transfusion than CMV (rare), syphilis (no transfusion-transmitted cases reported in >40 years), or HIV-1/2.
[QCMLS 2021: 1.6.3.8] [AABB Tech Manual 2020, p198, 203]

416. **d** Blood from a family member may be homozygous for a shared HLA haplotype, allowing donor lymphocytes to engraft in the recipient and cause transfusion-associated GVHD.
[QCMLS 2021: 1.6.3.2] [AABB Tech Manual 2020, p649-650]

417. **d** Laboratory testing for HTLV I/II is based on detection of IgG antibody by ELISA or ChLIA.
[QCMLS 2021: 1.1.2.1] [AABB Tech Manual 2020, p179; Harmening 2019, p326-327]

418. **a** Nucleic acid assays (TMA or PCR) are currently performed for the detection of HBV, HIV-1, HCV and WNV. Other serologic screening assays looking for the presence of antibody or antigen rely on enzyme immunosorbent assays or chemiluminescent immunoassays.
[QCMLS 2021: 1.1.2.1] [AABB Tech Manual 2020, p179-180; Harmening 2019, p326-327]

419. **b** While symptoms including fever and chills and hypotension may occur with other types of adverse effects of transfusion, hemolysis found only in the posttransfusion sample along with back pain or flank pain are highly suggestive of an acute hemolytic transfusion reaction. Acute hemolytic transfusion reactions occur within 24 hours of the transfusion event. Blood bank personnel then follow standard operating procedures, which will include, but not limited to, repeat ABO and Rh typing, DAT, antibody detection testing and crossmatching. There must also be a review of clerical information.
[QCMLS 2021: 1.6.3.1] [AABB Tech Manual 2020, p634, 638; AABB BB/TS Standards 2022, p95-96]

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420. **d** Hypothermia is caused when there is rapid infusion of RBC blood components that have been stored at 1°C-6°C. This can occur during a massive transfusion protocol or during the transfusion of neonates with resulting cardiac arrythmia. Administering RBC components with a specially-designed inline blood warmer will prevent this type of reaction.
[QCMLS 2021: 1.2.3.5] [AABB Tech Manual 2020, p646, 648, 676]

D. Apheresis and Extracorporeal Circulation

421. **b** For patients with thrombotic thrombocytopenic purpura (TTP), TPE using plasma is accepted as first-line therapy and is generally performed daily. The use of plasma prevents depletion of coagulation factors.
[QCMLS 2021: 1.6.4.1] [AABB Tech Manual 2020, p705-706]

422. **b** Myasthenia gravis is an autoimmune disease treated with TPE to remove autoantibodies. TPE is accepted as first-line therapy for this disease.
[QCMLS 2021: 1.6.4.1] [AABB Tech Manual 2020, p714, 716; Harmening 2019, p408]

423. **c** Patients with blast counts >100,000/µL develop these symptoms due to microvascular stasis. Using leukocytapheresis to decrease cell counts to <50,000 blasts/µL will help mitigate this patient's symptoms.
[QCMLS 2021: 1.6.4.2] [AABB Tech Manual 2020, p721; Harmening 2019, p409]

424. **d** The most common use of therapeutic plasmapheresis is to remove plasma abnormalities, such as pathological antibodies, immune complexes, or cryoglobulins.
[QCMLS 2021: 1.6.4.1] [AABB Tech Manual 2020, p713-714]

425. **b** Macroglobulinemia, also known as Waldenström macroglobulinemia, is a syndrome with IgM monoclonal paraprotein. Since IgM protein is intravascular, plasma exchange provides symptomatic relief.
[QCMLS 2021: 1.6.4.1] [AABB Tech Manual 2020, p720; Harmening 2019, p408]

426. **a** Extracorporeal photopheresis is used to treat diseases such as graft-versus-host-disease, cutaneous T-cell lymphoma and to inhibit rejection of solid organ transplants. Leukocytes from replicating patient cells are collected and treated with 8-methyloxypsoralen and exposed to UV light to inhibit DNA replication in leukocytes. The cells are then reinfused into the patient.
[QCMLS 2021: 1.6.4.3] [AABB Tech Manual 2020, p723-724]

427. **b** Selective adsorption is used for removing LDL cholesterol in patients with famial hypercholesterolemia. The patient's plasma is passed through a column containing compounds that will remove the LDL.
[QCMLS 2021: 1.6.4.4] [AABB Tech Manual 2020, p724-727]

428. **d** Erythrocytapheresis (RBC exchange) removes sickled RBCs and replaces them with donor RBCs to increase Hemoglobin A and O_2 delivery to the tissues, while reducing Hemoglobin S. This prevents complication including acute chest syndrome and stroke.
[QCMLS 2021: 1.6.4.2] [AABB Tech Manual 2020, p722-723]

E. Blood Administration and Patient Blood Management

429. **a** AABB Standards require that the transfusionist and one other individual (or an electronic identification system) positively identify the recipient and match the blood component to the recipient using 2 independent identifiers.
[QCMLS 2021: 1.6.5] [AABB BB/TS Standards 2022, p50; AABB Tech Manual 2020, p542-543]

430. **d** Release of blood components for patient transfusion requires verification by blood bank personnel of 2 independent patient identifiers, (ie, patient name and identification number), ABO group and Rh type, the donor unit's ABO group and Rh type, donation identification number, expiration date (if applicable, time), results of the crossmatch interpretation, special transfusion requirements, if applicable, final visual inspection of the product, and time and date of the unit release.
[QCMLS 2021: 1.5.1.5] [AABB Tech Manual 2020, p542-543; AABB Standards 2022, p47]

431. **d** One reason to quarantine blood components before transfusion is hemolysis of the red cells. This is an indication of contamination or improper storage.
[QCMLS 2021: 1.8.1.5] [AABB Tech Manual 2020, p164, 543]

432. **a** Vital signs (BP, pulse, respiration rate and temperature) must be taken and recorded on all patients receiving a blood component before the transfusion starts, during transfusion and after the transfusion has finished.
[QCMLS 2021: 1.6.5] [AABB Tech Manual 2020, p547-548; AABB BB/TS Standards 2022, p51]

433. **c** Strategies for intraoperative patient blood management include intraoperative blood recovery to reuse shed patient blood after washing and filtering, normovolemic hemodilution, and point-of-care testing and transfusion algorithms to manage data-driven transfusion decisions.
[QCMLS 2021: 1.6.6.1.2] [AABB Tech Manual 2020, p594-595; Harmening 2019, p564-567]

434. **d** A major cause of transfusion-associated fatalities is transfusion of blood to the wrong patient. Accurate identification of the recipient and comparison to the donor unit can help prevent such errors. Other errors associated with transfusion fatalities include those related to patient pretransfusion blood sample collections and misidentification.
[QCMLS 2021: 1.6.5] [AABB Tech Manual 2020, p542-543, 636; Harmening 2019, p368]

435. **c** Required documentation, in addition to the signed patient consent for transfusion, includes the transfusion order, component name and donation identification number, donor ABO and Rh, date and time of transfusion, vital signs taken before, during and after transfusion, amount of blood transfused, identification of transfusionist and any transfusion-related adverse events.
[QCMLS 2021: 1.6.5] [AABB BB/TS Standards 2022, p51; AABB Tech Manual 2020, p537, 547-548]

436. **d** Immediately before a transfusion is started, AABB Standards require verification of the donation identification number and donor ABO and Rh type. The following must also be verified: interpretation of crossmatch tests (if performed), special transfusion requirements are met (if applicable), and the unit has not expired.
[QCMLS 2021: 1.6.5] [AABB BB/TS Standards 2022, p49-50; AABB Tech Manual 2020, p544-545]

437. **b** Autologous donations have declined dramatically since the 1990s, but individuals with an antibody to a high-incidence antigen, where units may be difficult to find, are ideal candidates for preoperative autologous donation.
[QCMLS 2021: 1.6.6.1.1] [AABB Tech Manual 2020, p136; Harmening 2019, p365]

438. **c** Blood that is shed during high-blood-loss surgery can be recovered and given back to the patient. After the blood is collected, washed, and centrifuged, the red cells are returned to the patient.
[QCMLS 2021: 1.6.6.1.2] [AABB Tech Manual 2020, p594-595]

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Chapter 2

Chemistry

*The following items have been identified generally as appropriate for those preparing for both the MLS and MLT examinations. Items that are appropriate for the MLS examination **only** are marked with MLS ONLY.*

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168 Answers with Explanations
- 168 I. General Chemistry
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- 192 IV. Special Chemistry

I. General Chemistry

A. Carbohydrates

1. Following overnight fasting, hypoglycemia in non-diabetic adults is defined as a glucose of:

   a ≤70 mg/dL (≤3.9 mmol/L)
   b ≤60 mg/dL (≤3.3 mmol/L)
   c ≤55 mg/dL (≤3.0 mmol/L)
   d ≤45 mg/dL (≤2.5 mmol/L)

2. The results shown in this table are from a 21-year-old patient with a back injury who appears otherwise healthy:

test	result
whole blood glucose	77 mg/dL (4.2 mmol/L)
serum glucose	88 mg/dL (4.8 mmol/L)
CSF glucose	56 mg/dL (3.1 mmol/L)

   The best interpretation of these results is that:

   a the whole blood and serum values are expected but the CSF value is elevated
   b the whole blood glucose value should be higher than the serum value
   c all values are consistent with a normal healthy individual
   d the serum and whole blood values should be identical

3. The preparation of a patient for standard glucose tolerance testing should include:

   a a high carbohydrate diet for 3 days
   b a low carbohydrate diet for 3 days
   c fasting for 48 hours prior to testing
   d bed rest for 3 days

4. If a fasting glucose is 90 mg/dL, which of these 2-hour postprandial glucose results would most closely represent normal glucose metabolism?

   a 55 mg/dL (3.0 mmol/L)
   b 100 mg/dL (5.5 mmol/L)
   c 180 mg/dL (9.9 mmol/L)
   d 260 mg/dL (14.3 mmol/L)

5. A healthy person with a blood glucose of 80 mg/dL (4.4 mmol/L) would have a simultaneously determined cerebrospinal fluid glucose value of:

   a 25 mg/dL (1.4 mmol/L)
   b 50 mg/dL (2.3 mmol/L)
   c 100 mg/dL (5.5 mmol/L)
   d 150 mg/dL (8.3 mmol/L)

6. A 25-year-old man became nauseated and vomited 90 minutes after receiving a standard 75 g carbohydrate dose for an oral glucose tolerance test. The best course of action is to:

   a give the patient a glass of orange juice and continue the test
   b start the test over immediately with a 50 g carbohydrate dose
   c draw blood for glucose and discontinue test
   d place the patient in a recumbent position, reassure him and continue the test

7. Cerebrospinal fluid for glucose assay should be:

   a refrigerated
   b analyzed immediately
   c heated to 56°C
   d stored at room temperature after centrifugation

8. Which of these 2-hour postprandial glucose values demonstrates unequivocal hyperglycemia diagnostic for diabetes mellitus?

   a 160 mg/dL (8.8 mmol/L)
   b 170 mg/dL (9.4 mmol/L)
   c 180 mg/dL (9.9 mmol/L)
   d 200 mg/dL (11.0 mmol/L)

9. Serum levels that define hypoglycemia in preterm or low birth weight infants are:

   a the same as adults
   b lower than adults
   c the same as a normal full-term infant
   d higher than a normal full-term infant

10. A 45-year-old woman has a fasting serum glucose concentration of 95 mg/dL (5.2 mmol/L) and a 2-hour postprandial glucose concentration of 105 mg/dL (5.8 mmol/L). The statement which best describes this patient's fasting serum glucose concentration is:

    a normal; reflecting glycogen breakdown by the liver
    b normal; reflecting glycogen breakdown by skeletal muscle
    c abnormal; indicating diabetes mellitus
    d abnormal; indicating hypoglycemia

11. Pregnant women with symptoms of thirst, frequent urination or unexplained weight loss should have which of the following tests performed?

    a tolbutamide test
    b lactose tolerance test
    c epinephrine tolerance test
    d glucose tolerance test

12. In the fasting state, the arterial and capillary blood glucose concentration varies from the venous glucose concentration by approximately how many mg/dL (mmol/L)?

    a 1 mg/dL (0.05 mmol/L) higher
    b 5 mg/dL (0.27 mmol/L) higher
    c 10 mg/dL (0.55 mmol/L) lower
    d 15 mg/dL (0.82 mmol/L) lower

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13. The conversion of glucose or other hexoses into lactate or pyruvate is called:

a glycogenesis
b glycogenolysis
c gluconeogenesis
d glycolysis

14. Which of the following values obtained during a glucose tolerance test is diagnostic of diabetes mellitus?

a 2-hour specimen = 150 mg/dL (8.3 mmol/L)
b fasting plasma glucose = 126 mg/dL (6.9 mmol/L)
c fasting plasma glucose = 110 mg/dL (6.1 mmol/L)
d 2-hour specimen = 180 mg/dL (9.9 mmol/L)

15. Monitoring long-term glucose control in patients with adult onset diabetes mellitus can best be accomplished by measuring:

a weekly fasting 7 AM serum glucose
b glucose tolerance testing
c 2-hour postprandial serum glucose
d HbA_1c

16. A patient with type I, insulin-dependent diabetes mellitus has the results shown in this table:

test	patient	reference range
fasting blood glucose	150 mg/dL (8.3 mmol/L)	70-110 mg/dL (3.9-6.1 mmol/L)
HbA_1c	8.5%	4.0-6.0%
fructosamine	2.5 mmol/L	2.0-2.9 mmol/L

After reviewing these test results, the technologist concluded that the patient is in a:

a "steady state" of metabolic control
b state of flux, progressively worsening metabolic control
c improving state of metabolic control as indicated by fructosamine
d state of flux as indicted by the fasting glucose level

17. Total glycosylated hemoglobin levels in a hemolysate reflect the:

a average blood glucose levels of the past 2-3 months
b average blood glucose levels for the past week
c blood glucose level at the time the sample is drawn
d HbA_1c level at the time the sample is drawn

18. Which of these glycosylated hemoglobins is recommended by the ADA guidelines for testing diabetic patients?

a HbA_1a
b HbA_2
c HbA_1b
d HbA_1c

19. A patient with hemolytic anemia will:

a show a decrease in glycated Hb value
b show an increase in glycated Hb value
c show little or no change in glycated Hb value
d demonstrate an elevated HbA_1

20. In using ion-exchange chromatographic methods, falsely increased levels of HbA_1c might be demonstrated in the presence of:

a iron deficiency anemia
b pernicious anemia
c thalassemias
d HbS

21. An increase in serum acetone is indicative of a defect in the metabolism of:

a carbohydrates
b fat
c urea nitrogen
d uric acid

22. An infant with diarrhea is being evaluated for a carbohydrate intolerance. His stool yields a positive copper reduction test and a pH of 5.0. It should be concluded that:

a further tests are indicated
b results are inconsistent—repeat both tests
c the diarrhea is not due to carbohydrate intolerance
d the tests provided no useful information

23. Blood samples are collected at the beginning of an exercise class and after thirty minutes of aerobic activity. Which of the following would be most consistent with the post-exercise sample?

a normal lactic acid, low pyruvate
b low lactic acid, elevated pyruvate
c elevated lactic acid, low pyruvate
d elevated lactic acid, elevated pyruvate

24. What is the best method to diagnose lactase deficiency?

a H_2 breath test
b plasma aldolase level
c LD level
d D-xylose test

25. MLS ONLY The different water content of erythrocytes and plasma makes true glucose concentrations in whole blood a function of the:

a hematocrit
b leukocyte count
c erythrocyte count
d erythrocyte indices

26. In a specimen collected for plasma glucose analysis, sodium fluoride:

a serves as a coenzyme of hexokinase
b prevents reactivity of non glucose reducing substances
c precipitates proteins
d inhibits glycolysis

27. Which of these serum constituents is unstable if a blood specimen is left standing at room temperature for 8 hours before processing?

a cholesterol
b triglyceride
c creatinine
d glucose

28. MLS ONLY One international unit of enzyme activity is the amount of enzyme that will, under specified reaction conditions of substrate concentration, pH and temperature, cause utilization of substrate at the rate of:

a 1 mol/min
b 1 mmol/min
c 1 µmol/min
d 1 nmol/min

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29. MLS ONLY In spectrophotometric determination, which of the following is the formula for calculating the absorbance of a solution? A=abc

a (absorptivity × light path)/concentration
b (absorptivity × concentration)/light path
c absorptivity × light path × concentration
d (light path × concentration)/absorptivity

30. The most specific method for the assay of glucose in all body fluids utilizes:

a hexokinase
b glucose oxidase
c glucose-6-phosphatase
d glucose dehydrogenase

31. MLS ONLY Which of the following is an example of a glucose-specific colorimetric method?

a alkaline ferricyanide
b glucose oxidase
c hexokinase
d o-toluidine

32. MLS ONLY Increased concentrations of ascorbic acid inhibit chromogen production in which of the following commonly-used glucose methods?

a ferricyanide
b ortho-toluidine
c glucose oxidase (peroxidase)
d hexokinase

33. MLS ONLY In the hexokinase method for glucose determination, the actual end product measured is the:

a amount of hydrogen peroxide produced
b NADH produced from the reduction of NAD
c amount of glucose combined with bromcresol purple
d condensation of glucose with an aromatic amine

34. At midmorning, blood glucose levels fall and stimulate the secretion of which hormone?

a cortisol
b epinephrine
c glucagon
d insulin

35. Deficiency of this enzyme is the most frequent cause of galactosemia:

a glucose-6-phosphatase
b galactose-1-phosphate uridyl transferase
c galactokinase
d uridine diphosphate 4 epimerase

B. Lipids

36. MLS ONLY High levels of which lipoprotein class are associated with decreased risk of accelerated atherosclerosis?

a chylomicrons
b VLDL
c LDL
d HDL

37. The most consistent analytical error involved in the routine determination of HDL is caused by:
MLS ONLY
a incomplete precipitation of LDL
b coprecipitation of HDL and LDL
c inaccurate protein estimation of HDL
d a small concentration of apoB-containing lipoproteins after precipitation

38. If the LDL is to be calculated by the Friedewald formula, what are the 2 measurements that need to be carried out by the same chemical procedure?
MLS ONLY
a total cholesterol and HDL
b total cholesterol and triglyceride
c triglyceride and chylomicrons
d apolipoprotein A and apolipoprotein B

39. The chemical composition of HDL corresponds to:
MLS ONLY
a trigyceride 60%, cholesterol 15%, protein 10%
b trigyceride 10%, cholesterol 45%, protein 25%
c trigyceride 5%, cholesterol 15%, protein 50%
d trigyceride 85%, cholesterol 5%, protein 2%

40. In familial hypercholesterolemia, the hallmark finding is an elevation of:
MLS ONLY
a low-density lipoproteins
b chylomicrons
c high-density lipoproteins
d apolipoprotein A_1

41. Premature atherosclerosis can occur when which of the following becomes elevated?
MLS ONLY
a chylomicrons
b prostaglandins
c low-density lipoproteins
d high-density lipoproteins

42. Transportation of 60-75% of the plasma cholesterol is performed by:
MLS ONLY
a chylomicrons
b very low-density lipoproteins
c low-density lipoproteins
d high-density lipoproteins

43. Which of the following diseases results from a familial absence of high density lipoprotein?
MLS ONLY
a Krabbe disease
b Gaucher disease
c Tangier disease
d Tay-Sachs disease

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44. A 1-year-old girl with a hyperlipoproteinemia and lipoprotein lipase deficiency has the lipid profile results shown in this table:

test	result
cholesterol	300 mg/dL (7.77 mmol/L)
LDL	increased
HDL	decreased
triglycerides	1200 mg/dL (13.56 mmol/L)
chylomicrons	present

A serum specimen from this patient that is refrigerated overnight would most likely appear:

a clear
b cloudy
c creamy layer over cloudy serum
d creamy layer over clear serum

45. MLS ONLY Which of the following lipid results would be expected to be falsely elevated on a serum specimen from a non-fasting patient?

a cholesterol
b triglyceride
c HDL
d LDL

46. MLS ONLY A 9-month-old boy from Israel has gradually lost the ability to sit up, and develops seizures. He has an increased amount of a phospholipid called GM_2-ganglioside in his neurons, and he lacks the enzyme hexosaminidase A in his leukocytes. These findings suggest:

a Neimann-Pick disease
b Tay-Sachs disease
c phenylketonuria
d Hurler syndrome

47. Apolipoprotein A1 is the major component of which lipoprotein?

a Chylomicrons
b HDL
c LDL
d VLDL

48. The lipoprotein that transports cholesterol away from the tissues is:

a HDL
b IDL
c LDL
d VLDL

49. MLS ONLY A fasting serum sample from an asymptomatic 43-year-old woman is examined visually and chemically with the results shown in this table:

test	result
initial appearance of serum	milky
appearance of serum after overnight refrigeration	cream layer over turbid serum
triglyceride level	2000 mg/dL (22.6 mmol/L)
cholesterol level	550 mg/dL (14.25 mmol/L)

This sample contains predominantly:

a chylomicrons, alone
b chylomicrons and very low-density lipoproteins (VLDL)
c very low-density lipoproteins (VLDL) and low-density lipoproteins (LDL)
d high-density lipoproteins (HDL)

50. Chylomicrons are present in which of the following dyslipidemias?

a familial hypercholesterolemia
b hypertriglyceridemia
c deficiency in lipoprotein lipase activity
d familial hypoalphalipoproteinemia

51. MLS ONLY The function of the major lipid components of the very low-density lipoproteins (VLDL) is to transport:

a cholesterol from peripheral cells to the liver
b cholesterol and phospholipids to peripheral cells
c exogenous triglycerides
d endogenous triglycerides

52. Turbidity in serum suggests elevation of:

a cholesterol
b total protein
c chylomicrons
d albumin

53. MLS ONLY A lipemic serum is separated and frozen at –20°C for assay at a later date. One week later, prior to performing an assay for triglycerides, the specimen should be:

a warmed to 37°C and mixed thoroughly
b warmed to 15°C and centrifuged
c transferred to a glycerated test tube
d discarded and a new specimen obtained

54. As part of a hyperlipidemia screening program, the results shown in this table are obtained from a 25-year-old woman 6 hours after eating:

test	result
triglycerides	260 mg/dL (2.86 mmol/L)
cholesterol	120 mg/dL (3.12 mmol/L)

Which of the following is the **best** interpretation of these results?

a both results are normal, and not affected by the recent meal
b cholesterol is normal, but triglycerides are elevated, which may be attributed to the recent meal
c both results are elevated, indicating a metabolic problem in addition to the nonfasting state
d both results are below normal despite the recent meal, indicating a metabolic problem

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55. Blood is collected in a serum separator tube on a patient who has been fasting since midnight. The time of collection is 7 AM. The laboratory test that should be recollected is:

a triglycerides
b iron
c LD
d sodium

56. Which of the following is the formula for calculating absorbance given the percent transmittance (%T) of a solution?

a $1 - \log(\%T)$
b $\log(\%T)+2$
c $2 \times \log(\%T)$
d $2 - \log(\%T)$

57. The substance that is measured to estimate the serum concentration of triglycerides by most methods is:

a phospholipids
b glycerol
c fatty acids
d pre-beta lipoprotein

58. Which of the following methods for quantitation of high-density lipoprotein is most suited for clinical laboratory use?

a Gomori procedure
b homogeneous
c column chromatography
d agarose gel electrophoresis

C. Heme Derivatives

59. Hemoglobin A_1c represents:

a valine substitution for glutamine at the 6th position of the beta chain of hemoglobin
b ketone formation due to hydrolysis of the alpha and beta C-terminus forming an abnormal hemoglobin
c glycosylation of valine in the polypeptide N-terminus of normal adult hemoglobin
d glucose reduction in the presence of oxygen at the N-terminus of the polypeptide chains of hemoglobin

60. The principle of the occult blood test depends upon the:

MLS ONLY
a coagulase ability of blood
b oxidative power of atmospheric oxygen
c hydrogen peroxide in hemoglobin
d peroxidase-like activity of hemoglobin

61. A breakdown product of hemoglobin is:

MLS ONLY
a lipoprotein
b bilirubin
c hematoxylin
d Bence Jones protein

62. Hemoglobin S can be separated from hemoglobin D by:

a electrophoresis on a different medium and acidic pH
b hemoglobin A_2 quantitation
c electrophoresis at higher voltage
d Kleihauer-Betke acid elution

63. On electrophoresis at alkaline pH, which of the following is the slowest migrating hemoglobin?

a HbA
b HbS
c HbC
d HbF

64. The hemoglobin that is resistant to alkali (KOH) denaturation is:

MLS ONLY

a A
b A_2
c C
d F

65. The bilirubin results shown in the table are obtained on a patient:

day	result
1	4.3 mg/dL (73.5 µmol/L)
2	4.6 mg/dL (78.7 µmol/L)
3	4.5 mg/dL (77.0 µmol/L)
4	2.2 mg/dL (37.6 µmol/L)
5	4.4 mg/dL (75.2 µmol/L)
6	4.5 mg/dL (77.0 µmol/L)

Given that the controls are within range each day, what is a probable explanation for the result on day 4?

a no explanation necessary
b serum, not plasma, was used for testing
c specimen had prolonged exposure to light
d specimen was hemolyzed

66. Urobilinogen is formed in the:

a kidney
b spleen
c liver
d intestine

67. In bilirubin determinations, the purpose of adding a concentrated caffeine solution or methyl alcohol is to:

MLS ONLY

a allow indirect bilirubin to react with color reagent
b dissolve conjugated bilirubin
c precipitate protein
d prevent any change in pH

68. If the total bilirubin is 3.1 mg/dL (53.0 µmol/L) and the conjugated bilirubin is 2.0 mg/dL (34.2 µmol/L), the unconjugated bilirubin is:

MLS ONLY

a 0.5 mg/dL (8.6 µmol/L)
b 1.1 mg/dL (18.8 µmol/L)
c 2.2 mg/dL (37.6 µmol/L)
d 5.1 mg/dL (87.2 µmol/L)

69. The principle of the tablet test for bilirubin in urine or feces is:

MLS ONLY

a reaction between bile and 2,4-dichloronitrobenzene to make a yellow color
b liberation of oxygen by bile to oxidize orthotolidine to make a blue-purple color
c chemical coupling of bile with a diazonium salt to make a brown color
d chemical coupling of bilirubin with a diazonium salt to make a purple color

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70. Ten heparinized plasma samples are assayed for bilirubin by 1) Jendrassik-Grof method on analyzer 1, and by 2) the Evelyn-Malloy method on analyzer 2, in a method comparison study. The results are all 10-20% higher from analyzer 2 most likely because:

a caffeine-benzoate stabilizer in analyzer 2 prevented falsely decreased results as found in analyzer 1
b fibrinogen in the plasma samples caused falsely decreased results in the Jendrassik-Grof method
c alcohol reagents in method 2 cause precipitation of proteins and increased background turbidity
d hemolysis caused greater interference in method 1 than in method 2

71. Serial bilirubin determinations performed by Jendrassik-Grof method are shown in this table:

day	collected	assayed	result
1	7 AM	8 AM	14.0 mg/dL (239.4 µmol/L)
2	7 AM	6 PM	9.0 mg/dL (153.9 µmol/L)
3	6 AM	8 PM	15.0 mg/dL (256.5 µmol/L)

The best explanation for the results is:

a sample on day 2 has mild hemolysis and hemoglobin deterioration
b sample on day 2 has exposure to light
c sample on day 2 shows normal day to day variation
d reagent deterioration is evident on day 3

72. MLS ONLY In the liver, bilirubin is converted to:

a urobilinogen
b urobilin
c bilirubin-albumin complex
d bilirubin diglucuronide

73. MLS ONLY In which of the following disease states is conjugated bilirubin a major serum component?

a biliary obstruction
b hemolysis
c neonatal jaundice
d erythroblastosis fetalis

74. Kernicterus is an abnormal accumulation of bilirubin in:

a heart tissue
b brain tissue
c liver tissue
d kidney tissue

75. MLS ONLY In which of the following conditions does decreased activity of glucuronyl transferase result in increased unconjugated bilirubin and kernicterus in neonates?

a Gilbert disease
b Rotor syndrome
c Dubin-Johnson syndrome
d Crigler-Najjar syndrome

76. MLS ONLY A 21-year-old man with nausea, vomiting, and jaundice has these laboratory findings:

test	patient value	reference range
total serum bilirubin	8.5 mg/dL (145.4 µmol/L)	0-1.0 mg/dL (0.0-17.1 µmol/L)
conjugated serum bilirubin	6.1 mg/dL (104.3 µmol/L)	0-0.5 mg/dL (0.0-8.6 µmol/L)
urine urobilinogen	increased	
fecal urobilinogen	decreased	
urine bilirubin	positive	
AST	300 U/L	0-50 U/L
alkaline phosphatase	170 U/L	0-150 U/L

These can best be explained as representing:

a unconjugated hyperbilirubinemia, probably due to hemolysis
b unconjugated hyperbilirubinemia, probably due to toxic liver damage
c conjugated hyperbilirubinemia, probably due to hepatocellular disease
d conjugated hyperbilirubinemia, probably due to hepatocellular obstruction

77. A stool specimen that appears black and tarlike should be tested for the presence of:

a occult blood
b fecal fat
c trypsin
d excess mucus

78. MLS ONLY What substance gives feces its normal color?

a uroerythrin
b urochrome
c urobilin
d urobilinogen

79. MLS ONLY A condition in which erythrocyte protoporphyrin is increased is:

a acute intermittent porphyria
b iron deficiency anemia
c porphyria cutanea tarda
d acute porphyric attack

80. MLS ONLY The definitive diagnosis for hereditary coproporphyria (HCP) is a markedly increased:

a urine δ-aminolevulinic acid (ALA)
b urine porphobilinogen (PBG)
c fecal coproporphyrin III
d erythrocyte protoporphyrin

81. MLS ONLY A fresh urine sample is received for analysis for "porphyrins" or "porphyria" without further information or specifications. Initial analysis should include:

a porphyrin screen and quantitative total porphyrin
b quantitative total porphyrin and porphobilinogen screen
c porphyrin and porphobilinogen screen
d porphobilinogen screen and ion-exchange analysis for porphobilinogen

82. MLS ONLY Which of the following enzymes of heme biosynthesis is inhibited by lead?

a δ-aminolevulinate dehydratase
b porphobilinogen synthase
c uroporphyrinogen synthase
d bilirubin synthetase

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83. In amniotic fluid, the procedure used to detect hemolytic disease of the newborn is:

a measurement of absorbance at 450 nm
b creatinine
c lecithin/sphingomyelin ratio
d estriol

84. Hemoglobin S can be separated from hemoglobin D by which of the following methods?

a citrate agar gel electrophoresis at pH 5.9
b thin-layer chromatography
c alkali denaturation
d agarose gel electrophoresis at pH 8.4

85. Before unconjugated bilirubin can react with Ehrlich diazo reagent, which of the following must be added?

a acetone
b ether
c distilled water
d caffeine

86. The most widely used methods for bilirubin measurement are those based on the:

a Jaffe reaction
b Schales and Schales method
c 8-hydroxyquinoline reaction
d Jendrassik-Grof method

87. In the Evelyn-Malloy method for the determination of bilirubin, the reagent that is reacted with bilirubin to form a purple azobilirubin is:

a dilute sulfuric acid
b diazonium sulfate
c sulfobromophthalein
d diazotized sulfanilic acid

88. In the Jendrassik-Grof method for the determination of serum bilirubin concentration, quantitation is obtained by measuring the green color of:

a azobilirubin
b bilirubin glucuronide
c urobilin
d urobilinogen

89. In the Jendrassik-Grof reaction for total bilirubin, alkaline tartrate is added to:

a form diazo bilirubin, a reddish chromogen
b eliminate many spectrophotometric interferences
c act as an accelerator
d react with δ-bilirubin

90. The laboratory results shown in this table are obtained on a patient suspected to have liver disease.

test	patient value
total bilirubin	9.5 mg/dl
unconjugated bilirubin	8.5 mg/dl
urine bilirubin	negative
urine urobilinogen	increased

The type of jaundice would be classified as:

a prehepatic
b viral hepatic
c cirrhotic hepatic
d posthepatic

91. The laboratory results shown in this table are obtained on a patient suspected to have liver disease.

test	patient value
total bilirubin	9.5 mg/dL
unconjugated bilirubin	1.5 mg/dL
urine bilirubin	positive
urine urobilinogen	decreased

The type of jaundice would be classified as:

a prehepatic
b viral hepatic
c cirrhotic hepatic
d posthepatic

92. A quantitative measurement of urobilinogen uses the following reagent:

a diazo
b DMSO
c Erhlich
d ferric chloride

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II. Proteins & Enzymes

A. Enzymes

93. The results the biochemical profile shown in this table are most consistent with:

MLS ONLY

test	patient value	reference range
total protein	7.3 g/dL (73 g/L)	6.0-8.0 g/dL (60-80 g/L)
albumin	4.1 g/dL (41 g/L)	3.5-5.0 g/dL (35-50 g/L)
calcium	9.6 mg/dL (2.4 mmol/L)	8.5-10.5 mg/dL (2.1-2.6 mmol/L)
phosphorus	3.3 mg/dL (1.06 mmol/L)	2.5-4.5 mg/dL (0.80-1.45 mmol/L)
glucose	95 mg/dL (5.2 mmol/L)	65-110 mg/dL (3.6-6.1 mmol/L)
BUN	16 mg/dL (5.71 mmol/L)	10-20 mg/dL (3.57-7.14 mmol/L)
uric acid	6.0 mg/dL (356.9 µmol/L)	2.5-8.0 mg/dL (148.7-475.8 µmol/L)
creatinine	1.2 mg/dL (106.1 µmol/L)	0.7-1.4 mg/dL (61.9-123.8 µmol/L)
total bilirubin	3.7 mg/dL (63.3 µmol/L)	0.2-0.9 mg/dL (3.4-15.4 µmol/L)
alkaline phosphatase	275 U/L	30-80 U/L
lactate dehydrogenase	185 U/L	100-225 U/L
AST	75 U/L	10-40 U/L

a viral hepatitis
b hemolytic anemia
c common bile duct stone
d chronic active hepatitis

94. The most specific enzyme test for acute pancreatitis is:

a acid phosphatase
b trypsin
c amylase
d lipase

95. Which of the following enzymes are used in the diagnosis of acute pancreatitis?

a amylase (AMS)
b aspartate aminotransferase (AST)
c gamma-glutamyl transferase (GGT)
d lactate dehydrogenase (LD)

96. Which of the following enzymes catalyzes the conversion of starch to glucose and maltose?

a malate dehydrogenase (MD)
b amylase (AMS)
c creatine kinase (CK)
d isocitric dehydrogenase (ICD)

97. Which of the following sets of results would be consistent with macroamylasemia?

MLS ONLY

a normal serum amylase and elevated urine amylase values
b increased serum amylase and normal urine amylase values
c increased serum and urine amylase values
d normal serum and urine amylase values

98. In acute pancreatitis, the following results would be expected:
MLS ONLY
a LD isoenzyme 1 elevates higher than LD isoenzyme 2 within 72 hours
b lipase elevates within 4-8 hours of an attack, peaks at 24 hours and normalizes in 48 hours
c amylase and lactate rise successively in the first 24 hours and stay elevated 4 days
d amylase and lipase elevate in a few hours, but amylase normalizes within 3 days

99. What enzyme is measured in the reaction of maltoheptaose-nitrophenol producing glucose fragments and 4-nitrophenol?

a amylase
b hexokinase
c pyruvate kinase
d alkaline phosphatase

100. Aspartate amino transferase (AST) is characteristically elevated in diseases of the:

a liver
b kidney
c intestine
d pancreas

101. Amino transferase enzymes catalyze the:

a exchange of amino groups and sulfhydryl groups between alpha-amino and sulfur-containing acids
b exchange of amino and keto groups between alpha-amino and alpha-keto acids
c hydrolysis of amino acids and keto acids
d reversible transfer of hydrogen from amino acids to coenzyme

102. Aspartate aminotransferase (AST) and alanine aminotransferase (ALT) are both elevated in which of the following diseases?

a muscular dystrophy
b viral hepatitis
c pulmonary emboli
d infectious mononucleosis

103. A significant source of which of the following enzymes within RBCs should be considered during interpretation of and sources of interference in serum enzyme analysis?

a AST
b ALT
c GGT
d CK

104. Malic dehydrogenase is added to the aspartate aminotransaminase (AST) reaction to catalyze the conversion of:
MLS ONLY
a alpha-ketoglutarate to aspartate
b alpha-ketoglutarate to malate
c aspartate to oxalacetate
d oxalacetate to malate

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105. The results in this table are most consistent with:

MLS ONLY

enzyme	result
alkaline phosphatase	slight increase
aspartate amino transferase	marked increase
alanine amino transferase	marked increase
gamma-glutamyl transferase	slight increase

a acute hepatitis
b chronic hepatitis
c obstructive jaundice
d liver hemangioma

106. Which of the following clinical disorders is associated with the greatest elevation of lactate dehydrogenase isoenzyme 1?

a pneumonia
b glomerulonephritis
c pancreatitis
d pernicious anemia

107. The enzyme, which exists chiefly in skeletal muscle, heart, and brain, is grossly elevated in active muscular dystrophy, and rises early in myocardial infarction is:

a lipase
b transaminase
c lactate dehydrogenase
d creatine kinase

108. The enzyme present in almost all tissues that may be separated by electrophoresis into 5 components is:

a lipase
b transaminase
c creatine kinase
d lactate dehydrogenase

109. A common cause of a falsely increased LD_1 fraction of lactic dehydrogenase is:

a specimen hemolysis
b liver disease
c congestive heart failure
d older specimen

110. The presence of which of the following isoenzymes indicates acute myocardial damage?

a CKMM
b CKMB
c CKBB
d none

111. In which of the following conditions would a **normal** level of creatine kinase be found?

a acute myocardial infarct
b hepatitis
c progressive muscular dystrophy
d intramuscular injection

112. Of the following diseases, the one most often associated with elevations of lactate dehydrogenase isoenzymes 4 and 5 on electrophoresis is:

a liver disease
b hemolytic anemia
c myocardial infarction
d pulmonary edema

113. When myocardial infarction occurs, the first enzyme to become elevated is:

a CK
b LD
c AST
d ALT

114. A scanning of a CK isoenzyme fractionation revealed 2 peaks: a slow cathodic peak (CKMM) and an intermediate peak (CKMB). A possible interpretation for this pattern is:

a brain tumor
b muscular dystrophy
c myocardial infarction
d viral hepatitis

115. MLS ONLY An electrophoretic separation of lactate dehydrogenase isoenzymes that demonstrates an elevation in LD1 and LD2 in a "flipped" pattern is consistent with:

a myocardial infarction
b viral hepatitis
c pancreatitis
d renal failure

116. MLS ONLY A large decrease in the α_1 peak in serum electrophoresis is indicative of:

a acute phase reaction
b antitrypsin deficiency
c chronic inflammation
d nephrotic syndrome

117. A 10-year-old child is admitted to pediatrics with an initial diagnosis of skeletal muscle disease. The best confirmatory tests would be:

a creatine kinase and isocitrate dehydrogenase
b gamma-glutamyl transferase and alkaline phosphatase
c aldolase and creatine kinase
d lactate dehydrogenase and malate dehydrogenase

118. In the immunoinhibition phase of the CKMB procedure:

a M subunit is inactivated
b B subunit is inactivated
c MB is inactivated
d BB is inactivated

119. The presence of increased CKMB activity on a CK electrophoresis pattern is most likely found in a patient suffering from:

a acute muscular stress following strenuous exercise
b malignant liver disease
c myocardial infarction
d severe head injury

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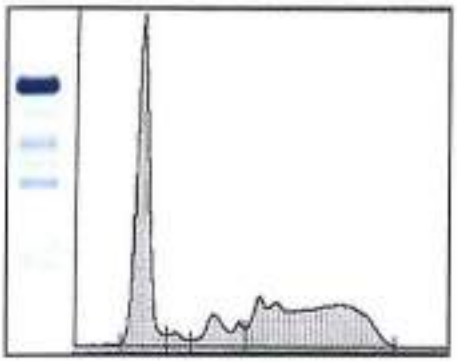
120. The electrophoresis pattern shown in this illustration and table is most likely caused by:



test	percentage	patient	reference range
total protein	100	6.6 g/L	6.4-8.3
albumin	45	3.0 g/L	3.9-5.1
α_1 globulin	8	0.5 g/L	0.2-0.4
α_2 globulin	9	0.5 g/L	0.4-0.8
β-γ globulin	38	2.5 g/L	0.5-1.0 β 0.6-1.3 γ

a acute inflammation
b hepatic cirrhosis
c nephrotic syndrome
d plasma cell neoplasm

121. Increased serum lactic dehydrogenase activity due to elevation of fast fraction (1 and 2) on electrophoretic separation is caused by:

a nephrotic syndrome
b hemolytic anemia
c pancreatitis
d hepatic damage

122. A serum sample drawn in the emergency room from a 42-year-old man yielded the laboratory results in this table:

test	patient	reference range
CK	185 U/L	15-160 U/L
AST	123 U/L	0-48 U/L
CKMB	6 U/L	2-12 U/L

Which of these conditions might account for these values?

a crush injury to the thigh
b cerebrovascular accident
c pulmonary infarction
d early acute hepatitis

123. The results in this table are most consistent with:

enzyme	result
alkaline phosphatase	marked increase
aspartate amino transferase	slight increase
alanine amino transferase	slight increase
gamma-glutamyl transferase	marked increase

a acute hepatitis
b osteitis fibrosa
c chronic hepatitis
d obstructive jaundice

124. The results in this table are most consistent with:

enzyme	result
alkaline phosphatase	slight increase
aspartate aminotransferase	slight increase
alanine aminotransferase	slight increase
gamma-glutamyl transferase	slight increase

a acute hepatitis
b chronic hepatitis
c obstructive jaundice
d liver hemangioma

125. What specimen preparation is used to perform the alkaline phosphatase isoenzyme determination?

a serum is divided into 2 aliquots, one is frozen and the other is refrigerated
b serum is divided into 2 aliquots, one is heated at 56°C and the other is unheated
c no preparation is necessary since the assay uses EDTA plasma
d protein-free filtrate is prepared first

126. Regan isoenzyme has the same properties as alkaline phosphatase that originates in the:

a skeleton
b kidney
c intestine
d placenta

127. The **most** heat labile fraction of alkaline phosphatase is obtained from:

a liver
b bone
c intestine
d placenta

128. The most sensitive enzymatic indicator for liver damage from ethanol intake is:

a alanine aminotransferase (ALT)
b aspartate aminotransferase (AST)
c gamma-glutamyl transferase (GGT)
d alkaline phosphatase

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129. What is the enzyme measured in the reaction of glycylglycine + γ-glutamyl-p-nitroaniline in which an increase in absorbance at 410 nm is detected due to formation of p-nitroaniline?

a glutamic oxaloacetate transferase
b γ-glutamyl transferase
c γ-globulinase
d glycine decarboxylase

130. Cholinesterase levels are generally assayed to aid in diagnosis of:

a pancreatitis
b methamphetamine overdose
c organophosphate poisoning
d hepatobiliary disease

131. Isoenzyme assays are performed to improve:

a precision
b accuracy
c sensitivity
d specificity

132. The protein portion of an enzyme complex is called the:

a apoenzyme
b coenzyme
c holoenzyme
d proenzyme

133. Which of the following chemical determinations may be of help in establishing the presence of seminal fluid?

a lactic dehydrogenase (LD)
b isocitrate dehydrogenase (ICD)
c acid phosphatase
d alkaline phosphatase

134. MLS ONLY Which of the following enzyme substrates is the most specific for prostatic acid phosphatase for quantitative endpoint reactions?

a p-nitrophenylphosphate
b thymolphthalein monophosphate
c beta-naphthol-phosphate
d beta-glycerophosphate

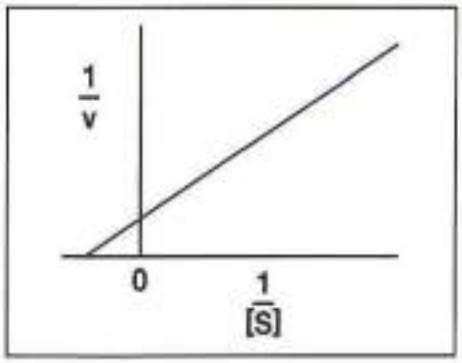
135. MLS ONLY Lactate dehydrogenase, malate dehydrogenase, isocitrate dehydrogenase, and glucose-6-phosphate dehydrogenase all:

a are liver enzymes
b are cardiac enzymes
c catalyze oxidation-reduction reactions
d are class III enzymes

136. Which of the following is the Henderson-Hasselbalch equation?

a $pK_a = pH + \log([acid]/[salt])$
b $pK_a = pH + \log([salt]/[acid])$
c $pH = pK_a + \log([acid]/[salt])$
d $pH = pK_a + \log([salt]/[acid])$

137. MLS ONLY The figure shows the reciprocal of the measured velocity of an enzyme reaction plotted against the reciprocal of the substrate concentration.



True statements about this figure include:

a the intercept of the line on the abscissa (x-axis) can be used to calculate the V_{max}
b the straight line indicates that the enzyme reaction proceeds according to zero order kinetics
c the intercept on the abscissa (x-axis) can be used to calculate the Michaelis-Menten constant
d the fact that the substrate concentration is plotted on both sides of the zero point indicates that the reaction is reversible

138. The absorbance readings shown in this table are taken in an assay of LD levels in which the total volume is 1.0 mL, sample volume is 10 µL, and the molar absorptivity of the product NADH measured at 340 nm in a 1 cm cuvet is 6220 Abs/mol/L. What is the LD enzyme activity in IU/L?

time (seconds)	Abs
20	0.023
40	0.046
80	0.089

a 148
b 370
c 740
d 1061

139. In the assay of lactate dehydrogenase, which of the following products is actually measured?

a NADH
b ATP
c lactic acid
d pyruvic acid

140. In the assay of lactate dehydrogenase (LD), the reaction is dependent upon which of the following coenzyme systems?

a NAD/NADH
b ATP/ADP
c Fe^{2+}/Fe^{3+}
d Cu/Cu^{2+}

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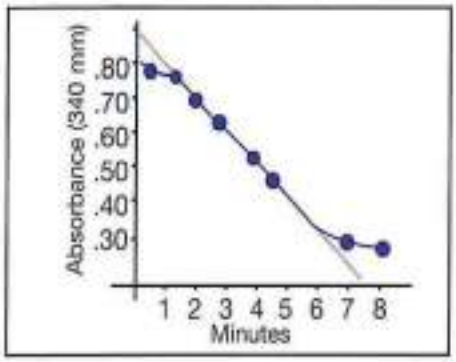
141. This illustration represents the change in absorbance at 340 nm over a period of 8 minutes in an assay for lactate dehydrogenase



True statements about this figure include:

a the reaction follows zero-order kinetics between 5 and 8 minutes
b the reaction is proceeding from lactate to pyruvate
c nonlinearity after 6 minutes is due to substrate exhaustion
d the change in absorbance is due to reduction of NAD to NADH

142. In competitive inhibition of an enzyme reaction, the:

MLS ONLY

a inhibitor binds to the enzyme at the same site as does the substrate
b inhibitor often has a chemical structure different to that of the substrate
c activity of the reaction can be decreased by increasing the concentration of the substrate
d activity of the reaction can be increased by decreasing the temperature

143. The most common cause of rapid nonlinearity following the timed lag phase in an enzyme kinetic assay is resolved by:

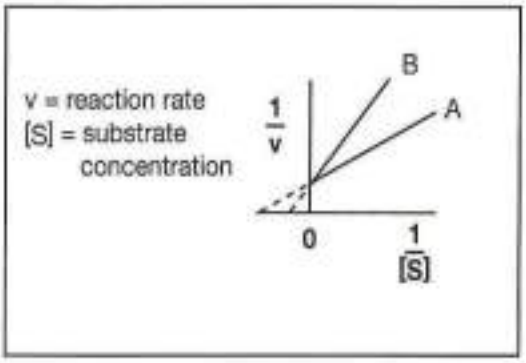
MLS ONLY

a addition of more coenzyme such as NADH to the reaction following the timed lag phase.
b preventing extraneous protein from binding to the E-S complex by making a protein free filtrate.
c decreasing the sample volume to dilute the enzyme so that substrate remains in excess during the reaction.
d eliminating metallic ions that may be making structural changes and inhibition to the enzyme active site.

144. The International Federation for Clinical Chemistry (IFCC) recommends the use of methods such as the Bessey-Lowry-Brock method for determining alkaline phosphatase activity. The substrate used in this type of method is:

a monophosphate
b phenylphosphate
c disodium phenylphosphate
d para-nitrophenylphosphate

145. The illustration represents a Lineweaver-Burk plot of 1/v vs 1/[S] in an enzyme reaction and the assumptions shown should be made: (MLS ONLY)



The enzyme concentration is the same for reactions A and B
The substrate concentration is in excess for reactions A and B
Reaction A occurs under ideal conditions

Which of the following statements about reaction B is true?

a it illustrates noncompetitive inhibition
b it illustrates competitive inhibition
c it illustrates neither competitive nor noncompetitive inhibition
d it could be the result of heavy metal contamination

B. Proteins & Other Nitrogen-Containing Compounds

146. The main function of serum albumin in the peripheral blood is to:

a maintain colloidal osmotic pressure
b increase antibody production
c increase fibrinogen formation
d maintain blood viscosity

147. In a pleural effusion caused by *Streptococcus pneumoniae*, the ratio of pleural fluid protein to the serum protein would likely be: (MLS ONLY)

a >0.6
b >0.5
c <0.3
d <0.7

148. The first step in analyzing a 24-hour urine specimen for quantitative urine protein is:

a subculture the urine for bacteria
b add the appropriate preservative
c screen for albumin using a dipstick
d measure the total volume

149. When performing glucose and protein analysis on spinal fluid, it is recommended to: (MLS ONLY)

a perform the test as usual
b concurrently analyze a blood sample
c immediately centrifuge the specimen
d dilute the specimen with deionized water

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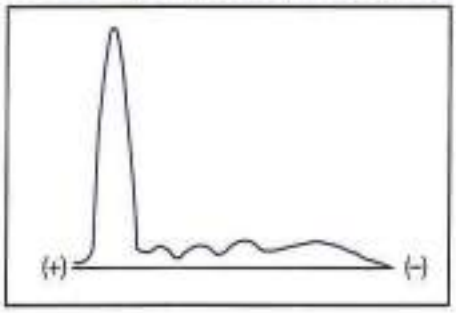
150. The direction in which the proteins migrate (ie, toward anode or cathode) during electrophoretic separation of serum proteins, at pH 8.6, is determined by: MLS ONLY

- **a** the ionization of the amine groups, yielding a net positive charge
- **b** the ionization of the carboxyl groups, yielding a net negative charge
- **c** albumin acting as a zwitterion
- **d** the density of the gel layer

151. The protein that has the highest dye-binding capacity is: MLS ONLY

- **a** albumin
- **b** alpha globulin
- **c** beta globulin
- **d** gamma globulin

152. The serum protein electrophoresis pattern shown is obtained on cellulose acetate at pH 8.6. MLS ONLY



Identify the serum protein fraction on the left of the illustration.

- **a** gamma globulin
- **b** albumin
- **c** alpha-1 globulin
- **d** alpha-2 globulin

153. The biuret reaction for the analysis of serum protein depends on the number of: MLS ONLY

- **a** free amino groups
- **b** free carboxyl groups
- **c** peptide bonds
- **d** tyrosine residues

154. The principle of the Biuret reaction is the result of the formation of a complex of:

- **a** $CuSO_4$ with peptide bonds
- **b** $NaNO_3$ with protein chains
- **c** BCG with albumin
- **d** chromogen with amino acids

155. In protein electrophoresis, using agarose gel medium and pH 8.6 buffer, an unexpected protein band migrates between the beta and gamma region could most likely be due to: MLS ONLY

- **a** oligoclonal proteins from multiple sclerosis
- **b** fibrinogen from incomplete clotting
- **c** bis-albumin
- **d** alpha-1 antitrypsin

156. The relative migration rate of proteins on agarose gel is based on: MLS ONLY

- **a** molecular weight and concentration
- **b** surface charge and molecular size
- **c** charge-to-mass ratio
- **d** molecular sieving

157. The order of migration in serum protein electrophoresis at pH 8.6, beginning with the fastest migration is as follows: MLS ONLY

a albumin, alpha-1 globulin, alpha-2 globulin, beta globulin, gamma globulin
b alpha-1 globulin, alpha-2 globulin, beta globulin, gamma globulin, albumin
c albumin, alpha-2 globulin, alpha-1 globulin, beta globulin, gamma globulin
d gamma globulin, beta globulin, alpha-2 globulin, alpha-1 globulin, albumin

158. Which of the following amino acids is associated with sulfhydryl group? MLS ONLY

a cysteine
b glycine
c serine
d tyrosine

159. Maple syrup urine disease is characterized by an increase in which of the following urinary amino acids? MLS ONLY

a phenylalanine
b tyrosine
c valine, leucine, and isoleucine
d cystine and cysteine

160. Increased serum albumin concentrations are seen in which of the following conditions? MLS ONLY

a nephrotic syndrome
b acute hepatitis
c chronic inflammation
d dehydration

161. The data in this table is obtained from a cellulose acetate protein electrophoresis scan:

test	result
albumin area	75 units
gamma globulin area	30 units
total area	180 units
total protein	6.5 g/dL (65 g/L)

The gamma globulin content in g/dL is:

a 1.1 g/dL (11 g/L)
b 2.7 g/dL (27 g/L)
c 3.8 g/dL (38 g/L)
d 4.9 g/dL (49 g/L)

162. A patient is admitted with biliary cirrhosis. If a serum protein electrophoresis is performed, which of the following globulin fractions will be most often be elevated? MLS ONLY

a alpha-1
b alpha-2
c beta
d gamma

163. Which of the following serum protein fractions is most likely to be elevated in patients with nephrotic syndrome? MLS ONLY

a alpha-1 globulin
b albumin
c alpha-2 globulin and beta globulin
d beta globulin and gamma globulin

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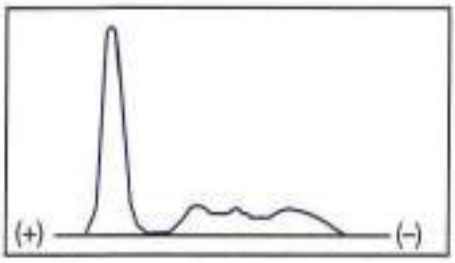
164. The electrophoresis pattern in the illustration and table is consistent with:

MLS ONLY

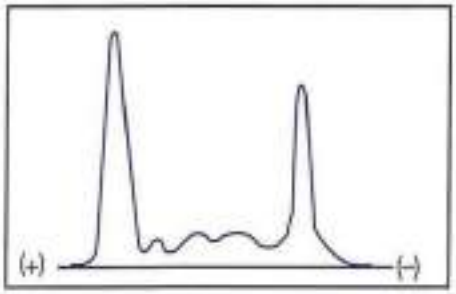


test	patient value	reference range
total protein	7.3 g/dL (73 g/L)	6.0-8.0 g/dL (60-80 g/L)
albumin	4.2 g/dL (42 g/L)	3.6-5.2 g/dL (36-52 g/L)
alpha-1	0.0 g/dL (0 g/L)	0.1-0.4 g/dL (1-4 g/L)
alpha-2	0.9 g/dL (9 g/L)	0.4-1.0 g/dL (4-10 g/L)
beta	0.8 g/dL (8 g/L)	0.5-1.2 g/dL (5-12 g/L)
gamma	1.4 g/dL (14 g/L)	0.6-1.6 g/dL (6-16 g/L)

a cirrhosis
b monoclonal gammopathy
c polyclonal gammopathy (eg, chronic inflammation)
d alpha-1 antitrypsin deficiency; severe emphysema

165. The serum electrophoresis pattern in the illustration and table is consistent with

MLS ONLY

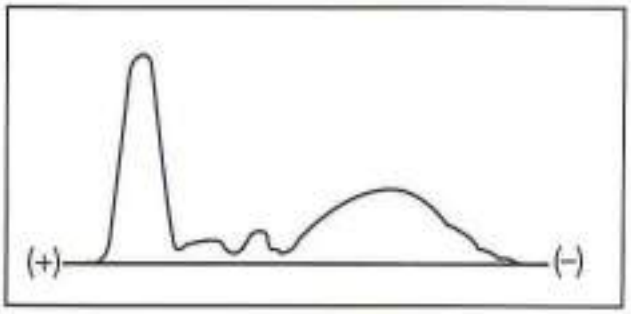


test	patient value	reference range
total protein	8.9 g/dL (89 g/L)	6.0-8.0 g/dL (60-80 g/L)
albumin	4.8 g/dL (48 g/L)	3.6-5.2 g/dL (36-52 g/L)
alpha-1	0.3 g/dL (3 g/L)	0.1-0.4 g/dL (1-4 g/L)
alpha-2	0.7 g/dL (7 g/L)	0.4-1.0 g/dL (4-10 g/L)
beta	0.8 g/dL (8 g/L)	0.5-1.2 g/dL (5-12 g/L)
gamma	2.3 g/dL (23 g/L)	0.6-1.6 g/dL (6-16 g/L)

a cirrhosis
b acute inflammation
c monoclonal gammopathy
d polyclonal gammopathy (eg, chronic inflammation)

166. The electrophoresis pattern in the illustration and table is consistent with:

MLS ONLY



test	patient value	reference range
total protein	6.1 g/dL (61 g/L)	6.0-8.0 g/dL (60-80 g/L)
albumin	2.3 g/dL (23 g/L)	3.6-5.2 g/dL (36-52 g/L)
alpha-1	0.2 g/dL (2 g/L)	0.1-0.4 g/dL (1-4 g/L)
alpha-2	0.5 g/dL (5 g/L)	0.4-1.0 g/dL (4-10 g/L)
beta	1.2 g/dL (12 g/L)	0.5-1.2 g/dL (5-12 g/L)
gamma	1.9 g/dL (19 g/L)	0.6-1.6 g/dL (6-16 g/L)

This pattern is consistent with:

a cirrhosis
b acute inflammation
c polyclonal gammopathy (eg, chronic inflammation)
d alpha-1 antitrypsin deficiency; severe emphysema

167. A characteristic of the Bence Jones protein that is used to distinguish it from other urinary proteins is its solubility:

MLS ONLY

a in ammonium sulfate
b in sulfuric acid
c at 40-60°C
d at 100°C

168. The electrophoretic pattern of plasma sample as compared to a serum sample shows a:

MLS ONLY

a broad prealbumin peak
b sharp fibrinogen peak
c diffuse pattern because of the presence of anticoagulants
d decreased globulin fraction

169. At a pH of 8.6 the gamma globulins move toward the cathode, despite the fact that they are negatively charged. What is this phenomenon called?

MLS ONLY

a reverse migration
b molecular sieve
c endosmosis
d migratory inhibition factor

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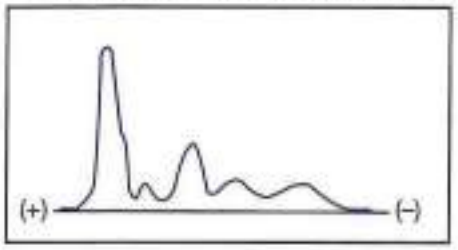
170. The electrophoresis pattern in the illustration and table is consistent with:

MLS ONLY



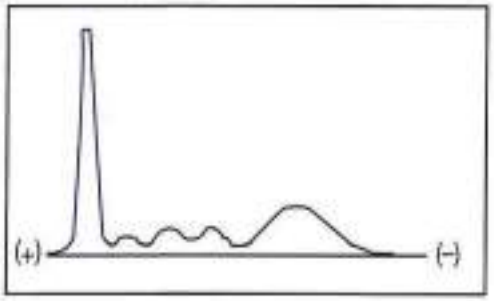
test	patient value	reference range
total protein	7.8 g/dL (78 g/L)	6.0-8.0 g/dL (60-80 g/L)
albumin	3.0 g/dL (30 g/L)	3.6-5.2 g/dL (36-52 g/L)
alpha-1	0.4 g/dL (4 g/L	0.1-0.4 g/dL (1-4 g/L)
alpha-2	1.8 g/dL (18 g/L)	0.4-1.0 g/dL (4-10 g/L)
beta	0.5 g/dL (5 g/L)	0.5-1.2 g/dL (5-12 g/L)
gamma	1.1 g/dL (11 g/L)	0.6-1.6 g/dL (6-16 g/L)

The serum protein electrophoresis pattern is consistent with:

a cirrhosis
b acute inflammation
c polyclonal gammopathy (eg, chronic inflammation)
d alpha-1 antitrypsin deficiency; severe emphysema

171. The electrophoresis pattern in the illustration and table is consistent with:

MLS ONLY



test	patient value	reference range
total protein	8.5 g/dL (85 g/L)	6.0-8.0 g/dL (60-80 g/L)
albumin	4.3 g/dL (43 g/L)	3.6-5.2 g/dL (36-52 g/L)
alpha-1	0.3 g/dL (3 g/L)	0.1-0.4 g/dL (1-4 g/L)
alpha-2	0.7 g/dL (7 g/L)	0.4-1.0 g/dL (4-10 g/L)
beta	0.9 g/dL (9 g/L)	0.5-1.2 g/dL (5-12 g/L)
gamma	2.3 g/dL (23 g/L)	0.6-1.6 g/dL (6-16 g/L)

a nephrotic syndrome
b monoclonal gammopathy
c polyclonal gammopathy (eg, chronic inflammation)
d alpha-1 antitrypsin deficiency; severe emphysema

172. Analysis of CSF for oligoclonal bands is used to screen for which of the following disease states? MLS ONLY

a multiple myeloma
b multiple sclerosis
c myasthenia gravis
d von Willebrand disease

173. The identification of Bence Jones protein is best accomplished by: MLS ONLY

a a sulfosalicylic acid test
b urine reagent strips
c immunofixation electrophoresis
d immunoelectrophoresis

174. To assure an accurate ammonia level result, the specimen should be:

a incubated at 37°C prior to testing
b spun and separated immediately, tested as routine
c spun, separated, iced, and tested immediately
d stored at room temperature until tested

175. Erroneous ammonia levels can be eliminated by all of the following except:

a assuring water and reagents are ammonia-free
b separating plasma from cells and performing test analysis as soon as possible
c drawing the specimen in a prechilled tube and immersing the tube in ice
d storing the specimen protected from light until the analysis is done

176. A critically ill patient becomes comatose. The physician believes the coma is due to hepatic failure. The assay most helpful in this diagnosis is: MLS ONLY

a ammonia
b ALT
c AST
d GGT

177. A serum sample demonstrates an elevated result when tested with the Jaffe reaction. This indicates:

a prolonged hypothermia
b renal functional impairment
c pregnancy
d arrhythmia

178. In order to prepare 100 mL of 15 mg/dL BUN (5.35 mmol/L) working standard from a stock standard containing 500 mg/dL (178.5 mmol/L) of urea nitrogen, the amount of stock solution that should be used is: MLS ONLY

a 3 mL
b 5 mL
c 33 mL
d 75 mL

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179. In an assay of LD levels 10 µL of sample is added to reagent to make a total volume of 1 mL. What is the dilution?

a 1/10
b 1/90
c 1/100
d 1/1000

180. A patient with glomerulonephritis is most likely to present with the following serum results:

MLS ONLY

a creatinine decreased
b calcium increased
c phosphorous decreased
d BUN increased

181. The principle excretory form of nitrogen is:

MLS ONLY

a amino acids
b creatinine
c urea
d uric acid

182. In the Jaffe reaction, creatinine reacts with:

MLS ONLY

a alkaline sulfasalazine solution to yield an orange-yellow complex
b potassium iodide to yield a reddish-purple complex
c sodium nitroferricyanide to yield a reddish-brown color
d alkaline picrate solution to yield an orange-red complex

183. Creatinine clearance is used to estimate the:

a tubular secretion of creatinine
b glomerular secretion of creatinine
c renal glomerular and tubular mass
d glomerular filtration rate

184. A blood creatinine value of 5.0 mg/dL (442.0 µmol/L) is most likely to be found with which of the following blood values?

a osmolality: 292 mOsm/kg
b uric acid: 8 mg/dL (475.8 µmol/L)
c urea nitrogen: 80 mg/dL (28.56 mmol/L)
d ammonia: 80 µg/dL (44 µmol/L)

185. Technical problems encountered during the collection of an amniotic fluid specimen causes doubt as to whether the specimen is amniotic in origin. Which of the following procedures would best establish that the fluid is amniotic in origin?

MLS ONLY

a measurement of absorbance at 450 nm
b creatinine measurement
c lecithin/sphingomyelin ratio
d human amniotic placental lactogen (HPL)

186. Which of the following represents the end product of purine metabolism in humans?

MLS ONLY

a AMP and GMP
b DNA and RNA
c allantoin
d uric acid

187. The troponin complex consists of:

MLS ONLY

a troponin T, calcium and tropomyosin
b troponin C, troponin I and troponin T
c troponin I, actin, and tropomyosin
d troponin C, myoglobin, and actin

188. The presence of C-reactive protein in the blood is an indication of:

MLS ONLY

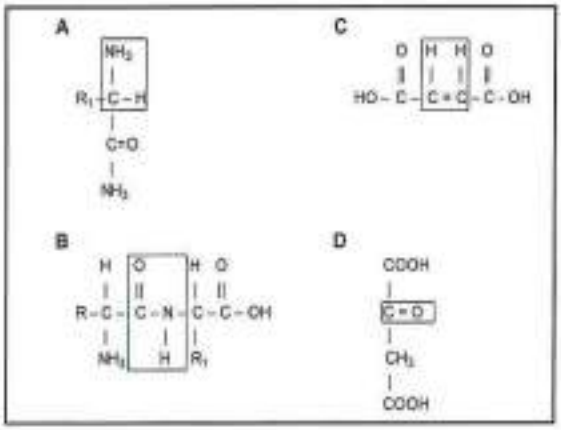
a a recent streptococcal infection
b recovery from a pneumococcal infection
c an inflammatory process
d a state of hypersensitivity

189. Oligoclonal bands are present on electrophoresis of concentrated CSF and also on concurrently tested serum of the same patient. The proper interpretation is:

MLS ONLY

a diagnostic for primary CNS tumor
b diagnostic for multiple sclerosis
c CNS involvement by acute leukemia
d need further testing to rule out multiple sclerosis

190. Which labeled part of this figure is an example of a peptide bond?



a A
b B
c C
d D

191. 90% of the copper present in the blood is bound to:

a transferrin
b ceruloplasmin
c albumin
d cryoglobulin

192. Which of the following determinations is useful in prenatal diagnosis of open neural tube defects?

MLS ONLY

a amniotic fluid alpha-fetoprotein
b amniotic fluid estriol
c maternal serum estradiol
d maternal serum estrone

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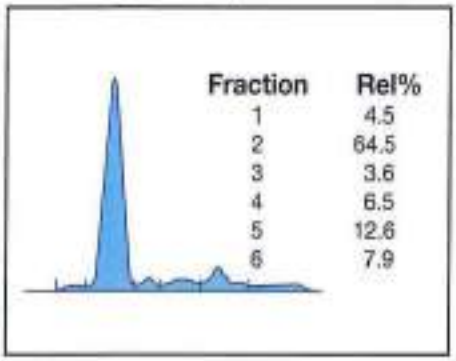
193. The electrophoresis pattern in the illustration is consistent with a(n):

MLS ONLY



a normal serum protein pattern
b normal CSF protein pattern
c abnormal serum protein pattern
d abnormal CSF protein pattern

194. Serum haptoglobin is:

MLS ONLY

a decreased in patients with tissue injury and neoplasia
b increased in patients with prosthetic heart valves
c decreased in hemolytic anemia
d increased in *in vitro* hemolysis

195. "Laboratory A" measures maternal serum alpha-fetoprotein (MSAFP) at 16-18 weeks' gestation as a screen for fetal disorders. The 16-week MSAFP median for Lab A is 32 µg/L. A 37-year-old woman has an MSAFP level of 34 µg/L at her 16th week. This result is consistent with:

MLS ONLY

a a normal MSAFP level for 16 weeks' gestation
b possible neural tube defect, including spina bifida
c possible multiple birth (ie, twins)
d possible trisomy disorder, including Down syndrome

196. In developing the reference for a new EIA for CEA, the range for the normal population is broader than that published by the vendor. Controls are acceptable with a narrow coefficient of variation. This may be explained by:

MLS ONLY

a positive interference by another tumor marker
b population skewed to a younger age
c improper temperature control during assay
d inclusion of nonsmokers and smokers in the study population

197. Clinical assays for tumor markers are most important for:

a screening for the presence of cancer
b monitoring the course of a known cancer
c confirming the absence of disease
d identifying patients at risk for cancer

198. Detection of which of the following substances is most useful to monitor the course of a patient with testicular cancer?

a alpha-fetoprotein
b carcinoembryonic antigen
c prolactin
d testosterone

199. Increased concentrations of alpha-fetoprotein (AFP) in adults are most characteristically associated with:

a hepatocellular carcinoma
b alcoholic cirrhosis
c chronic active hepatitis
d multiple myeloma

200. Carcinoembryonic antigen (CEA) is most likely to be produced in a malignancy involving the:

a brain
b testes
c bone
d colon

201. Which of the following is useful in the detection and management of carcinoma of the prostate?

a total prostate-specific antigen
b prostatic acid phosphatase
c human chorionic gonadotropin
d alpha-fetoprotein

202. A very high PSA level is expected; however, a low level is obtained. The explanation could be:

a African-American patient
b Hook effect
c smoking patient
d collected after digital rectal exam

203. Which of the following statements most correctly describes the utility of clinical laboratory assays for tumor markers?

a tumor markers are useful to screen asymptomatic patients for tumors
b tumor markers are highly specific
c tumor markers indicate the likelihood of an individual developing a tumor
d tumor markers are useful in tracking the efficacy of treatment

204. Cancer antigen 125 (CA 125) is a tumor marker associated with:

a breast carcinoma
b colon cancer
c lung cancer
d ovarian and endometrial carcinoma

205. In addition to carcinoma of the prostate, elevated prostate-specific antigen (PSA) can occur due to:

a aspirin therapy
b exogenous steroid use
c benign prostatic hyperplasia
d statin therapy (cholesterol lowering drug)

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206. In monitoring glomerular function, which of the following tests has the highest sensitivity?

IMLS ONLY

a urine sodium
b BUN/creatinine ratio
c creatinine clearance
d urea clearance

207. A 24-hour urine specimen (total volume = 1136 mL) is submitted to the laboratory for quantitative urine protein. Calculate the amount of protein excreted per day, if the total protein is 52 mg/dL.

a 591 mg
b 487 mg
c 220 mg
d 282 mg

208. Given the data in this table, the patient's creatinine clearance, in mL/min, is:

test	result
urine creatinine	90 mg/dL (7956 µmol/L)
serum creatinine	0.90 mg/dL (79.6 µmol/L)
patient's total body surface	1.73 m^2 (average = 1.73 m^2)
total urine volume in 24 hours	1500 mL

a 104
b 124
c 144
d 150

209. A 45-year-old male of average height and weight (ie, body surface area of 1.73 m^2) is admitted to the hospital for renal function studies. The laboratry results are shown in this table:

test	result
urine creatinine	120 mg/dL (10680 µmol/L)
serum creatinine	1.5 mg/dL (132.6 µmol/L)
total urine volume in 24 hours	1800 mL

Calculate the creatinine clearance for this patient in mL/min.

a 100
b 144
c 156
d 225

210. The creatinine clearance (mL/min) is equal to:

a urinary creatinine (mg/L)/[volume of urine (mL/min) × plasma creatinine (mg/L)]
b [urinary creatinine (mg/L) × volume (mL/min)] ÷ plasma creatinine (mg/L)
c urinary creatinine (mg/L) ÷ [volume of urine (mL/hour) × plasma creatinine (mg/L)]
d [urinary creatinine (mg/L) × volume (mL/hour)] ÷ plasma creatinine (mg/L)

211. An adult diabetic with renal complications has the laboratory results shown in this table:

MLS ONLY

test	result
sodium	133 mEq/L (133 mmol/L)
glucose	487 mg/dL (26.8 mmol/L)
BUN	84 mg/dL (30.0 mmol/L)
creatinine	5 mg/dL (442.0 µmol/L)

On the basis of these results, the calculated serum osmolality is:

a 266 mOsm/kg
b 290 mOsm/kg
c 323 mOsm/kg
d 709 mOsm/kg

212. These laboratory results are obtained in a creatinine clearance evaluation:

MLS ONLY

test	result
urine concentration	84 mg/dL
urine volume	1440 mL/24 hr
serum concentration	1.4 mg/dL
body surface area	1.60 m^2 (average = 1.73 m^2)

The creatinine clearance in mL/min is:

a 6
b 22
c 60
d 65

213. In the International System of Units, serum urea is expressed in millimoles per liter (mmol/L).

MLS ONLY

substance	chemical formula	atomic weight
urea	NH_2CONH_2	N = 14, C = 12, O = 16, H = 1

A serum urea nitrogen concentration of 28 mg/dL would be equivalent to what concentration of urea in SI units?

a 4.7 mmol/L
b 5.0 mmol/L
c 10.0 mmol/L
d 20.0 mmol/L

214. Stray light can be detected in a spectrophotometer by utilizing a:

MLS ONLY

a mercury vapor lamp
b holmium oxide glass
c potassium dichromate solution
d sharp cutoff filter

215. When separating serum proteins by cellulose acetate electrophoresis, using Veronal buffer at pH 8.6, beta globulin migrates:

a faster than albumin
b slower than gamma globulin
c faster than gamma globulin
d faster than alpha-2 globulin

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216. What is the proper pH for the buffered solution used to perform serum protein electrophoresis?

a 5.6
b 7.6
c 8.6
d 9.6

217. The buffer pH most effective at allowing amphoteric proteins to migrate toward the cathode in an electrophoretic system would be:

a 4.5
b 7.5
c 8.6
d 9.5

218. A double albumin band seen on serum protein electrophoresis indicates:

MLS ONLY
a severe liver disease
b bisalbuminemia
c acute inflammation
d hemolytic anemia

219. Which of the following serum proteins migrate with the beta-globulins on cellulose acetate at pH 8.6?

MLS ONLY
a ceruloplasmin
b hemoglobin
c haptoglobin
d C3 component of complement

220. Most chemical methods for determining total protein utilize which of the following reactions?

a molybdenum blue
b ferri-ferrocyanide
c resorcinol-HCl
d biuret

221. Bromcresol purple at a pH of 5.2 is used in a colorimetric method to measure:

a albumin
b globulin
c Bence Jones protein
d immunoprotein

222. A 55-year-old man with diabetes has a urinary albumin excretion of 40 mg/d. According to the National Kidney Foundation, the urine albumin be checked in these patients every:

a 3 months
b 6 months
c 12 months
d 24 months

223. A 22-year-old patient has trace urinary protein by dipstick method and a urine albumin excretion rate of 50 µg/min (normal is < 20). This correlates with the term:

a hematuria
b massive proteinuria
c microalbuminuria/albuminuria
d oliguria

224. Microalbuminuria/albuminuria is determined by measuring urinary levels of:

a fructosamine
b ratio of albumin to creatinine
c pre-albumin
d beta-2 microglobulin

225. A patient with a normal BUN and serum creatinine but increased serum and urinary uric acid levels most likely has:

a dehydration
b gout
c nephrotic syndrome
d renal failure

III. Acid-Base, Blood Gases & Electrolytes

A. Acid-Base Determinations (Including Blood Gases)

226. The expected blood gas results for a patient in chronic renal failure would match the pattern of:

a metabolic acidosis
b respiratory acidosis
c metabolic alkalosis
d respiratory alkalosis

227. Severe diarrhea causes:

a metabolic acidosis
b metabolic alkalosis
c respiratory acidosis
d respiratory alkalosis

228. The patient's results in this table are compatible with which of clinical conditions listed?

test	result
pH:	7.18
pO_2	86 mm Hg
pCO_2	60 mm Hg
O_2 saturation	92%
HCO_3	21 mEq/L (21 mmol/L)
TCO_2	23 mEq/L (23 mmol/L)
base excess:	–8.0 mEq/L (–8.0 mmol/L)

a fever
b uremia
c emphysema
d dehydration

229. The total CO_2 concentration is comprised of which components?

a $H_2CO_3 + HCO_3$
b $pCO_2 + HCO_3$
c $H_2CO_3 + pCO_2$
d $tCO_2 + H_2CO_3$

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230. An emphysema patient suffering from fluid accumulation in the alveolar spaces is likely to be in what metabolic state?

a respiratory acidosis
b respiratory alkalosis
c metabolic acidosis
d metabolic alkalosis

231. At blood pH 7.40, what is the ratio of bicarbonate to carbonic acid?

a 15:1
b 20:1
c 25:1
d 30:1

232. Carbonic acid concentration can be calculated as follows:

a $HCO_3/2.8$
b HCO_3/ H_2CO_3
c $pCO_2 \times 0.03$
d $tCO_2 \times 0.03$

233. The reference range for the pH of arterial blood measured at 37°C is:

a 7.28-7.34
b 7.33-7.37
c 7.35-7.45
d 7.45-7.50

234. A 68-year-old man arrives in the emergency room with a glucose level of 722 mg/dL (39.7 mmol/L) and serum acetone of 4+ undiluted. An arterial blood gas from this patient is likely to be:

a low pH
b high pH
c low pO_2
d high pO_2

235. A patient is admitted to the emergency room in a state of metabolic alkalosis. Which of the following would be consistent with this diagnosis?

a high TCO_2, increased HCO_3
b low TCO_2, increased HCO_3
c high TCO_2, decreased H_2CO_3
d low TCO_2, decreased H_2CO_3

236. A person suspected of having metabolic alkalosis would have which of the following laboratory findings?

a CO_2 content and pCO_2 elevated, pH decreased
b CO_2 content decreased and pH elevated
c CO_2 content, pCO_2 and pH decreased
d CO_2 content and pH elevated

237. Metabolic acidosis is described as a(n):

a increase in CO_2 content and pCO_2 with a decreased pH
b decrease in CO_2 content with an increased pH
c increase in CO_2 with an increased pH
d decrease in CO_2 content and pCO_2 with a decreased pH

238. Respiratory acidosis is described as a(n):

a increase in CO_2 content and pCO_2 with a decreased pH
b decrease in CO_2 content with an increased pH
c increase in CO_2 content with an increased pH
d decrease in CO_2 content and pCO_2 with a decreased pH

239. A common cause of respiratory alkalosis is:

a vomiting
b starvation
c asthma
d hyperventilation

240. Acidosis and alkalosis are best defined as fluctuations in blood pH and CO_2 content due to changes in:

a Bohr effect
b O_2 content
c bicarbonate buffer
d carbonic anhydrase

241. A blood gas sample is sent to the lab on ice, and a bubble is present in the syringe. The blood had been exposed to room air for at least 30 minutes. The following change in blood gases will occur:

a CO_2 content increased/pCO_2 decreased
b CO_2 content and pO_2 increased/pH increased
c CO_2 content and pCO_2 decreased/pH decreased
d pO_2 increased/HCO_3 decreased

242. The results in this table are most compatible with:

serum electrolytes	result	arterial blood	result
sodium	136 mEq/L (136 mmol/L)	pH	7.32
potassium	4.4 mEq/L (4.4 mmol/L)	pCO_2	79 mm Hg
chloride	92 mEq/L (92 mmol/L)		
bicarbonate	40 mEq/L (40 mmol/L)		

a respiratory alkalosis
b respiratory acidosis
c metabolic alkalosis
d metabolic acidosis

243. The most important buffer pair in plasma is the:

a phosphate/biphosphate pair
b hemoglobin/imidazole pair
c bicarbonate/carbonic acid pair
d sulfate/bisulfate pair

244. Most of the carbon dioxide present in blood is in the form of:

a dissolved CO_2
b carbonate
c bicarbonate ion
d carbonic acid

245. In respiratory acidosis, a compensatory mechanism is:

a increased respiration rate
b decreased ammonia formation
c increased blood pCO_2
d increased plasma bicarbonate concentration

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246. Specimens from a patient with chronic lung disease show the following results:

serum electrolytes	result
bicarbonate	39 mEq/L (39 mmol/L)

arterial blood	result
pH	7.32
pCO_2	78 mm Hg
pO_2	89 mm Hg

These results indicate:

a metabolic compensation by renal retention of bicarbonate
b metabolic compensation by renal excretion of bicarbonate
c respiratory compensation by retention of CO_2
d combined retention by excretion of CO_2

247. A serum sample from an unconscious patient in the emergency department has following laboratory results:

analyte	result	reference range
Na	137 mEq/L (139 mmol/L)	
glucose	100 mg/dL	
BUN	18 mg/dL	
anion gap	19	<12
osmolality	301 mOsm/Kg	285-295

This indicates the need to investigate for:

a chronic respiratory disease
b milk-alkali syndrome
c methanol or other organic poisoning
d renal compensation for respiratory alkalosis

248. Blood received in the laboratory for blood gas analysis must meet which of the following requirements?

a on ice, thin fibrin strands only, no air bubbles
b on ice, no clots, fewer than 4 air bubbles
c on ice, no clots, no air bubbles
d room temperature, no clots, no air bubbles

249. Arterial blood that is collected in a heparinized syringe but exposed to room air would be most consistent with the changes in which of the specimens shown in this table?

specimen	pO_2	pCO_2	pH
a	elevated	decreased	elevated
b	decreased	elevated	decreased
c	unchanged	elevated	unchanged
d	decreased	decreased	decreased

a specimen a
b specimen b
c specimen c
d specimen d

250. Specimens for blood gas determination should be drawn into a syringe containing:

MLS ONLY

a no preservative
b heparin
c EDTA
d oxalate

251. Unless blood gas measurements are made immediately after sampling, *in vitro* glycolysis of the blood causes a:

a rise in pH and pCO_2
b fall in pH and a rise in pO_2
c rise in pH and a fall in pO_2
d fall in pH and a rise in pCO_2

252. An arterial blood specimen submitted for blood gas analysis is obtained at 8:30 AM but is not received in the laboratory until 11 AM. The technologist should:

a perform the test immediately upon receipt
b perform the test only if the specimen is submitted in ice water
c request a venous blood specimen
d request a new arterial specimen be obtained

253. If the pK_a is 6.1, the CO_2 content is 25 mM/L, the salt equals the total CO_2 content minus the carbonic acid; the carbonic acid equals $0.03 \times pCO_2$ where $pCO_2 = 40$ mm Hg, it may be concluded that:

MLS ONLY

a $pH = 6.1 + \log[(40-0.03)/(0.03)]$
b $pH = 6.1 + \log[(25-0.03)/(0.03)]$
c $pH = 6.1 + \log[(25-1.2)/(1.2)]$
d $pH = 6.1 + \log[(1.2)/(1.2-25)]$

254. The bicarbonate and carbonic acid ratio is calculated from an equation by:

MLS ONLY

a Siggaard-Andersen
b Gibbs-Donnan
c Natelson
d Henderson-Hasselbalch

255. Calculate the blood pH given a pCO_2 of 60 mm Hg and a bicarbonate of 18 mmol/L:

MLS ONLY

a 6.89
b 7.00
c 7.10
d 7.30

256. Normally the bicarbonate concentration is about 24 mEq/L and the carbonic acid concentration is about 1.2; pK = 6.1, log 20 = 1.3. Using the equation

$$pH = pK + \log[salt]/[acid],$$

calculate the pH.

a 7.28
b 7.38
c 7.40
d 7.42

257. An electrode has a silver/silver chloride anode and a platinum wire cathode. It is suspended in KCl solution and separated from the blood to be analyzed by a selectively permeable membrane. Such an electrode is used to measure which of the following?

a pH
b pCO_2
c pO_2
d HCO_3

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258. Hydrogen ion concentration (pH) in blood is usually determined by means of which of these electrodes?

a silver
b glass
c platinum
d platinum-lactate

259. In a pH meter reference electrodes may include:

a silver-silver chloride
b quinhydrone
c hydroxide
d hydrogen

260. Amperometry is the principle of the:

IMLS ONLY

a pCO_2 electrode
b pO_2 electrode
c pH electrode
d ionized calcium electrode

261. Most automated blood gas analyzers directly measure:

a pH, HCO_3, and % O_2 saturation
b pH, pCO_2, and pO_2
c HCO_3, pCO_2, and pO_2
d pH, pO_2, and % O_2 saturation

262. Blood pCO_2 may be measured by:

a direct colorimetric measurement of dissolved CO_2
b a self-contained potentiometric electrode
c measurement of CO_2-saturated hemoglobin
d measurement of CO_2 consumed at the cathode

263. Which blood gas electrode is composed of a semi-permeable membrane, a silver/silver chloride reference electrode and glass electrode?

a pO_2
b % O_2 Sat
c pCO_2
d HCO_3

B. Electrolytes

264. Select the test which evaluates renal tubular function.

a IVP
b creatinine clearance
c osmolality
d microscopic urinalysis

265. A patient had the serum results shown in this table:

test	result
Na^+	140.mEq/L (140.mmol/L)
K^+	4.0 mEq/L (4.0 mmol/L)
glucose	95.mg/dL (5.2 mmol/L)
BUN	10.mg/dL (3.57 mmol/L)

Which osmolality is consistent with these results?

a 188
b 204
c 270
d 390

266. The degree to which the kidney concentrates the glomerular filtrate can be determined by:

a urine creatine
b serum creatinine
c creatinine clearance
d urine to serum osmolality ratio

267. Osmolal gap is the difference between:

a the ideal and real osmolality values
b calculated and measured osmolality values
c plasma and water osmolality values
d molality and molarity at 4°C

268. Quantitation of Na^+ and K^+ by ion-selective electrode is the standard method because:

a dilution is required for flame photometry
b there is no lipoprotein interference
c of advances in electrochemistry
d of the absence of an internal standard

269. What battery of tests is most useful in evaluating an anion gap of 22 mEq/L (22 mmol/L)?

a Ca^{2+}, Mg^{2+}, PO_4^{3-} and pH
b BUN, creatinine, salicylate and methanol
c AST, ALT, LD and amylase
d glucose, CK, myoglobin and cryoglobulin

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270. A patient with myeloproliferative disorder has the laboratory results shown in this table:

test	result
HGB	13 g/dL (130 mmol/L)
HCT	38%
WBC	30 × 10^3/μL (30 × 10^9/L)
platelets	1000 × 10^3/μL (1000 × 10^9/L)
serum Na^+	140 mEq/L (140 mmol/L)
serum K^+	7 mEq/L (7 mmol/L)

The serum K^+ should be confirmed by:

a repeat testing of the original serum
b testing freshly drawn serum
c testing heparinized plasma
d atomic absorption spectrometry

271. Serum "anion gap" is increased in patients with:

a renal tubular acidosis
b diabetic alkalosis
c metabolic acidosis due to diarrhea
d lactic acidosis

272. The anion gap is useful for quality control of laboratory results for:

a amino acids and proteins
b blood gas analyses
c sodium, potassium, chloride, and total CO_2
d calcium, phosphorus and magnesium

273. The buffering capacity of blood is maintained by a reversible exchange process between bicarbonate and:

a sodium
b potassium
c calcium
d chloride

274. Which of the following electrolytes is the chief plasma cation whose main function is maintaining osmotic pressure?

a chloride
b calcium
c potassium
d sodium

275. A potassium level of 6.8 mEq/L (6.8 mmol/L) is obtained. Before reporting the results, the first step the technologist should take is to:

a check the serum for hemolysis
b rerun the test
c check the age of the patient
d do nothing, simply report out the result

276. The solute that contributes the most to the total serum osmolality is:

a glucose
b sodium
c chloride
d urea

277. A sweat chloride result of 55 mEq/L (55 mmol/L) and a sweat sodium of 52 mEq/L (52 mmol/L) are obtained on a patient who has a history of respiratory problems. The best interpretation of these results is:

a normal
b normal sodium and an abnormal chloride test should be repeated
c abnormal results
d intermediate results

278. Which of the following is true about direct ion selective electrodes for electrolytes?

a whole blood specimens are acceptable
b elevated lipids cause falsely decreased results
c elevated proteins cause falsely decreased results
d elevated platelets cause falsely increased results

279. Sodium determination by indirect ion-selective electrode is falsely decreased by:

a elevated chloride levels
b elevated lipid levels
c decreased protein levels
d decreased albumin levels

280. A physician requested that electrolytes on a multiple myeloma patient specimen be run by direct ISE and not indirect ISE because:

a excess protein binds Na in indirect ISE
b Na is falsely increased by indirect ISE
c Na is falsely decreased by indirect ISE
d excess protein reacts with diluent in indirect ISE

281. Which percentage of total serum calcium is nondiffusible protein bound?

a 80-90%
b 51-60%
c 40-50%
d 10-30%

282. The best method for ionized calcium involves the use of:

a valinomycin incorporated into a semipermeable membrane that allows for change in current
b ion-selective electrode that detects change in potential when Ca^{2+} binds reversibly to the membrane
c 8-hydroxyquinoline selective membrane that binds with Ca^{2+} to prevent other ions which may change potential
d biuret reaction that removes protein prior to binding with arsenazo ions to cause change in voltage

283. The regulation of calcium and phosphorous metabolism is accomplished by which of the following glands?

a thyroid
b parathyroid
c adrenal glands
d pituitary

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284. A hospitalized patient is experiencing increased neuromuscular irritability (tetany). Which of the following tests should be ordered immediately?

a calcium
b phosphate
c BUN
d glucose

285. Which of the following is most likely to be ordered in addition to ionized calcium to determine the cause of tetany?

a magnesium
b phosphate
c sodium.
d vitamin D

286. A reciprocal relationship exists between:

a sodium and potassium
b calcium and phosphate
c chloride and CO_2
d calcium and magnesium

287. Fasting serum phosphate concentration is controlled primarily by the:

a pancreas
b skeleton
c parathyroid glands
d small intestine

288. A low concentration of serum phosphorus is commonly found in:

a patients who are receiving carbohydrate hyperalimentation
b chronic renal disease
c hypoparathyroidism
d patients with pituitary tumors

289. These serum laboratory results are obtained:

decreased albumin
decreased calcium
increased creatinine
increased phosphorus
increased magnesium

The results are most compatible with:

a multiple myeloma
b primary hyperparathyroidism
c chronic renal failure
d secondary hyperparathyroidism

290. Total iron-binding capacity measures the serum iron transporting capacity of:

a hemoglobin
b ceruloplasmin
c transferrin
d ferritin

291. The first step in the quantitation of serum iron is:

a direct reaction with appropriate chromogen
b iron saturation of transferrin
c free iron precipitation
d separation of iron from transferrin

292. A patient's blood is drawn at 8 AM for a serum iron determination. The result is 85 µg/dL (15.2 µmol/L). A repeat specimen is drawn at 8 PM; the serum is stored at 4°C and run the next morning. The result is 40 µg/dL (7.2 µmol/L). These results are most likely due to:

MLS ONLY

a iron deficiency anemia
b improper storage of the specimen
c possible liver damage
d the time of day the second specimen is drawn

293. A low serum iron with low iron binding capacity is most likely associated with:

MLS ONLY

a iron deficiency anemia
b renal damage
c anemia of chronic infection
d septicemia

294. Decreased serum iron associated with increased TIBC is compatible with which of the following disease states?

MLS ONLY

a anemia of chronic infection
b iron deficiency anemia
c chronic liver disease
d nephrosis

295. A patient has the laboratory results shown in this table:

MLS ONLY

test	patient value	reference range
serum iron	250 µg/dL (44.8 µmol/L)	60-150 µg/dL (10.7-26.9 µmol/L)
TIBC	250 µg/dL (62.7 µmol/L)	300-350 µg/dL (53.7-62.7 µmol/L)

The best conclusion is that this patient has:

a normal iron status
b iron deficiency anemia
c chronic disease
d iron hemochromatosis

296. Serum and urine copper levels are assayed on a hospital patient with the results shown in this table:

MLS ONLY

test	patient value	reference range
serum Cu	20 µg/dL (3.1 µmol/L)	70-140 µg/dL (11.0-22.0 µmol/L)
urine Cu	83 µg/dL (13.0 µmol/L)	<40 µg/dL (<63 µmol/L)

These results are most consistent with:

a normal copper levels
b Wilms tumor
c Wilson disease
d Addison disease

297. After a difficult venipuncture requiring prolonged application of the tourniquet, the serum K^+ is found to be 6.8 mEq/L (6.8 mmol/L). The best course of action is to:

MLS ONLY

a repeat the test using the same specimen
b adjust the value based on the current serum Na^+
c repeat the test using freshly drawn serum
d cancel the test

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298. MLS ONLY Serum from a patient with metastatic carcinoma of the prostate is separated from the clot and stored at room temperature. The laboratory results obtained are shown in this table:

test	patient value	reference range
Ca^{++}	10.8 mg/dL (2.7 mmol/L)	8.8-10.3 mg/dL (2.2-2.6 mmol/L)
LD	420 U/L	50-150 U/L
acid phosphatase	0.1 U/L	0-5.5 U/L

The technician should repeat the:

a LD using diluted serum
b acid phosphatase with freshly drawn serum
c LD with fresh serum
d tests using plasma

299. MLS ONLY The osmol gap is defined as measured Osm/kg minus the calculated Osm/kg. Normally, the osmol gap is less than:

a 10
b 20
c 40
d 60

300. MLS ONLY In the atomic absorption method for calcium, lanthanum is used:

a as an internal standard
b to bind calcium
c to eliminate protein interference
d to prevent phosphate interference

301. Which of the following methods is susceptible to the solvent displacing effect that results in falsely decreased electrolyte values?

a indirect ion-selective electrodes
b direct ion-selective electrodes
c alkaline electrophoretic separation of ions
d fluorescence

302. An automated method for measuring chloride which generates silver ions in the reaction is:

a coulometry
b mass spectroscopy
c chromatography
d polarography

303. Coulometry is often used to measure:

a chloride in sweat
b the pH in saliva
c bicarbonate in urine
d ammonia in plasma

304. Valinomycin enhances the selectivity of the electrode used to quantitate:

a sodium
b chloride
c potassium
d calcium

305. Magnesium carbonate is added in an iron binding capacity determination in order to:

a allow color to develop
b precipitate protein
c bind with hemoglobin iron
d remove excess unbound iron

306. Which of the following calcium procedures utilizes lanthanum chloride to eliminate interfering substances?

a o-cresolphthalein complexone
b precipitation with chloranilic acid
c chelation with EDTA
d atomic absorption spectrophotometry

307. The osmolality of a urine or serum specimen is measured by a change in the:

a freezing point
b sedimentation point
c midpoint
d osmotic pressure

308. Which of the following applies to cryoscopic osmometry?

a temperature at equilibrium is a function of the number of particles in solution
b temperature plateau for a solution is horizontal
c freezing point of a sample is absolute
d initial freezing of a sample produces an immediate solid state

IV. Special Chemistry

A. Endocrinology

309. A patient has the following test results:

increased serum calcium levels
decreased serum phosphate levels
increased levels of parathyroid hormone

This patient most likely has:

a hyperparathyroidism
b hypoparathyroidism
c nephrosis
d steatorrhea

310. TSH is produced by the:

a hypothalamus
b pituitary gland
c adrenal cortex
d thyroid

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311. A patient has the thyroid profile results shown in this table:

test	result
total T_4	decreased
free T_4	decreased
thyroid peroxidase antibody	positive
TSH	increased

This patient most probably has:

a hyperthyroidism
b hypothyroidism
c a normal thyroid
d Graves disease

312. A 45-year-old woman complains of muscle weakness, racing heart and sweating. Total and free T_4 are abnormally high. If the TSH showed a marked decrease, this would be consistent with:

a Graves disease
b an adenoma of the thyroid
c panhypopituitarism
d primary hypothyroidism

313. The majority of thyroxine (T_4) is converted into the more biologically active hormone:

a thyroglobulin
b thyroid-stimulating hormone (TSH)
c triiodothyronine (T_3)
d thyrotropin-releasing hormone

314. A 2-year-old child with a decreased serum T_4 is described as being somewhat dwarfed, stocky, overweight, and having coarse features. Of the following, the most informative additional laboratory test would be the serum:

a thyroxine-binding globulin (TBG)
b thyroid-stimulating hormone (TSH)
c triiodothyronine (T_3)
d cholesterol

315. MLS ONLY Assays for free T_4 measure hormone not bound to thyroxine-binding prealbumin, thyroxine-binding globulin and:

a thyrotropin-releasing hormone
b albumin
c free T_3
d thyroid-stimulating hormone

316. The recommended initial thyroid function test for either a healthy, asymptomatic patient or a patient with symptoms which may be related to a thyroid disorder is:

a free thyroxine (free T_4)
b thyroid-stimulating hormone (TSH)
c total thyroxine (T_4)
d triiodothyronine (T_3)

317. The screening test for congenital hypothyroidism is based upon the level of:

a free T_4
b thyroid-binding globulin
c thyroid-releasing hormone
d total thyroxine (T_4)

318. Which of the set of results shown in the table is consistent with primary hypothyroidism (eg, Hashimoto thyroiditis):

Result	TSH	free T_4	antithyroglobulin antibody
a	decreased	decreased	positive
b	increased	increased	positive
c	normal	decreased	negative
d	increased	decreased	positive

a result a
b result b
c result c
d result d

319. A 68-year-old female patient tells her physician of being "cold all the time" and recent weight gain, with no change in diet. The doctor orders a TSH level, and the laboratory reports a value of 8.7 µU/ mL (8.7 IU/L) (reference range = 0.5-5.0 µU/mL [0.5-5.0 IU/L]). This patient most likely has:

a primary hypothyroidism
b Graves disease
c a TSH-secreting tumor
d primary hyperthyroidism

320. Which of the following is secreted by the placenta and used for the early detection of pregnancy?

a follicle-stimulating hormone (FSH)
b human chorionic gonadotropin (HCG)
c luteinizing hormone (LH)
d progesterone

321. MLS ONLY During pregnancy, the form of estrogen that predominates and may be useful in prenatal screening is:

a estradiol
b estriol
c estrone
d pregnanediol

322. The hormone that triggers ovulation is:

a follicle-stimulating hormone (FSH)
b luteinizing hormone (LH)
c thyroid-stimulating hormone (TSH)
d human chorionic gonadotropin (HCG)

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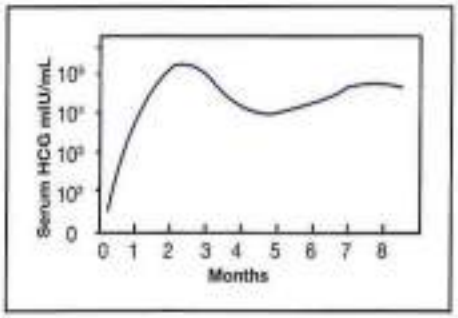
323. The HCG levels shown in the this figure most probably represent::

MLS ONLY



a trophoblastic disease following miscarriage at 4 months
b normal pregnancy
c development of trophoblastic disease
d miscarriage at 2 months with retained placenta

324. During a normal pregnancy, quantitative human chorionic gonadotropin (HCG) levels peak how many weeks after the last menstrual period?

a 2-4
b 8-10
c 14-16
d 18-20

325. Which of the following steroids is an adrenal cortical hormone?

a angiotensinogen
b aldosterone
c epinephrine
d growth hormone

326. What common substrate is used in the biosynthesis of adrenal steroids, including androgens and estrogens?

a cortisol
b catecholamines
c progesterone
d cholesterol

327. The biologically most active, naturally occurring androgen is:

MLS ONLY

a androstenedione
b cortisol
c epiandrosterone
d testosterone

328. Plasma for cortisol determinations is collected at 7 AM, after waking the patient, and at 10 PM that evening. The cortisol level of the morning sample is higher than the evening sample. This is consistent with:

a a normal finding
b Cushing syndrome
c Addison disease
d hypopituitarism

329. The major action of angiotensin II is:

a increased pituitary secretion of vasopressin
b increased vasoconstriction
c increased parathormone secretion by the parathyroid
d decreased adrenal secretion of aldosterone

330. The urinary excretion product measured as an indicator of epinephrine production is:

a dopamine
b dihydroxyphenylalanine (DOPA)
c homovanillic acid
d vanillylmandelic acid (VMA)

331. Test methodology for monitoring parathyroid glandular function in diagnosis of hypo- or hyperparathyroidism should be focused on assessing:

a intact PTHrP molecule
b the C-terminal PTH
c intact PTH amino acids 1-84
d N-terminal PTH amino acids 1-7

332. MLS ONLY The most common form (95%) of congenital adrenal hyperplasia is 21-hydroxylase deficiency, which is detected by elevated plasma:

a cortisol
b aldosterone
c 17-OH-progesterone
d 11-deoxycortisol

333. A diagnosis of primary adrenal insufficiency requires demonstration of:

a decreased urinary 17-keto- and 17-hydroxysteroids
b decreased cortisol production
c impaired response to ACTH stimulation
d increased urinary cortisol excretion after metyrapone

334. The screen for adrenal cortical hyperfunction with the greatest sensitivity and specificity is:

a 24-hour urine free cortisol
b plasma cortisol
c urinary 17-hydroxycorticosteroids
d plasma corticosterone

335. MLS ONLY A patient has signs and symptoms suggestive of acromegaly. The diagnosis would be confirmed if the patient had which of the following?

a an elevated serum phosphate concentration
b a decreased serum growth hormone releasing factor concentration
c no decrease in serum growth hormone concentration 90 minutes after oral glucose administration
d an increased serum somatostatin concentration

336. MLS ONLY Estrogen and progesterone receptor assays are useful in identifying patients who are likely to benefit from endocrine therapy to treat which of the following?

a ovarian cancer
b breast cancer
c endometriosis
d amenorrhea

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337. Which of the following sample collections would give an accurate assessment of potential excess cortisol production (hypercortisolism)?

a collect a plasma sample as a baseline, and another 1 hour after administration of metyrapone
b collect a plasma sample at 8 AM only
c collect a 24-hour urine free cortisol
d collect a plasma sample at 8 AM and at 8 AM the next day

338. MLS ONLY How is primary hypocortisolism (Addison disease) differentiated from secondary hypocortisolism (of pituitary origin)?

a adrenal corticotropic hormone (ACTH) is decreased in primary and elevated in secondary
b adrenalcorticotropic hormone (ACTH) is elevated in primary and decreased in secondary
c low aldosterone and hypoglycemia present with secondary hypocortisolism
d normal cortisol levels and blood pressure with primary hypocortisolism

339. Aldosterone is released by the adrenal cortex upon stimulation by:

a renin
b angiotensinogen
c angiotensin I
d angiotensin II

340. MLS ONLY 24-hour homovanillic acid (HVA) is usually ordered to help in diagnosis of:

a Cushing disease
b malignant neuroblastoma
c Conn disease
d Graves disease

341. A chemiluminescent EIA:

a measures absorption of light
b is less sensitive than radioisotopic reactions
c is monitored by the use of a gamma counter
d is quantitated by the amount of light produced by the reaction

B. Vitamins & Nutrition

342. MLS ONLY Which of the following substances is the biologically active precursor of a fat soluble vitamin?

a biotin
b retinol
c folic acid
d ascorbic acid

343. MLS ONLY Serum concentrations of vitamin B_{12} are elevated in:

a pernicious anemia in relapse
b patients on chronic hemodialysis
c chronic granulocytic leukemia
d Hodgkin disease

344. MLS ONLY Absorption of vitamin B_{12} requires the presence of:

a intrinsic factor
b gastrin
c secretin
d folic acid

345. Night blindness is associated with deficiency of which of the following vitamins?

a A
b C
c niacin
d thiamine

346. Pernicious anemia is associated with deficiency of vitamin:

a A
b B_{12}
c niacin
d thiamine

347. This vitamin is water soluble:

a A
b C
c D
d K

348. Rickets is associated with deficiency of which of the following vitamins?

a B_1
b C
c niacin
d D

349. Pellagra is associated with deficiency of which of the following vitamins?

a A
b B_1
c thiamine
d niacin

350. Assay of transketolase activity in blood is used to detect deficiency of:

MLS ONLY

a thiamine
b folic acid
c ascorbic acid
d riboflavin

C. Therapeutic Drug Monitoring

351. Blood specimens for digoxin assays should be obtained between 8 hours or more after drug administration because:

MLS ONLY

a tissue and serum levels need to reach equilibrium
b serum digoxin concentration will be falsely low prior to 6 hours
c all of the digoxin is in the cellular fraction prior to 6 hours
d digoxin protein-binding interactions are minimal prior to 6 hours

352. A drug has a half-life of 6 hours. If a dose is given every 6 hours, a steady-state drug level would usually be achieved in:

a 3-5 hours
b 10-12 hours
c 24-42 hours
d 48-50 hours

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353. MLS ONLY Free therapeutic drug levels are usually higher when serum protein concentrations are below normal. In which of the following conditions would this most likely occur?

a acute inflammation
b nephrotic syndrome
c pregnancy
d multiple myeloma

354. MLS ONLY Which of the following factors is relevant to therapeutic drug monitoring (TDM) of the aminoglycosides and vancomycin antibiotics?

a intestinal malabsorption
b nephrotoxicity
c ototoxicity
d declining renal function

355. The drug procainamide is prescribed to treat cardiac arrhythmia. What biologically active liver metabolite of procainamide is often measured simultaneously?

a phenobarbitol
b quinidine
c N-acetyl procainamide (NAPA)
d lidocaine

356. A carbonate salt used to control manic-depressive disorders is:

a digoxin
b acetaminophen
c lithium
d phenytoin

357. An antiepileptic (or anticonvulsant) used to control seizure disorders is:

a digoxin
b acetaminophen
c lithium
d phenytoin

358. A drug that relaxes the smooth muscles of the bronchial passages is:

a acetaminophen
b lithium
c phenytoin
d theophylline

359. A cardiac glycoside that is used in the treatment of congenital heart failure and arrhythmias by increasing the force and velocity of myocardial contraction is:

a digoxin
b acetaminophen
c lithium
d phenytoin

360. Lithium therapy is widely used in the treatment of:

a hypertension
b hyperactivity
c aggression
d manic-depressive (bipolar) disorder

361. An active metabolite of amitriptyline is:

MLS ONLY

a nortriptyline
b protriptyline
c butriptyline
d norbutriptyline

362. Phenobarbital is a metabolite of:

MLS ONLY

a primidone
b phenytoin
c amobarbital
d secobarbital

363. Reverse phase high-performance liquid chromatography is being increasingly utilized in therapeutic drug monitoring. The term reverse phase implies that the column eluant is:

MLS ONLY

a pumped up the column
b more polar than the stationary phase
c always nonpolar
d less polar than the stationary phase

D. Toxicology

364. Which of the following elevates carboxyhemoglobin?

MLS ONLY

a nitrite poisoning
b exposure to carbon monoxide
c sulfa drug toxicity
d sickle cell anemia

365. The reason carbon monoxide is so toxic is because it:

MLS ONLY

a is a protoplasmic poison
b combines with cytochrome oxidase
c has 200 times the affinity of oxygen for hemoglobin binding sites
d sensitizes the myocardium

366. Cocaine is metabolized to:

a carbamazepine
b codeine
c hydrocodone
d benzoylecgonine

367. The metabolite 11-nor-tetrahydrocannabinol-9-COOH can be detected by immunoassay 3-5 days after a single use of:

a methamphetamine
b cocaine
c benzodiazepine
d marijuana

368. A 3-year-old child is evaluated for abdominal pain and anorexia by a physician. A CBC reveals a hemoglobin of 9.8 g/dL (98 g/L) and basophilic stippling of the RBCs. The doctor should order further tests to check for poisoning from:

a arsenic
b iron
c mercury
d lead

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369. Zinc protoporphyrin or free erythrocyte protoporphyrin measurements are useful to assess blood concentrations of: MLS ONLY

- **a** lead
- **b** mercury
- **c** arsenic
- **d** beryllium

370. A salicylate level is performed to detect toxicity caused by ingestion of excess:

- **a** acetaminophen
- **b** aspirin
- **c** ibuprofen
- **d** pseudoephedrine

371. Testing for the diagnosis of lead poisoning in children should include:

- **a** erythrocyte protoporphyrin (EPP)
- **b** urine delta-aminolevulinic acid
- **c** whole blood lead
- **d** zinc protoporphyrin (ZPP)

372. Upon development of a thin-layer chromatogram for drug analysis all drug spots (including the standards) had migrated with the solvent front. The most probable cause for this would be: MLS ONLY

- **a** environmental temperature too warm
- **b** incorrect aqueous to nonaqueous solvent mixture
- **c** too much sample applied
- **d** chromatogram dried too quickly

373. To detect barbiturate abuse when analyzing urine specimens, immunoassay is the method of choice for screening. The method of choice for confirmation is:

- **a** nephelometry
- **b** thin-layer chromatography
- **c** gas chromatography/mass spectrometry
- **d** ultraviolet absorption spectroscopy

374. The typical specimen collected for forensic toxicity of drugs of abuse is:

- **a** random urine
- **b** serum
- **c** whole blood
- **d** hair follicle

375. The best initial specimen to assess for acetaminophen toxicity severity is:

- **a** 24-hour urine
- **b** 2-hour whole blood
- **c** random urine
- **d** 4-hour plasma

376. Drug metabolites are measured in a separation technique in a column containing small diameter beads coated with a liquid and with liquid pumped through under high pressure. This describes the principle of:

- **a** high-pressure focused electrophoresis
- **b** high-performance liquid chromatography
- **c** gas liquid chromatography
- **d** mass spectrophotometry-mass spectrometry

I. General Chemistry

A. Carbohydrates

1. **c** Diagnosis of hypoglycemia in non-diabetic adults. If blood glucose levels are measured while the person is experiencing those symptoms and found to be ≤45 mg/dL (≤2.5 mmol/L) in a woman or ≤55 mg/dL (≤3.0 mmol/L) in a man.
[QCMLS 2021: 2.1.6.6] [Bishop 2022, p275]

2. **c** Fasting whole blood is 10-15% lower than plasma glucose.
[QCMLS 2021: 3.3.5.1] [Tietz 2019, p384]

3. **a** GTT diet preparation. Perform in the AM after 3 days of unrestricted diet containing at least 150 g of carbohydrates/day.
[Tietz 2019, p614]

4. **b** Normal 2-hour postprandial value should return to approximately fasting state at 2 hours postprandial.
[QCMLS 2021: 2.1.5.1.9] [Tietz 2019, p615]

5. **b** CSF glucose is approximately 60% of blood glucose.
[QCMLS 2021: 3.3.5.1] [Tietz 2019, p386]

6. **c** GTT is discontinued when the patient vomits the glucose solution; however, the partial information is useful.
[Tietz 2019, p614]

7. **b** Glycolysis may occur more quickly on glucose in CSF due to presence of bacteria or cells so it should be analyzed immediately.
[QCMLS 3.3.3.2] [Tietz 2019, p384]

8. **d** A glucose level of ≥200 mg/L gives an unequivocal diagnosis of diabetes mellitus postprandially.
[QCMLS 2021: 2.1.5.1.9] [Tietz 2019, p613]

9. **b** Age effect on glucose and hypoglycemia in infants is determined at much lower levels. Neonates have lower glucose levels than adults. Additionally, the serum levels that define hypoglycemia is lower in preterm infants when compared to full terms infants.
[Tietz 2019, 361]

10. **a** Glucagon breaks down glycogen after an individual has fasted to prevent a rapid decline in the blood glucose.
[QCMLS 2021: 2.1.4.1.2] [Tietz 2019, p606]]

11. **d** A 2-step method for determining gestational diabetes is recommened.
[QCMLS 2021: 2.1.5.1.5] [Tietz 2019, p614]

12. **b** In fasting state, arterial/capillary glucose concentration is 2-5 mg higher than venous blood.
[Tietz 2019, p384]

13. **d** Definition of glycolysis.
[QCMLS 2021: 2.1.3.1] [Tietz 2019, p380]

14. **b** A fasting glucose of ≥126 is diagnostic of diabetes mellitus.
[QCMLS 2021: 2.1.5.1.9] [Tietz 2019, p613]

15. **d** HbA_1c represents average glucose over time and is the best predictor.
[QCMLS 2021: 2.1.8.2.7] [Tietz 2019, p620]

16. **c** Fructosamine is a glycated protein which may be used in place of HBA_1c in patients with hemoglobin variants. There is not agreement on the clinical utility of fructosamine.
[Tietz 2019, p623]

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17. **a** The rate of formation of glycated hemoglobin is directly proportional to the concentration of glucose in the blood, therefore, the glycated hemoglobin concentration is an average for glucose values over the preceding 8 to 12 weeks.
[QCMLS 2021: 2.1.8.2.7] [Tietz 2019, p620]

18. **d** HbA_1c is the major fraction of glycated hemoglobin.
[QCMLS 2021: 2.1.8.2.7] [Tietz 2019, p620]

19. **a** Glycated hemoglobin is directly related to the life of RBCs as it is the average glucose values over the lifespan of a RBC, approximately 120 days.
[QCMLS 2021: 2.1.8.2.7] [Tietz 2019, p620]

20. **d** HbS & HbC yield falsely decreased levels in ion exchange chromatography methods.
[QCMLS 2021: 2.1.8.2.7.2] [Tietz 2019, p621]

21. **a** Increase in ketones is found when there is a decreased use of carbohydrates, such as in diabetes.
[QCMLS 2021: 2.1.5.1.6.1] [Tietz 2019, p620]

22. **a** Copper reduction reaction detects many reducing substances.
[QCMLS 2021: 2.1.8.2.5] [Tietz 2019, p386]

23. **d** After strenuous exercise, lactate and pyruvate may increase due to increased carbohydrate metabolism in the skeletal muscle.
[QCMLS 2021: 2.1.3.1, 2.1.5.1.6.2] [Tietz 2019, p382-383]

24. **a** Diagnosis of lactase deficiency by H_2 breath test based on bacteria acting on unabsorbed disaccharides.
[Tietz 2019, p732]

25. **a** Glucose is measured in the water of the sample and not the cells. Water content is much higher in plasma than in whole blood so hematocrit, the ratio of cells to plasma, affects glucose result. Glucose is approximately 11% lower in whole blood due to presence of RBCs in the sample that displace the water. If the leukocyte count is high and serum or plasma is not separated from blood cells quickly, glucose will be consumed and falsely lower.
[Tietz 2019, p384, Bishop 2022, p277]

26. **d** Sodium fluoride exerts its preservative action by inhibiting the enzyme systems involved in glycolysis and thus prevents red or white blood cells from consuming the glucose and falsely lowering the amount in the first few minutes It will not protect glycolysis in the first hour. It does not react in the hexokinase method either enhancing or inhibiting activity nor does it react with non-glucose reducing methods.
[QCMLS 2021: 2.1.8.1] [Bishop 2022, p277]

27. **d** Glucose decreases at a rate of 5-7% per hour in whole blood at room temperature. Glycolysis will continue until the specimen is processed by centrifugation, and serum and plasma are separated from the cellular components of blood. Cholesterol, triglycerides and creatinine are not as unstable in serum left at room temperature over an 8-hour period.
[QCMLS 2021: 2.1.8.1] [Tietz 2019, p383; Bishop 2022, p277]

28. **c** In 1961, the enzyme commission recommended the adoption of an international unit (IU) of enzyme activity. The IU was defined as the amount of enzyme that would convert 1 µmol of substrate per minute under standard conditions. 1 IU = 1 µmol/min.
[QCMLS 2021: 2.6.4] [Bishop 2022, p235]

29. **c** A = abc is mnemonic for Absorbance = absorptivity × light path × concentration.
[Tietz 2019, p130]

30. **a** Hexokinase is the method most specific to both forms of glucose and the basis for the reference method and should be the method of choice for urine due to fewer interferences. Glucose oxidase/peroxidase method is the second most specific method mainly due to potential interferences in the peroxidase step and is commonly employed in automated methods as well as dipstick screening methods in routine urinalysis. Glucose dehydrogenase is suitable for POCT devices.
[QCMLS 2021: 2.1.8.2.4] [Tietz 2019, p383-385]]

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31. **b** Other distractors are not glucose-specific methods. Hexokinase is specific for other isomers of glucose. Ferricyanide was a nonspecific method for reducing substances method. o-toluidine is a condensation method and reacts with hemoglobin (found in feces), mannose and other sugars as well as glucose so not specific. It was used in the past for CSF glucose methods.
[QCMLS 2021: 2.1.8.2.1] [Tietz 2019, p385]

32. **c** Ascorbic acid interferes with the peroxidase step. This is a limitation of the glucose oxidase (peroxidase) reaction making it not as specific overall, as the hexokinase method. Other sugars react in the o-toluidine method but ascorbic acid is not generally thought to interfere significantly. The ferricyanide method is affected by many substances since it is nonspecific for reducing substances so ascorbic acid would also interfere but it is an obsolete method.
[QCMLS 2021: 2.1.8.2.1] [Tietz 2019, p385]

33. **b** The principle of the hexokinase method is the production of NADPH from NADP in a coupled enzymatic reaction involving hexokinase, ATP and glucose-6-phosphate dehydrogenase and NADP. Hydrogen peroxide is produced in the first step of the coupled glucose oxidase/peroxidase method. O-toluidine condenses glucose with an aromatic amine. Bromcresol purple binds with albumin to form a spectrophotometric product.
[QCMLS 2021: 2.1.8.2.4] [Tietz 2019, 384-85]

34. **c** Glucagon is released in fasting states and increases blood glucose levels by increasing glycogenolysis from the liver and increasing glyconeogenesis.
[QCMLS 2.1.4.1.2] [Tietz 2019, p606]

35. **b** The deficiency of galactose-1-phosphate uridyl transferase, which causes Type 1 glycogen storage disease, is the most common form of galactosemia.
[QCMLS 2.1.7.3] [Tietz 2019, p895]

B. Lipids

36. **d** High-density lipoprotein (HDL) is the smallest and most dense lipoprotein. Its role in lipid metabolism involves removing cholesterol from the peripheral cells and transporting it to the liver for further metabolism. Because of these actions, HDL is thought to be anti-atherogenic. Increased levels of LDL, VLDL and chylomicrons are associated with atherosclerosis.
[QCMLS 2021: 2.2.5, 2.2.4.1.4] [Tietz 2019, p411]

37. **d** Serum HDL has been routinely measured indirectly by a 2-step procedure. Precipitation of all of the non-HDL lipoproteins with a polyanion-divalent cation combination reagent and centrifugation to obtain the supernatant containing only HDL. The cholesterol bound to HDL is measured as HDL. One challenge regarding the method has been the selection of a precipitating reagent that would precipitate the ApoB containing lipoproteins. Dextran sulfate with magnesium has proven to be very effective.
[QCMLS 2021: 2.2.6.3] [Tietz 2019, p415]

38. **a** In the indirect measurement of LDL using the Friedewald equation, values are needed for the total cholesterol, HDL cholesterol and triglyceride. Because LDL and HDL are measured based on their cholesterol content, it is necessary to determine the total cholesterol and HDL cholesterol using the same cholesterol procedure following the precipitation step in the HDL method.
[QCMLS 2021: 2.2.6.4] [Tietz 2019, p415]

39. **c** High-density lipoprotein (HDL) is the smallest and most dense of the lipoproteins. This is evidenced by its lipid content of 20% and protein concentration of 50%.
[Bishop 2022, p291]

40. **a** Familial hypercholesterolemia is a genetic condition characterized by elevated serum cholesterol levels. In homozygotes and heterozygotes, the elevated cholesterol is associated with an increased LDL level. The lack or deficiency of the LDL receptors prevents the metabolism of LDL cholesterol, resulting in an increased LDL level.
[QCMLS 2021: 2.2.5.1] [Bishop 2022, p302]

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41. **c** Atherosclerosis is characterized by a thickening and hardening of the arterial walls by cholesterol plaques in the lining of the arteries. Elevated levels of cholesterol are associated with the development of the plaques. One of the roles of LDL is to transport cholesterol esters to the cells for metabolism. Elevated LDL levels are also associated with development of atherosclerosis.
[QCMLS 2021: 2.2.4.1.2] [Bishop 2022, p302]

42. **c** Low-density lipoprotein (LDL) transports about 70% of the total plasma cholesterol. HDL transports a smaller portion of the cholesterol through reverse-cholesterol transport from peripheral cells. Chylomicrons and VLDL transport triglycerides.
[QCMLS 2021: 2.2.4.1.2] [Tietz 2019, p401]

43. **c** Tangier disease results from a defect in the catabolism of Apo A-I, an essential apoprotein for HDL. In homozygotes, the plasma level for HDL is practically zero. It is a rare autosomal disorder. The reduced HDL levels result from increased HDL catabolism. Other more common causes of hypoalphalipoproteinemia include LCAT deficiency and mutations of the *APOA1* gene.
[QCMLS 2021: 2.2.5.3] [Tietz 2019, p409]

44. **d** The accumulation of chylomicrons is evident in blood samples of patients with familial chylomicronemia syndrome (FCS), which is sometimes known as lipoprotein lipase deficiency (LPLD), or Fredrickson type 1 hyperlipoproteinemia. Lipid analysis using overnight refrigeration involves incubating the sample at 4°C overnight. The chylomicrons, which have the lowest density of the lipoproteins, present as a thick homogenous cream layer and may be observed floating at the plasma surface. This typifies familial hyperchylomicronemia due to deficiency in lipoprotein lipase activity, which is responsible for breaking down triglycerides.
[QCMLS 2021: 2.2.5.1, 2.2.5.6, 2.2.1.4.3.4] [Tietz 2019, p403]

45. **b** Food intake can cause a transient increase in the triglyceride level by 50%. The LDL and HDL levels may be decreased depending on the fat content of the meal. Total cholesterol does not usually change significantly after a meal.
[Tietz 2019, p415]

46. **b** Tay-Sachs disease is a rare inherited disorder characterized by the near-total deficiency of the enzyme N-acetyl-beta-hexosaminidase A. The enzyme is responsible for the hydrolysis of the beta (1,4)-glycosidic bond between N-acetylgalactosamine and galactose in GM2 ganglioside. Neimann-Pick disease and Hurler syndrome are lysosomal disorders as is Tay-Sachs. Phenylketonuria results from an absent enzyme, but is an inborn error of metabolism.
[Bishop 2022, p756, Tietz 2019, p408]

47. **b** Apolipoprotein A1 acts as a receptor site on HDL.
[QCMLS 2.2.1.5] [Tietz 2019, p402]

48. **a** HDL serves to transport cholesterol away from the tissues and provides protection against cardiac disease.
[QCMLS 2.2.3.2.5] [Tietz 2019, p405]

49. **b** Mixed hyperlipoproteinemia or type V hyperlipoproteinemia occurs primarily in adulthood and is characterized by markedly elevated triglycerides, elevated very low-density lipoproteins (VLDL) and chylomicrons. Because of the markedly increased triglyceride level, the specimen integrity is milky, and overnight refrigeration shows a creamy layer over turbid serum due to the chylomicrons and triglycerides.
[QCMLS 2021: 2.2.5] [Tietz 2019, p403-404]

50. **c** Lipoprotein lipase hydrolyzes triglycerides and chylomicrons during normal lipid metabolism. A deficiency in lipoprotein lipase results in markedly increased serum chylomicrons and triglycerides.
[QCMLS 2021: 2.2.5] [Tietz 2019, p407-408]

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51. **d** In the endogenous pathway for lipid metabolism, the hepatocytes can synthesize triglycerides from carbohydrates and fatty acids. The triglycerides are packaged in VLDL, and ultimately delivered to the circulation in that form. Exogenous triglycerides are transported primarily by chylomicrons. HDL transports cholesterol from peripheral cells to the liver. LDL transports cholesterol and phospholipids to peripheral cells.
[QCMLS 2021: 2.2.1.4.1] [Bishop 2022, p287, 292]

52. **c** Elevated levels of chylomicrons in serum or plasma will result in a turbid specimen. The large size of the chylomicron will reflect the light, causing a turbid appearance.
[Bishop 2022, p292]

53. **a** Frozen samples should be allowed to thaw slowly at room temperature (22°C) or in a 37°C water bath and should then be mixed thoroughly before analysis.
[Tietz 2019, p219]

54. **b** A high-fat diet increases the serum concentrations of triglycerides. Fasting overnight for 10-14 hours is the optimal interval for fasting around which to standardize blood collections, including lipids.
[QCMLS 2021: 2.2.6.6] [Tietz 2019, p39]

55. **a** Triglycerides are affected by recent ingestion (<8 hour) of food. Fasting overnight for 10-14 hours is the optimal interval for fasting around which to standardize blood collections, including lipids. Iron is subject to circadian rhythm and is best analyzed on a specimen drawn in the morning, however a specific time of fasting is not as critical. LD and sodium are not affected by recent ingestion of food.
[QCMLS 2021: 2.2.6.6] [Tietz 2019, p39]

56. **d** Because the following relationship is true, A = light stopped and T = light passed through, A and T are inversely related. They are also logarithmically related, because the absorption of light is a logarithmic function. $A = -\log (I/I_0) = \log(100\%) - \log(\%T)$. Log of 100% = 2.
[Bishop 2022, p117]

57. **b** There are several enzymatic methods for measuring serum triglyceride. The first step of the coupled reactions involves the hydrolysis of triglyceride by lipase to produce glycerol and fatty acids. Glycerol is a reactant in 1 of 2 enzymatic sequences for the final measurement of triglycerides. Fatty acids are not easily measured. Pre-betalipoprotein is the source of a large portion of endogenous triglycerides, but is not produced in enzymatic methods of triglyceride analysis. Phospholipids are a separate from triglycerides although similar. They contain 2 esterified fatty acids and a phospholipid group such as choline on the glycerol molecule. Thus, they are more complex than triglycerides and not a by-product of enzymatic hydrolysis with lipase.
[QCMLS 2021: 2.2.6.6] [Bishop 2022, p307]

58. **b** The high-volume HDL method is homogenous or direct. This method uses antibodies or complexing agents, such as cyclodextrin, to mask or consume non-HDL cholesterol lipids but does not require a separation step. The indirect (2-3 step) method required separate pre-treatment and centrifugation to remove interferants and non-HDL cholesterol lipids. The Gomori procedure has been used with acid phosphatase or uric acid but is more currently associated with trichrome histological staining of muscle tissue. Column chromatography is a purification step prior to separation by ultracentrifugation, a research method for lipoprotein separation. Agarose gel electrophoresis has been used historically for separation of lipoproteins including HDL but is tedious, laborious and not used routinely.
[QCMLS 2021: 2.2.6.3] [Tietz 2019, p415]

C. Heme Derivatives

59. **c** HbA_1c is the addition of glucose (glycation) to the N-terminal valine of the B-chain of HbA.
[QCMLS 2021: 2.1.8.2.7] [Tietz 2019, p520]

60. **d** Hemoglobin has endogenous peroxidase activity that is capable of oxidizing guaiac in the presence of hydrogen peroxide to a blue product.
[QCMLS 2021: 2.11.2] [Tietz 2019, p85]

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61. **b** Hemoglobin metabolism breakdown product is bilirubin.
[QCMLS 2021: 2.4.2.1] [Bishop 2022, p590]

62. **a** Agarose electrophoresis at pH 6.4 using a citrate buffer will separate HbS from HbD.
[Tietz 2019, p505]

63. **c** At alkaline pH, hemoglobins migrate according to electrical charge, with HbH moving the fastest (closest to the anode) and HbC moving the slowest.
[Tietz 2019, p505, 507-508]

64. **d** Fetal hemoglobin may be quantitated using NaOH or KOH based on the principle that it is resistant to alkali denaturation.
[Bishop 2022, p592]

65. **c** Light-exposed bilirubin is structurally altered (oxidized), and decreases 30-50% per hour.
[QCMLS 2021: 2.4.7] [Bishop 2022, p498]

66. **d** Unconjugated bilirubin is reduced by intestinal microbes to form urobilinogen.
[QCMLS 2021: 2.4.2.1] [Tietz 2019, p519]

67. **a** Principle of diazo reaction with unconjugated bilirubin: reaction with caffeine solution or methyl alcohol makes the unconjugated bilirubin soluble, allowing it to react with the diazo reagent.
[QCMLS 2021: 2.4.7] [Tietz 2019, p521]

68. **b** Total bilirubin = (direct) conjugated bilirubin + (indirect) unconjugated bilirubin = 3.1 – 2.0 = 1.1.
[QCMLS 2021: 2.4.3.6, 2.4.7] [Bishop 2022, p498]

69. **d** In urine, conjugated bilirubin is reacted with diazo tablet to form a purple color.
[QCMLS 2021: 2.4.7] [Tietz 2019, p521]

70. **c** Jendrassik-Grof (method 2) is not affected by turbidity because it uses a serum blank.
[QCMLS 2021: 2.4.7] [Bishop 2022, p498]

71. **b** Light-exposed bilirubin is reduced 30-50% per hour, while hemolysis may interfere depending on the method.
[QCMLS 2021: 2.4.7] [Bishop 2022, p498]

72. **d** Bilirubin is conjugated with glucuronic acid to produce bilirubin diglucuronide.
[QCMLS 2021: 2.4.2.1] [Tietz 2019, p518]

73. **a** Intrahepatic biliary atresia in the newborn can also cause conjugated hyperbilirubinemia, while in the adult it is likely due to biliary obstruction or hepatitis.
[QCMLS 2021: 2.4.4.4] [Tietz 2019, p520]

74. **b** Definition of kernicterus: High concentrations of unconjugated bilirubin that passes the immature blood-brain barrier of the newborn.
[QCMLS 2021: 2.4.3.2] [Tietz 2019, p519]

75. **d** There are several forms of Crigler-Najjar syndrome, but they each result in abnormal metabolism of bilirubin due to absence or deficiency of UDP-glucuronyltransferase.
[QCMLS 2021: 2.4.6.1] [Tietz 2019, p520]

76. **c** The results point to hepatocellular disorder and highly elevated AST but near normal alkaline phosphatase, typical of acute hepatitis.
[QCMLS 2021: 2.4.4.2] [Tietz 2019, p711]

77. **a** Bleeding from upper GI causes digested blood to appear as dark and tarry in feces. Mucus and fecal fat may be visible in feces as an oily appearance.
[QCMLS 3.7.3.1] [Henry 2022, p345, Tietz 2019, p85]

78. **c** Bile pigments (uro-, meso-, stercobilin) are found in feces. The precursors are urobilinogens, which are oxidized to form the bile pigments.
[QCMLS 3.7.3.1] [Bishop 2022, p490]

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79. **b** Correlation of disorder and high RBC protoporphyrins including zinc protoporphyrin (ZPP). In iron deficiency anemia, the Zn competes with Fe in developing RBCs, thus increasing the amount of ZPP in RBCs.
[QCMLS 2021: 2.3.3.3] [Tietz 2019, p528]

80. **c** HCP is caused by deficiency of coproporphyrinogen oxidase, which causes accumulation of coproporphyrin III preferentially in feces.
[QCMLS 2021: 2.3.3.1] [Tietz 2019, p531]

81. **c** Rapid initial screening by qualitative or semi-quantitative methods of urine porphobilinogen should precede complex testing.
[QCMLS 2021: 2.3.4] [Tietz 2019, p533]

82. **a** Lead inhibits 3 enzymes in the heme biosynthesis pathway: δ-aminolevulinic acid dehydratase (ALAD), coproporphyrinogen oxidase, and ferrochelatase. Other enzymes in the hematopoietic pathway are not affected by lead.
[QCMLS 2021: 2.3.2.2] [Tietz 2019, p597-598, Bishop 2022, p637-638]

83. **a** The procedure "change in absorbance of amniotic fluid at 450 nm" is used to detect bilirubin, which is associated with hemolytic disease of newborn (HDN). Creatinine is measured in amniotic fluid to rule out contamination with urine. L/S ratio is to detect fetal lung maturity. Estriol is the predominant estrogen secreted during pregnancy, but is not clinically useful for detecting HDN.
[Bishop 2022, p692] [QCMLS 3.2.6.2]

84. **a** HbD migrates with HbS in alkaline electrophoresis but migrates with HbA in citrate acid electrophoresis. Alkaline denaturation is used in detecting and differentiating HbF from other forms.
[Tietz 2019, p505]

85. **d** Caffeine is added to accelerate the reaction of unconjugated bilirubin to the diazo reagent. Ethanol, dyphylline, and surfactants have also been used, but not acetone or ether.
[QCMLS 2021: 2.4.7] [Tietz 2019, p521, Bishop 2022, p498]

86. **d** Jendrassik-Groff is the modified Evelyn-Malloy method and is most commonly used as bilirubin methods. The Jaffe reaction is used in creatinine analysis and Schales and Schales is a classic chloride method, while 8-hydroxyquinoline is used in the CPC method of calcium to remove interference from Mg^{2+}.
[QCMLS 2021: 2.4.7] [Tietz 2019, p521]

87. **d** The diazo method of Evelyn-Malloy involves bilirubin reacting with diazotized sulfanilic acid to form azobilirubin.
[QCMLS 2021: 2.4.7] [Bishop 2022, p498]

88. **a** In the Jendrassik-Grof method for bilirubin measurement, the addition of caffeine plus diazotized sulfanilic acid and the serum produces azobilirubin. Ascorbic acid, alkaline tartrate and dilute HCl are added to the reaction mixture to stabilize and create the azobilirubin. The blue-green azobilirubin is measured. Bilirubin glucuronide, urobilin, and urobilinogen are intermediaries in bilirubin metabolism. They are not measured by this method.
[QCMLS 2021: 2.4.7] [Tietz 2019, p521]

89. **b** Alkaline tartrate raises the pH and converts the azobilirubin from a reddish purple chromogen, with similar absorbances (450-560 nm) as hemoglobin and other interfering substances, to a bluish chromogen, which absorbs at 600 nm.
[QCMLS 2021: 2.4.7] [Bishop 2022, p499]

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90. **a.** In prehepatic jaundice, there is an issue before the bilirubin gets to the liver. This is usually seen as increased RBC breakdown, such as in hemolytic disease or neonatal physiologic (due to increased destruction of RBCs due to HbF). The elevation of unconjugated bilirubin (that is elevated total bilirubin with normal direct bilirubin) occurs when the rate of production exceeds the rate of conjugation.
[QCMLS 2.4.4.1] [Tietz 2019, p520]

91. **d.** Posthepatic jaundice is most often due to a blockage of the bile duct either by a gall stone or by a tumor. Total and direct bilirubin are both increased as there is no place for the bilirubin to go to so it spills out into to blood stream. Urobilinogen in either decreased or absent, depending on whether there is a partial or complete blockage.
[Tietz 2019, p520]

92. **c.** The reagent used in the measurement of urobilinogen is paradimethylaminobenzaldehyde, also known as Erhlich reagent, which forms a red color when reacted with urobilinogen and sodium acetate.
[QCMLS 2.4.8] [Tietz 2019, p515]

II. Proteins & Enzymes

A. Enzymes

93. **c** An elevated alkaline phosphatase level is associated with cholestatic-hepatic biliary obstruction, increased alkaline phosphatase more so than AST.
[QCMLS 2021: 2.4.4.3, 2.4.5] [Tietz 2019, p706]

94. **d** There is an increase in the serum levels of amylase and lipase in acute pancreatitis. However, the elevated level of lipase persists longer than amylase. Elevated levels of lipase and amylase are seen in other intra-abdominal conditions, but the frequency of elevations is less with lipase than amylase.
[QCMLS 2021: 2.6.13.1] [Bishop 2022, p251-254]

95. **a** Amylase and lipase are hydrolases involved in the breakdown of starch and glycogen, and lipid metabolism, respectively. Both enzymes are primarily located in the pancreas. Disorders of the pancreas are characterized by elevated levels of the enzymes. 5'-NT, GGT, AST, and LD are elevated in liver and hepatobiliary diseases.
[QCMLS 2021: 2.6.13.1] [Bishop 2022, p251]

96. **b** In the amyloclastic, saccharogenic, and chromogenic methods for measurement of amylase, the substrate, starch, is converted to glucose and maltose.
[QCMLS 2021: 2.6.12.1] [Bishop 2022, p252]

97. **b** Macroamylasemia is an asymptomatic condition that results when the amylase molecule and immunoglobulins combine to form a complex. The complex is too large to be filtered across the glomerulus. Lack of renal clearance leads to an increased serum amylase and a decreased urine amylase.
[QCMLS 2021: 2.6.12.1] [Bishop 2022, p251, 575]

98. **d** Amylase and lipase are present primarily in the pancreas. Pancreatitis results in the release of the enzymes into the serum, with amylase returning to normal in 2-3 days but lipase persisting. Beta-hydroxybutyrate is measured for diabetic acidosis and LD isoenzymes are evaluated for disorders involving the heart and liver.
[QCMLS 2021: 2.6.12.1] [Bishop 2022, p252]

99. **a** Amylase is measured by amyloclastic methods where the breakdown of starch is the principle for measurement, or by saccharogenic methods where formation of sugars from starch breakdown is measured.
[QCMLS 2.6.12.2] [Bishop 2022, p251; Tietz 2019, p326]

100. **a** Aspartate aminotransferase (AST) is involved in the transfer of an amino group between aspartate and alpha-keto acids. AST is present in several tissues, with its highest concentrations in cardiac tissue, liver and skeletal muscle. Depending on the type of liver disease, the levels may be 100× the upper limits of normal (ULN).
[QCMLS 2021: 2.6.7.1] [Bishop 2022, p244]

101. **b** Aspartate aminotransferase (AST) belongs to the class of transferase enzymes. Specifically, AST catalyzes the transfer of an amino group from aspartate to α-ketoglutarate forming oxaloacetate and glutamate.
[QCMLS 2021: 2.6.7.1] [Bishop 2022, p244]

102. **b** The transferases, alanine aminotransferase (ALT) and aspartate aminotransferase (AST) are located primarily in the liver. Elevated serum levels of the enzymes are seen in hepatocellular disorders. The levels may be 100× the upper limit of normal. The ALT level is usually higher than AST. Increased levels of AST are also seen in infectious mononucleosis and muscular dystrophy, but ALT is not elevated in the clinical disorders.
[QCMLS 2021: 2.6.7.1] [Bishop 2022, p245]

103. **a** Aspartate aminotransferase (AST) is nonspecific for liver and found widely in many tissues including RBCs, so that sources of error from hemolysis can confuse interpretation of AST.
[QCMLS 2021: 2.6.7.2] [Bishop 2022, p244]

104. **d** In the coupled reaction of AST measurement, malate dehydrogenase catalyzes the oxidation of oxaloacetate to malate in the indicator reaction.
[QCMLS 2021: 2.6.7.2] [Bishop 2022, p244]

105. **a** In acute hepatocellular disorders, the serum levels of AST and ALT can be 100× the upper limit of normal. Slight increases of the enzyme activities are seen in chronic hepatitis, hemangioma, and obstructive jaundice.
[QCMLS 2021: 2.6.7.1] [Bishop 2022, p244-245, 501]

106. **d** Elevated serum levels of LD (up to 50× the upper limit of normal) are seen with pernicious anemia. The ineffective erythropoiesis results in the release of large quantities of LD1 and LD2. Increased levels of LD1 and LD2 may be seen in renal disease, but the increase is not as great as for pernicious anemia. Slight increases of LD3 are seen in pulmonary conditions and pancreatitis.
[QCMLS 2021: 2.6.6.1] [Bishop 2022, p242-243]

107. **d** Creatine kinase (CK) catalyzes the reversible phosphorylation of creatine. The highest levels of the enzyme are found in skeletal muscle, heart muscle, and brain tissue. Increased serum enzyme activity is present in diseases involving the listed muscles and tissue. Lipase is measured for acute pancreatitis; the transaminase and lactate dehydrogenase (LD) are not markedly increased in muscular dystrophy.
[QCMLS 2021: 2.6.5.1] [Tietz 2019, p315]

108. **d** Lactate dehydrogenase (LD) catalyzes the interconversion of lactic and pyruvic acids. Electrophoretically, using agarose or cellulose acetate medium, LD can be separated into 5 isoenzymes, LD1-LD5. CK and lipase have 3 isoenzymes; AST has 2.
[QCMLS 2021: 2.6.6.1] [Bishop 2022, p242-243]

109. **a** Lactate dehydrogenase is widespread in many tissues including heart, liver and red blood cells. Hemolysis is a preanalytical/preexamination error that releases LD and other intracellular biochemicals into the serum or plasma and would falsely increase LD activity.
[QCMLS 2021: 2.6.6.2] [Bishop 2022; p243]

110. **b** The 3 CK isoenzymes are CK1 or CKBB, CK2 or CKMB, CK3 or CKMM. CKMB is primarily located in myocardial tissue. Damage to the myocardial tissue will cause an elevation of the CKMB level.
[QCMLS 2021: 2.6.5.1] [Bishop 2022, 237-241]

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111. **b** Creatine kinase (CK) is located in brain tissue and heart and skeletal muscle. Diseases involving the tissue site will increase the level of the enzyme activity. CK activity is not increased in hepatitis.
[QCMLS 2021: 2.6.5.1] [Bishop 2022, p237-241]

112. **a** Elevations of serum LD4 and LD5 fractions are seen in liver and skeletal muscle diseases because the isoenzymes are located in the tissues. LD1 and LD2 are elevated in hemolytic anemia and myocardial infarction. Increased levels of LD3 are observed in pulmonary edema.
[QCMLS 2021: 2.6.5.1] [Bishop 2022, p242-243]

113. **a** After an acute myocardial infarction (AMI), CK activity increases 4-6 hours after the symptoms, peaks at 12-24 hours and returns to normal within 48-72 hours. AST increases 6-8 hours after the infarction. Elevated levels of LD are noted 12-24 hours after the symptoms. ALT activity does not increase with a AMI.
[QCMLS 2021: 2.6.5.1] [Bishop 2022, p239]

114. **c** Although, CKMB activity is more specific for the myocardium, CKMM is present in both the skeletal and heart muscles. An increase of the isoenzyme activity may occur after an acute myocardial infarction. Only 1 peak would be present for a brain tumor and muscular dystrophy; no peaks would be present for hepatitis because the liver is not a tissue source of CK.
[QCMLS 2021: 2.6.5.1] [Bishop 2022, p239]

115. **a** The major LD isoenzymes in the serum of healthy persons are LD2, accounting for 29-39% of the total activity and LD1=14-26% of enzyme activity. In a myocardial infarction the pattern is changed. The activity of LD1 is greater than LD2 andthe ratio of LD1 to LD2 is >1. This is known as a flipped pattern. The normal ratio is 0.45-0.74.
[QCMLS 2021: 2.6.6.1] [Bishop 2022, p243]

116. **b** α_1 antitrypsin comprises 90% of the α_1 band on protein electrophoresis.
[QCMLS 2.5.9.3.1] [Tietz 2019, p308]

117. **c** Increased levels of aldolase and CK are seen with skeletal muscle disease. The magnitude of the elevation is dependent on the type of skeletal muscle disease.
[QCMLS 2021: 2.6.5.1] [Tietz 2019, p220]

118. **a** In the immunoinhibition technique for CKMB determination, antibodies are directed against the M and B units of the enzymes. Anti-M inhibits all M activity but not B activity. CK activity is measured before and after inhibition. The activity remaining after inhibition is a result of the B subunit for BB and MB activity.
[QCMLS 2021: 2.6.5.2] [Bishop 2022, p241]

119. **c** Of the 3 CK isoenzymes, CKMB is located in the myocardial. The fraction is elevated with an acute myocardial infarction (AMI). CKMM is elevated in acute muscular stress following strenuous exercise. CKBB is increased in brain injury.
[QCMLS 2021: 2.6.5.1] [Bishop 2022, p239-241]

120. **b** The decreased albumin and a lack of resolution between the β and γ globulins called a "β-γ bridge" and is seen in liver cirrhosis. It is caused by decreased transferrin and increased β-lipoproteins and immunoglobulins. The liver is the site of albumin synthesis and the primary site of transferrin synthesis.
[QCMLS 2.5.11] [Tietz 2019, p308]

121. **b** The LD1 and LD2 fractions are increased in hemolytic anemia due to the intramedullary hemolysis. LD5 is increased with hepatic damage. LD3 may be increased with acute pancreatitis. The LD isoenzyme pattern in renal disease is very similar to a normal pattern except for the higher absolute values.
[QCMLS 2021: 2.6.6.1] [Bishop 2022, p242-243]

122. **a** Elevation of the levels of CK and AST is seen in muscle damage due to the crush injury to the thigh. AST levels can increase up to 4-8× the upper limit of normal. Cerebrovascular accident and pulmonary infarction have increased CKBB levels. In acute hepatitis, the AST level may be 100× the upper limit of normal.
[QCMLS 2021: 2.6.5.1, 2.6.7.1] [Bishop 2022, p238, 244]

123. **d** Obstructive jaundice is characterized by an increased ALP—3× the upper limit of normal—and a marked increase in GGT. The aminotransferases are slightly elevated owing to the fact that they are sensitive for acute hepatocellular conditions.
[QCMLS 2021: 2.6.9.1, 2.6.11] [Bishop 2022, p236, 245]

124. **b** Chronic hepatitis is a chronic inflammation of the hepatocytes that persists for at least 6 months. The serum enzyme levels may be variable depending on the condition. ALT, AST, and ALP may be increased by 2× the upper limit of normal. GGT is slightly increased.
[QCMLS 2021: 2.6.7.2, 2.6.8.1, 2.6.9.1, 2.6.11] [Bishop 2022, p501-503]

125. **b** The heat activation method of ALP isoenzyme separation involves heating an aliquot of the serum sample at 56°C for 10 minutes. An untreated aliquot of the sample along with the heated one are assayed for ALP activity.
[QCMLS 2021: 2.6.9.1] [Bishop 2022, p247]

126. **d** The Regan isoenzyme is an abnormal ALP isoenzyme. The carcinoplacental ALP has properties similar to the placental enzyme, in that it is also heat stable (65°C, 30 min). It has been detected in lung, breast, ovarian and colon cancer.
[QCMLS 2021: 2.6.9.1] [Bishop 2022, p246]

127. **b** The major serum ALP isoenzymes are located in the liver, bone, intestine and placenta. Bone ALP is the least heat stable, followed by the liver, intestinal, and placental fractions in increasing order of stability.
[QCMLS 2021: 2.6.9.1] [Bishop 2022, p247]

128. **c** GGT levels are elevated in alcoholism. The levels may range from 2-3× the upper limit of normal. ALT, AST, and ALP may be increased depending on the alcohol damage to the liver.
[QCMLS 2021: 2.6.11] [Bishop 2022, p249]

129. **b** The substrate for GGT is glycylglycine and γ-glutamyl-p-nitroaniline with formation of a colored product, p-nitroaniline, as a result of the GGT activity. Glutamic oxaloacetate transferase is more commonly known as aspartate transaminase and utilizes aspartate and α-ketoglutarate as the substrate.
[QCMLS 2.6.11.2] [Bishop 2022, p249]

130. **c** Lipase and amylase are used in diagnosis of pancreatitis, while ALP and GGT are commonly measured to assess hepatobiliary disease. Organophosphate poisoning is assessed by measuring pseudocholinesterase.
[QCMLS 2.6.14] [Tietz 2019, p315]

131. **d** Isoenzymes are multiple forms of an enzyme that possess the ability to catalyze a reaction, but differ in structure. For enzymes located in many tissue sites, an increased total enzyme activity cannot be associated with a specific clinical disorder. However, since the isoenzyme fractions are located in various tissue sources, measurement of the different fractions are considered a more specific indicator of various disorders than total levels.
[Bishop 2022, p238-239; Tietz 2019, p218]

132. **a** The holoenzyme is the active system formed by a protein portion called the apoenzyme and a cofactor, which can be an activator if inorganic and a coenzyme if organic.
[Tietz 2019, p313]

133. **c** Approximately 20-30% of the seminal fluid is prostatic fluid. The composition of the prostatic fluid is acid phosphatase, citric acid, and proteolytic enzymes. The activity of prostatic acid phosphatase may be measured in seminal fluid for medicolegal cases involving rape.
[QCMLS 2021: 2.6.10.1] [Bishop 2022, p247-248]

134. **b** Thymolphthalein monophosphate is the most specific substrate of choice for quantitative endpoint reactions; however, p-nitrophenylphosphate is the preferred substrate for continuous monitoring.
[QCMLS 2021: 2.6.10.2] [Bishop 2022, p247-248]

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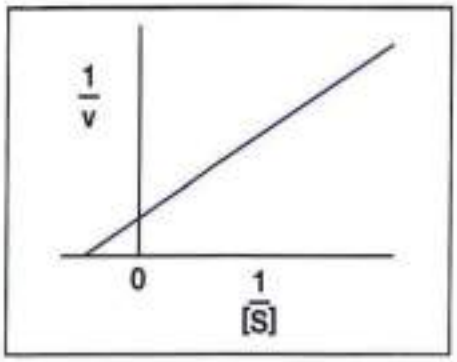
135. **c** All of the enzymes are dehydrogenases, which are oxidoreductases. The oxidoreductases catalyze oxidation reduction reaction between 2 substrates. The enzymes may be located in the liver and the heart; however, the enzymes are in class 1 and not class 3 according to the Enzyme Commission of the IUB system.
[Bishop 2022, p227]

136. **d** The Henderson-Hasselbalch equation describes the derivation of pH as a measure of acidity (using the acid dissociation constant, pK_a) in biological and chemical systems.
[QCMLS 2021: 2.8.2] [Bishop 2022, p350]

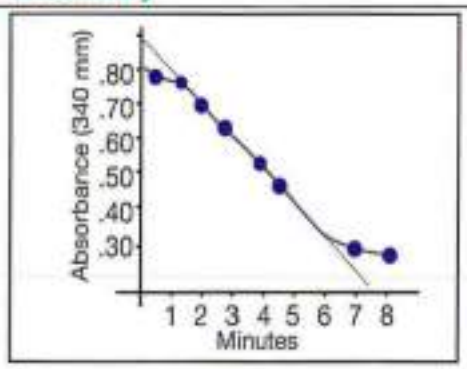


137. **c** Reciprocal of substrate concentration that produces 1/2 the maximal velocity (K_m) is displayed as the intercept of the x-axis on a Lineweaver-Burk transformation. $1/V_{max}$ is determined from 0,0 point up to the ordinate, y-axis. This is a double-reciprocal plot of the original hyperbolic curve and only when $1/V_{max}$ is reached, is the line representing zero order kinetics.
[QCMLS 2021: 2.6.3.1.2] [Bishop 2022, p233-234]

138. **d** In determining enzyme activity a stable change in absorbance per minute is measured and then entered into the equation for calculation of IU. In this example,
ΔAbs/min = 0.089 – 0.023/60 seconds = 0.23.
(0.023 ΔAbs/min/6220 Abs/mol/L) × 10^6 µmol/mol × 1000Δ l/10 Δ L = 1061 IU/L.
[Tietz 2019 p33]

139. **a** NADH is the final spectrophotometric product measured in the preferred LD lactate to pyruvate in alkaline pH chemical reaction. The reverse reaction produces NAD but is has lower linearity due to substrate exhaustion and potential interference from outside sources of NAD. Lactic acid (lactate) is a substrate and pyruvic acid is a product in this reaction but it is easier to measure absorbance change at 340 nm of NAD to NADH because fewer sources of interference are found at that wavelength. ATP plays a role in other enzyme analyses.
[QCMLS 2021: 2.6.7.2] [Tietz 2019, p327]

140. **a** Nicotinamide adenine dinucleotide is the coenzyme system for the LD assay. NADH is the reduced form and NAD is the oxidized form. The coenzymes serve as a substrate for dehydrogenases reactions. ATP/ADP and Cu/Cu^{2+} are not coenzymes. Fe^{2+}/Fe^{3+} is an activator for enzymatic reactions but not key in LD.
[QCMLS 2021: 2.6.7.2] [Tietz 2019, p327]

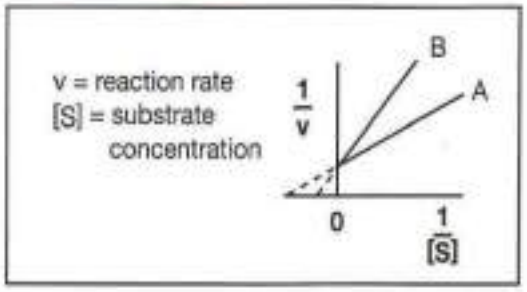


141. **c** In the continuous monitoring method of the measurement of LD activity, the decrease in absorbance at each time interval indicates that the product NAD is formed (pyruvate substrate converted to lactate) is constant up to 6 minutes. After 6 minutes, the substrate concentration is limited, and there is a decrease in the rate of product formation as indicated by the lack of linearity.
[QCMLS 2021: 2.6.3.1] [Bishop 2022, p233-234]

142. **a** Competitive inhibitors bind at the active site of enzyme for the substrate and compete with the substrate for binding sites because of similar chemical structure/configuration. Increasing the substrate concentration will minimize competitive inhibition because it binds preferentially to the active site of the enzyme. In general decreasing temperature will decrease activity of the enzyme reaction but substrate has a bigger effect on competitive inhibition.
[Bishop 2022, p232]

143. **c** Extremely elevated enzyme levels in the sample exhaust substrate concentration before linearity is reached or verified. This is more common than presence of inhibitors. Diluting the sample enzyme level by decreasing the amount of sample will allow for zero order kinetics to be achieved and optimum assay conditions.
[QCMLS 2021: 2.6.3.1, 2.6.3.2] [Bishop 2022, p234]

144. **d** Alkaline phosphatase catalyzes the hydrolysis of para-nitrophenyl phosphate, forming phosphate and free 4-nitrophenyl (4-npp) which, under alkaline conditions, has a very intense yellow color. IFCC recommended methods use 4-npp as the substrate. Nitroblue tetrazolium (NBT) is used with the alkaline phosphatase substrate in western blotting methods and 5-bromo-4-chloro-3-indolyl phosphate is used in immunohistology methods.
[QCMLS 2021: 2.6.9.2] [Bishop 2022, p246]



145. **b** In competitive inhibition, the binding of the substrate is affected; thus, the apparent K_m will be higher while the V_{max} remains the same. In noncompetitive inhibition, the x intercept ($-1/K_m$ or amount of substrate needed to reach V_{max}) may be the same but the time to reach V_{max} is different as indicated by the different y intercept ($1/V_{max}$). In uncompetitive inhibition both V_{max} and K_m are different for substrate and the inhibitor.
[Bishop 2022, p232]

B. Proteins & Other Nitrogen-Containing Compounds

146. **a** Albumin represents >50% of the total mass of protein.
[QCMLS 2021: 2.5.9.2.2] [Bishop 2022, p177-178, Tietz 2019, p299]

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147. **b** Exudate (inflammation or some cancers within pleural cavity. (1) ratio of pleural fluid to serum protein >0.5, (2) a pleural fluid LD value >2/3 the upper limit of normal serum LD, or (3) ratio of pleural fluid to serum LD >0.6).
[QCMLS 2021: 3.4.3] [Bishop 2022, p588-89]

148. **d** The volume of a timed urine is measured and reported as the mass of protein excreted during the 24-hour period.
[Bishop 2022, p195]

149. **b** Note color and characterics and verify that the sample is only for chemistry analysis before centrifuging. CSF should reflect glucose and plasma levels, thus should test for these analytes in plasma at the same time as testing the CSF.
[Bishop 2022, p27]

150. **b** Electrophoresis is used to separate proteins based on the charge that they carry at a certain pH. Proteins are made up of both ionizable amine and carboxyl groups, and can become either negative or positively charged depending on the pH of the solution.
[QCMLS 2021: 2.5.11] [Bishop 2022, p188, Tietz 2019, p168, 307]

151. **a** Intensity of staining is proportional to the amount of protein in the band.
[QCMLS 2021: 2.5.11] [Tietz 2019, p307]

152. **b** Concentration of albumin in serum: Represents >50% of the total mass of protein.
[QCMLS 2021: 2.5.11] [Tietz 2019, p299]

153. **c** Principle of biuret reaction: copper ions form complexes with peptide bonds in proteins.
[QCMLS 2021: 2.5.8] [Tietz 2019, p306]

154. **a** The biuret reaction uses alkaline copper sulfate solution to react with the peptide bonds in the proteins.
[QCMLS 2.5.8] [Tietz 2019, p306]

155. **b** Fibrinogen is found in plasma because it has not been used in blood clotting.
[QCMLS 2021: 2.5.9, t2.12] [Bishop 2022, p189 Tietz 2019, p172, 306]

156. **c** The pore size on the agarose gel allows proteins to pass through, allowing for separate based on their charge-to-mass ratio only.
[QCMLS 2021: 2.16.3] [Tietz 2019, p172]

157. **a** The pore size on the agarose gel allows proteins to pass through, allowing for separate based on their charge-to-mass ratio only.
[QCMLS 2021: 2.16.3] [Tietz 2019, p172, 306]

158. **a** The cysteine amino acid contains a thio (–SH) side chain.
[QCMLS 2021: 2.5.2] [Tietz 2019, p292]

159. **c** Maple syrup disease is caused by a deficiency of the α-keto acid dehydrogenase complex (BCKDC).
[QCMLS 2021: 2.5.2] [Tietz 2019, p883, 889]

160. **d** Elevated albumin levels usually indicated dehydration and are not clinically significant.
[QCMLS 2021: 2.5.9.2.3] [Bishop 2022, p178]

161. **a** Since the total area and the values of the total protein, total and γ-globulin areas are given, a simple A ratio and proportion procedure would be used to calculate the γ-globulin content in g/dL. (γ globulin/total area) × TP = 30 units/180 units × 6.5 = 1.1 g/dL.
[Tietz 2019, p170]

162. **d** Elevated polyclonal γ-globulins with β-γ bridging due to increased IgA.
[QCMLS 2021: 2.5.11] [Bishop 2022, p189]

163. **c** Due to loss of albumin in nephrotic syndrome, increased α-2 globulin is a compensatory mechanism.
[QCMLS 2021: 2.5.11] [Bishop 2022, p189]

164. **d** Alpha-1 antitrypsin deficiency would be seen on electrophoresis where there is a decreased or absent alpha-1 peak.
[QCMLS 2021: 2.5.11] [Bishop 2022, p189, 192]

165. **c** Abnormal protein of a malignant plasma cell will be seen on electrophoresis as a monoclonal gammopathy.
[QCMLS 2021: 2.5.11, f2.7-f2.10] [Bishop 2022, p189]

166. **a** Abnormal liver function such as cirrhosis causes beta-gamma bridging.
[QCMLS 2021: 2.5.11] [Bishop 2022, p189]

167. **d** Bence Jones protein can be detected based on precipitation in urine heated to 40°C to 60°C and then redissolution when further heated to 100°C. However, IFE is more specific and more sensitive.
[QCMLS 2021: 2.5.11] [Henry 2022, p962]

168. **b** Anticoagulant prevents clotting, therefore fibrinogen is present.
[QCMLS 2021: 2.5.11] [Tietz 2019, p175]

169. **c** With a pH 8.0-9.0, proteins take on a negative charge, ie, a negative ion cloud forms. This ion cloud moves in the opposite direction to the cathode. This phenomenon is called electroendosmosis or endosmosis.
[Tietz 2019, p168]

170. **b** An increase or decrease in proteins called acute phase reactants are seen in the alpha-2 region. CRP is one of the strongest proteins to increase in response in inflammation.
[QCMLS 2021: 2.5.11, f2.6] [Bishop 2022, p175, 189]

171. **c** An increased on the gamma region indicates a polyclonal increase in plasma immunoglobulins (ie, IgG) and are the normal response to infection.
[QCMLS 2021: 2.5.11, f2.6] [Tietz 2019, p190, Bishop 2022, p305]

172. **b** An increase in autoimmune IgG with 3 oligoclonal bands, occurs in demyelinating diseases of the CNS, especially multiple sclerosis.
[QCMLS 2021: 2.5.11] [Tietz 2019, p309, Bishop 2022, p192]

173. **c** immunofixation electrophoresis can be performed to distinguish between monoclonal free kappa or lambda light chains and immunoglobulins.
[QCMLS 2021: 2.5.11, f2.14] [Bishop 2022, p196]

174. **c** Specimens should be immediately be placed on ice after collection as the concentration of ammonia in whole blood will increase rapidly due to deamination of amino acids.
[Bishop 2022, p221]

175. **d** Specimens should be immediately be placed on ice after collection as the concentration of ammonia in whole blood will increase rapidly due to deamination of amino acids.
[Bishop 2022, p221]

176. **a** Hyperammonemia can detect encephalopathy in hepatic failure and Reyes syndrome.
[Bishop 2022, p220]

177. **b** In the Jaffe reaction, creatinine reacts with picrate ion in an alkaline medium to yield an orange-red complex. Creatinine is a test for kidney function.
[QCMLS 2021: 2.7.3.2] [Tietz 2019, p220, Bishop 2022, p218]

178. **a** The equation C1V1 = C2V2 can be rearranged to solve for V2 = C1V1/C2, so 15 mg/dL × 100 mL / 500 mg/dL = 3 mL.
[QCMLS 2021: 2.6.4] [Tietz 2019, p117]

179. **c** First convert both volumes to the same unit. 1 milliliter = 1000 microliters. A dilution = sample volume/total volume where total volume = sample volume + diluent volume. 10/1000 = 1/100.
[Tietz 2019, p117]

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180. **d** Causes of increased BUN include glomerular nephritis, chronic renal failure, and tubular necrosis.
[QCMLS 2021: 2.7.5] [Tietz 2019, p369, Bishop 2022, p211]

181. **c** Nitrogen, a metabolic product of protein catabolism is converted to urea for excretion.
[QCMLS 2021: 2.7.3.1] [Tietz 2019, p367, Bishop 2022, p209]

182. **d** In the Jaffe reaction, creatinine reacts with picrate ion in an alkaline medium to yield an orange-red complex. Creatinine is a test for kidney function.
[QCMLS 2021: 2.7.3.2] [Tietz 2019 p220, Bishop 2022, p218]

183. **d** Glomerular filtration rate is estimated by filtration markers such as creatinine clearance.
[QCMLS 2021: 2.7.4] [Bishop 2022, p32]

184. **c** Abnormal results of creatinine and urea nitrogen correlate with abnormal kidney function.
[QCMLS 2021: 2.7.4] [Tietz 2019, p369, 783]

185. **b** Maternal urine instead of amniotic fluid can be identified by measuring creatinine or urea.
[QCMLS 2021: 2.7.4] [Bishop 2022, 592]

186. **d** Uric acid is the major product in the metabolism of purine bases.
[QCMLS 2021: 2.7.3.3] [Tietz 2019, p369]

187. **b** Troponin complex (TIC triple complex) consists of troponin T, troponin I, and troponin C.
[QCMLS 2021: 2.5.9.5.6] [Bishop 2022, p184]

188. **c** CRP is one of the strongest proteins to increase in response in inflammation.
[QCMLS 2021: 2.5.11, f2.6] [Bishop 2022, p175, 189]

189. **d** An increase in autoimmune IgG with 3 oligoclonal bands, occurs in demyelinating diseases of the CNS, especially multiple sclerosis.
[QCMLS 2021: 2.5.11] [Tietz 2019, p309, Bishop 2022, p192]

190. **b** Structural definition of a peptide bond. An amide type of covalent bond that links 2 amino acids from C1 on one amino acid to N2 on a second amino acid.
[QCMLS 2021: 2.5.1] [Bishop 2022, p164, Tietz 2019, p295]

191. **b** Ceruloplasmin, an α-2 glycoprotein synthesized in the liver, binds 90% of the copper in the body. The other 10% is bound to albumin.
[QCMLS 2021: 2.5.9, t2.12, 2.5.9.4.3] [Bishop 2022, p179]

192. **a** α-fetoprotein is the embryonic form of albumin. Elevated maternal serum AFP is associated with neural tube defects. Decreased levels are associated with trisomy 18 and 21.
[QCMLS 2021: 2.5.9.3.2] [Bishop 2022, p178]

193. **b** Normal CSF includes prealbumin, the fraction at 4.5%.
[QCMLS 2021: 2.5.11, f2.4] [Bishop 2022, p197, Henry 2022, p280-81]

194. **c** Haptoglobin is decreased in hemolytic anemia and other causes of *in vivo* hemolysis because it binds strongly with free hemoglobin. Several genotypes exist that may correlate with nephrotic syndrome so phenotyping may be useful for that and for rare deficiency states.
[QCMLS 2021: 2.5.9.4.2] [Bishop 2022 p179]

195. **a** MoM calc = 34/32 = 1.06 (or <2). This represents normal maternal AFP levels. An increased level could indicate neural tube defect, but multiple fetuses should be ruled out. A decreased maternal AFP level could indicate Down syndrome.
[Bishop 2022, p593]

196. **d** Increased CEA levels are seen in patients with liver damage, heavy smokers and following radiation and chemotherapy. The normal population used would not likely have another tumor marker in many or any of the samples to cause interference and age has not presented a marked variation in results. Temperature fluctuations would affect the reference materials including controls, thereby invalidating the test results.
[QCMLS 2021: 2.11.2] [Bishop 2022, p715]

197. **b** Tumor markers are useful for monitoring therapy, detecting recurrence and aiding in prognosis of tumors, but are not useful for screening the general population for cancer or confirming the absence of disease They are not sensitive or specific enough for diagnosis but may aid in the diagnosis in that they are generally set up to be more sensitive than specific and indicate when more testing is needed. Tumor markers are used in conjunction with clinical information; histology results in diagnosis.
[QCMLS 2021: 2.11] [Bishop 2022, p710]

198. **a** Most testicular tumors are germ cell tumors which are characterized by elevated serum levels of alpha-fetoprotein (AFP). Measurement of serum AFP is used in the diagnosis, therapy and follow-up of testicular cancer. The carcinoembryonic antigen is a marker for colon, GI or lung cancer. The serum levels of testosterone and prolactin are not increased in testicular cancer due to the embryonic nature of testicular tumors.
[QCMLS 2021: 2.11.7] [Bishop 2022, p713]

199. **a** Alpha-fetoprotein (AFP) is an oncofetal glycoprotein marker for hepatocellular carcinoma. Elevated levels of AFP (<200 µg/L) are seen in hepatitis and cirrhosis. However, in hepatocellular carcinoma, the levels can be greater than 1000 µg/L.
[QCMLS 2021: 2.11.7] [Bishop 2022, p713]

200. **d** The carcinoembryonic antigen (CEA) is a marker for colon, gastrointestinal and lung cancer. Elevated serum levels of CEA are primarily seen with colon cancer. Although the levels may be increased in individuals with benign conditions, the level of CEA elevation is greater for colon cancer. Brain cancer may be associated with CKBB, testicular cancer is most associated with AFP or hCG and bone cancer is associated with alkaline phosphatase.
[QCMLS 2021: 2.11.2] [Tietz 2019, p340, 345]

201. **a** tPSA (and free PSA) along with digital rectal exam is the recommended screen for prostate cancer in males over 50 years of age. Prostatic acid phosphatase is not sensitive for screening and not specific as bone may be a source of this enzyme, although it has been used historically. HCG and AFP are better screens for testicular cancer or especially choriocarcinoma.
[QCMLS 2021: 2.11.1] [Bishop 2022, p248, 717]

202. **b** A falsely low level could be due to the Hook effect, in which all reagent antibody is saturated with the antigen. PSA levels may be higher than expected due to release from a digital rectal exam of the prostate or in African Americans.
[Bishop 2022 p717]

203. **d** Markers are good for monitoring therapy and detecting recurrence of tumors. Because markers are not highly specific or sensitive enough, they are not good for screening asymptomatic patients in the general population or predicting the likelihood of future tumor development. Oncogenes may be better predictors of probability of tumor development.
[QCMLS 2021: 2.11] [Bishop 2022, p710]

204. **d** CA 125 has a 90% PPV for ovarian cancer. CA 19-9 is associated with pancreatic cancer and CA 19-9 and CEA are associated with colon cancer. CA 15-3 and CA 27-29 and receptor tumor markers are associated with breast cancer. CEA is also associated with lung cancer.
[QCMLS 2021: 2.11.4] [Bishop 2022, p709, 713]

205. **c** PSA can be elevated due to BPH, prostate infection, irritation following DRE and even after recent ejaculation. It is not particularly associated with aspirin, statin or corticosteroid use.
[QCMLS 2021: 2.11.1] [Bishop 2022, p717]

206. **c** Creatinine clearance offers the highest sensitivity in monitoring glomerular function (filtration) of the tests listed. BUN/creatinine ratio can help to determine if elevated urea is due to pre-renal or renal conditions but is not specific to glomerular function. Sodium is more an indication of tubular function of reabsorption or excretion of sodium and water.
[QCMLS 2021: 2.7.4] [Bishop 2022, p32]

207. **a** 52 mg/dL = 0.52 mg/mL. Therefore 0.52 mg/mL × 1136 mL = 591 mg.
[Bishop 2022, 195]

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208. **a** Creatinine clearance = (urine creatinine × urine volume [mL/min]/serum creatinine) × 1.73 / total body surface.
[QCMLS 2021: 2.7.4] [Bishop 2022, p217]

209. **a** Creatinine clearance = (urine creatinine × urine volume [mL/min]/serum creatinine) × 1.73 / total body surface.
[QCMLS 2021: 2.7.4] [Bishop 2022, p217]

210. **b** Renal clearance tests are used to assess kidney function. Renal clearance of a substance is a rate measurement that expresses the volume of blood cleared of that substance (typically creatinine) per unit of time. The unit for the clearance is mL/ min. To calculate creatinine clearance, the following information is required: Serum concentration [S], urine concentration [U] (Note: the serum and urine concentration must be in the same units, for example, mg/L, mmol/L or mg/dL), and volume of urine excreted per minute (V) (volume of urine collected divided by the time period in minutes within 24 hours equal to 1440 minutes). The following formula can then be used: clearance (uncorrected for body mass) = [U] × V/[S].
[QCMLS 2021: 2.7.4] [Bishop 2022, p217]

211. **c** Calculated osmolality (mOsm/kg) = (2 × sodium [mEq/L]) + (glucose [mg/dL]/18) + (BUN [mg/dL]/ 2.8). (133 × 2) + (487/18) + (84/2.8) = 323 mOsm/kg.
[QCMLS 2021: 2.8.3] [Bishop 2022, p320]

212. **d** Creatinine clearance = (urine creatinine × urine volume [mL/min]/serum creatinine) × 1.73 / total body surface. (84 mg/dL × 1440 mL/1440 min/1.4 mg/dL) × 1.73/1.60 = 65 mL/min.
[QCMLS 2021: 2.7.4] [Bishop 2022, p217]

213. **c** MW of urea = 60; MW of N 14 (2 × 14 in urea N). Converting urea N (BUN) to urea is achieved by multiplying UN × 2.14 to equal urea since MW of urea (60)/MW of 2 N (28) = *x* g of urea/1 g of urea N. 28 mg/dL BUN × 2.14 = 60 mg/dL urea.
60 mg/dL × 1 mmol/ 60 mg × 10 dL/L = 10 mmol/L.
[QCMLS 2021: 2.7.3.1] [Bishop 2022, p210]

214. **d** Stray light is any wavelength of light other than that transmitted through the monochromator by the intended sample. It may be due to scratches or dust on the optics. It can be detected by sharp cut-off filters. The mercury vapor lamp or holmium glass filter are used to check the accuracy of the monochromator providing the accurate wavelength. Potassium dichromate solution is used to verify the accuracy of absorbance provided by the photometer.
[Bishop 2022, p196]

215. **c** At pH 8.6, all proteins become anionic due to their carboxyl terminus and migrate to the anode. The protein fractions are named based on the order of their protein electrophoresis migration: albumin, alpha-1, alpha-2, beta, and gamma globulins.
[QCMLS 2021: 2.5.11] [Bishop 2022, p188]

216. **c** An alkaline buffer (pH 8.6) is used in protein electrophoresis to ensure that the all the serum proteins carry a net negative charge.
[QCMLS 2021: 2.5.11] [Bishop 2022, p188]

217. **a** Proteins are amphoteric substances; that is, they contain acidic and basic groups. Their overall (net) charge is highly positive at low pH values, neutral at a particular pH, and negative at still more alkaline pH values. At a pH of 4.5, the positively charged proteins will migrate toward the cathode in an electrophoretic system. Reminder: The cathode is the negatively-charged electrode.
[Tietz 2019, p168]

218. **b** Severe liver disease and acute inflammation usually depict a decreased quantity of albumin while bisalbuminemia is a rare and insignificant genetic defect causing a double albumin band during separation by electrophoresis.
[QCMLS 2021: 2.5.11] [Bishop 2022, p178]

219. **d** The C3 component of complement migrates with beta globulins (beta-2) on electrophoresis. Ceruloplasmin migrates in the alpha-2 region while haptoglobin migrates at alpha-2 zone. Hemoglobin should not be present in serum but if it is, it migrates between alpha-2 and beta-1 zone.
[QCMLS 2021: 2.5.7, t2.11] [Bishop 2022, p180]

220. **d** The total protein method most commonly used, due to ease of automation and accuracy, is Biuret; molybdenum blue is a key reagent in phosphorus analysis. Ferri-ferrocyanide is the reagent in an obsolete method for glucose analysis while resorcinol-HCl is used in fatty acid analysis.
[QCMLS 2021: 2.5.6] [Tietz 2019, p306]

221. **a** The common albumin method uses bromcresol purple at pH 5.2 as a dye binding method. Bromcresol green dye is also used. Globulins do not react well due to the pH used and the higher affinity of the dyes for albumin. Globulins are measured individually with specific immunoassay methods such as immunonephelometry. Bence Jones protein are immunoglobulin fragments of light chains and are detected by urine protein electrophoresis and immunofixation.
[QCMLS 2021: 2.5.9.2.4] [Tietz 2019, p301]

222. **b** According to the American Diabetes Association, for patients with type 2 diabetes, urine albumin should be assessed annually, starting from the point of diagnosis.
[QCMLS 2021: 2.1.5.1.9] [Tietz 2019, p626]

223. **c** Microalbuminuria/albuminuria defined as excretion of 30-300 mg of albumin/24 hr or 20-200 µg/min or 30-300 µg/mg creatinine on 2 of 3 urine collections.
[QCMLS 2021: 2.1.5.1.9] [Tietz 2019, p362; Bishop 2022, p283]

224. **b** Trace amounts of albumin measured in urine is termed microalbuminuria/albuminuria. Given the challenges of complete collection of a 24-hour urine, a random urine albumin is measured along with urine creatinine to determine if the urine specimen is adequate to assess albumin levels.
[Bishop 2022, p283]

225. **b** Increased uric acid levels are seen with gout.
[QCMLS 2021: 2.7.3.3] [Tietz 2019, p371-372]

III. Acid-Base, Blood Gases & Electrolytes

A. Acid-Base Determinations (Including Blood Gases)

226. **a** Reduced excretion of acids due to decreased ammonia formation, decreased Na^+-H^+ exchange, and decreased GFR.
[QCMLS 2021: 2.8.3, t2.16] [Tietz 2019, p693]

227. **a** Excessive loss of bicarbonate, Na^+, and K^+ due to diarrhea can cause acidosis.
[QCMLS 2021: 2.8.3, t2.16] [Tietz 2019, p694]

228. **c** Reduced lung capacity to eliminate CO_2 can cause respiratory acidosis.
[QCMLS 2021: 2.16.5] [Tietz 2019, p696]

229. **a** Total CO_2 is comprised of carbonic acid (H_2CO_3) and bicarbonate (HCO_3^-).
[QCMLS 2.8.2] [Tietz 2019, p686]

230. **a** Reduced lung capacity to eliminate CO_2 can cause respiratory acidosis.
[QCMLS 2021: 2.16.5] [Tietz 2019, p696]

231. **b** Normal ratio bicarbonate is 25 mmol/L to 1.25 mmol/L of carbonic acid resulting in a 20:1 ratio.
[QCMLS 2021: 2.8.3] [Tietz 2019, p689]

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232. **c** H_2CO_3 is measured as dissolved pCO_2. $H_2CO_3 = pCO_2 \times 0.03$.
[QCMLS 2.8.2] [Tietz 2019, p689]

233. **c** Arterial pH reference range.
[QCMLS 2021: 2.8.3] [Tietz 2019, p688]

234. **a** Ketones accumulate in diabetic ketoacidosis and cause a decrease in biocarbonate and result in a high anion gap and metabolic acidosis (ie, decreased pH).
[QCMLS 2021: 2.8.3] [Tietz 2019, p693]

235. **a** HCO_3^- is increased, with respiratory compensation, which results in high TCO_2 in metabolic alkalosis.
[QCMLS 2021: 2.8.3] [Tietz 2019, p695]

236. **d** HCO_3^- and pH is increased, with respiratory compensation, which results in high TCO_2 in metabolic alkalosis.
[QCMLS 2021: 2.8.3] [Tietz 2019, p695]

237. **d** In metabolic acidosis, HCO_3^- levels are decreased due to either the bicarbonate being used as a buffer against acid build-up or bicarbonate being lost through the urine or GI tract, causing decreased pH.
[QCMLS 2021: 2.8.3] [Tietz 2019, p692]

238. **a** Respiratory acidosis results when there is a decrease in pulmonary elimination of CO_2, resulting in an increase in pCO_2 and decreased in pH.
[QCMLS 2021: 2.8.3] [Tietz 2019, p696]

239. **d** Respiratory alkalosis caused by a decreased in pCO_2 due to hyperventilation.
[QCMLS 2021: 2.8.3] [Tietz 2019, p696]

240. **c** The body normally produces acids, which are eliminated by the lungs and kidneys with the blood serving as a buffer system. When the buffering system is disrupted, acidosis or alkalosis occurs.
[QCMLS 2021: 2.8.3] [Tietz 2019, p688]

241. **d** When exposed to air, the CO_2 content and pCO_2, and pO_2 and pH of blood, will rise. This is because the pO_2 of room air is higher than the pO_2 of arterial blood.
[QCMLS 2021: 2.8.3] [Tietz 2019, p433]

242. **b** Electrolyte/blood gas values in respiratory acidosis. Acidosis is determined when an arterial blood sample has a pH of <7.35 and an increased total CO_2 and pCO_2.
[QCMLS 2021: 2.8.3] [Tietz 2019, p688, 696]

243. **c** The bicarbonate/carbonic acid pair is the most important buffer pair in plasma with a normal ratio of 20:1.
[QCMLS 2021: 2.8.2] [Tietz 2019, p689]

244. **c** Bicarbonate, the major component of CO_2 in blood, is usually defined as the total of HCO_3^-, carbonate ion and CO_2.
[QCMLS 2021: 2.8.2] [Tietz 2019, p688]

245. **d** The kidneys are the main compensatory mechanism in respiratory acidosis where there is increased Na^+/H exchange, increased ammonia formation, and increased bicarbonate reabsorption.
[QCMLS 2021: 2.8.3] [Tietz 2019, p696]

246. **a** This patient exhibits a slight decrease in pH, increased bicarbonate, and an increased pCO_2. Given the bicarbonate buffer system predominating in the blood, pH is directly related to bicarbonate levels and inversely related to carbonic acid as measured by pCO_2. These results indicate a respiratory acidosis with metabolic compensation by renal retention of bicarbonate.
[QCMLS 2021: 2.8.3] [Bishop 2022 p356-357]

247. **c** Calculated osmolality = (2 × 137) + (100 / 18) + (18 / 2.8) = 286. Osmolal gap = 301 – 286 = 15 (normal <10), which indicates possible organic acids such as methanol, ethanol, propylene glycol, or ethylene glycol. Increased anion gap indicates metabolic acidosis with increased osmolal gap.
[QCMLS 2021: 2.8.3, 2.9.1.4] [Bishop 2022, p319]

248. **c** There are several preanalytical/preexamination interferences with blood gases, including presence of air bubbles, which will falsely increase pO_2 and decrease pCO_2. Room temperature allows for cellular metabolism to lower pH and pO_2, and raise pCO_2.
[QCMLS 2021: 2.8.3] [Tietz 2019, p433-436]

249. **a** The presence or exposure of excess gas (oxygen) in the syringe used to collect blood gas specimens will cause diffusion of carbon dioxide out of the specimen, oxygen into the specimen, and an increase in pH. Specimens not maintained in ice water nor analyzed immediately experience cellular glycolysis, which causes increase in carbon dioxide, decrease in oxygen, and lower pH.
[QCMLS 2021: 2.8.3] [Bishop 2022, p365]

250. **b** Arterial specimens are best collected anaerobically with lyophilized heparin anticoagulant in sterile syringes.
[QCMLS 2021: 2.8.3] [Tietz 2019, p433, Bishop 2022, p365]

251. **d** If not immediately analyzed, failure to adequately chill blood gas specimens will allow glucose metabolism, which increases pCO_2 and lowers pH.
[QCMLS 2021: 2.8.3] [Tietz 2019, p434]

252. **d** Arterial blood gases even when maintained in ice water and anaerobic conditions deteriorate within an hour due to diffusion through the plastic syringe.
[QCMLS 2021: 2.8.3] [Bishop 2022, p365]

253. **c** pH = 6.1 + log(salt/acid); salt = total carbon dioxide content – carbonic acid.
[QCMLS 2021: 2.8.2] [Bishop 2022, p350]

254. **d** $pH = pK_a + \log(salt/acid)$; salt = bicarbonate; acid = carbonic acid.
[QCMLS 2021: 2.8.2] [Bishop 2022, p353]

255. **c** Solve using the Henderson-Hasselbalch equation. $pH = pK' + \log([HCO_3]/[.03 \times pCO_2])$, where pK', the negative log of the dissociation constant for dissolved $CO_2 + H_2CO_3 = 6.1$ and 0.03 is the solubility coefficient for CO_2 gas. $pH = 6.1 + \log(18 / [.03 \times 60]) = 6.1 + \log(18 / 1.8)$ $pH = 6.1 + \log(10)$. Since $\log(10) = 1$, pH = 7.1.
[QCMLS 2021: 2.8.2] [Bishop 2022, p353]

256. **c** Given the values of bicarbonate, carbonic acid, and the pK, the pH can be easily calculated using the Henderson-Hasselbalch equation. The Henderson-Hasselbalch equation describes the derivation of pH as a measure of acidity (using the acid dissociation constant, pK_a) $pH = 6.1 + \log(24 / 1.2)$; pH = 6.1 + 1.3 = 7.40 in biological and chemical systems.
[QCMLS 2021: 2.8.2] [Tietz 2019, p431]

257. **c** This describes the components of the pO_2 electrode. The pH electrode is glass of composition selective for H^+ and is measured versus a silver/silver chloride internal reference electrode, while the pCO_2 electrode is a modified pH electrode with a selective polymer membrane covering allowing for carbon dioxide diffusion. Bicarbonate is measured after conversion of total carbon dioxide in an enzymatic reaction to form a spectrophotometric analyte or measured with an indirect ISE method similar to the pCO_2 electrode.
[QCMLS 2021: 2.16.5] [Tietz 2019, p155-156]

258. **b** The pH electrode, a glass electrode, contains a specially designed thin piece of glass as a membrane. The glass membrane is made of silicon dioxide, added oxides, and various metals. The membrane is selectively sensitive to hydrogen ions. A silver electrode may be part of the polarographic electrode for oxygen or coulometric electrode for chloride. Platinum electrodes are often the cathode where oxygen is reduced in the amperometric method for oxygen.
[QCMLS 2021: 2.16.5] [Tietz 2019, p147-148, 154-155]

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259. **a** The reference pH electrode is often constructed of Ag and AgCl. Quinhydrone has been used in analytical chemistry redox half cell for measuring pH, but not used in clinical laboratories. The glass sensing electrode detects hydrogen ions (H^+) rather than hydrogen (H_2).
[QCMLS 2021: 2.16.5] [Bishop 2022, p103-4]

260. **b** The pO_2 electrode functions on the amperometric principle, in which the measurement of electrical current occurs at a constant voltage (or potential). pH is measured with an ion-selective glass electrode, while pCO_2 is measured by gas-sensing electrode with pH detection. Ca^{2+} is measured with ion-selective electrode fitted with a selective membrane.
[QCMLS 2021: 2.16.5] [Bishop 2022, p107]

261. **b** pH, pCO_2, and pO_2 are directly measured by modern blood gas analyzers; other parameters, including HCO_3^- and oxygen saturation, are calculated.
[QCMLS 2021: 2.16.5] [Bishop 2022, p360]

262. **b** The pCO_2 electrode is a self-contained potentiometric cell. CO_2 gas from the sample or calibration matrix diffuses through the selective membrane and dissolves in the internal electrolyte layer. Carbonic acid is formed and dissociates, shifting the pH of the bicarbonate solution in the internal layer. This shift is related to the carbon dioxide in the sample.
[QCMLS 2021: 2.16.5] [Tietz 2019, p151]

263. **c** The pCO_2 electrode uses a gas-permeable membrane covering a glass electrode with internal Ag/AgCl reference for detecting change in potential due to H^+ produced in the bicarbonate/carbonic acid reaction. It is a modified pH electrode. Oxygen saturation can be measured photometrically or calculated. Bicarbonate is usually measured after conversion of total carbon dioxide in an enzymatic reaction to form a spectrophotometric analyte.
[QCMLS 2021: 2.16.5] [Tietz 2019, p151]

B. Electrolytes

264. **c** The best tests for renal tubular function when investingating diabetes insipidus are serum and urine osmolality.
[QCMLS 2021: 2.7.4.2] [Tietz 2019, p783]

265. **c** Osmolality empirical calculation is 1.86[Na^+] + glucose/18 +urea/2.8 + 9.
[QCMLS 2021: 2.8.3] [Tietz 2019, p428]

266. **d** Serum (>300 mOsm/kg) and urine osomality (<300 mOsm/kg) determinations can be diagnostic of diabetes insipidus.
[QCMLS 2021: 2.9.7.3]

267. **b** The definition of an osmolal gap is the difference between the observed/measured and calculated osmolalities in serum analysis.
[QCMLS 2021: 2.9.7.3] [Tietz 2019, p420]

268. **c** Ion selective electrode standard Na^+/K^+.
[QCMLS 2021: 2.9.4.3] [Tietz 2019, p148]

269. **b** The anion gap is due to unmeasured anions (eg, proteins, SO_4^{2-}, $H_2PO_4^{2-}$) that are present in plasma.
[QCMLS 2021: 2.9.7.1] [Tietz 2019, p693]

270. **c** Patients with very high WBC or platelet counts can result in a pseudohyperkalemia due to their rupture during the clotting process.
[QCMLS 2021: 2.9.4.3] [Tietz 2019, p41, 442]

271. **d** Lactic acidosis causes an actual increased anion gap due to acidosis.
[QCMLS 2021: 2.9.7.1] [Tietz 2019, p694]

272. **c** The anion gap is the sum of Cl^- and HCO_3^- subtracted from the Na^+, the difference, or "gap," averaging 12 mmol/L in healthy subjects.
[QCMLS 2021: 2.9.7.1] [Tietz 2019, p693]

273. **d** The HCO_3^- is exchanged for plasma Cl^- in the erythrocyte to maintain electroneutrality. This is called the chloride shift.
[QCMLS 2021: 2.8.3] [Tietz 2019, p690-692]

274. **d** Na^+ as the major cation in plasma is responsible for maintaining osmotic pressure.
[QCMLS 2021: 2.9.1.4] [Tietz 2019, p421]

275. **a** K^+ is released from the red cells during hemolysis, affecting the serum measurement.
[QCMLS 2021: 2.9.4.3] [Tietz 2019, p84]

276. **b** Osmolality is determined by the number of particles of solute rather than their size or charge. Sodium, as the major serum cation, is the solute that contributes half of serum osmolality. Chloride, while the major anion, exists in smaller concentrations (99-109 mEq/L) than sodium (136-145 mEq/L) in normal serum, so it contributes less to osmolality than does sodium. Glucose, urea, and other ions are present in much smaller concentrations in serum.
[QCMLS 2021: 2.9.1.4, 2.9.3, 2.9.5.1] [Tietz 2019, p427]

277. **d** Sweat chloride reference intervals, recommended for all patients regardless of age, are: ≤29 mmol/L: CF unlikely; 30-59 mmol/L: intermediate; ≥60 mmol/L: indicative of CF.
[Tietz 2019, p425]

278. **a** Direct ISE methods are NOT affected by the electrolyte exclusion effect and are primarily used in whole blood measurements.
[QCMLS 2021: 2.9.4.3] [Tietz 2019, p423]

279. **b** Indirect ISE methods are affected by the electrolyte exclusion effect, caused by the lipid in the sample which results in falsely decreased values.
[QCMLS 2021: 2.9.4.3] [Tietz 2019, p423]

280. **c** Indirect ISE methods are affected by the electrolyte exclusion effect, caused by the protein in the sample which results in falsely decreased values.
[QCMLS 2021: 2.9.4.3] [Tietz 2019, p423]

281. **c** About 40% of Ca^{2+} is bound to albumin.
[QCMLS 2021: 2.10.1] [Bishop 2022, p331]

282. **b** Ionized Ca^{2+} measurment uses ISE with special membranes that selectively bind Ca^{2+} ions.
[QCMLS 2021: 2.10.1.6] [Bishop 2022, p335]

283. **b** Parathyroid glands regulate Ca^{2+} and PO_4^{3-} metabolism by the excretion of PTH.
[QCMLS 2021: 2.10.1.1] [Bishop 2022, p332]

284. **a** Symptoms of tetany can occur with hypocalciumemia.
[QCMLS 2021: 2.10.1.4] [Bishop 2022, p334]

285. **a** Hypomagnesium can cause hypocalciumemia, which in turn can cause tetany.
[QCMLS 2021: 2.10.2.2] [Bishop 2022, p333]

286. **b** Reciprocal relationship of Ca^{2+} and phosphate.
[QCMLS 2021: 2.10.1.1] [Bishop 2022, p332]

287. **c** PTH increases phosphorus excretion in response to change in calcium levels.
[QCMLS 2021: 2.10.1.1] [Bishop 2022, p332, 341, 471]

288. **a** The most common cause of low phosphate is in patients who are receiving long-term parenteral treatment.
[QCMLS 2021: 2.10.1.5] [Bishop 2022, p341]

289. **c** Mineral results in chronic renal disease. In chronic renal disease, the vitamin D activation decreases, causing increased phosphorus and decreased calcium concentrations.
[QCMLS 2021: 2.10.1.1, 2.10.1.5] [Bishop 2022, p333, Tietz 2019, p862]

290. **c** Transferrin is the principal plasma transport protein for iron.
[QCMLS 2021: 2.10.2.1.4] [Tietz 2019, p302]

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291. **d** Iron must be released from transferrin before it is reduced from Fe^{3+} to Fe^{2+} and complexed with a chromogen, such as bathophenanthroline or ferrozine.
[QCMLS 2021: 2.10.2.1.4] [Tietz 2019, p516]

292. **d** The time at which a specimen is obtained is important for blood constituents such as iron that undergo marked diurnal variation.
[QCMLS 2021: 2.10.2.1.4] [Tietz 2019, p80]

293. **c** Release of iron from the cells is not adequate for heme formation.
[QCMLS 2021: 2.10.2.1.3] [Tietz 2019, p512]

294. **b** Iron deficiency anemia is characterized by a decreased iron level and increased transferrin and TIBC levels.
[QCMLS 2021: 2.10.2.1.3] [Bishop 2022, p632]

295. **d** Hemochromatosis is characterized by a high iron level and a decreased TIBC.
[QCMLS 2021: 2.10.2.1.3] [Bishop 2022, p682]

296. **c** Toxic levels of copper in the urine but low levels in the serum suggest Wilson disease in which there is a deficiency in the transport protein, ceruloplasmin. Ceruloplasmin, made by the liver, is the primary serum copper-bearing protein. Wilms tumor is a nephroblastoma that affects kidney function, while Addison disease is associated with hypocortisolism, hypoglycemia, and electrolyte disturbances.
[QCMLS 2021: 2.5.9.4.3] [Bishop 2022, p179]

297. **c** Use of a tourniquet for over 1-3 minutes can cause elevation in potassium as well as protein and albumin, calcium, and hemoglobin. Redrawing the specimen with shorter tourniquet time is preferred. Repeating the test with the same sample will likely give the same result since it appears to be a preanalytical/preexamination error rather than an analytical error. Pumping of the fist will falsely increase K^+ as well.
[QCMLS 2021: 2.9.4.2] [Tietz 2019, p80, 422]

298. **b** Serum for acid phosphatase measurement should not be stored at room temperature without further preservation. This analyte requires special collection (citrate 10 g/L) and storage (frozen) conditions to help stabilize the pH at about 6.2.
[QCMLS 2021: 2.6.10.2] [Tietz 2019, p226, 329]

299. **a** The difference between the actual osmolality commonly measured by freezing point depression and the calculated osmolality is referred to as the osmol gap. This is used to screen for unmeasured osmotically active substances, including volatile alcohols. Normally, the osmol gap is <10 mOsm/kg by most opinions especially after correction for ethanol.
[QCMLS 2021: 2.8.3] [Tietz 2019, p566]

300. **d** In calcium analysis by AAS, lanthanum is added to bind with phosphate, thereby preventing interference by the formation of calcium phosphate.
[QCMLS 2021: 2.10.1.6] [Tietz 2019, p83, 135, 749]

301. **a** The electrolyte exclusion effect applies only to indirect methods because the sample is diluted prior to analysis. This effect is caused by the solvent displacing effect of high concentrations of lipid and protein in the sample resulting in falsely-decreased values. The exclusion effect due to hyperlipidemia or hyperproteinemia is not an issue with direct ISE or spectrophotometric methods.
[QCMLS 2021: 2.9.3.1.1] [Tietz 2019, p423]

302. **a** Coulometry is an electrochemical technique used to measure the amount of electricity passing between 2 electrodes in an electrochemical cell. An application of coulometry is the titration of chloride with silver ions generated by electrolysis from a silver wire at the anode. Polarography is also an electrochemical technique, but measures current. Mass spectroscopy and chromatography are separation techniques.
[QCMLS 2021: 2.10.5] [Tietz 2019, p159]

303. **a** Coulometry is still used for chloride determinations in body fluids, such as sweat. However, chloride ion-selective electrodes (ISE) are commonly used today.
[QCMLS 2021: 2.16.5] [Bishop 2022, p107, 329]

304. **c** Analyzers fitted with ion-selective electrodes (ISE) usually contain potassium electrodes with liquid ion-exchange membranes that incorporate valinomycin. Valinomycin is a neutral carrier (ionophore) that binds potassium in the center of a ring of oxygen atoms. Sodium is measured with a selective glass electrode, while chloride is measured coulometrically. Ionized calcium is measured with a polymer membrane ISE, but it does not contain valinomycin.
[QCMLS 2021: 2.16.5] [Tietz 2019, p149, 159]

305. **d** The total iron binding capacity (TIBC) is the amount of iron that transferrin and other minor iron binding proteins are capable of binding. In the measure of the TIBC, the molecules are saturated with iron (since they are only 30% saturated normally). Magnesium carbonate is used to remove the excess unbound by adsorption. In an acid pH, the transferrin releases iron (but does not precipitate the transferrin protein) so that reduced iron (Fe^{2+}) can complex with the dye, ferrozine. Circulating plasma iron bound to transferrin is measured rather than iron bound to hemoglobin.
[QCMLS 2021: 2.10.2.1.4] [Tietz 2019, p516]

306. **d** Atomic absorption spectrophotometry (AAS) measures calcium by detecting its atomic absorption of electromagnetic radiation. One limitation of this method is the non-spectral interference that occurs when phosphates are present and complex with calcium. The use of lanthanum chloride with the method has prevented the interference. Lanthanum chloride competes for the phosphate. O-cresolphthalein complexone uses dilute acid and 8-hydroxyquinoline to remove interferences. EDTA chelates out calcium and magnesium.
[QCMLS 2021: 2.10.1.6] [Tietz 2019, p83, 135, 749]

307. **a** Osmometry of serum and other body fluids is commonly measured by freezing-point depression, using a freezing point osmometer. Sodium provides the largest contribution to osmolality, the concentration of dissolved particles in mmol/Kg of plasma or serum water. Measuring osmotic pressure is time consuming and subject to errors so freezing point or vapor depression are used most readily in the clinical lab. Sedimentation is used in sedimentation rate of red blood cells in whole blood to nonspecifically assess for presence of inflammatory proteins in the plasma.
[QCMLS 2021: 2.9.7.2] [Bishop 2022, p111]

308. **a** The osmolality of a solution does not depend on the kind of particles but only on the number of particles, therefore it is called a colligative property. In cryoscopic osmometry, the sample is frozen using a process to determine at what temperature freezing occurs and that is compared to standard freezing point of a pure solution. Thus, freezing point depression is the principle and it is not absolute, but changes based on number of particles in solution.
[QCMLS 2021: 2.9.7.3] [Bishop 2022, p112]

IV. Special Chemistry

A. Endocrinology

309. **a** Parathyroid glands regulate Ca^{2+} and PO_4^{3-} metabolism by the excretion of PTH. In hyperparathyroidism, the Ca^{2+} is increased and PO_4 levels are decreased.
[QCMLS 2021: 2.10.1.4] [Bishop 2022, p332]

310. **b** TSH is produced by pituitary gland, while TRH is produced in the hypothalamus and thyroxine is produced in the thyroid gland.
[QCMLS 2021: 2.12.1.4.2] [Bishop 2022, p382]

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311. **b** Increased TSH, decreased free T_4,and total T_4, and positive microsomal Ab are consistent with primary hypothyroidism. A patient with hyperthyroidism, including Graves disease, would have increased free and total T_4. Hyperthyroidism is also called thyrotoxicosis.
[QCMLS 2021: 2.12.1.5.4] [Bishop 2022, p404]

312. **a** The patient's symptoms and TSH point to primary hyperthyroidism and if the free T_4 is elevated this is consistent with primary hyperthyroidism, likely due to Graves disease, a type of thyrotoxicosis. Panhypopituitarism is associated with low thyroid hormone levels due to low levels of TSH and other pituitary hormones.
[QCMLS 2021: 2.12.1.5.1] [Bishop 2022 p401, 405-406]

313. **c** T_3 is more biologically active; 80% of T_4 is converted into T_3.
[QCMLS 2021: 2.12.1.4.1, 2.12.1.4.3] [Bishop 2022, p398]

314. **b** Congenital hypothyroidism presents with very low thyroid hormones and is best confirmed by serum TSH.
[QCMLS 2021: 2.12.1.5.6] [Bishop 2022, p404, 748]

315. **b** >99% of T_3 and T_4 are bound to proteins: thyroxine-binding prealbumin, thyroxine-binding globulin, and albumin. TRH and TSH are hormones that stimulate the pituitary and thyroid glands to produce thyroxine and triiodothyronine.
[QCMLS 2021: 2.12.1.10] [Bishop 2022, p398]

316. **b** TSH is the American Thyroid Association's recommended screening test due to the extreme sensitivity of testing methods for detecting endocrine changes. It is useful for monitoring thyroid therapy and to detect subclinical hypo- or hyperthyroidism.
[QCMLS 2021: 2.12.1.10] [Tietz 2019, p819]

317. **d** Current recommendations for neonatal screening of thyroid function is with blood spot samples tested for either TSH or free T_4. Total thyroxine levels were measured in the past, but are affected by thyroid hormone protein binding issues.
[QCMLS 2021: 2.12.1.5.5] [Bishop 2022, p748]

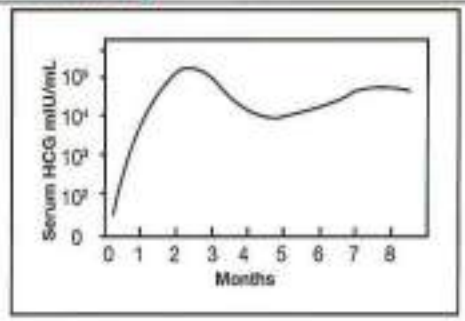
318. **d** Hashimoto thyroiditis is a type of chronic lymphocytic thyroiditis resulting in autoimmune destruction of the thyroid gland, characterized by decreased total and free thyroxine and increased levels of TSH and titer of anti-TPO, anti-thyroglobulin and possibly anti-TSH receptor antibodies. Graves disease is also often positive for anti-TPO but is associated with elevated free T_4 and decreased TSH levels.
[QCMLS 2021: 2.12.1.5.7] [Bishop 2022, p404]

319. **a** Age, sex, and physical complaint, with elevated TSH point to primary hypothyroidism. TSH-secreting tumors are quite rare. Graves disease is a type of primary hyperthyroidism, and TSH would be decreased.
[QCMLS 2021: 1.12.1.5.4] [Bishop 2022, p400]

320. **b** hCG is the primary marker for early pregnancy. FSH and LH are pituitary hormones necessary for ovulation and the menstrual cycle.
[QCMLS 2021: 2.12.5.1] [Bishop 2022, p608]

321. **b** During pregnancy, the largest fraction of estrogen in urine is estriol; however, unconjugated estriol has the most clinical utility in fetal screening.
[Tietz 2019, p864, 873]

322. **b** Luteinizing hormone triggers ovulation, which then turns the follicle into the corpeus luteum.
[QCMLS 2021: 2.12.5.1] [Tietz 2019, p834]



323. b The chart represents hCG levels in normal pregnancy; the peak occurs at 2-2.5 months.
[QCMLS 2021: 2.12.5.1] [Tietz 2019, p869]

324. b Serum hCG levels peak at 8-10 weeks after the last menstrual period (LMP).
[QCMLS 2021: 2.12.5.1] [Tietz 2019, p868]

325. b Aldosterone is a hormone produced by the adrenal cortex, therefore called an adrenal corticosteroid. Angiotensinogen is a protein produced in the liver; growth hormone is produced in the pituitary; and epinephrine is an amine hormone produced in the adrenal medulla.
[QCMLS 2021: 2.12.2.1.4] [Bishop 2022, p377, 385]

326. d All adrenal steroid hormones (including cortisol and progesterone) are enzymatically derived from cholesterol. Catecholamines are not steroid hormones.
[QCMLS 2021: 2.12.5.1] [Bishop 2022, p377]

327. d Testosterone is the most biologically active androgen in the embryonic stage and later effects sperm production and secondary sex characteristics. Cortisol is a corticosteroid. Dehydroepiandrosterone (DHEA) and androstenedione are made in small amounts by the testes.
[QCMLS 2021: 2.12.5.2] [Bishop 2022, p449]

328. a The normal variation of serum cortisol is higher at 8 AM than 4 PM, which are often the standard times used to measure cortisol. This assumes sleep patterns at night. These may be altered in disorders such as Cushing or Addison disease, as well as in hypopituitarism.
[QCMLS 2021: 2.12.2.2] [Tietz 2019, p780, 791]

329. b Angiotensin II is a vasoconstrictor and stimulates the adrenal cortex to produce aldosterone. Vasopressin is released when osmoreceptors in the hypothalamus are stimulated by hyperosmolarity. Parathyroid hormone's main action is to regulate calcium, phosphorus, and magnesium levels by multi-organic effect. While angiotensin II is part of the renin-angiotensin system, aldosterone production and secretion by the adrenal gland is actually stimulated by angiotensin II so levels increase and cause renal reabsorption of sodium and increased blood volume.
[QCMLS 2021: 2.7.2.1] [Tietz 2019, p655]

330. d Vanillylmandelic acid (VMA) is the major metabolite of both epinephrine and norepinephrine. VMA is measured in a 24-hour urine specimen. Homovanillic acid is a metabolite of dopamine. DOPA is a precursor to dopamine, made from tyroside. Dopamine is an intermediate in the production of norepinephrine and epinephrine.
[QCMLS 2021: 2.11.11.2] [Bishop 2022, p440]

331. c Intact parathyroid hormone (84 amino acids with C and N terminal ends) relates best to secretory function of the parathyroid gland. PTHrP is a tumor marker, and fragments of PTH often correlate with renal failure.
[Bishop 2022, p476]

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332. c 95% of congenital adrenal hyperplasia is associated with a deficiency of 21-hydoxylase. Increased 17-OH progesterone is seen if measured by the laboratory. Cortisol and 11-deoxycortisol are increased in Cushing disease, and require 17-hydroxylase to be produced. Aldosterone is increased in congenital adrenal hyperplasia caused by 17-hydroxylase deficiency.
[QCMLS 2021: 2.12.2.1.5] [Bishop 2022, p430]

333. c ACTH stimulation tests, using synthetic ACTH, will differentiate primary from other causes of adrenal insufficiency. Synthetic ACTH will not cause the adrenal gland to respond in primary insufficiency. Diagnosis may be suspected, but needs further testing with decreased 8 AM cortisol level and increased baseline ACTH level. 17-keto- and 17-hydroxysteroids will also often be decreased with adrenal insufficiency but not confirmed without ACTH stimulation. A decreased urinary and serum cortisol level after metyrapone is given will confirm secondary adrenal insufficiency.
[QCMLS 2021: 2.12.2.1.2] [Bishop 2022, p433]

334. a The 24-hour urine free cortisol is the most sensitive and specific screen for hypercortisolism but must be confirmed on a second occasion. Likewise, plasma cortisol after overnight dexamethasone suppression is diagnostic. Urinary 17 hydroxycorticosteroids or plasma corticosterone are not diagnostic for adrenal cortical hyperfunction and not commonly ordered.
[QCMLS 2021: 2.12.2.1.1] [Bishop 2022, p435]

335. c A single measurement of growth hormone (GH) is not diagnostic. Following an overnight fast, a 100 g oral glucose load will cause a large drop in serum growth hormone in a normal individual, but will not suppress in patients with acromegaly, which with screening also show increased GH levels. Elevated phosphate is associated with decreased parathyroid hormone. Decreased growth hormone releasing factor may be present but not easily detected. Increased somatostatin is associated with inhibition of GH and TSH but not easily measured nor diagnostic for acromegaly.
[QCMLS 2021: 2.12.4] [Bishop 2022, p417-418]

336. b About 55-60% of patients whose breast tumors demonstrate estrogen receptors (ER) respond well to endocrine therapy. Ovarian cancer can be monitored with CA-125 tumor marker, while FSH and LH are often used to monitor the menstrual cycle.
[Bishop 2022, p712]

337. c Due to circadian variation, the 24-hour UFC is an accurate measurement of active forms of cortisol. While a 8 AM and 4 PM plasma cortisol is an effective screen for hypercortisolism, it is not diagnostic. Administration of metyrapone is used to test pituitary secretion of ACTH to rule out secondary adrenal insufficiency.
[QCMLS 2021: 2.12.2.1.1] [Bishop 2022, p434]

338. b Primary hypocortisolism = decreased cortisol/elevated ACTH; secondary hypocortisolism = decreased cortisol/decreased ACTH. Hypoglycemia and hypoaldosteronism may be associated with both forms of hypocortisolism. Cortisol levels are expected to be decreased in both primary and secondary hypocortisolism.
[QCMLS 2021: 2.12.2.1.2] [Bishop 2022, p431-433]

339. d Angiotensin II directly stimulates the adrenal cortex to release aldosterone. While angiotensin II is part of the renin-angiotensin system, aldosterone production and secretion by the adrenal gland is actually stimulated by angiotensin II so levels increase and cause renal reabsorption of sodium and increased blood volume. Renin is produced by the kidney with enzyme activity that acts upon angiotensinogen, a glycoprotein produced in the liver. This forms angiotensin I, which circulates and is converted to ATA II by angiotensin converting enzyme (ACE) in the pulmonary circulation.
[QCMLS 2021: 2.7.2.1] [Bishop 2022, p318]

340. **b** HVA is a metabolic by-product of adrenal medullary catecholamines, which are produced in excessive amounts in malignant neuroblastoma. Cushing disease is associated with adrenal cortical adenomas producing excessive corticosteroids. Graves disease is associated with hyperthyroidism.
[QCMLS 2.12.3] [Bishop 2022 p439]

341. **d** Chemiluminescent labels are based on the emission, rather than absorption, of light produced during a chemical reaction. These labels are very useful because they are ultrasensitive and provide very low levels of detection (1×10^{-18} to 1×10^{-21} mol/L) with little or no background interference. Unlike radioisotopic reactions, they do not produce harmful emissions and are not monitored by gamma counters.
[QCMLS 2021: 2.16.4] [Tietz 2019, p246]

B. Vitamins & Nutrition

342. **b** Retinol is one of the 3 biologically active forms of fat-soluble vitamin A.
[QCMLS 2021: 2.15.1] [Bishop 2022, p690]

343. **c** In chronic myeloid leukemia, the serum vitamin B_{12} exists in the bound form and the binding capacity for added B_{12} is increased, so serum concentrations are elevated. The other choices are causes of B_{12} deficiency.
[QCMLS 2021: 2.15.1, t2.31]

344. **a** Intestinal absorption of vitamin B_{12} requires the intrinsic factor.
[QCMLS 2021: 2.15.1, t2.31] [Bishop 2022, p696]

345. **a** A deficiency of vitamin A leads to night blindness, and if prolonged, total blindness. Profound vitamin C deficiency is scurvy; B_1 (thiamine) is beriberi and niacin deficiency is pellagra.
[QCMLS 2021: 2.15.1, t2.29] [Tietz 2019, p472; Bishop 2022, p691]

346. **b** Vitamin B_{12} is not absorbed well in pernicious anemia due to lack of intrinsic factor resulting in neuropathy, macrocytic anemia and decreased levels of vitamin B_{12}.
[QCMLS 2.15.2] [Bishop 2022, p697]

347. **b** Based on their chemical makeup, vitamins A, D, E, and K are not water soluble, while vitamin C and the B vitamins are water soluble.
[QCMLS 2.15.1] [Bishop 2022, p690]

348. **d** A deficiency of vitamin D in children leads to rickets.
[QCMLS 2021: 2.15.1, t2.31] [Bishop 2022, p692]

349. **d** A deficiency of niacin may be seen with chronic alcoholism, and is known as pellagra.
[Bishop 2022, p690]

350. **a** Erythrocyte transketolase activity is decreased in thiamine deficiency. Low values of it have also been found in chronic alcoholism. Folate functions metabolically as coenzyme in single carbon transfer reactions. It plays a role in nerve, growth and hematologic function. Riboflavin deficiency is associated with decreased glutathione reductase activity. Ascorbic acid, a reducing substance, is associated with many functions, and its deficiency is termed scurvy.
[Bishop 2022, p693]

C. Therapeutic Drug Monitoring

351. **a** Intestinal absorption of digoxin is variable, and tissue uptake is slow; therefore, serum levels are measured 8 hours after administration to permit tissue and serum levels to equilibrate. The serum level at 6 hours will not be falsely low, but also will not correlate well with the pharmacologic value in the tissue due to cellular uptake. Likewise, it takes 8-10 hours for the cellular uptake to correlate with serum value, so prior to 6 hours the digoxin is not at the cellular level. In circulation, 25% of digoxin is protein-bound.
[QCMLS 2021: 2.14.2.1] [Bishop 2022, p631]

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352. **c** A steady-state therapeutic drug level is achieved between 5 and 7 doses. Many variables affect when steady state is achieved.
[QCMLS 2021: 2.14.1] [Bishop 2022, p630]

353. **b** Low serum protein means less of a drug is bound to protein. This may occur due to nephrotic syndrome, which causes significant protein loss and hypoalbuminemia. Acute inflammation, pregnancy and multiple myeloma are all associated with relative or actual hyperproteinemia.
[QCMLS 2021: 2.7.5.1.2] [Bishop 2022, p185, 625]

354. **a** The aminoglycoside antibiotics (eg, gentamicin and tobramycin) are associated with toxicity of the ears (ototoxicity) and proximal convoluted tubules of the kidneys (nephrotoxicity). Since they are poorly absorbed in the GI tract, these antibiotics are always administered as IV or IM so intestinal malabsorption will not affect blood levels of these drugs.
[Bishop 2022, p635-636]

355. **c** NAPA is the active metabolite of procainamide. Primidone is the inactive form of phenobarbital, but may be measured along with or instead of phenobarbital.
[QCMLS 2021: 2.14.2.2] [Bishop 2022, p634]

356. **c** Lithium (carbonate) is used to treat manic depression or bipolar disorder. Digoxin is a cardioactive drug; phenytoin is used in epileptic seizure control; and acetaminophen is a nonprescription nonsteroidal anti-inflammatory drug (OTC NSAID).
[QCMLS 2021: 2.14.2.5] [Bishop 2022, p640-641]

357. **d** Phenytoin (trade name Dilantin®) is an anticonvulsant therapeutic drug used to treat seizure disorders.
[Bishop 2022, p637]

358. **d** The action of the drug theophylline is bronchodilation and smooth muscle relaxation.
[Bishop 2022, 644]

359. **a** Digoxin at therapeutic serum levels (0.5-1.5 ng/mL) improves cardiac muscle contraction and rhythm.
[QCMLS 2021: 2.14.2.1] [Bishop 2022, p632-633]

360. **d** Lithium (carbonate) is used to treat manic depression or bipolar disorder.
[QCMLS 2021: 2.14.2.5] [Bishop 2022, p640-641]

361. **a** Nortriptyline is an active metabolite of amitriptyline, which must be included in analysis for tricyclic antidepressants (TCAs). Desipramine is an active metabolite of imipramine. Doxepin metabolizes to nordoxepin.
[Bishop 2022, p666]

362. **a** Primidone is an inactive proform of phenobarbital.
[Bishop 2022, p636-637]

363. **b** The stationary phase is nonpolar, while the mobile phase is a polar buffer. This means that the eluant moving off the column is also polar. Retention of an analyte on a reversed-phase column depends on the relative amounts of polar and nonpolar character of the analyte. Retention on the reversed-phase packing material is favored by increased nonpolar content of the analyte, whereas residence in the mobile phase leading to early elution from the column is favored by an increased content of polar functionalities present on the analyte.
[QCMLS 2021: 2.16.7] [Bishop 2022, p113]

D. Toxicology

364. **b** Normal Hb is changed to abnormal derivative CO-Hb when exposed to carbon monoxide. Nitrate changes the oxidation state of iron to form methemoglobin.
[QCMLS 2021: 2.13.1.2, 2.13.5] [Tietz 2019, p601]

365. **c** CO prevents heme iron from binding with oxygen, and increases the oxygen affinity for hemoglobin, causing a shift to the left in the oxygen-hemoglobin dissociation curve.
[QCMLS 2021: 2.13.5] [Tietz 2019, p567]

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366. **d** The primary metabolite of cocaine is benzoylecgonine, which is produced by the liver and eliminated in the urine. Benzoylecgonine is detected in drugs of abuse screens for cocaine. Many screening methods for natural opiates (opium, morphine, and codeine) or synthetic opioids (heroin, hydromorphone, and oxycodone) detect codeine or morphine metabolites.
[Bishop 2022, p665]

367. **d** 11-nor-THC-COOH is the urinary metabolite of cannabinoids (marijuana and hashish). Ecgonines are metabolites of cocaine while benzodiazepines have many metabolites such as nordiazepam for diazepam.
[Bishop 2022, p665]

368. **d** Lead interferes with directly with heme synthesis, which on a CBC may present as a decreased hemoglobin, with basophilic stippling of the red blood cells. These findings in a child may indicate lead toxicity. Whole blood lead is the recommended test; but urine delta-aminolevulinic acid and RBC zinc protoporphyrin are also useful assays. Iron overload causes liver, pancreatic and cardiac damage. Both mercury and arsenic toxicities are multi-organic affecting GI, renal, CNS systems, but may have hematologic effects as well.
[QCMLS 2021: 2.3.2.2] [Bishop 2022, 657-658]

369. **a** Erythrocyte zinc protoporphyrin is a useful screen for lead toxicity. Mercury or arsenic levels may be specifically tested in whole blood or 24-hour urine. Iron levels are determined directly by serum iron and TIBC levels with supportive diagnostic information provided by CBC, ferritin and transferrin levels.
[QCMLS 2021: 2.3.4.5, 2.3.2.2.1] [Bishop 2022, p658-659]

370. **b** Salicylate levels are used to determine if aspirin (acetylsalicylic acid) toxicity is present. Toxic serum or plasma levels are generally >300 µg/mL. Acetaminophen is also an OTC NSAID with trade name of Tylenol®, while ibuprofen is marketed under the trade names Motrin® or Advil®. Pseudoephedrine is an OTC decongestant.
[QCMLS 2021: 2.13.8] [Bishop 2022, p661]

371. **c** Measurement of whole blood lead is the recommended test for children. In adults higher lead levels are significant; therefore, other methods (eg, erythrocyte protoporphyrin and delta-aminolevulinic acid) are acceptable for adults.
[QCMLS 2021: 2.10.2.5] [Tietz 2019, p156]

372. **b** Principles of adsorption and selectivity in thin-layer chromatography. Excessive heat or drying of the chromatogram often causes evaporation of the solvent and poor migration. Too much sample may cause poor resolution, but does not cause fast migration.
[QCMLS 2021: 2.16.7] [Bishop 2022, p115; Tietz 2019, p198]

373. **c** In practice, a positive screening result for barbiturates obtained by immunoassay is confirmed by gas chromatography/mass spectrometry analysis of the urine specimen. Nephelometry is used more frequently with protein quantification, while TLC and UV absorption spectrophotometry are labor-intensive and rarely used methods due to their impracticality.
[QCMLS 2021: 2.16.7] [Tietz 2019, p574]

374. **a** Urine is typically used to screen for drugs of abuse due to the convenience of sample and its ability to detect recent drug exposure.
[QCMLS 2.13.1] [Bishop 2022, p662]

375. **d** Overdose of acetaminophen is associated with hepatotoxicity. Degree of anticipated hepatotoxicity can be assessed by measuring plasma initially at 4 hours or within 24 hours of ingested dosage and applying the result in µg/mL to the Rumack-Matthew nomogram.
[QCMLS 2.13.4] [Tietz 2019, p571]

376. **b** In electrophoresis, separation of molecules takes place in a gel medium rather than in a column and does not involve high pressure but voltage to enhance separation.
Gas liquid chromatography is similar to HPLC, but separation occurs due to a carrier gas (eg, argon or helium) being pumped through the column containing small diameter beads.
Mass spectrometry is a type of detector used in GC or HPLC to provide definitive identification of chemicals within the samples.
[QCMLS 2.16.7] [Bishop 2022, p115-16]

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Chapter 3

Urinalysis & Body Fluids

*The following items have been identified generally as appropriate for those preparing for both the MLS and MLT examinations. Items that are appropriate for the MLS examination **only** are marked with MLS ONLY.*

199 Questions
199 I. Urinalysis
229 II. Body Fluids

243 Answers with Explanations
243 I. Urinalysis
267 II. Body Fluids

I. Urinalysis

A. Physical

1. A patient with uncontrolled diabetes mellitus will most likely have:

   a pale yellow urine with a high specific gravity
   b dark yellow urine with a high specific gravity
   c pale yellow urine with a low specific gravity
   d dark yellow urine with a low specific gravity

2. While performing an analysis of a baby's urine, the laboratorian notices the specimen has a "mousy" odor. Of the following substances that may be excreted in urine, the one that most characteristically produces this odor is:

   a phenylpyruvic acid
   b acetone
   c coliform bacilli
   d porphyrin

3. An ammonia-like odor is characteristically associated with urine from patients with:

   a phenylketonuria
   b viral hepatitis
   c a bacterial infection
   d a yeast infection

4. Urine that develops a "port wine" or deep red color after standing may contain:

   a melanin
   b porphyrins
   c bilirubin
   d urobilinogen

5. Which of the following collection methods would yield the most sterile urine sample?

   a random
   b catheterization
   c suprapubic aspiration
   d clean-catch midstream

6. Urine from a 50-year-old patient with severe hepatic disease is reported as having an "amber" or dark brown color. The color of the sample is most likely caused by the presence of:

   a biliverdin
   b bilirubin
   c pyridium
   d melanin

7. The clarity of a urine sample should be determined after:

   a the sample has been centrifuged
   b thorough mixing of the specimen
   c the addition of 3% sulfosalicylic acid
   d the specimen is heated to body temperature

8. A bright orange urine from a 24-year-old patient with cystitis likely contains:

   a bilirubin
   b pyridium
   c rifampin
   d ammonia

9. After standing, a urine sample that develops a black coloration would most likely contain:

   a bile pigments
   b porphyrinogens
   c homogentisic acid
   d red blood cells

10. The yellow color of urine is primarily due to the presence of:

    a urochrome
    b melanin
    c bilirubin
    d stercobilin

11. Following a severe crush injury, a patient is transported to the emergency room where a blood sample and a urine sample are collected. The patient's urine sample appears reddish-brown, which may be due to the presence of:

    a stercobilin
    b porphyrins
    c myoglobin
    d fresh blood

12. A random urine specimen collected shortly after the patient ate lunch appeared cloudy. Results of the reagent test strip were normal. The most likely cause of the sample's turbidity is the presence of:

    a bacteria
    b white blood cells
    c amorphous urates
    d amorphous phosphates

13. In which of the following metabolic diseases will urine turn dark brown to black upon standing?

    a phenylketonuria
    b alkaptonuria
    c maple syrup urine disease
    d diabetic ketoacidosis

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14. If testing cannot be performed within one hour of collection, urine samples should be:

    a frozen
    b refrigerated
    c acidified with HCl
    d discarded down the sink

15. Measurement of urine specific gravity aids in the evaluation of the kidneys' ability to:

    a filter the plasma
    b concentrate the urine
    c produce erythropoietin
    d excrete waste products

16. Osmolality is a measure of:

    a all dissolved particles, including ions
    b only undissociated molecules
    c the total salt concentration
    d only ionic compounds

17. A urine sample has a high specific gravity by refractrometry, but normal value when measured by reagent test strip. An increased presence of which of the following might explain the discrepancy between these measurements?

    a protein
    b sodium
    c hydrogen
    d ketones

18. If an ambulatory, adult patient is suspected of having a urinary tract infection, which of the following collection methods would be most appropriate to obtain a urine sample?

    a random
    b catheterization
    c suprapubic aspiration
    d clean-catch midstream

19. A urine's specific gravity by reagent test strip is directly proportional to its:

    a color and clarity
    b ionic solutes
    c total volume
    d cellular content

20. Isosthenuria is associated with a specific gravity that is fixed around:

    a 1.001
    b 1.010
    c 1.020
    d 1.040

21. Which of the following collection types should be performed to obtain a highly concentrated urine specimen?

    a random
    b first morning
    c post-prandial
    d 24-hour timed

22. A deficiency in arginine vasopressin (antidiuretic hormone [ADH]) is associated with:

a glucosuria
b proteinuria
c hyposthenuria
d bilirubinuria

23. When using a refractometer to measure urine concentration, the laboratorian must correct for which of the following in their calculations?

a temperature
b pressure
c glucose
d volume

24. Which of the following collection methods would account for diurnal variation when quantitatively measuring urinary analytes?

a random
b first morning
c 24-hour timed
d two-hour timed

25. The method of choice for determining urine concentration following administration of x-ray contrast dye is:

a osmometry
b refractometry
c spectrophotometry
d densitometry

26. Which of the following urinary parameters may be measured to assess renal tubular function?

a creatinine
b specific gravity
c urea nitrogen
d total volume

27. Refractive index is a comparison of light:

a velocity in solutions to light velocity in solids
b velocity in air to light velocity in solutions
c scattering by air to light scattering by solutions
d scattering by particles in solution

28. The presence of biliverdin may cause a urine sample to appear:

a black
b dark red
c blue-green
d bright orange

29. A urine sample collected from a 5-day-old baby has a noticeably sweet odor reminiscent of caramel or burnt sugar. This odor is most commonly associated with:

a cystinuria
b alkaptonuria
c phenylketonuria
d maple syrup urine disease

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30. A random urine sample from a patient with diabetes insipidus is most likely to appear:

a pale yellow
b yellow
c dark yellow
d dark brown

31. Patients that produce more than 3000 mL (3 L) of urine in a 24-hour period are described as having:

a anuria
b oliguria
c nocturia
d polyuria

32. A urine sample with specific gravity of 1.003 will likely appear:

a amber
b yellow
c colorless
d dark yellow

B. Chemical

33. After receiving a timed urine for quantitative analysis, the laboratorian must first:

a subculture the urine for bacteria
b add the appropriate preservative
c screen for albumin using a dipstick
d measure and record the total volume

34. A falsely low result for urobilinogen may occur if the urine specimen is:

a exposed to light
b adjusted to a neutral pH
c cooled to room temperature
d collected in a nonsterile container

35. Which of the following urine results is most likely to be affected by prolonged light exposure?

a pH
b protein
c ketones
d bilirubin

36. The results in this table are obtained on a urine specimen collected at 8:00 AM:

test	result
color	yellow
clarity	hazy
glucose	250 mg/dL
bilirubin	negative
ketones	40 mg/dL
specific gravity	1.010
blood	negative
pH	6.0
protein	100 mg/dL
urobilinogen	0.2 mg/dL
nitrite	positive
leukocyte esterase	moderate

If the sample is stored unrefrigerated and retested at 12:00 PM, which of the following test results would likely be decreased due to the storage conditions?

a protein
b glucose
c ketones
d nitrite

37. Which of the following would be affected by allowing a urine specimen to remain at room temperature for 3 hours before analysis?

a pH
b protein
c occult blood
d specific gravity

38. The glucose test pad on a test strip may yield a false-positive result in the presence of:

a bleach
b lactose
c galactose
d ascorbic acid

39. Which of the following confirmatory tests has historically been used to semi-quantitatively determine the presence of urinary ketones?

a SSA
b Ictotest®
c Acetest®
d Clinitest®

40. The occult blood test pad on a test strip may yield a false-positive result in the presence of:

a bleach
b protein
c hemoglobin
d ascorbic acid

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41. Which of the following can lead to a false-negative urine protein reading?

a presence of albumin
b presence of mucus
c a concentrated urine
d a dilute urine

42. The pH of a urine specimen is related to its concentration of free:

a sodium ions
b hydrogen ions
c calcium ions
d magnesium ions

43. After standing at room temperature, a urine pH will typically increase due to bacterial production of:

a nitrite
b urease
c esterase
d nitrate

44. Urine reagent test strips should be stored in a tightly-sealed container in/on the:

a freezer (–20°C)
b refrigerator (5°C)
c benchtop (20°C)
d incubator (37°C)

45. The principle of the reagent test strip for urine protein depends on:

a reduction of copper sulfate
b the protein error of indicators
c reaction with Ehrlich reagent
d a double-sequential enzyme reaction

46. The urine reagent test strip pad for protein is most sensitive to:

a albumin
b hemoglobin
c paraproteins
d mucoproteins

47. Patients that take high doses of vitamin C may have a false-negative result for which of the following urine reagent test strip analytes?

a pH
b ketones
c bilirubin
d specific gravity

48. Which of the following reagents is embedded on the reagent test strip pad for ketones?

a p-arsanilic acid
b acetoacetic acid
c Ehrlich reagent
d sodium nitroprusside

49. Which of the following reagents is used to enhance the reaction of acetone with sodium nitroprusside in the tablet test for ketones?

a lactose
b galactose
c ascorbic acid
d glacial acetic acid

50. A reagent test strip pad impregnated with a stabilized diazonium salt, such as diazotized 2,4-dichloroaniline, will yield a positive reaction in an acid medium with:

a bilirubin
b ketones
c hemoglobin
d urobilinogen

51. Which of the following substances may interfere with the reagent test strip pad for leukocyte esterase and yield a false-negative result when present in high concentrations?

a lactose
b ketones
c protein
d bilirubin

52. The principle of the reagent test strip for urobilinogen is based on its reaction with:

a p-arsanilic acid
b sodium nitroprusside
c diazotized 2,4-dichloroaniline
d p-dimethylaminobenzaldehyde

53. When employing the urine reagent test strip method for protein, a false-positive result may occur in the presence of:

a large amounts of glucose
b x-ray contrast media
c Bence Jones proteins
d a highly alkaline urine

54. Which of the following substances would most likely be found in the urine of a patient with anorexia nervosa?

a protein
b ketones
c glucose
d bilirubin

55. The presence of ketones in a random urine sample from a 2-year-old child would most likely be associated with:

a prolonged vomiting
b a hemolytic event
c a urinary tract infection
d biliary tract obstruction

56. A patient's urinalysis revealed a positive bilirubin and a normal urobilinogen level. These results are associated with:

a hemolytic disease
b biliary tract obstruction
c hepatic disease
d urinary tract infection

57. A urine specimen with an elevated urobilinogen concentration but a negative bilirubin result may indicate the patient has:

a gallstones
b viral hepatitis
c hemolytic anemia
d liver cirrhosis

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58. Microscopic analysis of a urine specimen yields a moderate amount of red blood cells in spite of a negative result for occult blood using a reagent strip. The presence of which of the following may explain this discrepancy?

a bleach
b vitamin C
c salicylates
d hemoglobin

59. The principle of the reagent test strip for microalbumin is based on:

a reaction with a diazonium salt
b the protein error of indicators
c an immunochemical reaction
d reaction with p-arsanilic acid

60. Measurement of which of the following may be performed to account for the patient's hydration status and renal function?

a creatinine
b hydrogen
c sodium
d glucose

61. The reagent test pad for pH contains which of the following?

a diazotized 2,4-dichloroaniline
b methyl red and bromthymol blue
c glucose oxidase and peroxidase
d tetrabromphenol blue and a buffer

62. Which of the following tablet tests has historically been used to rule out a false-positive bilirubin result achieved by reagent test strip?

a Acetest®
b Clinitest®
c Ictotest®
d guaiac test

63. A clear, red-brown urine specimen resulted in a positive reaction for blood on the reagent test strip, but no red blood cells were seen during microscopic examination. These results most likely indicate the presence of:

a nitrates
b pyridium
c porphyrins
d myoglobin

64. A double-sequential enzyme reaction employing glucose oxidase and peroxidase is used to measure which of the following urinary analytes by reagent test strip?

a pH
b nitrites
c glucose
d ketones

65. Development of a pink color on the absorbent mat surrounding an Ictotest® tablet should be reported as:

a invalid
b positive
c negative
d nonreactive

66. The absorbent mat surrounding an Ictotest® tablet develops a dark purple color. This reaction indicates the presence of which of the following?

a protein
b glucose
c ketones
d bilirubin

67. A urine specimen analyzed for glucose by a glucose oxidase reagent test strip may yield a falsely low or negative value in the presence of:

a bleach
b galactose
c ascorbic acid
d hydrogen peroxide

68. The surface of an Acetest® tablet develops a dark purple color. This reaction indicates a large presence of which of the following in the patient's urine sample?

a protein
b glucose
c ketones
d bilirubin

69. A urinalysis performed on a sample from a 2-week-old infant with diarrhea showed a negative reaction with the glucose oxidase reagent test strip. Historically, a copper reduction tablet test would have been performed to check the urine sample for the presence of:

a glucose
b galactose
c bilirubin
d ketones

70. A patient suspected of having recurrent urinary tract infections provides a clean-catch midstream urine sample. A positive value for which of the following reagent test strip results would best support this diagnosis?

a glucose
b occult blood
c urobilinogen
d leukocyte esterase

71. The occult blood pad of a reagent test strip yields an orange background with several green spots. This indicates the specimen likely contains:

a myoglobin
b intact RBCs
c hemoglobin
d ascorbic acid

72. A reagent test strip pad for occult blood is reported as positive, but microscopic examination did not reveal the presence of red blood cells. This patient's condition can be termed:

a oliguria
b hematuria
c hemoglobinuria
d hemosiderinuria

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73. A patient seen at an urgent care facility for lower back pain and a fever has the urinalysis results shown in this table:

test	result
color	dark yellow
clarity	cloudy
glucose	negative
bilirubin	negative
ketones	negative
specific gravity	1.010
blood	small
pH	6.0
protein	30 mg/dL
urobilinogen	0.2 mg/dL
nitrite	negative
leukocyte esterase	moderate

Microscopic examination shows the presence of 15-20 RBC/HPF, 30-40 WBC/HPF, 3+ bacteria, and 2-3 renal tubular epithelial cells/HPF. The discrepancy between the "negative" result for nitrite on the reagent test strip and the presence of bacteria on microscopy may be best explained by:

a failure to mix the specimen before centrifuging
b failure to test the sample within 1 hour of collection
c presence of an oxidizing detergent in the specimen container
d the presence of a non-nitrate-reducing organism

74. Which of the following reagent test strip pads will yield a positive reaction when the sample contains hemoglobin or myoglobin?

a protein
b ketones
c occult blood
d leukocyte esterase

75. The results of a urinalysis on a first morning specimen are shown in this table:

test	result
color	dark yellow
clarity	hazy
glucose	negative
bilirubin	negative
ketones	negative
specific gravity	1.025
blood	negative
pH	8.5
protein	negative
urobilinogen	1.0 mg/dL
nitrite	negative
leukocyte esterase	negative

After viewing uric acid crystals in the centrifuged urine sediment, the laboratorian should question which of the results?

a pH
b protein
c glucose
d specific gravity

76. A false-negative result may occur for the nitrite reagent test strip pad if the sample:

a contains a nitrate-reducer
b is contaminated with skin flora
c is contaminated with a cleaning agent
d does not incubate in the bladder long enough

77. The nitrite reagent test strip pad is most useful in the identification of a bacterial urinary tract infection when evaluated in combination with which of the following test pads?

a pH
b occult blood
c specific gravity
d leukocyte esterase

78. Despite their presence in a sample, the leukocyte esterase reagent test strip pad would yield a negative result for which of the following types of white blood cells?

a monocytes
b eosinophils
c lymphocytes
d neutrophils

79. Reagent test strip pads for ketones primarily measure:

a acetone
b cholesterol
c acetoacetic acid
d beta-hydroxybutyric acid

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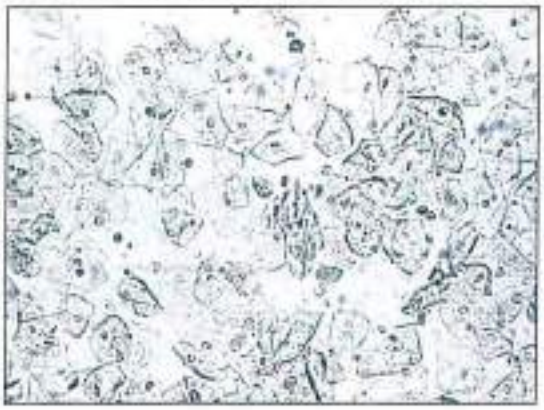
C. Microscopic

80. Microscopic examination of cellular elements in a urine sediment, such as RBCs, WBCs, bacteria, and epithelial cells, should be enumerated using:

a low power (10x objective)
b high dry power (40x objective)
c oil immersion (100x objective)
d phase contrast microscopy

81. The centrifuged sediment for a random urine sample collected from a 17-year-old female patient is shown in the photomicrograph.



The reagent test strip is negative for leukocyte esterase and nitrite. The elements seen in the sediment are most likely the result of:

a contamination
b improper preservation
c a urinary tract infection
d chronic glomerulonephritis

82. Oval fat bodies are:

a hyaline casts that contain lipids
b squamous epithelial cells that contain lipids
c transitional epithelial cells that contain lipids
d renal tubular epithelial cells that contain lipids

83. Microscopic examination of a urine sediment reveals the presence of ghost cells which would most likely be associated with a:

a pH of 6.5
b specific gravity 1.000
c ketone concentration of 5 mg/dL
d glucose concentration of 1000 mg/dL

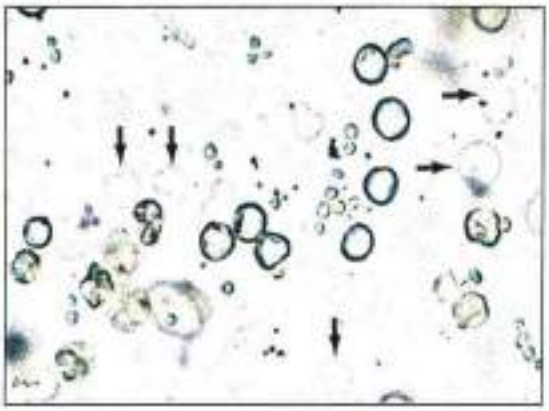
84. Glitter cells are a microscopic finding of:

a red blood cells in hypertonic urine
b red blood cells in hypotonic urine
c white blood cells in hypertonic urine
d white blood cells in hypotonic urine

85. Which of the following cells is most commonly associated with a non-clean-catch specimen and/or vaginal contamination?

a white blood cells
b renal epithelial cells
c squamous epithelial cells
d transitional epithelial cells

86. Identify the elements indicated by the arrows in the photomicrograph:



a clue cells
b ghost cells
c glitter cells
d oval fat bodies

87. Identify the type of cast seen in the photomicrograph:

a broad
b hyaline
c cellular
d granular

88. The presence of which of the following is associated with bacterial vaginosis?

a clue cells
b ghost cells
c glitter cells
d oval fat bodies

89. Which of the following elements is most likely to be seen in a urine sediment as a result of catheterization?

a hyaline casts
b oval fat bodies
c transitional epithelial cells
d renal tubular epithelial cells

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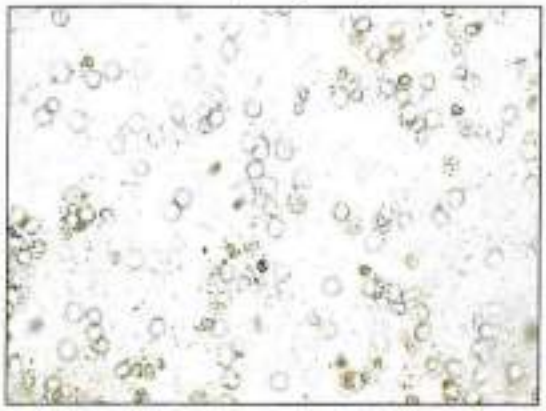
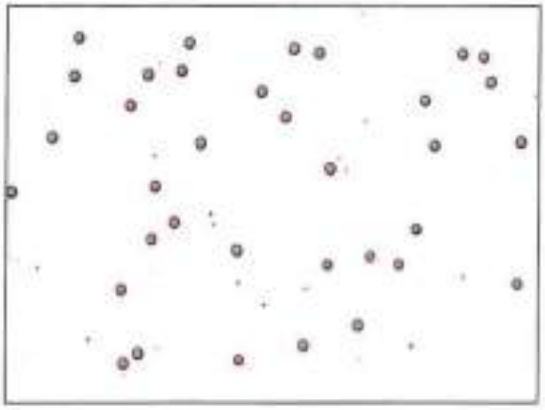
90. What is the most likely diagnosis given this microscopic finding?



a cystitis
b pyelonephritis
c glomerulonephritis
d nephrotic syndrome

91. Identify the elements seen in the photomicrograph:



a red blood cells
b urothelial cells
c white blood cells
d tyrosine crystals

92. Urinary casts are primarily composed of:

a albumin
b globulins
c paraprotein
d uromodulin

93. Examination for casts should be performed:

a after adding glacial acetic acid to the sediment
b under high dry power (40x obj.) with bright light
c after adding Sternheimer-Malbin stain to the supernatant
d using subdued lighting and low power (10x obj.) magnification

94. Which of the following casts is most likely to be found in a urine sample from an apparently healthy person?

a fatty
b waxy
c hyaline
d granular

95. Which of the following casts is most indicative of renal failure/end stage renal disease?

a waxy
b cellular
c granular
d hyaline

96. Shortly after collection of a random urine sample, the laboratorian performed a microscopic urinalysis and reported the results shown in the table:

test	result
WBC	10-13/HPF
RBC	2-6/HPF
hyaline casts	5-7/LPF
bacteria	1+

The urine sediment is reevaluated by a trainee 2 hours later and similar values were reported for the WBC, RBC, and bacteria, but they found no hyaline casts. The most probable explanation for this discrepancy is that the casts dissolved due to a(n):

a increase in urine pH
b decrease in urine pH
c increase in specimen temperature
d decrease in specimen temperature

97. Which of the following aids in differentiating a spherical transitional cell from a round renal tubular cell?

a round renal tubular cells are larger
b spherical transitional cells are larger
c renal tubular cell nuclei are eccentric
d spherical transitional cell nuclei are eccentric

98. The presence of which of the following urine microscopic constituents would best differentiate between cystitis and pyelonephritis?

a WBCs
b RBCs
c bacteria
d WBC casts

99. Renal tubular epithelial cell casts are most indicative of:

a pyelonephritis
b tubular necrosis
c glomerulonephritis
d nephrotic syndrome

100. Hyaline casts found in the urine of a football player admitted to the hospital following a concussion are most likely the result of:

a dehydration and urinary stasis
b trauma to the blood-brain barrier
c significant trauma to the kidneys
d excessive ingestion of electrolytes

101. Which of the following casts most frequently appears to have broken or serrated edges?

a fatty
b waxy
c hyaline
d granular

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102. To distinguish between a clump of WBCs and a WBC cast, it is important to observe:

a a positive nitrite reaction
b a positive leukocyte esterase reaction
c the presence of a cast matrix
d the presence of free-floating WBCs

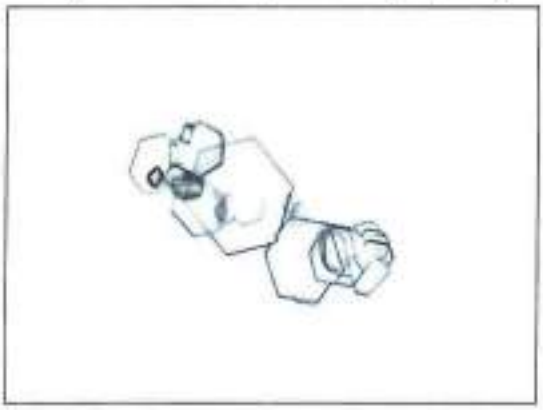
103. Renal tubular epithelial cells are most likely to be confused with:

a oil droplets
b white blood cells
c sulfonamide crystals
d squamous epithelial cells

104. Prior to reporting a red blood cell cast, it is important to observe:

a hyaline casts
b granular casts
c free-floating RBCs
d increased white blood cells

105. The crystals seen in the photomicrograph may indicate the patient has:



a cystinuria
b tyrosinemia
c galactosemia
d maple syrup urine disease

106. The presence of red blood cell casts correlates best with which of the following conditions?

a cystitis
b pyelonephritis
c glomerulonephritis
d nephrotic syndrome

107. A white precipitate in a refrigerated urine specimen with pH of 7.5 would most likely be:

a uric acid crystals
b amorphous urates
c amorphous phosphates
d triple phosphate crystals

108. Which of the following is an abnormal crystal described as a hexagonal plate?

a cystine
b tyrosine
c leucine
d cholesterol

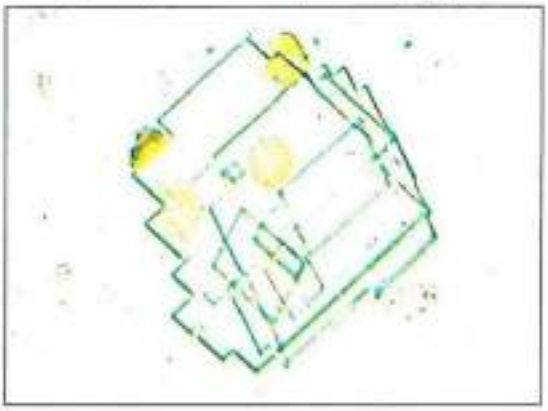
109. The primary component of most renal calculi is:

a uric acid
b cholesterol
c calcium oxalate
d ammonium biurate

110. A clear, yellow urine is stored at 5°C as testing will be delayed. When removed from the refrigerator, the sample appears cloudy. This is most likely due to the presence of:

a bacteria
b hyaline casts
c white blood cells
d amorphous crystals

111. Identify the crystals in the photomicrograph:



a cystine
b uric acid
c isoleucine
d cholesterol

112. Which of the following crystals may be found in an acidic urine?

a ammonium biurate
b calcium phosphate
c calcium oxalate
d triple phosphate

113. Using polarized light microscopy, which of the following urinary elements is birefringent?

a cholesterol
b triglycerides
c fatty acids
d neutral fats

114. Which of the following crystals appear as colorless, fine, silky needles?

a leucine
b tyrosine
c cholesterol
d hemosiderin

115. Which of the following crystals would most likely be present in an amber-colored urine yielding a positive bilirubin?

a cystine
b cysteine
c tyrosine
d uric acid

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116. Following ingestion of ethylene glycol (antifreeze), which of the following crystals may be found in the urine?

a yellow-brown spherules
b colorless, dumbbells/rings
c colorless, rosettes/rhomboids
d flat plates with notched corners

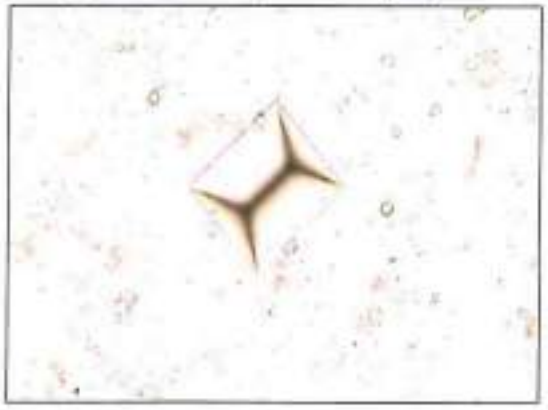
117. Cholesterol crystals will most likely be observed in urine with a:

a pH of 8.0
b protein of 4+
c ketone of 40 mg/dL
d glucose of 500 mg/dL

118. The finding of a large amount of uric acid crystals in a urine specimen from a 6-month-old baby may indicate:

a viral hepatitis
b ethylene glycol toxicity
c increased purine metabolism
d improper sample collection

119. The crystal seen in the photomicrograph would be categorized as:

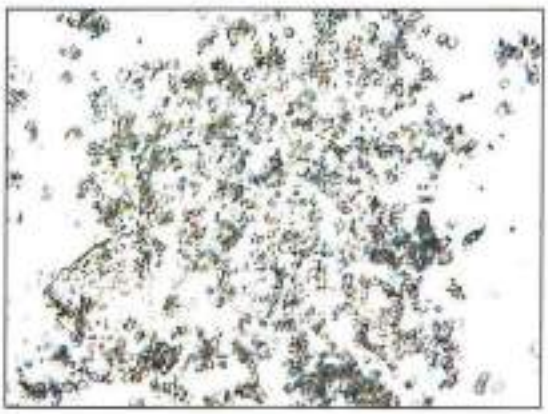


a an acidic crystal that is considered pathologic
b an alkaline crystal that is considered pathologic
c an acidic crystal that is considered nonpathologic
d an alkaline crystal that is considered nonpathologic

120. Identify the crystals in the photomicrograph:

a tyrosine
b uric acid
c calcium oxalate
d ammonium biurate

121. The photomicrograph is from the sediment of an alkaline urine:



In order to perform the microscopic examination, the laboratorian might consider:

a requesting the patient collect a new sample
b diluting the urine sediment with normal saline
c adding a few drops of acetic acid to the sediment
d mixing equal portions of the sediment and warm water

122. Polarized light might be used to differentiate between:

a hyaline and waxy casts
b uric acid and cystine crystals
c red blood cells and white blood cells
d squamous and transitional epithelial cells

123. Which of the following contaminants has a dimpled center and can appear as a Maltese cross under polarized light?

a oil droplets
b air bubbles
c glass shards
d starch crystals

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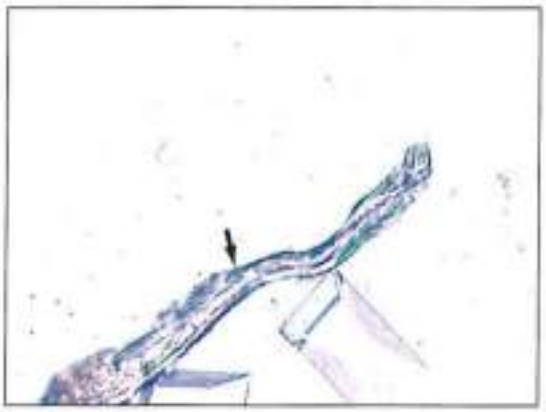
124. Which of the following crystals appear as colorless, rhombic plates or rosettes when present in a urine sample?

a leucine
b uric acid
c cholesterol
d triple phosphate

125. Identify the microscopic element indicated by the arrow in the photomicrograph:



a cloth fiber
b waxy cast
c hyaline cast
d granular cast

126. A newer laboratorian is having trouble differentiating between red blood cells, oil droplets, and yeast during microscopic analysis of urine sediments. To aid in differentiation, they add a few drops of glacial acetic acid to the sediment which will:

a lyse the yeast cells
b lyse the red blood cells
c dissolve the oil droplets
d crenate the red blood cells

127. When identifying urinary crystals, which of the following reagent test strip results would be most helpful?

a pH
b nitrite
c protein
d specific gravity

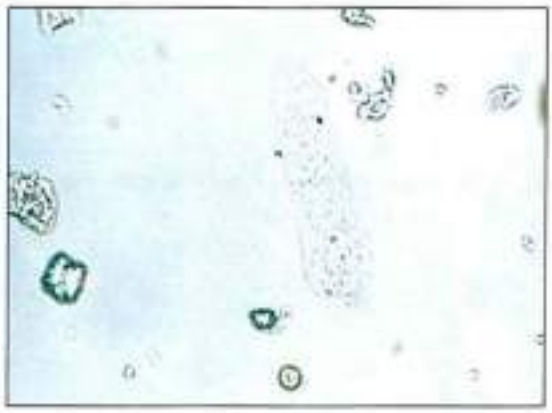
128. Bacteria are considered significant in the urine sediment when the:

a nitrite is negative
b protein is positive
c specimen is dark yellow
d sample contains WBCs

129. Which of the following must exhibit rapid motility in order to be identified in a urine sediment?

a yeast
b bacteria
c spermatozoa
d *Trichomonas*

130. Identify the type of cast seen in the photomicrograph:



a waxy
b cellular
c hyaline
d granular

131. Which of the following reagent test strip pads would most likely yield a positive result consistent with the photomicrograph?

a ketones
b glucose
c specific gravity
d nitrite

132. The sediment of a urine specimen with a reagent strip glucose of 250 mg/dL (13.8 mmol/L), pH of 5.5, and negative nitrite shows the presence of small, ovoid cells that are budding. These results are most consistent with a urinary tract infection by:

a *Escherichia coli*
b *Candida albicans*
c *Trichomonas vaginalis*
d *Enterobius vermicularis*

133. Which of the following urinary crystals appear as yellow-brown spheres with concentric circles when viewed microscopically?

a leucine
b bilirubin
c triple phosphate
d ammonium biurate

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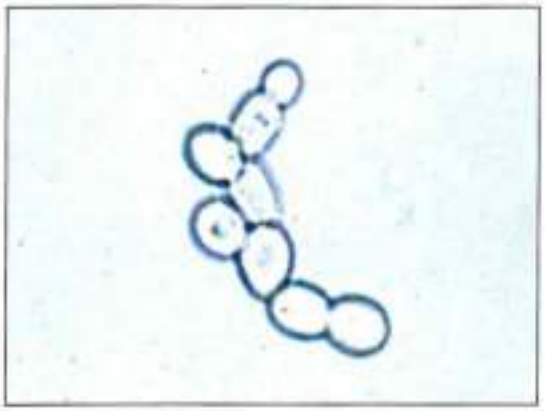
134. Identify the elements in the photomicrograph:



a yeast
b red blood cells
c white blood cells
d renal tubular epithelial cells

D. Renal Physiology

135. The normal renal threshold for glucose in an adult is approximately:

a 50 mg/dL (2.8 mmol/L)
b 100 mg/dL (5.5 mmol/L)
c 160 mg/dL (8.8 mmol/L)
d 300 mg/dL (16.5 mmol/L)

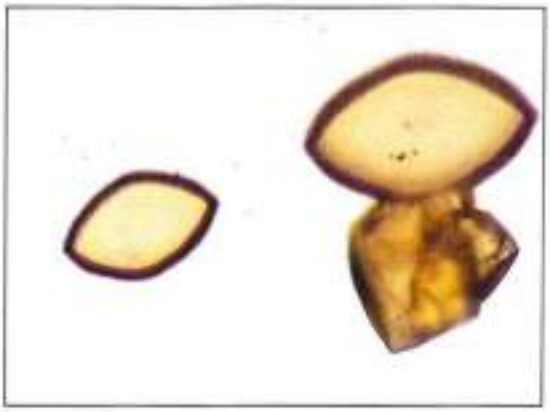
136. If the total volume of urine excreted in a 24-hour period by an adult patient is 300 mL, their condition would be termed:

a anuria
b oliguria
c polyuria
d dysuria

137. The normal glomerular filtration rate is:

a 10 mL/min
b 120 mL/min
c 660 mL/min
d 1,200 mL/min

138. A urine sediment from a 52-year-old patient is shown in the photomicrograph:



The elements seen in the sediment may indicate the patient has:

a gout
b cystitis
c cystinuria
d glomerulonephritis

139. A physician wants to determine if an abdominal fluid collected during surgery is urine. Which of the following analytes would normally be present in the urine of an apparently healthy patient?

a protein
b glucose
c bilirubin
d creatinine

140. Arginine vasopressin (antidiuretic hormone [ADH]) regulates the reabsorption of:

a water
b glucose
c calcium
d potassium

141. Which of the following components would be present in the filtrate as a result of glomerular damage?

a urea
b RBCs
c glucose
d amino acids

142. Polyuria is often associated with:

a viral hepatitis
b tubular damage
c diabetes mellitus
d acute glomerulonephritis

143. Cessation of urine flow is termed:

a anuria
b dysuria
c diuresis
d azotemia

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144. Ketonuria is associated with increased metabolism of:

a lipids
b proteins
c amino acids
d carbohydrates

145. Bilirubinuria may be associated with:

a strenuous exercise
b destruction of platelets
c hepatobiliary obstruction
d acute interstitial nephritis

146. Hepatic disorders may result in the presence of which of the following types of urinary crystals?

a cystine
b cholesterol
c bilirubin
d uric acid

147. A patient with renal tubular acidosis would most likely excrete a urine with a:

a low pH
b high pH
c neutral pH
d variable pH

148. Glycosuria may be due to:

a hypoglycemia
b renal tubular dysfunction
c increased renal reabsorption
d increased glomerular filtration rate

149. The area of the nephron that is impermeable to water is the:

a proximal convoluted tubule
b descending loop of Henle
c ascending loop of Henle
d distal convoluted tubule

150. The urinary tract structures primarily responsible for renal concentration are the:

a renal pelvis
b renal papillae
c cortical nephrons
d juxtamedullary nephrons

151. A random urine sample collected from a 45-year-old male patient following a transfusion reaction yields the laboratory results in this table:

test	result	microscopic findings	
color	red	WBC	0-3/HPF
clarity	hazy	RBC	2-5/HPF
glucose	negative	casts/LPF	2-4 hyaline/LPF
bilirubin	negative	epithelial cells	3-5 renal tubular/HPF
ketones	negative		
specific gravity	1.010		
blood	large		
pH	7.5		
protein	negative		
urobilinogen	0.2 mg/dL		
nitrite	negative		
leukocytes	negative		

These results may indicate the patient has:

a renal glucosuria
b renal tubular acidosis
c acute tubular necrosis
d acute interstitial nephritis

152. Failure of the nephron to produce ammonia from glutamine will result in urine with a:

a high pH
b positive nitrite
c positive protein
d low specific gravity

E. Disease States

153. A random urine sample with an increased microalbumin concentration is predictive of:

a nephropathy
b hypertension
c diabetes insipidus
d nephrotic syndrome

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247. Acetylcholinesterase activity may be measured on amniotic fluid when a positive alpha-fetoprotein result is obtained to evaluate for:

a fetal lung maturity
b open neural tube defects
c respiratory distress syndrome
d hemolytic disease of the newborn

248. A mildly increased CSF lactate (25-30 mg/dL) is generally associated with:

a fungal meningitis
b multiple sclerosis
c multiple myeloma
d bacterial meningitis

249. A decreased CSF total protein concentration may indicate the occurrence of which of the following?

a meningitis
b carcinoma
c hemorrhage
d CSF leakage

250. An acholic or pale/clay-colored stool is characteristic of:

a steatorrhea
b maldigestion
c malabsorption syndrome
d a post-hepatobiliary obstruction

251. Results from a CSF cell differential for an adult patient suspected of having meningitis are listed in the table:

test	result
neutrophils	3%
lymphocytes	62%
monocytes	23%
eosinophils	12%

These results suggest:

a viral meningitis
b fungal meningitis
c bacterial meningitis
d tubercular meningitis

252. A vaginal swab is received in the laboratory for testing and immediately placed in sterile physiological saline to prepare a suspension. A wet mount is prepared and the laboratorian reports the presence of numerous clue cells. This most likely indicates the patient has:

a candidiasis
b trichomoniasis
c atrophic vaginitis
d bacterial vaginosis

253. A wet preparation is made from a fresh fecal specimen and stained with Wright stain. The laboratorian reports the presence of 3-5 WBC per high power field, noting that the cells are neutrophils. These results may indicate the patient has:

a gallstones
b maldigestion
c ulcerative colitis
d malabsorption

254. A white blood cell differential is performed on CSF from an adolescent patient suspected of having meningitis. The results shown in this table are reported:

test	result
neutrophils	89%
lymphocytes	7%
monocytes	3%
eosinophils	<1%

These results suggest:

a viral meningitis
b fungal meningitis
c bacterial meningitis
d tubercular meningitis

255. The presence of rice bodies in a synovial fluid is strongly associated with:

a gouty arthritis
b rheumatoid arthritis
c traumatic collection
d infection with *S. aureus*

256. A stool specimen that appears black and tarlike should be tested for the presence of:

a occult blood
b fecal fat
c trypsin
d excess mucus

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I. Urinalysis

A. Physical

1. **a** Patients with diabetes mellitus may be hyperglycemic due to a defect in production or function of insulin. An elevated blood glucose concentration leads to an increase in plasma osmolality, which, in turn, stimulates osmoreceptors in the hypothalamus to trigger thirst. Patients with diabetes mellitus have polydipsia causing them to drink large volumes of water, resulting in polyuria. The increased volume dilutes the concentration of urochrome (chromogen), making the urine pale yellow. Glucose is filtered at the glomerulus and actively reabsorbed in the renal tubules until the plasma renal threshold is met. Excess glucose, as occurs in hyperglycemia, will be excreted in the urine, leading to an elevated specific gravity.
[QCMLS 2021: 3.1.5.2] [Strasinger 2021, p94, 110]

2. **a** Phenylketonuria is an inborn error of metabolism in which the conversion of phenylalanine to tyrosine is disrupted, leading to accumulation of keto acids, such as phenylpyruvic acid. The keto acids are filtered at the kidneys and excreted in the urine causing a characteristic mousy or musty odor.
[QCMLS 2021: 3.1.2.2] [Strasinger 2021, p134, 237]

3. **c** Urea is a metabolic waste produced from the breakdown of proteins and amino acids in the liver and excreted in the urine. Some of the most common bacterial pathogens associated with urinary tract infections, such as *Proteus* and *Klebsiella* species, produce urease, an enzyme which breaks down urea into ammonia. When present, these bacteria lead to the conversion of urea into ammonia and development of the characteristic ammonia odor associated with a bacterial infection.
[QCMLS 2021: 3.1.2.2] [Strasinger 2021, p93, 134]

4. **b** Porphyrias are a class of diseases related to defects in the synthesis of hemoglobin. Some porphyrias result in the accumulation of porphobilinogen (PBG) which is water-soluble and excreted in the urine. Urinary PBG is oxidized when exposed to air leading the sample to develop a deep red or port wine color.
[QCMLS 2021: 3.1.2.1] [Strasinger 2021, p128, 243]

5. **c** Clean-catch midstream urine samples and those collected by catheterization should provide a sterile urine specimen, but there is risk of contamination if collection procedures are not followed correctly. Suprapubic aspiration would yield the most sterile urine sample as the specimen is collected directly from the bladder avoiding potential contamination from the skin or genital region.
[QCMLS 2021: 3.1.1.2] [Strasinger 2021, p96]

6. **b** Patients with hepatic disease/damage may leak conjugated bilirubin into the bloodstream, where it can be filtered and excreted via the kidneys. Bilirubin is a yellow colored pigment and may result in a dark yellow, brown, or amber color when present in urine.
[QCMLS 2021: 3.1.2.1] [Strasinger 2021, p127]

7. **b** The presence of formed and cellular elements will alter the clarity of a urine sample. These elements settle to the bottom of the container so the sample must be properly mixed prior to evaluating the sample's clarity.
[Strasinger 2021, p130]

8. **b** Pyridium, also known as phenazopyridine, is a drug commonly taken to treat symptoms associated with a bladder infection. The drug has a strong orange pigment resulting in urine samples with a bright orange color that can interfere with chemical tests relying on observation of a color reaction.
[QCMLS 2021: 3.1.2.1] [Strasinger 2021, p128]

9. **c** Alkaptonuria is an inborn error of phenylalanine metabolism, leading to accumulation of homogentisic acid (HGA) in the blood and connective tissues and excretion of HGA in urine. After standing, the urine sample becomes more alkaline and, if present, HGA will cause the sample to develop a dark brown or black color.
[QCMLS 2021: 3.1.2.1] [Strasinger 2021, p129, 239]

10. **a** Freshly voided urine normally ranges from pale yellow to dark yellow, due to the presence of yellow pigments (urochrome, uroerythrin, and urobilin) and the hydration status of the patient.
[QCMLS 2021: 3.1.2.1] [Strasinger 2021, p126]

11. **c** Myoglobin is a small oxygen-binding protein found in cardiac and skeletal muscle tissue. When muscle damage occurs, myoglobin can be released into circulation then filtered and excreted by the kidneys. Urine samples that contain myoglobin may appear reddish-brown.
[QCMLS 2021: 3.1.2.1] [Strasinger 2021, p128]

12. **d** Hydrochloric acid is released from the parietal cells in the stomach to aid in digestion. To maintain homeostasis, bicarbonate is released into the blood and filtered at the kidneys. This results in a more alkaline urine following a meal, which is a phenomenon known as the "alkaline tide." Due to the increase in urinary pH, alkaline crystals, such as amorphous phosphates, precipitate and lead to increased turbidity.
[Strasinger 2021, p142]

13. **b** Alkaptonuria is an inborn error of phenylalanine metabolism leading to accumulation of homogentisic acid (HGA) in the blood and connective tissues and excretion of HGA in urine. After standing, the urine sample becomes more alkaline and, if present, HGA will cause the sample to develop a dark brown or black color.
[QCMLS 2021: 3.1.2.1] [Strasinger 2021, p129, 239]

14. **b** Urine samples should be tested within 1-2 hours of collection to ensure results reflect the patient's status rather than changes that occur due to loss of sample integrity. If testing cannot be performed in a timely fashion, the sample should be refrigerated (2-8°C) to minimize cellular deterioration, bacterial proliferation, and changes in the chemical composition of the sample.
[QCMLS 2021: 3.1.1.1] [Strasinger 2021, p96]

15. **b** Water and analytes are filtered by the glomerulus and then selectively reabsorbed or secreted in the renal tubules controlling the final composition and concentration of the urine. Specific gravity is the ratio of the density of the solution compared to the density of an equal volume of distilled water. Urine density is influenced by the hydration state of the patient and the amount and mass of the solutes reabsorbed or secreted by the renal tubules.
[QCMLS 2021: 3.1.3.2] [Strasinger 2021, p131]

16. **a** An osmole is the molecular weight of a solute (g) divided by the number of ions or particles into which it dissociates in solution. Osmolality measures the number of dissociated particles in a solution regardless of their mass.
[Strasinger 2021, p132]

17. **a** Refractometers are used to measure urine concentration based on the refractive index of the sample. The refractive index is disproportionately affected by solutes with a large mass, such as glucose and proteins, which can lead to an overestimation of the sample's specific gravity. The reagent test strip reaction utilizes a polyelectrolyte resulting in measurement of only ionic solutes.
[QCMLS 2021: 3.1.3.2] [Strasinger 2021, p131, 160]

18. **d** Patients that are capable of collecting their own specimen should be asked to collect a clean-catch midstream urine sample when a urinary tract infection is suspected. If collected appropriately, this sample type provides a urine specimen that is less likely to be contaminated and can be used for bacteriological testing.
[QCMLS 2021: 3.1.1.2] [Strasinger 2021, p98]

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19. **b** The reagent test strip reaction for specific gravity is based on the change in the dissociation constant of a polyelectrolyte when maintained at an alkaline pH. The polyelectrolyte releases protons in proportion to the number of ionic solutes in the solution, which will increase the pH of the solution. This is visualized by using a pH indicator embedded in the test pad.
[QCMLS 2021: 3.1.3.2] [Strasinger 2021, p160]

20. **b** After passing through the glomerulus, the plasma filtrate has a specific gravity of 1.010. After water and analytes are reabsorbed and/or secreted in the renal tubules, the urine specific gravity may be increased (hypersthenuria) or decreased (hyposthenuria). An inability to effectively concentrate the urine results in a fixed specific gravity of 1.010, termed isosthenuria, which is associated with renal failure.
[Strasinger 2021, p131]

21. **b** A first morning specimen typically yields a highly concentrated urine sample as the intake of fluids is generally minimal during the nighttime hours.
[QCMLS 2021: 3.1.1.3] [Strasinger 2021, p86]

22. **c** Arginine vasopressin (AVP), also known as antidiuretic hormone (ADH), is a hormone produced in the hypothalamus and released by the posterior pituitary in response to increased plasma osmolality. AVP regulates the permeability of the distal convoluted tubule, effectively controlling the reabsorption of water. A deficiency of AVP results in decreased reabsorption of water and excretion of a dilute urine specimen with low specific gravity (hyposthenuria).
[QCMLS 2021: 3.1.5.1] [Strasinger 2021, p111, 131]

23. **c** Refractometers are used to measure urine concentration based on the refractive index of the sample, which is disproportionately affected by solutes with a large mass, such as proteins and glucose. When present, as evidenced by the chemical reagent test strip, corrections for glucose and protein must be calculated by subtracting 0.003 for each gram of protein and subtracting 0.004 for each gram of glucose present.
[QCMLS 2021: 3.1.3.2] [Strasinger 2021, p131]

24. **c** Analytes with diurnal variation are produced in variable amounts dependent on circadian rhythm. To account for diurnal variation, a 24-hour timed collection may be performed to obtain all urine excreted within that time period.
[QCMLS 2021: 3.1.1.3] [Strasinger 2021, p97]

25. **a** Large, iatrogenic molecules, such as dextran and x-ray contrast dye, can significantly increase specific gravity measurements obtained by refractometry. Osmometers measure urine concentration based on the colligative properties of the solution, such as freezing point depression, which are not influenced by the mass of the solutes.
[QCMLS 2021: 3.1.3.2] [Strasinger 2021, p117, 131]

26. **b** Specific gravity is the ratio of the density of the solution compared to the density of an equal volume of distilled water. Urine density is influenced by the hydration state of the patient and the amount and mass of the solutes reabsorbed or secreted by the renal tubules; therefore, specific gravity measurements are useful in evaluating the function of the renal tubules.
[QCMLS 2021: 2.7.4.2] [Strasinger 2021, p131]

27. **b** Refractometers are used to measure urine concentration based on the refractive index of the sample. The refractive index compares the velocity of light as it passes through the air with the velocity of light as it bends and passes through a liquid, as the liquid and constituent solutes slow down the light's velocity.
[QCMLS 2021: 3.1.3.2] [Strasinger 2021, p131]

28. **c** Hemoglobin is degraded into heme and globin by the reticuloendothelial system. Heme is then metabolized into biliverdin by heme oxygenase then reduced to bilirubin by biliverdin reductase. Bilirubin is light labile and urine samples should be protected from light. If exposed to light, bilirubin is photo-oxidized forming biliverdin which causes the sample to appear yellow-green or blue-green.
[QCMLS 2021: 3.1.2.1] [Strasinger 2021, p127]

29. **d** Maple syrup urine disease is an inborn error of metabolism in which leucine, isoleucine, and valine are converted into keto acids in the liver, but cannot be further metabolized due to an enzyme deficiency. The keto acids accumulate in the blood and tissues and are excreted in the urine causing a characteristically sweet odor reminiscent of maple syrup, caramel, or burnt sugar.
[QCMLS 2021: 3.1.2.2] [Strasinger 2021, p134, 240]

30. **a** Diabetes insipidus is associated with decreased production or lack of renal response to arginine vasopressin (AVP). AVP regulates the permeability of the distal convoluted tubule, effectively controlling the reabsorption of water. A deficiency or lack of response to AVP results in decreased reabsorption of water and increased urinary excretion. Patients with diabetes insipidus typically produce dilute urine specimens that appear colorless or pale-yellow.
[QCMLS 2021: 3.1.5.1] [Strasinger 2021, p226]

31. **d** The volume of urine excreted per day is dependent on several factors such as hydration status, renal function, hormonal control, and the use of diuretics. Healthy adults produce 1-2 L (1000-2000 mL) of urine per day. Oliguria refers to production of <400 mL/day, which can lead to anuria or cessation of urinary flow. Nocturia describes increased excretion of urine during the night and polyuria describes the production of >2.5-3 L in a 24-hour period.
[Strasinger 2021, p93]

32. **c** A specific gravity of 1.003 is at the lower end of the normal range indicating a dilute sample. Dilute samples are most likely to appear pale yellow or colorless.
[QCMLS 2021: 3.1.3.2] [Strasinger 2021, p126, 131]

B. Chemical

33. **d** A healthy adult produces 1-2 L of urine in a 24-hour period. Before any portion of a timed sample is removed for testing, the laboratorian should measure and record the total volume of the sample as the volume may be used to report out the rate of excretion for some analytes.
[QCMLS 2021: 3.1.1.3] [Strasinger 2021, p97]

34. **a** When exposed to light, urobilinogen is photo-oxidized to urobilin, which may result in a falsely low result for urobilinogen on the reagent test strip.
[QCMLS 2021: 3.1.3.9] [Strasinger 2021, p157]

35. **d** Bilirubin is rapidly photo-oxidized to biliverdin when exposed to light. Delayed testing and exposure to light may result in a falsely low result for bilirubin on the reagent test strip.
[QCMLS 2021: 3.1.3.8] [Strasinger 2021, p155]

36. **b** Urine samples should be tested within 1-2 hours of collection. If testing cannot be performed in a timely fashion, the sample should be capped and stored refrigerated (2-8°C) to minimize cellular deterioration, bacterial proliferation, and changes in the chemical composition of the sample. Samples that are not properly preserved may yield a falsely decreased glucose value as a result of continued glycolysis by cells and/or bacteria present in the sample.
[Strasinger 2021, p95, 149]

37. **a** Urine samples should be tested within 1-2 hours of collection or refrigerated. If testing is delayed and the sample is not properly preserved, the sample's pH may be falsely elevated as a result of the conversion of urea to ammonia by urease-producing bacteria or loss of carbon dioxide.
[Strasinger 2021, p95, 143]

38. **a** The glucose reagent test strip reaction is based on the oxidation of a chromogen embedded in the test pad. In the presence of an oxidizing agent, such as bleach, the chromogen will be oxidized generating a color change regardless of the presence of glucose, leading to a potentially false-positive result.
[QCMLS 2021: 3.1.3.4] [Strasinger 2021, p150]

39. **c** Acetest® tablets have historically been used to semi-quantitatively confirm the presence of ketones, including acetoacetate and acetone, in serum, urine, and other body fluids.
[QCMLS 2021: 3.1.3.5] [Strasinger 2021, p152]

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40. **a** The occult blood reagent test strip reaction is based on the oxidation of a chromogen embedded in the test pad. In the presence of a strong oxidizing agent, such as bleach, the chromogen will be oxidized generating a color change regardless of the presence of blood, leading to a potentially false-positive result.
[QCMLS 2021: 3.1.3.7] [Strasinger 2021, p153]

41. **d** A dilute urine may yield a falsely low or negative result for protein, as the concentration may be below the sensitivity of the test pad.
[QCMLS 2021: 2.7.4.1] [Strasinger 2021, p146]

42. **b** The pH scale describes the concentration of hydrogen ions in solution. The reagent test pad includes methyl red and bromthymol blue which interact with hydrogen ions in the sample to produce a visible color.
[QCMLS 2021: 2.1.8.2.8] [Strasinger 2021, p142]

43. **b** If the sample is not properly preserved, the sample's pH may increase as a result of the conversion of urea to ammonia by urease-producing bacteria or the loss of carbon dioxide.
[QCMLS 2021: 3.1.3.3] [Strasinger 2021, p95, 143]

44. **c** Reagent strips must be handled and stored properly to prevent deterioration by heat, light, moisture, and exposure to volatile compounds. The strips should be stored in a dark, tightly capped bottle at room temperature (20°C) to minimize exposure to light. A desiccant pack is included in the container to absorb moisture.
[Strasinger 2021, p141]

45. **b** Commercial reagent test strips for urine protein are based on the principle of the protein error of indicators. The test pad contains a buffer and a pH indicator. Proteins accept hydrogen ions from the indicator causing a change in pH and a resultant change in the test pad's color.
[QCMLS 2021: 3.1.3.6] [Strasinger 2021, p145]

46. **a** The reagent test strip reaction for the protein test pad is based on the principle of the protein error of indicators in which proteins accept hydrogen from the indicators producing a color change. The test pad is more sensitive to albumin than other proteins, as albumin contains more amino groups with which to bind hydrogen.
[QCMLS 2021: 3.1.3.6] [Strasinger 2021, p145]

47. **c** The presence of high amounts of ascorbic acid/vitamin C (25+ mg/dL) may lead to a falsely negative bilirubin result by combining with the diazonium salt and lowering the sensitivity of the bilirubin test pad reaction.
[QCMLS 2021: 3.1.3.8] [Strasinger 2021, p156]

48. **d** The reagent test strip pad for ketones is embedded with sodium nitroprusside and a buffer. If present, acetoacetic acid will react with the sodium nitroprusside in an alkaline medium to produce a purple color.
[QCMLS 2021: 3.1.3.5] [Strasinger 2021, p151]

49. **a** Acetest® tablets have historically been used to confirm questionable results of the ketone test pad on reagent test strips. The tablets contain sodium nitroprusside similar to the reagent test pads but include several additional chemicals to enhance the reaction including lactose, glycine, and disodium phosphate.
[QCMLS 2021: 3.1.3.5] [Strasinger 2021, p152]

50. **a** The reagent test strip reaction for bilirubin is based on the diazo reaction. The reagent test pad is embedded with 2,4-dichloroaniline diazonium salt or 2,6-chlorobenzene-diazonium-tetrafluoroborate. In an acid medium, bilirubin, if present, will react to produce an azo dye that ranges in color from tan to purple.
[QCMLS 2021: 3.1.3.8] [Strasinger 2021, p155]

51. **c** The presence of high amounts of protein (500+ mg/dL), glucose (3+ g/dL), oxalic acid, and ascorbic acid can reduce the sensitivity of the leukocyte esterase pad leading to falsely low or negative values.
[QCMLS 2021: 3.1.3.11] [Strasinger 2021, p160]

52. **d** The reagent test strip for urobilinogen is based on reaction with Ehrlich reagent (p-dimethylaminobenzaldehyde) or 4-methyloxybenzene-diazonium-tetrafluoroborate (diazo reaction) to generate a red color change.
[QCMLS 2021: 3.1.3.9] [Strasinger 2021, p157]

53. **d** The protein test pad must be held at an acidic pH to allow the protein's effect on the double indicators to lead to a resultant color change. Samples that are highly alkaline may neutralize the buffering capacity of the test pad and affect the indicators leading to a color change unrelated to the presence of proteins.
[Strasinger 2021, p146]

54. **b** Ketones are the byproducts of lipid metabolism. Patients that cannot metabolize carbohydrates effectively or do not have an adequate intake of carbohydrates, as may occur with anorexia nervosa, will metabolize available lipids as an energy source instead.
[QCMLS 2021: 3.1.3.5] [Strasinger 2021, p151]

55. **a** Patients experiencing prolonged vomiting are often dehydrated and do not have an adequate intake of food to utilize for energy. When glycogen stores are depleted, the body will metabolize fats as an alternative energy source, producing ketones, which are water-soluble and can be filtered at the kidneys for excretion in the urine.
[QCMLS 2021: 3.1.3.5] [Strasinger 2021, p151]

56. **b** Conjugated bilirubin is normally transported to the intestines, where it is converted into urobilinogen by intestinal normal flora and excreted in the feces. A biliary tract obstruction may inhibit the normal flow of conjugated bilirubin into the intestine causing it to enter the bloodstream. Conjugated bilirubin is water-soluble and will be filtered into the urine leading to a positive result during urinalysis. Since the obstruction impedes the flow of bilirubin into the intestine, less urobilinogen is formed. The reagent test pad cannot measure a decrease in urobilinogen therefore a normal value is likely to be reported. In contrast, conditions that lead to increased bilirubin production with no impairment to conjugation or transport, such as a hemolytic event or hepatocellular disease, will most likely result in an elevated urobilinogen.
[QCMLS 2021: t3.3] [Strasinger 2021, p154-155]

57. **c** Since the bilirubin result is negative, hepatocellular disease and obstruction of the biliary tract can be ruled out as these conditions would most likely cause a positive result due to leakage or overflow of bilirubin from the liver. Conditions that cause in vivo hemolysis and release of hemoglobin into circulation, such as hemolytic anemia or transfusion reaction, result in an increase of bilirubin and urobilinogen. Assuming adequate liver function and the absence of an obstruction, bilirubin will pass normally into the intestines to be converted into urobilinogen. A small amount of urobilinogen is normally reabsorbed into the bloodstream and filtered into the urine. With increased production of urobilinogen, the amount being reabsorbed will also increase resulting in an elevated urine urobilinogen value.
[QCMLS 2021: t3.3] [Strasinger 2021, p154-155]

58. **b** The occult blood test pad is embedded with a peroxide and a chromogen. If present, the heme moiety of RBCs has pseudoperoxidase activity and will oxidize the chromogen, generating a color change. The presence of high amounts (25+ mg/dL) of ascorbic acid/vitamin C, which is a strong reducer, can interfere with the occult blood test pad reaction leading to a falsely negative result.
[QCMLS 2021: 3.1.3.7] [Strasinger 2021, p153]

59. **c** Reagent test pads for microalbumin employ immunochemical techniques utilizing antihuman albumin antibodies.
[QCMLS 2021: 2.1.8.2.9] [Strasinger 2021, p146]

60. **a** Creatinine is enzymatically produced from creatine in muscle tissue to generate energy through the action of creatine phosphokinase. Creatinine is found in the bloodstream at relatively constant concentrations and is filtered at the kidneys making it a useful marker of glomerular function and can also be used to correct for the patient's hydration state.
[Strasinger 2021, p114, 147]

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61. **b** The reagent test strip pad for pH is embedded with two indicators, methyl red and bromothymol blue, to differentiate pH units across a broad range (5.0-8.5).
[QCMLS 2021: 3.1.3.3] [Strasinger 2021, p143]

62. **c** The reagent test pad for bilirubin can yield a false-positive result for urine samples with an abnormal color. Ictotest® tablets are more sensitive than the reagent test strip pads for bilirubin and have historically been used to confirm a positive result. If an interference is suspected, several drops of water can be added to the Ictotest® mat after adding patient sample to draw interfering substances into the mat leaving bilirubin on the surface.
[QCMLS 2021: 3.1.3.8] [Strasinger 2021, p156]

63. **d** Myoglobin, hemoglobin, and RBCs will yield a positive reaction on the occult blood test pad and are generally associated with a red or red-brown urine color. In the absence of RBCs during microscopic analysis, the positive result is likely due to the presence of hemoglobin or myoglobin, which cannot be viewed microscopically.
[QCMLS 2021: 3.1.2.1, 3.1.3.7] [Strasinger 2021, p128, 153]

64. **c** The reagent test strip reaction for glucose is based on a double sequential enzyme reaction utilizing glucose oxidase and peroxidase. Glucose is oxidized to gluconic acid and a peroxide is formed. In the presence of the peroxide, the peroxidase oxidizes the chromogen generating a color change.
[QCMLS 2021: 3.1.3.4] [Strasinger 2021, p149]

65. **c** Ictotest® tablets contain several chemicals including p-nitrobenzene-diazonium-p-toluenesulfonate. When present, bilirubin will react with chemicals from the tablet to produce a blue or purple color on the test mat. The appearance of colors other than blue or purple should be reported as a negative result.
[QCMLS 2021: 3.1.3.8] [Strasinger 2021, p156]

66. **d** When performing the Ictotest®, patient sample is added to the absorbent test mat, then a tablet is placed on top of the mat. Water is added to the tablet then allowed to wick off the surface onto the mat, where it can interact with the patient sample. When present, bilirubin will react with the chemicals from the tablet to produce a blue or purple color on the test mat. Any amount of blue or purple should be reported as a positive value.
[QCMLS 2021: 3.1.3.8] [Strasinger 2021, p156]

67. **c** The glucose test pad reaction is based on a double sequential enzyme reaction utilizing glucose oxidase and peroxidase. If present, glucose will be oxidized to gluconic acid and a peroxide will be formed. The peroxidase causes oxidation of the chromogen generating a color change which allows semi-quantitative reporting for glucose (mg/dL). The presence of a strong reducer, such as ascorbic acid (vitamin C), can interfere with the reaction leading to a falsely low or negative glucose result.
[QCMLS 2021: 3.1.3.4] [Strasinger 2021, p149]

68. **c** When performing the Acetest®, patient sample is added to the surface of a tablet. When present, ketones will react with the chemicals in the tablet, primarily sodium nitroprusside, to produce a purple color on the surface of the tablet. The reaction is reported semi-quantitatively as negative, small, moderate, or large depending on the strength of the color formation.
[QCMLS 2021: 3.1.3.5] [Strasinger 2021, p152]

69. **b** A copper reduction test such as Clinitest® may be performed to screen urine for the presence of reducing substances such as glucose, galactose, fructose, pentose and lactose. In the absence of glucose, as demonstrated by a negative reagent test strip value, the copper reduction test is primarily performed to screen samples for the presence of galactose associated with galactosemia, an inborn error of carbohydrate metabolism.
[QCMLS 2021: 3.1.3.4] [Strasinger 2021, p150]

70. **d** Patients with an active urinary tract infection are likely to have white blood cells, or leukocytes, present in their urine, which can be detected by the leukocyte esterase test pad.
[QCMLS 2021: 3.1.3.11] [Strasinger 2021, p159]

71. **b** Myoglobin, hemoglobin, and RBCs will yield a positive reaction on the occult blood test pad. Myoglobin and hemoglobin cause the test pad to develop a uniform color; whereas intact RBCs may cause a speckled or spotted pattern, since the reaction only occurs where the cells came in contact with the test pad and lysed.
[QCMLS 2021: 3.1.3.7] [Strasinger 2021, p153]

72. **c** Myoglobin, hemoglobin, and RBCs will yield a positive reaction on the occult blood test pad. In the absence of RBCs during microscopic analysis, the positive result is either due to presence of myoglobin or hemoglobin in the urine (myoglobinuria or hemoglobinuria).
[QCMLS 2021: 3.1.3.7] [Strasinger 2021, p153]

73. **d** The majority of urinary tract infections are caused by gram-negative bacteria that reduce nitrates into nitrites. However, there are several non-nitrate-reducing, gram-positive bacteria and yeasts that also cause urinary tract infections, which would account for the negative nitrite result despite visualization of bacteria microscopically.
[QCMLS 2021: 3.1.3.10] [Strasinger 2021, p158]

74. **c** The occult blood test pad is embedded with a peroxide and a chromogen. The heme component of hemoglobin and myoglobin demonstrates pseudoperoxidase activity, which will oxidize the chromogen in the presence of a peroxide. This generates a color change that is recorded semi-quantitatively.
[QCMLS 2021: 3.1.3.7] [Strasinger 2021, p153]

75. **a** The reported pH from the reagent test strip reveals an alkaline urine; however, uric acid crystals are only seen in acidic urines. The reagent test strip should be repeated to confirm the pH value and the laboratorian should take care to ensure excess urine is removed from between the test pads to avoid the chemicals bleeding from one pad to another. The laboratorian should also confirm that the samples have not been switched or mislabeled.
[QCMLS 2021: 3.1.3.1, 3.1.4.6.2] [Strasinger 2021, p143, 201]

76. **d** Bacteria must have sufficient time to convert nitrates into nitrites; insufficient incubation can result in a false-negative result. First morning collections are preferred due to the long incubation time in the bladder; but are not commonly performed due to the symptoms associated with a UTI. Allowing the urine to incubate in the bladder for at least 4 hours prior to collection will increase the likelihood of detecting nitrites.
[QCMLS 2021: 3.1.3.10] [Strasinger 2021, p158]

77. **d** There are several factors that influence the reliability of the nitrite test pad, eg, incubation time, whether or not the organism converts nitrates to nitrites, or if there is a sufficient concentration of nitrates. It may be useful to evaluate the results of the nitrite and the leukocyte esterase test pads in concert as the number of WBCs will increase in response to the urinary tract infection, regardless of the underlying pathogen.
[QCMLS 2021: 3.1.3.10, 3.1.3.11] [Strasinger 2021, p157]

78. **c** The reagent test strip pad for leukocyte esterase is embedded with an acid ester such as indoxylcarbonic acid ester and a diazonium salt. If present, leukocyte esterase will react with the acid ester to produce an aromatic compound, which will in turn react with the diazonium salt in an acid medium to produce an azo dye that is purple. Leukocyte esterase is found in granulocytic WBCs, ie, neutrophils, monocytes, eosinophils, and basophils, but is **not** present in lymphocytes.
[QCMLS 2021: 3.1.3.11] [Strasinger 2021, p159]

79. **c** Acetoacetic acid, acetone, and beta-hydroxybutyric acid are collectively referred to as "ketones." The reagent test strip pad for ketones is embedded with sodium nitroprusside (nitroferricyanide) and a buffer. If present, acetoacetic acid will react with the sodium nitroprusside in an alkaline medium to produce a purple color. Nitroprusside does not react with beta-hydroxybutyric acid and glycine must be present for the reaction to include acetone.
[QCMLS 2021: 3.1.3.5] [Strasinger 2021, p151]

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80. **b** Microscopic examination of the prepared urine sediment should be performed by observing at least 10 fields using both low power (10x) and high dry power (40x). The sample should first be evaluated at low power for the presence of casts and then evaluated using high dry power to identify and enumerate cellular and formed elements such as RBCs, WBCs, bacteria, epithelial cells, yeast and crystals.
[QCMLS 2021: 3.1.4.2] [Strasinger 2021, p169]

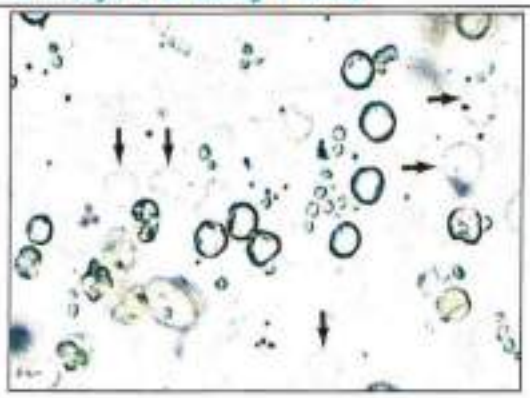
81. **a** The photomicrograph shows numerous squamous epithelial cells. Squamous epithelial cells line the vagina and female urethra and the distal portion of the male urethra. Their presence in urine is generally considered as having little to no pathological significance due to normal cellular sloughing; however, they may be present in increased numbers as a result of a genitourinary tract infection. The negative results for both nitrite and leukocyte esterase suggest that the patient is not experiencing an infection, and the presence of the squamous epithelial cells is more likely the result of a non-clean-catch specimen or contamination.
[QCMLS 2021: 3.1.4.3.1] [Strasinger 2021, p183]

82. **d** Renal tubular epithelial cells line the proximal and distal convoluted tubules as well as the collecting ducts of the nephron. The renal tubular epithelial cells absorb analytes from the urinary filtrate and secrete substances into the filtrate for excretion. In disorders that result in lipids passing into the filtrate, lipids are absorbed into the renal tubular epithelial cells. When the cells slough from the tubules, they appear in the urine and are designated as oval fat bodies.
[QCMLS 2021: 3.1.4.3.4] [Strasinger 2021, p186]

83. **b** Red blood cells present in an alkaline urine or a urine with low specific gravity (hypotonic) will absorb water through osmosis. This causes them to swell and eventually lyse, releasing their hemoglobin contents. The remaining red blood cell membranes appear as faint empty cells referred to as "ghost cells."
[QCMLS 2021: 3.1.4.4.1] [Strasinger 2021, p179]

84. **d** When present in an alkaline or hypotonic urine (low specific gravity), white blood cells will absorb water causing them to swell. Granules inside the swollen cells exhibit Brownian movement, which gives the appearance that the cell is glittering and so these cells are referred to as "glitter cells."
[QCMLS 2021: 3.1.4.4.2] [Strasinger 2021, p180]

85. **c** Squamous epithelial cells line the vagina and female urethra and the distal portion of the male urethra. Due to normal cellular sloughing, their presence in urine is generally considered to be the result of a non-clean-catch specimen or vaginal contamination
[QCMLS 2021: 3.1.4.3.1] [Strasinger 2021, p183]



86. **b** The photomicrograph shows numerous ghost cells. When present in an alkaline urine, the red blood cell membrane allows hemoglobin to leak out from the cell, resulting in the appearance of an empty cell membrane or a ghost cell.
[QCMLS 2021: 3.1.4.4.1] [Strasinger 2021, p179]

87. **d** Casts are cylindrical structures, composed primarily of uromodulin, that form within the lumen of the renal tubules. They are viewed using low power (10x) and appear as pale, smooth cylinders with parallel edges, rounded ends, and a low refractive index. Casts are classified according to the presence of an inclusion such as RBCs, WBCs, fats, epithelial cells, and coarse/fine granules. This photomicrograph shows a cast with fine granular inclusions, which would be reported as a granular cast.
[QCMLS 2021: 3.1.4.5.6] [Strasinger 2021, p190, 196]

88. **a** A primary cause of bacterial vaginosis is the bacterium *Gardnerella vaginalis*. Squamous epithelial cells that line the vagina may become covered with the coccobacilli, giving them a granular and irregular appearance. As these cells are sloughed, they may appear in the urine sediment and are referred to as "clue cells."
[QCMLS 2021: 3.1.4.3.5] [Strasinger 2021, p183]

89. **c** Transitional epithelial, or urothelial, cells line the renal pelvis, calyces, ureters, bladder, and proximal portion of the male urethra. These can be dislodged from the walls of the bladder as a result of catheterization and appear in the urine sediment singly, in pairs, or in clumps.
[QCMLS 2021: 3.1.4.3.2] [Strasinger 2021, p184]

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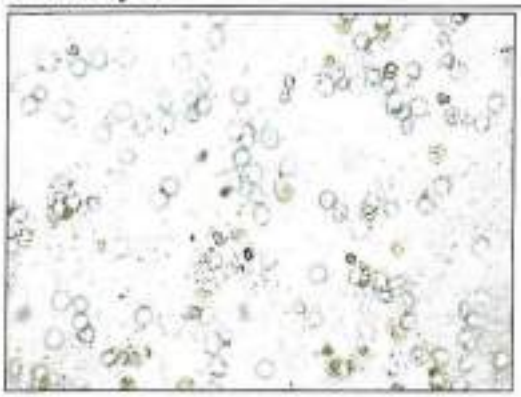
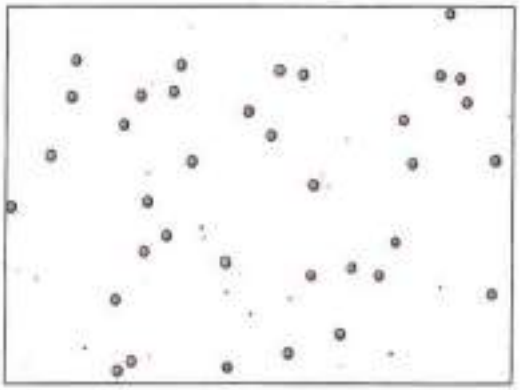


90. **a** The photomicrograph shows the presence of numerous white blood cells and significant bacteriuria. While the presence of 0-8 WBCs/HPF is considered normal, an increased presence, as seen in the pictured sediment, would be associated with inflammation of the genitourinary tract or an infection of the lower urinary tract (urethritis, cystitis) or upper urinary tract (pyelonephritis). Given the presence of bacteria but relative absence of renal epithelial cells and/or WBC casts, the most likely diagnosis would be a lower urinary tract infection involving the bladder (cystitis).
[QCMLS 2021: 3.1.4.4.2, 3.1.4.8.2] [Strasinger 2021, p182, 193]



91. **a** The photomicrograph shows the presence of several red blood cells. RBCs appear as small (6-8 µm), anucleated, biconcave discs that show a central pallor when the fine adjustment knob of the microscope is used to slightly adjust the plane being viewed. RBCs can be confused microscopically with several other urinary constituents such as yeast, small crystals, and fat globules as well as confounding artifacts such as air bubbles and oil droplets. Care should be taken to correlate the results of the chemical examination with the microscopic observations, as RBCs should produce a positive result on the occult blood pad of the reagent test strip.
[QCMLS 2021: 3.1.4.4.1] [Strasinger 2021, p178]

92. **d** Casts are primarily composed of uromodulin (Tamm-Horsfall proteins), but may also contain other proteins, such as albumin and immunoglobins. Uromodulin is continuously excreted by the renal tubular cells, but appears to be increased following exercise and stressful conditions. In conditions that promote urinary stasis, decreased urinary pH, and/or the presence of calcium or sodium, uromodulin aggregates into fibrils that mesh to form the matrix of casts.
[QCMLS 2021: 3.1.4.5] [Strasinger 2021, p190]

93. **d** Casts are much larger than other urinary constituents so they are observed using the low power (10x) objective and are often found along the edges of the coverslip. Given their relatively low refractive index, casts may not be visible under bright light. Subdued lighting should be used which can be accomplished by adjusting the settings of the condenser, iris, and/or rheostat. If casts are observed at low power, they should be evaluated using high dry power (40x) to determine the presence of inclusions (eg. WBC, RBC, bacteria) which is used to further classify the type of cast (eg, WBC cast).
[QCMLS 2021: 3.1.4.2, 3.1.4.5] [Strasinger 2021, p190]

94. **c** Hyaline casts are formed in the lumen of the renal tubules and are almost entirely composed of uromodulin. The presence of 0-2 hyaline casts/LPF is considered normal and may be present following strenuous exercise, heat exposure, dehydration, or stress in otherwise healthy adults.
[QCMLS 2021: 3.1.4.5.1] [Strasinger 2021, p192]

95. **a** Waxy casts are associated with significant urinary stasis as seen in chronic renal failure and are believed to be caused by degeneration of other casts (eg, hyaline, cellular, granular). Prolonged/significant urinary stasis allows casts to form in the renal tubules and/or collecting ducts, which may then be dislodged and viewed in the urinary sediment.
[QCMLS 2021: 3.1.4.5.7] [Strasinger 2021, p198]

96. **a** Urine samples should be tested within 1-2 hours of collection to minimize changes that occur to the chemical and cellular composition of the sample. Samples that contain urease-producing bacteria will demonstrate an increase in pH over time as urea is converted into ammonia. This results in an alkaline urine which promotes the dissolution of urinary casts.
[QCMLS 2021: 3.1.4.1] [Strasinger 2021, p190]

97. **c** Transitional epithelial (urothelial) cells are 20-30 µm and appear in several forms, including spherical, polyhedral, and caudate, but all forms have centrally-located nuclei. Renal epithelial cells vary in size (14-60 µm) and shape (columnar, round, oval, cuboidal) depending on where they originate in the renal tubules, but all typically have eccentric nuclei.
[QCMLS 2021: 3.1.4.3.2, 3.1.4.3.3] [Strasinger 2021, p183-184]

98. **d** Cystitis refers to an infection or inflammation of the bladder (eg, lower urinary tract infection); whereas, pyelonephritis is an infection or inflammation of the renal tubules (eg, upper urinary tract infection). White blood cells, red blood cells, and bacteria are common findings of both lower and upper urinary tract infections. The presence of WBC casts would indicate involvement of the renal tubules and aid in differentiating cystitis from pyelonephritis.
[QCMLS 2021: 3.1.4.5.4] [Strasinger 2021, p194]

99. **b** Renal tubular epithelial cells line the proximal and distal convoluted tubules as well as the collecting ducts of the nephron. Damage to the renal tubules, as may occur in tubular necrosis, can cause these cells to be dislodged. The cells may then be trapped in casts forming within the tubules, resulting in the formation of renal tubular epithelial casts.
[QCMLS 2021: 3.1.4.5.5] [Strasinger 2021, p196]

100. **a** The formation of hyaline casts is common following strenuous exercise, periods of dehydration, and situations in which decreased urinary flow occurs.
[QCMLS 2021: 3.1.4.5] [Strasinger 2021, p192]

101. **b** Waxy casts are believed to be the result of degeneration of other types of casts (eg, hyaline, cellular, granular) and are associated with extreme urinary stasis. Due to their extended time in the renal tubules, they are brittle and often contain notches, cracks, and jagged edges as the result of disintegration.
[QCMLS 2021: 3.1.4.5.7] [Strasinger 2021, p197]

102. **c** White blood cells frequently occur in clumps and can resemble a WBC cast. WBC casts indicate a more serious tubular infection, whereas WBC clumps can be seen in cystitis so it is important to properly differentiate them. To positively identify the structure as a WBC cast rather than a clump of cells, the laboratorian must determine that a cast matrix is present.
[QCMLS 2021: 3.1.4.5.4] [Strasinger 2021, p194]

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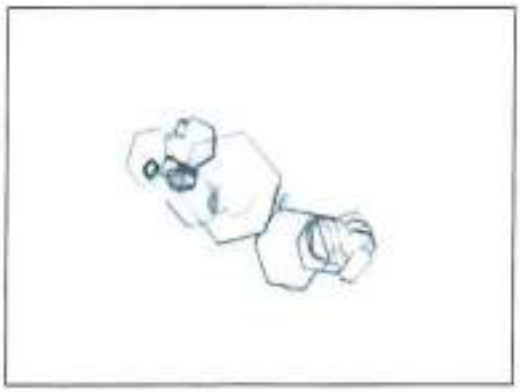
103. **b** Renal tubular epithelial cells vary in size (14-60 µm) and shape (columnar, round, oval, cuboidal), depending on where they originate in the renal tubules, and have eccentric nuclei. Given their similar size (10-12 µm), presence of a nucleus, and ratio of nucleus to cytoplasm content, white blood cells and round renal tubular epithelial cells can be challenging to differentiate. However, renal tubular epithelial cells are generally slightly larger than WBCs and their nuclei are eccentric, which may aid in differentiating these cell types. A positive leukocyte esterase result would also correlate with the presence of WBCs but does not rule out the presence of renal epithelial cells as both cell types may be present in the sample.
[QCMLS 2021: 3.1.4.3.3, 3.1.4.4.2] [Strasinger 2021, p182]

104. **c** RBC casts indicate serious glomerular disease so it is important to properly differentiate them from other types of casts. To positively identify an RBC cast, the laboratorian must also observe free-floating RBCs in the sediment and correlate the results with a positive occult blood value from the chemical examination.
[QCMLS 2021: 3.1.4.5.2] [Strasinger 2021, p193]



105. **a** The photomicrograph shows the presence of cystine crystals, which appear as colorless, hexagonal plates in acidic urines. Cystine crystals are associated with cystinuria, which is a metabolic disorder that prevents reabsorption of cystine in the renal tubules.
[QCMLS 2021: 3.1.4.7.4] [Strasinger 2021, p206]

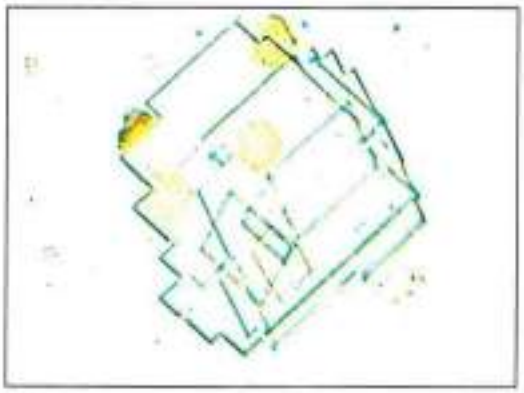
106. **c** RBCs normally should not be able to pass through the glomerular membrane. The presence of RBC casts indicates there is bleeding within the nephron, or that damage has occurred to the glomeruli and RBCs are entering the nephron inappropriately, as seen in glomerulonephritis.
[QCMLS 2021: 3.1.4.5.2] [Strasinger 2021, p192]

107. **c** As the temperature of a urine sample decreases, solutes are more likely to precipitate and form crystals. After refrigeration of an alkaline urine, amorphous phosphates may precipitate, leading to the appearance of a white precipitate that does not dissolve upon warming.
[QCMLS 2021: 3.1.4.7.3] [Strasinger 2021, p199, 204]

108. **a** Cystine crystals appear as colorless, hexagonal plates that frequently occur in clumps, leading to a layered or laminated appearance.
[QCMLS 2021: 3.1.4.7.4] [Strasinger 2021, p206]

109. **c** Approximately 75% of renal calculi are composed of calcium compounds such as calcium oxalate and calcium phosphate. Magnesium ammonium phosphate (struvite), uric acid, and cystine account for a large portion of the remaining 25% of renal calculi.
[QCMLS 2021: 2.7.5.3] [Strasinger 2021, p229]

110. **d** Solutes are more likely to precipitate and form crystals as the temperature of the solution decreases. After refrigeration, amorphous crystals, such as amorphous urates or amorphous phosphates, may precipitate and cause the sample to become cloudy. Amorphous urates will dissolve when the specimen is briefly warmed. Amorphous phosphates can be dissolved by the addition of acetic acid, but the acid will destroy other sediment constituents, such as RBCs, and should only be performed after viewing the sample for these other elements.
[QCMLS 2021: 3.1.4.6.3, 3.1.4.7.3] [Strasinger 2021, p199]



111. **d** The photomicrograph shows the presence of cholesterol crystals. Cholesterol crystals appear in acidic or neutral urines as colorless, rectangular plates with notched corners. They frequently occur in clumps, leading to a layered or laminated appearance.
[QCMLS 2021: 3.1.4.7.5] [Strasinger 2021, p206]

112. **c** Calcium oxalate crystals are most commonly found in acidic or neutral urines. Ammonium biurate, calcium phosphate, and triple phosphate are seen in alkaline urines.
[QCMLS 2021: 3.1.4.6.1] [Strasinger 2021, p202]

113. **a** Cholesterol is capable of polarizing light and demonstrates weak birefringence when viewed using a polarizing microscope.
[QCMLS 2021: 3.1.4.7.5] [Strasinger 2021, p206]

114. **b** Tyrosine crystals are colorless or yellow, fine needles that often appear in clusters or sheaves. They are associated with liver disease and inherited amino acid metabolic disorders such as tyrosinemia.
[QCMLS 2021: 3.1.4.7.7] [Strasinger 2021, p207]

115. **c** An amber-colored urine is associated with the presence of bilirubin, which is confirmed by the reaction on the reagent test strip. The presence of bilirubin in the urine sample suggests liver disease or damage. Tyrosine crystals and leucine crystals may be seen in patients with liver disease due to impaired amino acid metabolism.
[QCMLS 2021: 3.1.2.1, 3.1.4.7.7] [Strasinger 2021, p127, 207]

116. **b** Following ingestion of ethylene glycol/antifreeze, calcium oxalate crystals in their monohydrate form may be seen in acidic urines. Unlike the more commonly seen envelope-shaped dihydrate crystals, monohydrate calcium oxalate crystals appear as colorless ovals or dumbbells.
[QCMLS 2021: 3.1.4.6.1] [Strasinger 2021, p203-204]

117. **b** The presence of cholesterol crystals in urine is associated with lipiduria which occurs in nephrotic syndrome. In nephrotic syndrome, >3.5 g of protein are excreted per day in the urine, which would yield a positive result for the protein pad when tested using a reagent test strip.
[QCMLS 2021: 3.1.4.7.5] [Strasinger 2021, p206]

118. **c** The presence of a large amount of uric acid crystals is associated with increased metabolism of purines and nucleic acids as seen in patients with gout, Lesch-Nyhan syndrome, or those receiving chemotherapy.
[QCMLS 2021: 3.1.4.6.2] [Strasinger 2021, p202, 247]

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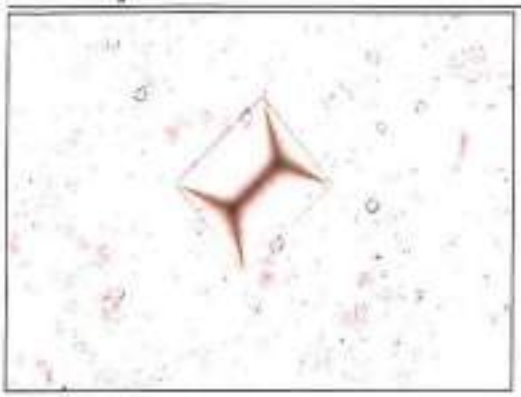


119. **d** The photomicrograph shows the presence of a triple phosphate crystal. Triple phosphate (ammonium magnesium phosphate) crystals appear in alkaline urines as colorless prisms, which are often described as "coffin lids." They are considered to be clinically insignificant (nonpathologic), but usually occur in samples with bacterial overgrowth, which results in an alkaline urine.
[QCMLS 2021: 3.1.4.7.1] [Strasinger 2021, p204]

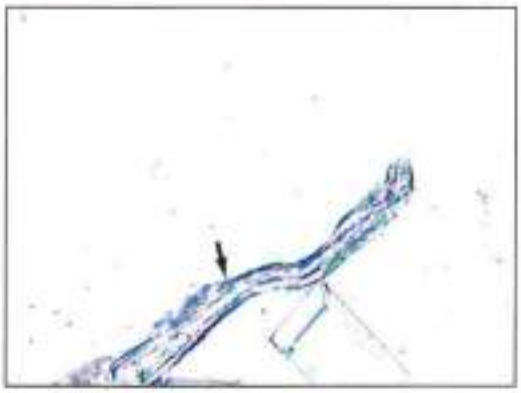
120. **b** The photomicrograph shows the presence of several uric acid crystals. Uric acid crystals appear in acidic urines as colorless or yellow-brown rhombic plates. They often appear in clusters referred to as "rosettes." They may be considered as a normal urinary crystal in small amounts, but large amounts are associated with increased purine metabolism and would be considered clinically significant.
[QCMLS 2021: 3.1.4.6.2] [Strasinger 2021, p202]

121. **c** The photomicrograph shows the presence of a large amount of amorphous crystals, which may inhibit visualization of other clinically significant elements. The crystals are likely amorphous phosphates because the sample has an alkaline pH. Amorphous phosphates appear as a white precipitant following centrifugation. They can be dissolved by adding a few drops of dilute acetic acid to the prepared sediment. This may allow visualization of other formed elements previously obscured by the crystals; however, the acid will also destroy other sediment constituents, such as RBCs, and should only be performed after viewing the sample for these other elements.
[QCMLS 2021: 3.1.4.7.3] [Strasinger 2021, p30, 199-200]

122. **b** Uric acid crystals most commonly appear as rhombic plates but can also have a 6-sided shape similar to cystine. Polarizing microscopy can be used to distinguish these 2 types of crystals as uric acid crystals are strongly birefringent, whereas cystine crystals generally do not polarize light.
[QCMLS 2021: 3.1.4.6.2, 3.1.4.7.4] [Strasinger 2021, p202, 206]

123. **d** Starch granules appear as spheres with a dimpled center and produce a Maltese cross under polarized light.
[Strasinger 2021, p210]

124. **b** Uric acid crystals most commonly appear in acidic urines as colorless or yellow-brown rhombic plates and often appear in clusters referred to as "rosettes." They can also appear as wedges and 4- or 6-sided plates that can be confused with cystine crystals.
[QCMLS 2021: 3.1.4.6.2] [Strasinger 2021, p202]



125. **a** The photomicrograph shows a cloth fiber, which is a contaminant in the urine sediment. Hair and cloth fibers are often mistaken for casts and care must be taken to differentiate them. Fibers are generally longer, more refractile, and will polarize light.
[Strasinger 2021, p210]

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126. b The addition of a few drops of glacial acetic acid would cause RBCs to lyse, but would not affect oil droplets and yeast. Care must be taken as the addition of the acid would also lyse other formed elements and should only be added to an aliquot of the urine sediment after evaluation of the other elements.
[QCMLS 2021: 3.1.4.4.1] [Strasinger 2021, p179]

127. a Precipitation of urinary crystals is influenced by several factors including: concentration of urine solutes (eg, calcium, uric acid), temperature, and the pH of the solution. When differentiating crystals with a similar appearance, knowing the pH of the sample may be of value as some crystals are more soluble in an acidic pH and will only precipitate in alkaline urines (eg, triple phosphate), while others are more soluble in an alkaline pH and precipitate in acidic urines (eg, uric acid).
[QCMLS 2021: 3.1.4.6, 3.1.4.7] [Strasinger 2021, p199]

128. d The bladder is normally a sterile environment and bacteria should not be present in the urine. Samples collected by catheterization and suprapubic aspiration are considered sterile, but other methods (eg, random, clean-catch, pediatric U-bag) are prone to contamination with bacteria from the skin and/or genital region. If testing is delayed, bacteria will continue to multiply in the sample and can generate a falsely positive nitrite. The presence of WBCs in the sample confirm the bacteria are present due to an active infection.
[QCMLS 2021: 3.1.4.4.2, 3.1.4.8.2, 3.1.5.6] [Strasinger 2021, p187]

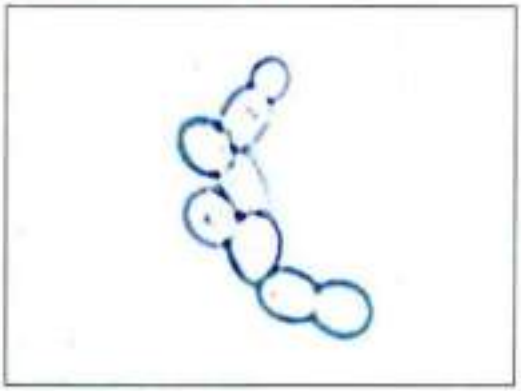
129. d *Trichomonas vaginalis* is a pear- or turnip-shaped flagellate primarily associated with vaginal infections. It has four anterior flagella and an undulating membrane, but may be mistaken for a WBC or renal tubular epithelial cell in urine samples. To differentiate these elements, the laboratorian should look for motility, which indicates the element is a flagellate.
[QCMLS 2021: 3.1.4.8.3] [Strasinger 2021, p189]

130. c The photomicrograph shows the presence of a hyaline cast (arrow). Hyaline casts form in the renal tubules and appear as pale, smooth cylinders with parallel sides and rounded ends. They may be straight or curved depending on the shape of the tubule in which they were formed.
[QCMLS 2021: 3.1.4.5.1] [Strasinger 2021, p192]

131. **d** The photomicrograph shows the presence of numerous rod-shaped bacteria (bacilli). Many of the bacteria that cause urinary tract infections convert nitrates into nitrites. Given sufficient incubation time in the bladder, these bacteria will convert the nitrates in the urine into nitrites, which will generate a positive reaction on the nitrite test pad of the reagent test strip.
[QCMLS 2021: 3.1.4.8.2, 3.1.3.10] [Strasinger 2021, p158, 187]

132. **b** Yeast appear as small, colorless ovoid cells similar to RBCs. The cells show budding that aids in identifying them as yeast, of which *Candida albicans* is a frequent cause of urinary tract infections. The ideal conditions for growth of *Candida albicans* include an acidic pH and the presence of glucose, which explains why patients with diabetes frequently have urinary tract infections caused by *Candida*.
[QCMLS 2021: 3.1.4.8.1] [Strasinger 2021, p189]

133. **a** Leucine crystals appear in acidic or neutral urines as yellow-brown spheres with concentric circles and radial striations. Their presence is associated with liver disease and inborn errors of metabolism (eg, MSUD).
[QCMLS 2021: 3.1.4.7.8] [Strasinger 2021, p206]



134. **a** The photomicrograph shows the presence of budding yeast. Yeast appear as small, colorless ovoid cells that are highly refractile. The presence of budding and branching aids in differentiating yeast cells from RBCs.
[QCMLS 2021: 3.1.4.8.1] [Strasinger 2021, p188]

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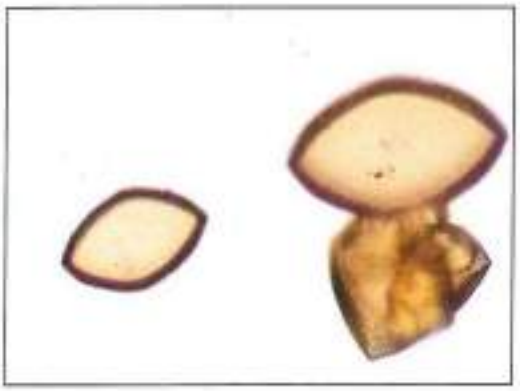
135. c After formation of the plasma ultrafiltrate by the glomerulus, substances are reabsorbed and secreted in the renal tubules. Tubular reabsorption occurs through active and passive transport mechanisms. Normally, substances are actively transported from the ultrafiltrate into the blood stream until the renal threshold is met. The renal threshold is the plasma concentration at which a substance, such as glucose, is no longer reabsorbed by the proximal convoluted tubules. The renal threshold for glucose ranges from 160-180 mg/dL (8.88-9.99 mmol/L).
[QCMLS 2021: 2.7.2.5] [Strasinger 2021, p110, 148]

136. b The volume of urine excreted per day is dependent on several factors, such as hydration status, renal function, hormonal control, and the use of diuretics. Healthy adults produce 1-2 L (1000-2000 mL) of urine per day. The prefix *oligo-* means scanty and *oliguria* refers to production of less than 400 mL/day. The prefix *an-* means not with *anuria* referring to cessation of urinary flow. *Poly-* means many and the term polyuria is used to describe the production of more than 2.5-3.0 L in a 24-hour period. Patients with urinary tract infections often experience pain during urination, which is termed dysuria (*dys-* means bad or ill).
[Strasinger 2021, p93]

137. b Plasma is filtered at the glomerulus at a rate of 120 mL/min forming the plasma ultrafiltrate. The rate of glomerular filtration can be evaluated using a clearance test in which the concentration of an analyte is measured in both a urine sample and a paired blood sample.
[QCMLS 2021: 2.7.2.2] [Strasinger 2021, p106, 115]



138. a The photomicrograph shows the presence of uric acid crystals. Uric acid crystals appear in acidic urines as colorless or yellow-brown rhombic plates and are associated with increased metabolism of purines and nucleic acids, as seen in patients with gout.
[QCMLS 2021: 3.1.4.6.2] [Strasinger 2021, p202]

139. d Creatinine is a waste product excreted via the kidneys. As such, urine normally contains a high concentation of creatinine, making it a useful measurement in determining if a fluid is urine. Normally, urine should be negative for protein, glucose, and bilirubin using a reagent test strip.
[QCMLS 2021: 2.7.3.1, 2.7.3.2] [Strasinger 2021, p93, 143, 148, 154]

140. a Arginine vasopressin (AVP), also known as antidiuretic hormone (ADH), is produced in the hypothalamus and secreted by the posterior pituitary when plasma osmolality and sodium concentrations are elevated. AVP regulates the permeability of the walls of the distal convoluted tubule and the collecting ducts to water. When the body is dehydrated (elevated plasma osmolality/increased sodium), AVP is released by the pituitary gland to increase the permeability of the walls and allow water to be reabsorbed into the bloodstream.
[QCMLS 2021: 2.7.2.3] [Strasinger 2021, p110]

141. **b** Plasma is filtered at the glomerulus forming a plasma ultrafiltrate. When the glomerular membrane is intact, only substances with a molecular weight <70,000 (eg, urea, glucose, amino acids) will pass through. Larger molecular weight substances and cells (eg, RBCs) cannot readily pass through the glomerular membrane and their presence in urine is associated with damage to the glomerulus.
[QCMLS 2021: 2.7.2.2] [Strasinger 2021, p105]

142. **c** Patients with diabetes mellitus often have elevated plasma glucose concentrations resulting in glucosuria. Glucose is an osmotically active substance that results in excretion of urine volumes >2.5-3.0 L per 24 hours (polyuria).
[QCMLS 2021: 3.1.5.2] [Strasinger 2021, p93]

143. **a** Healthy adults produce 1-2 L (1000-2000 mL) of urine per day, but the total volume is dependent on several factors such as hydration status, renal function, hormonal control, and the use of diuretics. The prefix *an-* means not, or without, and *anuria* refers to cessation of urinary flow.
[Strasinger 2021, p93]

144. **a** Acetoacetic acid, beta-hydroxybutyrate, and acetone, collectively referred to as ketones, are intermediate products of lipid (fat) metabolism. When carbohydrates cannot be used to generate energy, fats are used as an alternative energy source and ketones can be detected in the urine.
[QCMLS 2021: 3.1.3.5] [Strasinger 2021, p151]

145. **c** Bilirubin is a product of hemoglobin degradation. Conjugated bilirubin is normally transported to the intestines, where it is converted into urobilinogen by intestinal normal flora and excreted in the feces. A biliary tract obstruction may inhibit the normal flow of conjugated bilirubin into the intestine causing it to enter the bloodstream. Conjugated bilirubin is water-soluble and will be filtered into the urine, leading to a positive bilirubin result on the reagent test strip.
[QCMLS 2021: 3.1.3.8, 3.1.5.5] [Strasinger 2021, p154]

146. **c** Patients with hepatic disease or damage may leak conjugated bilirubin into the bloodstream, where it can be filtered and excreted via the kidneys. In acidic urines, bilirubin crystals appear as small clusters of fine, yellow-brown needles and are considered clinically significant.
[QCMLS 2021: 3.1.4.7.6, 3.1.5.5] [Strasinger 2021, p208]

147. **b** There are several forms of renal tubular acidosis (RTA), which is characterized by decreased ability to secrete hydrogen ions in the distal convoluted tubules (distal RTA) and/or decreased ability to reabsorb bicarbonate in the proximal convoluted tubules (proximal RTA). This leads to excretion of an alkaline urine despite the patient being in a state of metabolic acidosis.
[QCMLS 2021: 2.7.5.2.2] [Brunzel 2018, p220]

148. **b** Glucose is filtered at the glomerulus and reabsorbed by active transport mechanisms in the proximal convoluted tubules (PCT) up to the renal threshold. Damage and/or dysfunction of the PCT may cause decreased glucose reabsorption leading to inappropriate excretion of glucose in the urine (glycosuria) and a normal or low plasma glucose concentration.
[QCMLS 2021: 2.7.2.3, 3.1.5.3] [Strasinger 2021, p110, 226]

149. **c** Following significant reabsorption of water and solutes in the proximal convoluted tubule, the filtrate enters the descending limb of the loop of Henle. In this section of the renal tubules, water is passively reabsorbed due to the high osmotic gradient of the renal medulla, effectively concentrating the filtrate. To prevent excessive absorption of water, the walls of the ascending limb of the loop of Henle are impermeable to water.
[QCMLS 2021: 2.7.2.3] [Strasinger 2021, p110]

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150. **d** The juxtamedullary nephrons have long loops of Henle that allow the urinary filtrate to pass through the renal medulla; whereas cortical nephrons, located in the renal cortex, have short loops of Henle that do not reach the medulla. The renal medulla has a high osmotic gradient that promotes passive reabsorption of water in the descending loop of Henle leading to concentration of the filtrate.
[QCMLS 2021: 2.7.1.1] [Strasinger 2021, p106]

151. **c** Urine samples with a red appearance may contain RBCs, hemoglobin, myoglobin, or porphyrins. The red coloration may also be the result of certain drugs and/or foods. The positive occult blood reaction indicates the sample contains RBCs, hemoglobin, or myoglobin. Upon microscopic examination, presence of only a few RBCs suggests the sample contains hemoglobin or myoglobin. The presence of several renal tubular cells suggests damage to the renal tubules, as may occur in acute tubular necrosis (ATN). The most common causes of ATN are renal ischemia and exposure to nephrotoxins such as hemoglobin and myoglobin.
[QCMLS 2021: 3.1.2.1, 3.1.3.7, 3.1.4.3.3, 2.7.5.5] [Strasinger 2021, p128, 153, 165, 222]

152. **a** Ammonia is produced in the renal tubules from the amino acid glutamine. The ammonia reacts with a hydrogen ion in the filtrate forming an ammonium ion, which is excreted in the urine. Failure of the nephron to produce ammonia allows the hydrogen ions in the filtrate to be reabsorbed leading to metabolic acidosis and a high urinary pH (alkaline).
[QCMLS 2021: 2.8.3] [Strasinger 2021, p111]

E. Disease States

153. **a** Patients with hypertension and diabetes mellitus are at risk for developing kidney disease (nephropathy). The presence of small amounts of albumin in the urine, termed microalbuminuria, is used to detect early development of kidney disease.
[QCMLS 2021: 3.1.3.6] [Strasinger 2021, p145]

154. **a** The photomicrograph shows the presence of 2 spherical transitional epithelial (urothelial) cells in the urine sediment. Transitional epithelial cells are 20-30 µm in size and appear in several forms including spherical, polyhedral, and caudate. They have moderate cytoplasm and a centrally-located nucleus. Transitional epithelial cells line the renal pelvis, ureter, bladder, and a portion of the urethra; therefore, their presence in a urine sediment is associated with lower urinary tract infections, such as urethritis and cystitis.
[QCMLS 2021: 3.1.4.3.2] [Strasinger 2021, p183]

155. **a** Glucose is filtered at the glomerulus and reabsorbed by active transport mechanisms in the proximal convoluted tubules until the plasma renal threshold is met. Once the renal threshold is met, excess glucose is excreted in the urine, resulting in glucosuria. In hyperglycemic disorders, such as diabetes mellitus, blood glucose concentrations are elevated and excess glucose is excreted in the urine. Patients with diabetes mellitus often develop proteinuria as the disease progresses due to nephropathy.
[QCMLS 2021: 3.1.5.2] [Strasinger 2021, p144, 146]

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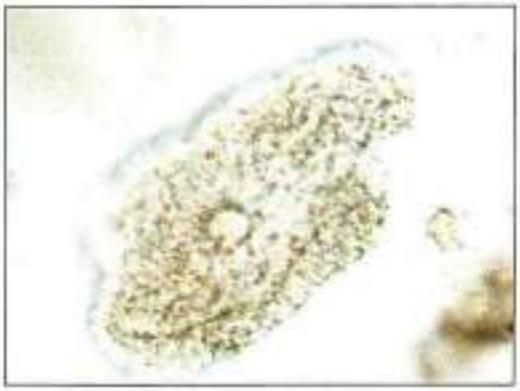
156. **b** After being filtered at the glomerulus, glucose in the filtrate is reabsorbed by active transport mechanisms in the proximal convoluted tubules (PCT) up to the renal threshold. Nearly all glucose will be reabsorbed from the filtrate, and it should not be present in urine. Damage and/or dysfunction of the PCT may cause decreased glucose reabsorption in the PCT, leading to inappropriate excretion of glucose in the urine (renal glycosuria) despite a normal or low plasma glucose concentration.
[QCMLS 2021: 2.7.2.3, 3.1.5.3] [Strasinger 2021, p110, 226]

157. **d** The presence of RBCs in the urine (hematuria), as evidenced by the positive occult blood test pad and visualization of RBCs microscopically, is associated with damage to the kidneys and/or genitourinary tract, such as trauma, kidney stones, glomerular disease, infection, and exposure to endogenous or exogenous toxins. The presence of RBC casts is primarily associated with glomerular diseases, such as glomerulonephritis, in which damage to the glomerular membrane allows cells and large proteins to inappropriately pass through the membrane into the renal tubules. In the renal tubules, the proteins contribute to the formation of casts which will adhere the RBCs forming RBC casts.
[QCMLS 2021: 3.1.4.4.1, 3.1.4.5.2] [Strasinger 2021, p153]

158. **b** The presence of a few uric acid crystals may be considered normal, but large amounts would be considered clinically significant as they are associated with increased purine metabolism, as occurs in gout, following chemotherapy, and in some metabolic disorders.
[QCMLS 2021: 3.1.4.6.2] [Strasinger 2021, p202]

159. **b** Damage to the electrical charge of the glomerular membrane, allowing the passage of high molecular weight proteins and lipids, occurs in nephrotic syndrome. This results in markedly increased urine protein levels and the appearance of fatty casts and oval fat bodies that are characteristic of nephrotic syndrome.
[QCMLS 2021: 3.1.3.6, 3.1.4.3.4, 3.1.4.5.8] [Strasinger 2021, p202]



160. **b** The photomicrograph shows the presence of a squamous epithelial cell with a rough or granular appearance, known as a "clue cell." The negative results for leukocyte esterase and nitrite rule out a urinary tract infection. Clue cells are squamous epithelial cells from the vagina that are covered with coccobacilli, *Gardnerella vaginalis*, which gives them a granular and irregular appearance. As these cells are sloughed, they may appear in the urine sediment and are considered clinically significant as they indicate bacterial vaginosis.
[QCMLS 2021: 3.1.4.3.5] [Strasinger 2021, p228; Brunzel 2018, p223]

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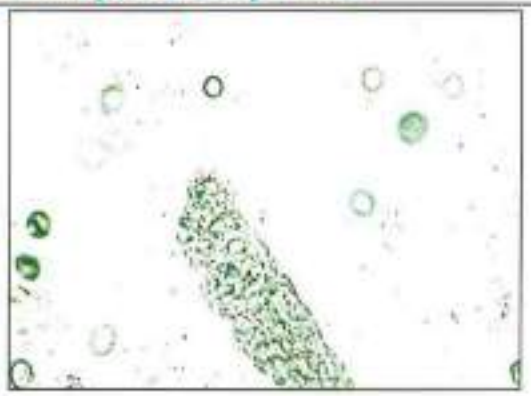
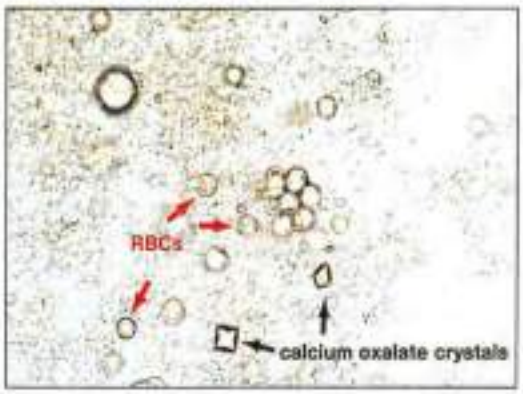
161. d The negative result for nitrite and lack of bacteria upon microscopic examination suggests that the patient does not have an active urinary tract infection and that the WBCs are present due to inflammation. Acute interstitial nephritis is caused by an allergic reaction, often occurring following organ transplantation or administration of a medication/drug that results in inflammation of the renal tubules. WBCs and WBC casts may be present in the urine sediment as a result of the inflammatory process.
[QCMLS 2021: 3.1.4.5.4] [Strasinger 2021, p228]

162. d A major cause of glomerular disorders, such as acute glomerulonephritis, is the deposition of immune complexes on the glomerular membrane. The resulting damage allows proteins and cells, such as RBCs, to pass through the membrane and into the renal tubules. In the renal tubules, the proteins contribute to the formation of casts, which will adhere the RBCs, forming RBC casts. The presence of RBCs in the urine may yield a red colored sample and generate a positive result for the occult test pad. Increased proteins (albumin) will cause a positive reaction on the protein test pad.
[QCMLS 2021: 3.1.2.1, 3.1.4.4.1, 3.1.4.5.2] [Strasinger 2021, p220]

163. b The photomicrograph shows the presence of tyrosine crystals, which appear in acidic urines as colorless or yellow, fine needles in clusters or sheaves. They are associated with liver disease and inherited amino acid metabolic disorders such as tyrosinemia.
[QCMLS 2021: 3.1.4.7.7] [Strasinger 2021, p207, 237]



164. d The photomicrograph shows the presence of RBCs and a granular cast. The presence of RBCs in the urine indicates damage to the kidneys (eg, the glomerular membrane) or the genitourinary tract. Granular casts are thought to be the result of degeneration of cellular casts, such as RBC casts, indicating a prolonged disease state (eg, chronic glomerulonephritis).
[QCMLS 2021: 3.1.4.4.1, 3.1.4.5.6] [Strasinger 2021, p180, 196]



165. b The photomicrograph shows the presence of RBCs (red arrows), calcium oxalate (dihydrate) crystals (black arrows), and amorphous debris. The majority of renal calculi are composed of calcium oxalate, and the presence of these crystals in the urine suggests that the patient may have a kidney stone. As the stone passes through the kidney and genitourinary tract, damage can occur, resulting in the hematuria and a positive occult blood result on the reagent test strip.
[QCMLS 2021: 3.1.4.4.1, 3.1.4.6.1] [Strasinger 2021, p203]

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II. Body Fluids

A. Physical

166. **a** CSF is a clear, colorless fluid. Specimens with a significantly increased amount of protein may appear cloudy, but the presence of lipids causes the sample to appear milky. When bacteria are present, the sample will likely have an increased number of WBCs, which would cause the sample to appear cloudy, but not milky. Glucose is soluble in CSF and would not contribute to the turbidity of the sample despite being present in elevated concentrations.
[QCMLS 2021: 3.3.4.2] [Strasinger 2021, p254]

167. **c** Normal synovial fluid is viscous due to its high concentration of hyaluronic acid. When expressed from a syringe (string test), it will form a string of 4-6 cm. If the fluid forms small discrete droplets, it is considered to have abnormally low viscosity, which is associated with inflammatory disorders. Hyaluronidase, produced by neutrophils and some bacteria, degrades the hyaluronic acid, leading to decreased viscosity.
[QCMLS 2021: 3.5.4.3] [Strasinger 2021, p296]

168. **a** Serous fluids are collected by inserting a sterile needle into the body cavity and aspirating several milliliters of the effusion. The procedures are named according to the cavity involved: thoracentesis (pleural), pericardiocentesis (pericardial), and paracentesis (peritoneal). The accumulation of fluid in the peritoneal cavity is referred to as "ascites," and it is collected by paracentesis.
[QCMLS 2021: 3.4.3, 3.4.4] [Strasinger 2021, p306, 314]

169. **a** Normal CSF is clear and colorless with viscosity similar to water. Samples that are opalescent appear hazy, which is usually due to the presence of lipids. Fluids that are pale-yellow, yellow, pink, or orange are described as being "xanthochromic." Xanthochromia generally indicates the presence of RBC degradation products from a bleed into the CNS. The color depends on the amount of blood and the length of time since the bleed occurred.
[QCMLS 2021: 3.3.4.1] [Strasinger 2021, p254]

170. **b** Sperm vitality, or viability, is evaluated by preparing a stained smear using eosin-nigrosin stain and observing the slide under oil immersion (100x objective). Dead or damaged sperm do not have an intact plasma membrane and will readily take up the stain, appearing pink or red against the purple background. Viable sperm have an intact membrane and will not take up the stain, appearing bluish-white against the purple background. One hundred sperm are viewed and classified as viable (white) or non-viable (pink/red). Specimens with <58% vitality are considered abnormal and may be associated with decreased infertility.
[QCMLS 2021: 3.6.7.3] [Strasinger 2021, p286]

171. **c** Amniotic fluid is normally colorless, but may also appear pale yellow or yellow later in pregnancy. A dark yellow appearance suggests the presence of bilirubin, which is associated with increased RBC destruction and hemolytic disease of the fetus and newborn (HDFN).
[QCMLS 2021: 3.2.5.1] [Strasinger 2021, p331]

172. **c** Amniotic fluid is normally colorless or pale yellow. Fluids that appear green may contain meconium. Meconium is formed in the fetal intestines from intestinal secretions and ingestion of amniotic fluid. It contains biliverdin which will give amniotic fluid a green coloration. Meconium is not normally excreted *in utero*, and its presence in amniotic fluid indicates fetal distress.
[QCMLS 2021: 3.2.5.1] [Strasinger 2021, p331]

173. **c** During bacterial vaginosis, microbes such as *Gardnerella vaginalis* proliferate, leading to increased production of polyamines. To evaluate for the increased presence of polyamines, an amine (whiff) test is performed on the vaginal secretion by placing a drop of the sample suspension onto a clean slide along with one drop of 10% potassium hydroxide (KOH). When present, polyamines are volatized into trimethylamines, producing a "fishy" odor.
[QCMLS 2021: 6.7.9] [Strasinger 2021, p361]

174. **c** Semen from a healthy adult should coagulate after ejaculation then liquefy within 30 minutes. Once the fluid is completely liquefied, the viscosity of the sample is evaluated by aspirating it into a Pasteur pipette and allowing it to fall under the influence of gravity. Normal seminal fluid is watery and forms discrete droplets. An abnormal viscosity is indicated by the formation of a string or thread >2 cm in length, and may be associated with the presence of anti-sperm antibodies.
[QCMLS 2021: 3.6.5.2, 3.6.5.4] [Strasinger 2021, p281]

175. **a** Normal semen coagulates shortly after ejaculation to form a coagulum. The coagulum should liquefy within 30 minutes of collection. Specimens that do not liquefy by 60 minutes are reported to have liquefaction time >60 minutes and are considered abnormal.
[QCMLS 2021: 3.6.5.2] [Strasinger 2021, p280]

B. Chemical

176. **a** CSF contains approximately 15-45 mg/dL of proteins. Similar to serum, albumin accounts for the majority of total proteins in CSF. Transthyretin is the second most prevalent CSF protein, with haptoglobin, ceruloplasmin, transferrin, tau transferrin, and immunoglobins comprising the remaining portion. Fibrinogen is not normally found in CSF and indicates contamination with peripheral blood when present.
[QCMLS 2021: 3.3.5.2] [Strasinger 2021, p266]

177. **d** To evaluate if a specimen is synovial fluid, the Ropes or mucin clot test can be performed by adding a solution of 2-5% acetic acid to an aliquot of the sample. The acid polymerizes the hyaluronic acid normally found in synovial fluid and forms a solid clot.
[QCMLS 2021: 3.5.5.1] [Strasinger 2021, p296]

178. **b** Urine and CSF can be ruled out based on the sample's protein concentration as these fluids would have concentrations in the mg/dL range, not g/dL. The sample is likely to be synovial fluid based on the presence of hyaluronate, because this is a component of synovial fluid, not pleural fluid.
[QCMLS 2021: 3.5.4.3] [Strasinger 2021, p295]

179. **a** Synovial fluid contains a large amount of hyaluronic acid. To evaluate if a specimen is synovial fluid, a few drops of 2-5% acetic acid can be added to an aliquot of the sample. The acid will polymerize the hyaluronic acid in synovial fluid, causing it to become turbid and form a solid clot.
[QCMLS 2021: 3.5.5.1] [Strasinger 2021, p295]

180. **c** Hyaluronic acid, also known as *hyaluronate*, is the principal mucin in synovial fluid and serves to lubricate the joint. A low hyaluronic acid concentration leads to decreased viscosity of the synovial fluid, and inflammation of the joint.
[QCMLS 2021: 3.5.4.3] [Strasinger 2021, p294]

181. **a** In guaiac-based fecal occult blood tests, guaiac (colorless chromogen) is oxidized in the presence of a peroxide (reagent) and a peroxidase to yield a blue color. False positives may occur if the patient ingests foods that produce peroxidase (eg, horseradish, broccoli, radishes, melons) or foods that contain animal blood, since the reaction is not specific for human hemoglobin. Patients that take aspirin frequently may have occult bleeding not associated with colorectal cancer, but acetaminophen is not known to cause bleeding. Ascorbic acid (vitamin C) is a strong reducing agent and, at high doses, can interfere with the test, leading to a false-negative result.
[QCMLS 2021: 3.7.4.1] [Strasinger 2021, p345]

182. **a** Fecal occult blood testing may be performed to screen for gastrointestinal bleeding and is commonly used to screen for colorectal cancer. Fecal occult blood tests employ several methods, including those that are guaiac-based (gFOBT), immunochemical (iFOBT or FIT), and porphyrin-based. In guaiac-based methods, guaiac is oxidized in the presence of a peroxide (reagent) and a peroxidase to yield a blue color. The heme moiety of hemoglobin has pseudoperoxidase activity and will result in a positive occult blood reaction.
[QCMLS 2021: 3.7.4.1] [Strasinger 2021, p345]

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183. **d** Normal CSF contains protein fractions similar to those found in serum, such as transthyretin (prealbumin), albumin, transferrin, haptoglobin, ceruloplasmin, IgG, and IgA. Normal CSF does not contain IgM, fibrinogen, or beta lipoprotein.
[QCMLS 2021: 3.3.5.2] [Strasinger 2021, p266]

184. **b** The IgG index is a comparison of the CSF IgG:serum IgG ratio to the CSF albumin:serum albumin ratio. It is used to determine if increased IgG in CSF is due to increased production in the CNS or contamination from a breach in the blood-brain barrier. An IgG index >0.70 suggests increased intrathecal production of immunoglobulins; whereas values <0.70 indicate damage to the blood-brain barrier.
[QCMLS 2021: 3.3.5.2] [Strasinger 2021, p267]

185. **a** Seminal fluid contains a high concentration of acid phosphatase. Measurement of the acid phosphatase activity of a fluid can allow for positive determination of the presence of seminal fluid.
[QCMLS 2021: 3.6.6.4] [Strasinger 2021, p287]

186. **a** Sweat is collected by stimulating an area of the skin with the cholinergic drug, pilocarpine nitrate. In a process known as iontophoresis, a small electric current is used to deliver the pilocarpine into the sweat glands. The sweat is collected into gauze or a plastic microbore tube and measured to determine the concentration of chloride. Abnormal sweat chloride concentrations are associated with cystic fibrosis.
[QCMLS 2021: 3.6.2] [Tietz 2019, p424]

187. **b** There are several types of meningitis including bacterial, tubercular, and fungal. Chemical and microscopic analysis of CSF can assist in differentiating them. CSF glucose concentrations are decreased in bacterial meningitis due to glycolysis by bacteria and WBCs (pleocytosis). CSF lactate concentrations are significantly elevated due to anaerobic glycolysis. If the blood-brain barrier were damaged, CSF albumin and total protein concentrations would be elevated due to inappropriate passage of plasma proteins into the CSF.
[QCMLS 2021: 3.3.5.1] [Strasinger 2021, p268]

188. **a** For patients with normal opening pressure, approximately 1-2 mL of CSF is collected into each of 3-4 sterile tubes. The tubes are labeled in the order of collection (1, 2, 3, 4) and sent to the laboratory for analysis. In general, tube 1 is used for chemical and serological testing as this tube is the most likely to contain peripheral blood or be contaminated from the collection itself. Tube 2 (or 4) is used for microbiological testing and tube 3 is used for cellular studies.
[QCMLS 2021: 3.3.3.1] [Strasinger 2021, p253]

189. **b** Demyelination of the myelin sheath, as occurs in multiple sclerosis (MS), results in the presence of myelin basic proteins (MBP) in the CSF. Measurement of MBP can be used to monitor disease progression and the effectiveness of treatment for patients with MS.
[QCMLS 2021: 3.3.5.2] [Strasinger 2021, p267]

190. **a** Pleural effusions are analyzed to determine their concentration of total proteins and lactate dehydrogenase activity, which is compared to their serum concentrations to classify the fluid as a transudate or exudate. Pleural exudates often have additional chemical testing, including analyses to determine their pH, glucose and triglyceride concentrations, and levels of adenosine deaminase and amylase activity. An abnormally low pH (<7.3) aids in identification of situations that require more aggressive treatments, such as placement of a chest-tube drain, in addition to administration of antibiotics in cases of pneumonia. Significantly low pH value (6.0) suggests an esophageal rupture has occurred, allowing the influx of gastric fluid.
[QCMLS 2021: 3.4.6.5] [Strasinger 2021, p307, 312]

191. **a** When collecting amniotic fluid by amniocentesis, there is a small chance the fluid collected may be urine, given the close proximity of the uterus and maternal bladder. Urine and amniotic fluid both appear as clear, colorless or yellow fluids, making it impossible to distinguish them based on physical examination alone. Chemical examinations, such as glucose, protein, urea and creatinine, may be performed to identify the fluid type. Amniotic fluid contains significant concentrations of glucose and protein, whereas maternal urine, in the absence of renal disease and diabetes mellitus, should be negative for these analytes. In contrast, urine contains significant concentrations of urea and creatinine that do not appear in amniotic fluid until late in pregnancy (37 weeks), when fetal renal function begins.
[QCMLS 2021: 3.2.6.1] [Strasinger 2021, p329]

192. **a** Lecithin (L) is a component of surfactants that coat pulmonary alveoli and provide stability to the alveoli as they expand and contract during respiration. It is produced in low amounts until the 35th week of gestation, when production rapidly increases to stabilize the fetal pulmonary alveoli. Sphingomyelin (S) is a lipid produced in constant amounts after the 26th week of gestation. Evaluation of the ratio of lecithin to sphingomyelin (L/S ratio) in an amniotic fluid provides information on the maturity of the fetal lungs and risk for respiratory distress syndrome (in the event of preterm delivery). An L:S ratio of <2.0 suggests fetal lung immaturity, whereas a ratio ≥2.0 indicates maturity.
[QCMLS 2021: 3.2.6.3] [Strasinger 2021, p333]

193. **a** Measurement of carcinoembryonic antigen (CEA) may be useful in evaluating pleural and peritoneal effusions from patients with previous history of or suspected to have a CEA-producing tumor.
[QCMLS 2021: 3.4.6.6] [Strasinger 2021, p312, 316]

194. **a** Fetal fibronection (fFN) is a glycoprotein found in the extramedullary matrix at the junction of the placenta and uterine wall. In normal pregnancy, fFN should not be detectable after 24-35 weeks' gestation. Increased concentrations (>50 µg/L) are associated with an increased risk for preterm delivery.
[QCMLS 2021: 3.9.1] [Strasinger 2021, p365]

195. **d** Albumin is not synthesized in the CSF, but is transported across the blood-brain barrier. The CSF/serum albumin index is used to evaluate the integrity of the blood-brain barrier, as damage would result in increased CSF albumin. An index <9.0 is considered normal. Indices >9.0 indicate damage has occurred to the blood-brain barrier allowing proteins to inappropriately enter the CSF.
[QCMLS 2021: 3.3.5.2] [Strasinger 2021, p267]

196. **d** Phosphatidylglycerol (PG) is a lipid component of pulmonary surfactants. Surfactants coat the alveoli and provide stability during respiration. PG is undetectable in amniotic fluid until the 35th week of gestation. Performing this test on amniotic fluid collected prior to 35 weeks is inappropriate and should be considered invalid.
[QCMLS 2021: 3.2.6.3] [Strasinger 2021, p333]

197. **a** A paired, fasting plasma specimen must be collected at the same time the arthrocentesis is performed in order to evaluate the difference in glucose concentrations between the fluids. In noninflammatory and hemorrhagic conditions, the difference is typically <20 mg/dL. Differences >20 mg/dL suggest an inflammatory condition, and those >40 mg/dL indicate sepsis.
[QCMLS 2021: 3.5.5.2] [Strasinger 2021, p300]

198. **b** The presence of "fresh" blood in the stool of a newborn must be evaluated to determine if the blood is from a gastrointestinal bleed in the newborn or if maternal blood was swallowed during the delivery. In the Apt test, a suspension is made from the stool and centrifuged, resulting in a pink supernatant that is transferred to 2 tubes. The second tube is alkalinized with sodium hydroxide and any resulting color change evaluated. Maternal hemoglobin (HbA) turns yellow or brown, while fetal hemoglobin (HbF) remains pink.
[QCMLS 2021: 3.7.4.2] [Brunzel 2018, p333]

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199. **c** For adult patients, a normal fasting plasma/serum glucose would be in the range of 74-100 mg/dL. Normally, fasting whole-blood glucose concentrations are approximately 10-12% lower than serum/plasma glucose concentrations. Glucose is actively transported across the blood-brain barrier, and CSF glucose concentrations are approximately 60-70% that of serum/plasma.
[QCMLS 2021: 3.3.5.1] [Strasinger 2021, p268;Tietz 2019, p384-386]

200. **b** Since glucose is actively transported across the blood-brain barrier, measurement of paired CSF and blood samples is performed. CSF glucose concentrations are normally approximately 60-70% that of the blood. If the blood glucose is reported as 80 mg/dL (4.4 mmol/L), the CSF glucose concentration would be expected to be approximately 48-56 mg/dL (2.6-3.1 mmol/L).
[QCMLS 2021: 3.3.5.1] [Strasinger 2021, p268]

201. **c** CSF testing should be performed immediately. If testing will be delayed, the tube(s) to be used for chemical and serological testing should be stored in the freezer to minimize glycolysis by any cellular elements present in the sample. The tube(s) for hematology should be refrigerated (up to 4 hours), and the tube(s) for microbiology should be kept at room temperature.
[QCMLS 2021: 3.3.3.2] [Strasinger 2021, 254]

C. Microscopic

202. **c** CSF is collected by inserting a sterile needle between the 3rd and 4th, or 4th and 5th, lumbar vertebrae and into the subarachnoid space. The first tube collected may be contaminated with peripheral blood and should not be used for cellular studies. The cell count should be performed on the last tube collected, because it will be the least likely to have been contaminated by peripheral blood. Cell counts are performed on a well-mixed sample; whereas a WBC differential would be performed on a stained, cytocentrifuged slide preparation. The addition of glacial acetic acid would lyse RBCs present in the sample.
[QCMLS 2021: 3.3.3.1] [Strasinger 2021, p254]

203. **c** Sperm motility is evaluated microscopically by preparing a wet mount and observing the slide under high dry power (40x objective). To evaluate sperm morphology, a prepared slide is stained using Giemsa, Wright, or Papanicolaou stain and observed using oil immersion (100x objective). Sperm vitality can be evaluated by preparing a stained smear using eosin-nigrosin stain and observing the slide using oil immersion. The concentration of sperm can be determined by performing a manual count on a well-mixed, diluted specimen loaded onto a cover-slipped Neubauer hemacytometer.
[QCMLS 2021: 3.6.7.2] [Strasinger 2021, p284]

204. **b** Monosodium urate (MSU) and calcium pyrophosphate dihydrate (CPPD) crystals are the most common crystals seen in synovial fluid; however, hydroxyapatite, cholesterol, corticosteroid and calcium oxalate crystals are also found. Ammonium biurate crystals are most often see in urine, not synovial fluid. MSU crystals appear as fine sharp needles with negative birefringence. CPPD crystals are rhomboid or square crystals with positive birefringence. Calcium oxalate crystals are octahedrals and demonstrate negative birefringence.
[QCMLS 2021: 3.5.5.3] [Strasinger 2021, p299]

205. **d** Using a Neubauer hemacytometer, the concentration of sperm in a seminal fluid can be determined by performing a manual count on a well-mixed, diluted aliquot of the sample. The same number of quadrants are counted on both sides of the hemacytometer, and the values should match within the laboratory's predefined criteria. The values are then averaged and used to calculate the sperm concentration (sperm/µL) using the formula: (cell average × dilution factor) divided by (number of quadrants counted × quadrant volume [µL]). Reference intervals for sperm concentration are most commonly reported in milliliters. To convert the calculated value from microliters to milliliters, the value is multiplied by 1,000. Alternately, if a standard dilution of 1:20 is performed and 5 RBC quadrants are counted in the large center square, the average number of sperm counted can be multiplied by 1,000,000 to yield the concentration per milliliter directly.
[QCMLS 2021: 3.6.7.5] [Strasinger 2021, p207-208]

206. **b** When the axes of monosodium urate crystals are aligned (parallel to) the polarizer, they appear yellow. When their axes are perpendicular to the polarizer, they appear blue.
[QCMLS 2021: 3.5.6.3] [Strasinger 2021, p299]

207. **d** Blood from a pathologic cause (eg, cerebral hemorrhage) would yield the same appearance in all of the tubes collected; whereas, a traumatic collection would affect the first tube collected more than subsequent ones. Since the CSF in tube 1 appears light pink and the fluid in tube 3 is colorless, it is likely tube 1 contains peripheral blood. This is also supported by the significantly different cell counts.
[QCMLS 2021: 3.3.4.1] [Strasinger 2021, p256]

208. **d** Blood from a traumatic lumbar puncture would affect the first tube collected more than subsequent ones, while blood from a pathological cause (eg, subarachnoid hemorrhage) would yield the same appearance in all of the tubes collected. Since all three tubes appear light pink and hazy, it is likely the patient has a subarachnoid hemorrhage.
[QCMLS 2021: 3.3.4.1] [Strasinger 2021, p256]

209. **a** The etched grid on an "improved" Neubauer hemacytometer delineates 9 quadrants in a 3×3 grid, each measuring 1 mm by 1 mm for an area of 1 mm^2. When a glass coverslip is seated on the ridges of the hemacytometer, a chamber with fixed depth of 0.1 mm is created. Each quadrant measures 1 × 1 × 0.1 mm for a volume of 0.1 mm^3 or 0.1 µL. The entire chamber measures 3 × 3 × 0.1 mm for a volume of 0.9 mm^3 or 0.9 µL.
[QCMLS 2021: 3.3.6.1] [Brunzel 2018, p355]

210. **c** Sperm motility is evaluated by preparing a wet mount and viewing the slide under high dry power (40x objective) within 1 hour of collection. Two hundred sperm are viewed and classified based on their speed and direction of movement. The World Health Organization (WHO) established a grading scale of 0 to 4, with 4 indicating rapid, straight-line movement and 0 indicating no movement. At least 50% of the evaluated sperm should have a rating of 2 or more. A simpler system for grading motility has also been established by the WHO and includes classifying sperm as having: progressive motility, nonprogressive motility, or no motility. At least 50% of the evaluated sperm should demonstrate motility (progressive or nonprogressive) to be considered normal.
[QCMLS 2021: 3.6.7.1] [Strasinger 2021, p282-283]

211. **a** After synovial fluid is collected using a sterile needle and syringe, it is transferred to appropriately labeled collection tubes based on the testing to be performed. In general, the first portion is placed into a plain red-top tube for chemical and immunologic testing. The second portion is placed into a tube with liquid EDTA or sodium heparin for microscopic studies, including a cell count and crystal identification. The final portion is placed into a sterile red-top tube or sterile tube with sodium heparin or sodium polyanethole sulfonate for microbiological studies.
[QCMLS 2021: 3.5.3] [Strasinger 2021, p295]

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212. **a** Sperm motility and agglutination are evaluated using a wet preparation and observing the slide under high dry power (40x objective). A slide stained with Giemsa, Wright, or Papanicolaou stain is used to evaluate sperm morphology under oil immersion (100x objective). Sperm vitality is evaluated on a stained smear prepared with eosin-nigrosin stain, using the 100x objective and oil immersion.
[QCMLS 2021: 3.6.7.4] [Brunzel 2018, p296]

213. **d** Monosodium urate (MSU) crystals and calcium pyrophosphate dihydrate (CPPD) crystals are found in synovial fluid samples from patients with gout and pseudogout, respectively. MSU crystals appear as fine sharp needles with strong negative birefringence, whereas CPPD crystals are rhomboid-shaped crystals with weak positive birefringence.
[QCMLS 2021: 3.5.6.3] [Strasinger 2021, p299]

214. **d** During routine semen analysis, the morphology of 200 sperm are evaluated using oil immersion and the 100x objective (1000x magnification). Normal sperm have 3 distinct components including a head, midpiece, and tail. Abnormalities may occur in 1 or more of these components, leading to poor ovum penetration (head abnormalities) and/or decreased motility (abnormalities of the midpiece and/or tail). Specimens with <14% of sperm demonstrating normal morphology may be associated with infertility.
[QCMLS 2021: 3.6.7.2] [Strasinger 2021, p283-284]

215. **d** After the cell count is performed for each side/chamber of the hemacytometer, the values are compared and should match within the laboratory's predefined acceptance criteria (generally 20% or less). If the counts have a difference greater than the acceptable limit, the hemacytometer should be cleaned with 70% alcohol, dried, and reloaded with a well-mixed specimen for a new count to be performed.
[QCMLS 2021: 3.3.6.1] [Brunzel 2018, p353]

216. **b** Steatorrhea is characterized by the excretion of >7 g of fecal fat per day, which is associated with disorders of malabsorption or maldigestion. Chemical and microscopic analyses may be performed on a stool sample to aid in differentiating between these 2 types of conditions. To qualitatively evaluate the presence of fecal fats (neutral fats, fatty acid salts, fatty acids, and cholesterol), 2 slides are viewed microscopically. One slide is pretreated with ethanol and the second slide is pretreated with acetic acid. Both slides are then stained with Sudan III, Sudan IV, or oil red O.
[QCMLS 2021: 3.7.5.1] [Strasinger 2021, p342, 344]

217. **c** Gout is associated with increased purine metabolism, which leads to elevated blood and synovial concentrations of uric acid. Uric acid precipitates in the joint(s) forming monosodium urate (MSU) crystals which causes the pain and swelling associated with gout.
[QCMLS 2021: 3.5.7.1] [Strasinger 2021, 297]

218. **a** Pseudogout is most often associated with degenerative arthritis and endocrine disorders that produce elevated serum calcium levels. Calcium pyrophosphate dihydrate (CPPD) crystals form in the synovial fluid and are associated with pseudogout.
[QCMLS 2021: 3.5.7.2] [Strasinger 2021, 297]

D. Physiology

219. **a** Tau transferrin is found only in CSF. When present in nasal or middle ear fluids, it confirms the presence of CSF, which aids in the diagnosis of CSF rhinorrhea and otorrhea, respectively.
[QCMLS 2021: 3.3.5.2] [Strasinger 2021, p266]

220. **c** Pulmonary surfactant is packaged into granules for storage starting at the 24th week of gestation. The granules enter amniotic fluid during the 26th week. Quantitative measurement of these granules, known as lamellar bodies, in amniotic fluid is used in the evaluation of fetal lung maturity and risk for respiratory distress syndrome associated with preterm delivery. The diameter of lamellar bodies is similar to that of small platelets; therefore, the lamellar body count (LBC) can be performed using the platelet channel of an automated hematology analyzer.
[QCMLS 2021: 3.2.6.3] [Strasinger 2021, p334]

221. **c** Bilirubin is a degradation product of hemoglobin. It is conjugated in the liver and converted into urobilinogen and stercobilinogen by bacteria in the intestines. Intestinal oxidation of stercobilinogen forms urobilin, which gives feces its normal orange-brown coloration.
[QCMLS 2021: 3.7.3.1] [Strasinger 2021, p343]

222. **b** Seminal fluid is gray-white, translucent, and should liquefy within 30 minutes. Complete collections yield a volume of 2-5 mL and fresh semen is slightly alkaline (7.2-7.8). Volumes <2 mL or >5 mL have been associated with infertility.
[QCMLS 2021: 3.2.5.3] [Strasinger 2021, p280]

223. **c** Albumin is not synthesized in the CSF, but is transported across the blood-brain barrier. The CSF/serum albumin index is used to evaluate the integrity of the blood-brain barrier, because damage would result in increased CSF albumin. The IgG index compares CSF/serum IgG to CSF/serum albumin to determine if high IgG concentrations are due to increased intrathecal damage to the blood-brain barrier. Total protein determinations are performed on CSF to evaluate the blood-brain barrier, but are not reported as a ratio. The fluid/serum lactate dehydrogenase ratio is used to determine if an effusion is a transudate or an exudate.
[QCMLS 2021: 3.3.5.2] [Strasinger 2021, p266, 307]

224. **b** Effusions are classified as transudates or exudates based on their physical, chemical, and cellular characteristics; which assists the clinician to identify and treat the underlying cause. Transudates are generally clear and pale yellow, while exudates appear cloudy and yellow, green, pink, etc. A paired blood sample is collected to allow for comparison of analyte concentrations (glucose, proteins, lactate dehydrogenase [LD]) between the fluid and serum. Transudates have a fluid/serum total protein ratio <0.5, fluid/serum LD ratio <0.6, and fluid glucose >60 mg/dL; while exudates have a fluid/serum total protein ratio >0.5, fluid/serum LD ratio >0.6, and fluid glucose <60 mg/dL. Additionally, exudates generally have higher WBC counts (>1000/µL). Chylous and pseudochylous effusions appear milky due to the presence of lipids such as cholesterol and/or triglycerides.
[QCMLS 2021: 3.4.5, 3.4.6.1, 3.4.6.2, 3.4.7.1, t3.6] [Strasinger 2021, p307-308]

225. **c** After a successful vasectomy, there should be no sperm present in the ejaculate (azoospermia). Occasionally a few sperm are present following vasectomy, but they should be nonmotile. The presence of a single motile spermatozoon indicates the vasectomy was unsuccessful.
[Strasinger 2021, p287]

226. **d** Amniotic fluid may be collected early in gestation (after 14 weeks' gestation) if the fetus is suspected of having a neural tube defect or genetic defect. If the fluid is collected to evaluate fetal lung maturity and risk for respiratory distress associated with preterm delivery, amniocentesis should not be performed before 20 weeks' gestation as all results will show lung immaturity.
[QCMLS 2021: 3.2.6.3] [Strasinger 2021, p329]

227. **d** When performing a lumbar puncture, if the opening pressure is low, only 1-2 mL of fluid will be collected. With low-volume specimens, testing must be prioritized to ensure a sterile specimen is used for microbiological testing. The ordering physician should be consulted to prioritize testing, but, in general, the Gram stain and culture will be performed first, followed by the cell count, and then chemical and immunological testing.
[QCMLS 2021: 6.7.6.3] [Brunzel 2018, p246]

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228. **a** Effusions with a milky appearance that persists after centrifugation may contain chyle. To differentiate if the specimen is chylous or pseudochylous, lipoprotein analysis is performed. Chylous specimens demonstrate an elevated triglyceride content and the presence of chylomicrons; whereas, pseudochylous specimens have low triglyceride content and chylomicrons are not present.
[QCMLS 2021: 3.4.6.2, 3.4.6.3, t3.7] [Strasinger 2021, p308, 314]

229. **c** ΔA450 determinations are performed on amniotic fluid to measure the concentration of bilirubin in the sample. Bilirubin is light labile, and samples should be protected from light. Amber-colored containers or aluminum foil should be used in the collection of amniotic fluid to prevent photo-oxidation of bilirubin. If the specimen is not immediately protected from light after collection and throughout transport, the loss of bilirubin may cause falsely low results.
[QCMLS 2021: 3.2.4.2, 3.2.6.2] [Strasinger, p330-331]

230. **b** CSF is normally clear and colorless. Fluids with a pink, yellow, or orange appearance are referred to as "xanthochromic." Xanthochromia generally indicates the presence of RBC degradation products from a bleed into the CNS. The color depends on the amount of blood and the length of time since the bleed occurred. A pink supernatant after centrifugation indicates the presence of hemoglobin. An orange color is associated with heavy hemolysis, and the presence of bilirubin will cause a yellow coloration.
[QCMLS 2021: 3.3.4.1] [Strasinger 2021, p256]

231. **d** CSF normally appears clear and colorless. Specimens that contain radiographic contrast media may appear oily; whereas the presence of WBCs, lipids, and proteins may cause the sample to appear hazy, cloudy, turbid, or milky.
[QCMLS 2021: 3.3.4.2] [Strasinger 2021, p256]

E. Disease States

232. **c** Pleural effusions are classified as transudates or exudates based on their physical, chemical, and cellular characteristics. Transudates are clear, pale yellow fluids that occur as a result of a noninflammatory process. Exudates are cloudy fluids, associated with an inflammatory process, that appear yellow, green, pink, etc. Collection of a paired blood sample allows for comparison of analyte concentrations between the fluid and serum. Transudates have a fluid/serum total protein ratio <0.5, fluid/serum lactate dehydrogenase (LD) ratio <0.6, and difference between fluid-serum glucose <30 mg/dL; whereas exudates have a fluid/serum total protein ratio >0.5, fluid/serum LD ratio >0.6, and fluid-serum glucose difference >30 mg/dL. Transudates generally have lower WBC counts (<1000/μL) than exudates.
[QCMLS 2021: 3.4.5, 3.4.6.1, 3.4.6.2, 3.4.7.1, t3.8] [Strasinger 2021, p307-308]

233. **b** Transudates occur as a result of a noninflammatory process associated with an increase in hydrostatic pressure or a decrease in oncotic pressure. Congestive heart failure can lead to a transudative effusion as a result of poor cardiac function. Transudates are clear, pale yellow fluids that have a fluid/serum total protein ratio <0.5, fluid/serum lactate dehydrogenase ratio <0.6, fluid glucose <60 mg/dL, and WBC count <1000/μL.
[QCMLS 2021: 3.4.3, 3.4.5, 3.4.6.1, 3.4.6.2, 3.4.7.1, t3.8] [Strasinger 2021, p307-308]

234. **a** A small amount of serous fluid normally fills the cavity between the cavity wall (the parietal membrane) and the organ (visceral membrane). The fluid serves as a cushion, lubricant, and transport media between the visceral and parietal membranes. An increase in the fluid, referred to as an "effusion," can be due to infection, inflammation, cancer, and defects in hydrostatic and/or colloidal pressure. Effusions are classified as transudates or exudates based on their physical, chemical, and cellular characteristics.
[QCMLS 2021: 3.4.3] [Strasinger 2021, p306]

235. **a** Serous fluid is collected by inserting a sterile needle into the body cavity and aspirating several milliliters of the effusion. The procedures are named according to the cavity involved: thoracentesis (pleural), pericardiocentesis (pericardial), and paracentesis (peritoneal). Collection by thoracentesis indicates the fluid is a pleural effusion. To classify the effusion as a transudate or exudate, evaluations of the fluid's physical, chemical, and cellular characteristics are performed along with chemical analysis of a paired blood sample. Fluids are classified as an exudate if their fluid/serum lactate dehydrogenase ratio is >0.6.
[QCMLS 2021: 3.4.4, 3.4.6.1, t3.8] [Strasinger 2021, p306-307]

236. **a** Lamellar bodies are produced by type II fetal pneumocytes around the 24th week of gestation and secreted into the amniotic fluid near the 26th week. Quantitative measurement of lamellar bodies in amniotic fluid is used in the evaluation of fetal lung maturity and risk for respiratory distress syndrome associated with preterm delivery.
[QCMLS 2021: 3.2.6.3] [Strasinger 2021, p334]

237. **b** Amniocentesis is an invasive procedure and should only be performed when absolutely necessary. Gestational age is generally determined based on the patient's last menstrual period and does not routinely require amniocentesis. An amniocentesis may be performed for an Rh-negative mother if hemolytic disease of the newborn is suspected (ΔA450 determination); however, this would be inappropriate for an Rh-positive mother. Measurement of folic acid concentrations in maternal serum would be more appropriate. In women with a high maternal serum alpha-fetoprotein (AFP), amniocentesis is used to collect fluid to detect levels of AFP and acetylcholinesterase. High levels are predictive of neural tube disorders, such as spina bifida and anencephaly.
[QCMLS 2021: 3.2.6.4] [Strasinger 2021, p332]

238. **d** Cystic fibrosis is an autosomal recessive disease affecting ion channels that results in abnormal electrolyte and mucous secretions. Patients with cystic fibrosis have elevated sweat chloride concentrations (≥60 mmol/L) and abnormally viscous secretions throughout the body.
[QCMLS 2021: 3.8.3] [Tietz 2019, p425]

239. **c** Cystic fibrosis is most commonly caused by a deletion of phenylalanine at position 508 (ΔF508) of the cystic fibrosis transmembrane conductance regulator (CFTR) protein. Trisomy 21 describes the presence of a third copy of chromosome 21, which is associated with Down syndrome. The Philadelphia chromosome is a gene translocation associated with acute myelocytic leukemia. Fragile X is caused by an increased number of nucleotide repeats leads to a form of intellectual disability.
[QCMLS 2021: 3.8.3] [Tietz 2019, p298]

240. **d** Increased intrathecal synthesis of IgG in the CNS is associated with neurologic disorders such as multiple sclerosis. CSF protein electrophoresis may be performed to evaluate the presence of oligoclonal bands in the gamma region, which indicate an increased amount of immunoglobulins such as IgG. To confirm the oligoclonal bands are the result of neurological disease rather than leakage of blood into the CNS, a serum protein electrophoresis should be performed at the same time. The presence of 2 or more oligoclonal bands in the CSF pattern that are not present in the serum pattern indicate a neurological disorder such as MS.
[QCMLS 2021: 3.3.5.2] [Strasinger 2021, p267]

241. **c** CSF from an adult patient contains 0-5 WBC/μL (primarily lymphocytes and monocytes). In general, patients with bacterial meningitis will have a high WBC count with a predominance of neutrophils; whereas lymphocytes and monocytes predominate in viral meningitis. Macrophages appear in the CSF when they are needed to remove cellular debris, such as degrading RBCs, indicating a previous hemorrhage.
[QCMLS 2021: 3.3.6.2] [Strasinger 2021, p257-263]

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242. **a** In bacterial meningitis, CSF glucose will be decreased due to bacterial and cellular glycolysis. Normal CSF glucose is approximately 60-70% of plasma glucose, which is 70-100 mg/dL when the patient is in a fasting state. The CSF total protein concentration is normally expected to be 15-45 mg/dL. Patients with bacterial meningitis often have a markedly elevated CSF total protein. Lactate is produced during anaerobic glycolysis and CSF concentrations are significantly elevated (>35 mg/dL) in bacterial meningitis. Sample A has a decreased glucose, markedly elevated total protein, and significantly elevated lactate concentration, which is consistent with bacterial meningitis.
[QCMLS 2021: 3.3.5.1, 3.3.5.2, 3.3.5.3] [Strasinger 2021, p266, 268]

243. **a** Ochronotic shards are small, dark, pepper-like granules that appear in synovial fluid from individuals with alkaptonuria. The granules are pieces of cartilage, pigmented by homogentisic acid, that have broken loose within the joint space.
[QCMLS 2021: 3.5.4.2] [Brunzel 2018, p277]

244. **a** Lactoferrin is a glycoprotein expressed by activated neutrophils. A positive lactoferrin value is associated with an inflammatory disorder such as chronic inflammatory bowel disease. Fecal occult blood testing is largely performed to screen for colorectal cancer but will also yield a positive result in the presence of blood from anywhere in the gastrointestinal tract. Calprotectin is found in neutrophils, monocytes, and macrophages and is qualitatively measured in fecal specimens to differentiate inflammatory bowel disease and inflammatory bowel syndrome. Steatorrhea is associated with malabsorption syndromes and indicates an increased amount of fecal fats.
[QCMLS 2021: 37.5.1] [Strasinger 2021, p342]

245. **a** Patients with cystic fibrosis have elevated sweat chloride concentrations. Sweat chloride determinations are performed by coulometric titration (also known as the Cotlove chloridometer technique, or chloridometry).
[QCMLS 2021: 3.8.2] [Tietz 2019, p425]

246. **d** Rh-negative mothers with an Rh-positive baby may develop anti-Rh antibodies if they are exposed to fetal RBCs. In subsequent pregnancies with an Rh-positive fetus, hemolytic disease of the fetus/newborn (HDFN) may occur, leading to destruction of fetal RBCs. When lysed, RBCs release hemoglobin, which is converted into bilirubin. When HDFN is suspected, a ΔA450 determination may be performed on amniotic fluid to measure the concentration of bilirubin in the sample. In HDFN, the concentration of bilirubin in the amniotic fluid is elevated relative to the severity of the disease.
[QCMLS 2021: 3.2.6.2, 3.2.8.1] [Strasinger 2021, p331]

247. **b** Acetylcholinesterase (AChE) is an enzyme found in the central nervous system, RBCs, skeletal muscle tissue, and fetal serum. When analyses of maternal serum and amniotic fluid yield elevated AFP results, the amniotic fluid should be tested to determine the AChE activity, as it is a more specific indicator for neural tube disorders.
[QCMLS 2021: 3.2.6.5] [Strasinger 2021, p332]

248. **a** CSF lactate concentrations are increased in meningitis and conditions that cause hypoxia in the CNS. Fungal and tubercular meningitis are associated with mildly increased lactate concentrations (>25 mg/dL) and and bacterial meningitis yields significantly elevated (>35 mg/dL) lactate concentrations.
[QCMLS 2021: 3.3.5.3] [Strasinger 2021, p268]

249. **d** Elevated CSF total protein concentrations may be associated with meningitis, cerebral hemorrhage, and neurologic diseases, such as MS, due to inappropriate passage of proteins through the blood-brain barrier or increased intrathecal production of immunoglobulins. A decreased total protein may be the result of CSF leakage, recent lumbar puncture, or dilution caused by rapid CSF production.
[QCMLS 2021: 3.3.5.2] [Strasinger 2021, p266]

250. d Feces is normally orange-brown due to the presence of bile pigments produced from bilirubin in the liver and intestines. In conditions where the biliary duct is partially or completely obstructed, bile pigments are not present and the stool has a pale or clay-colored appearance termed "acholic."
[QCMLS 2021: 3.7.3.1] [Strasinger 2021, p343]

251. b The WBC differential for a normal adult patient would be expected to yield 30-90% lymphocytes, 10-50% monocytes, and 0-6% neutrophils. Eosinophils are rarely present in CSF, but an increased presence is associated with parasitic and fungal infections.
[QCMLS 2021: 3.3.6.2] [Strasinger 2021, p259]

252. d Clue cells are squamous epithelial cells with at least 75% of their surface covered by coccobacilli. This makes the cell appear granular with an irregular border described as "shaggy." Their presence in vaginal secretions indicates bacterial vaginosis caused by *Gardnerella vaginalis*.
[QCMLS 2021: 6.7.9] [Strasinger 2021, p358]

253. c WBC are not normally present in feces. The presence of neutrophils suggests of an inflammatory condition in which the mucosal wall has been compromised, whereas the presence of lymphocytes is associated with noninflammatory conditions.
[QCMLS 2021: 3.7.5.2] [Strasinger 2021, p343]

254. c The WBC differential for a normal adolescent patient would be expected to yield 10-40% lymphocytes, 50-90% monocytes, and 0-10% neutrophils. Neutrophilic pleocytosis is most commonly associated with bacterial meningitis.
[QCMLS 2021: 3.3.6.2] [Strasinger 2021, p259]

255. b Rice bodies are white, free-floating particles that appear in synovial fluid. They are pieces of cartilage covered by fibrinous tissue that resemble polished, shiny grains of rice that vary in size. They may be seen in many arthritic conditions, but they are most commonly observed in association with rheumatoid arthritis.
[QCMLS 2021: 3.5.4.2] [Brunzel 2018, p277]

256. a Feces normally has an orange-brown color. Samples that appear black or tar-like, referred to as "melena," are associated with the presence of blood originating from the upper gastrointestinal tract, but may also be the result of medications or therapies (eg, coal ingestion). An occult blood test may be performed to evaluate if the sample contains blood.
[QCMLS 2021: 3.7.3.1, 3.7.4.1] [Strasinger 2021, p343, 345]

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Chapter 4

Hematology

*The following items have been identified generally as appropriate for those preparing for both the MLS and MLT examinations. Items that are appropriate for the MLS examination **only** are marked with MLS ONLY.*

279 Questions
- 279 I. Physiology
- 284 II. Disease States
- 310 III. Laboratory Testing

335 Answers with Explanations
- 335 I. Physiology
- 338 II. Disease States
- 350 III. Laboratory Testing

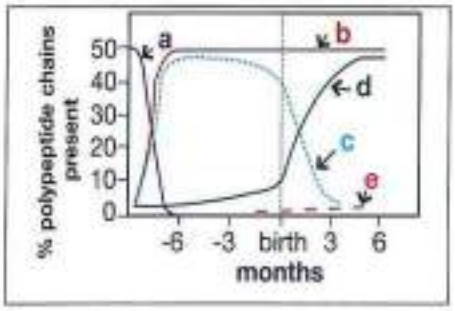
I. Physiology

A. Production

1. The majority of the iron in an adult is found as a constituent of:

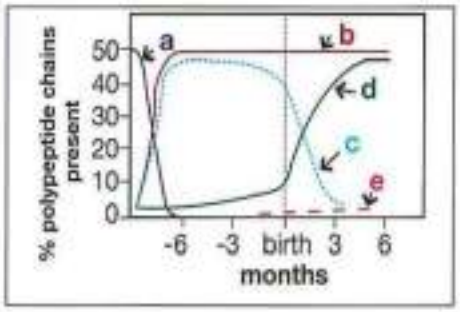
   a hemoglobin
   b hemosiderin
   c myoglobin
   d transferrin

2. Which curve in this figure represents the production of alpha polypeptide chains of hemoglobin?



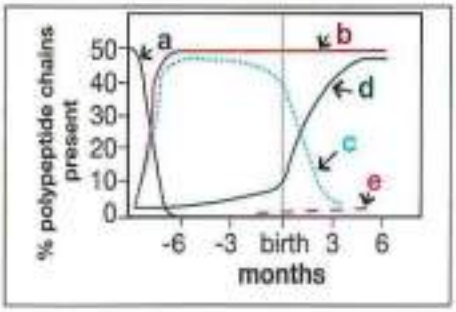
   a curve a
   b curve b
   c curve c
   d curve d

3. Which curve in this figure represents the production of beta polypeptide chains of hemoglobin?



a curve b
b curve c
c curve e
d curve d

4. Which curve in this figure represents the production of gamma polypeptide chains of hemoglobin?:



a curve a
b curve b
c curve c
d curve d

5. In order for hemoglobin to combine reversibly with oxygen, the iron must be:

a complexed with haptoglobin
b freely circulating in the cytoplasm
c attached to transferrin
d in the ferrous state

6. Which description best fits the Donath-Landsteiner antibody?

a IgM cold agglutinin
b biphasic IgM hemolysin
c IgG biphasic hemolysin
d IgG warm agglutinin

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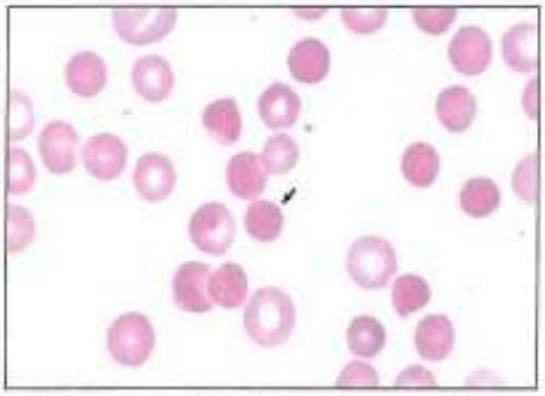
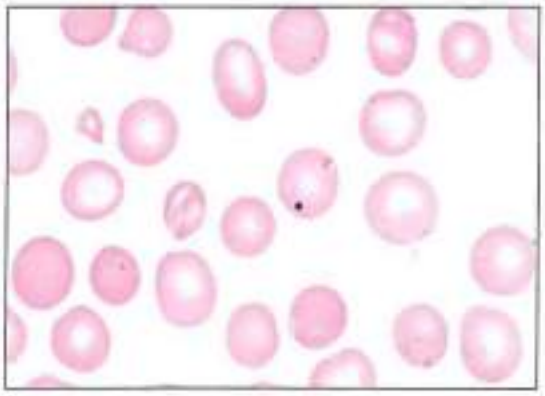
7. The abnormal erythrocyte in the center of this image (arrow) may result from which of the following processes?



a deficiencies of cellular membrane proteins
b absence of plasma lipoproteins
c defects in the cellular lipid bilayer
d deficiencies of cellular enzymes

8. What is the composition of the inclusion seen in this RBC?



a DNA
b RNA
c iron
d denatured hemoglobin

9. Which of the following ions is bound to hemoglobin in methemoglobin?

a Ca^{2+}
b Fe^{3+}
c Fe^{2+}
d Mg^{2+}

10. An increased amount of cytoplasmic basophilia in a blood cell indicates:

a increased cytoplasmic maturation
b decreased cytoplasmic maturation
c reduction in size of the cell
d decreased nuclear maturation

11. Specific (secondary) granules of the neutrophilic granulocyte:

a appear first at the myelocyte stage
b contain esterases
c are formed on the mitochondria
d are derived from azurophil (primary) granules

12. In normal adult bone marrow, the most common granulocyte is the:

a basophil
b myeloblast
c eosinophil
d metamyelocyte

13. Pluripotent hematopoietic stem cells are capable of producing:

a daughter cells of only one cell line
b only T lymphocytes and B lymphocytes
c erythropoietin, thrombopoietin, and leukopoietin
d lymphoid and myeloid stem cells

14. Which of the following cytokines is most responsible for eosinophil differentiation and release from the bone marrow?

a IL-1
b IL-2
c IL-4
d IL-5

15. Auer rods are characterized as:

a fused primary granules
b DNA precipitates
c denatured hemoglobin
d large cytoplasmic granules

16. Which of the following is characteristic of cellular changes as megakaryoblasts mature into megakaryocytes within the bone marrow?

a progressive decrease in overall cell size
b increasing basophilia of cytoplasm
c nuclear division without cytoplasmic maturation
d fusion of the nuclear lobes

17. Which of the following cells is the largest cell in the bone marrow:

a megakaryocyte
b histiocyte
c osteoblast
d mast cell

18. MLS ONLY Which one of the following is a true statement about megakaryocytes in a bone marrow aspirate?

a an average of 5-10 should be found in each low power field (10x)
b the majority of forms are the MK1 stage
c morphology must be determined from the biopsy section
d quantitative estimation is done using the 40x lens

B. Destruction

19. After the removal of red blood cells from the circulation hemoglobin is broken down into:

a iron, porphyrin, and amino acids
b iron, heme, and globin
c heme, protoporphyrin, and amino acids
d heme, hemosiderin, and globin

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C. Function

20. The main function of the hexose monophosphate shunt in the erythrocyte is to:

    a regulate the level of 2,3-DPG
    b provide reduced glutathione to prevent hemoglobin oxidation
    c prevent the reduction of heme iron
    d provide energy for membrane maintenance

21. In the normal adult, the spleen acts as a site for:

    a storage of red blood cells
    b production of red blood cells
    c synthesis of erythropoietin
    d removal of imperfect and aging cells

22. Cells for the transport of O_2 and CO_2 are:

    a erythrocytes
    b granulocytes
    c lymphocytes
    d thrombocytes

23. Erythropoietin acts to:

    a shorten the replication time of the granulocytes
    b stimulate RNA synthesis of erythroid cells
    c increase colony-stimulating factors produced by the B lymphocytes
    d decrease the release of marrow reticulocytes

24. Cells that produce antibodies and are capable of direct cytolysis:

    a erythrocytes
    b granulocytes
    c lymphocytes
    d thrombocytes

25. Phagocytosis is a function of:

    a erythrocytes
    b granulocytes
    c lymphocytes
    d thrombocytes

26. Which cells are involved in immediate hypersensitivity reactions?

    a eosinophils
    b basophils
    c plasma cells
    d reactive lymphocytes

27. A patient is on 100 mg of aspirin/day to prevent the formation of clots caused by platelets. The mechanism in which aspirin impairs platelet function is by:

    a inactivating cyclooxygenase which blocks thromboxane A2
    b impairs vWF by via GPIb/IX/V receptor
    c decreased amounts of arachidonic acid
    d inactivation of ADP and phospholipase A2

28. One of the major glands in an infant primarily responsible for producing lymphocytes is the:

    a thymus
    b adrenal
    c thyroid
    d pituitary

29. When iron in hemoglobin is in the +3 state, it is termed:

a sulfhemoglobin
b methemoglobin
c carboxyhemoglobin
d ferrihemoglobin

30. Phagocytosis in neutrophils can be described as a process to:

a defend against parasites
b mediate sensitivity reactions
c neutralize products from mast cells
d kill and degranulate bacteria

II. Disease States

A. Erythrocytes

31. The characteristic erythrocyte found in pernicious anemia is:

MLS ONLY

a microcytic
b spherocytic
c hypochromic
d macrocytic

32. Hemolysis in paroxysmal nocturnal hemoglobinuria (PNH) is:

MLS ONLY

a temperature-dependent
b complement-independent
c antibody-mediated
d caused by a red cell membrane defect

33. Which of the following is most closely associated with idiopathic hemochromatosis?

MLS ONLY

a iron overload in tissue
b target cells
c Cabot rings
d ringed sideroblasts

34. A patient with polycythemia vera who is treated by phlebotomy is most likely to develop a deficiency of:

a iron
b vitamin B_{12}
c folic acid
d erythropoietin

35. The direct antiglobulin test can help distinguish:

MLS ONLY

a inherited from acquired spherocytosis
b intravascular from extravascular hemolysis
c heterozygous from homozygous thalassemia
d sickle cell trait from sickle cell disease

36. The anemia of chronic inflammation is characterized by:

a decreased iron stores in the reticuloendothelial system
b decreased serum iron levels
c macrocytic erythrocytes
d increased serum iron binding capacity

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37. Factors commonly involved in causing anemia in patients with chronic renal disease include:

MLS ONLY

a marrow hypoplasia
b inadequate erythropoiesis
c vitamin B_{12} deficiency
d increased erythropoietin production

38. A 20-year-old woman with sickle cell anemia, whose usual hemoglobin concentration is 8 g/dL (80 g/L), develops fever, increased weakness and malaise. The hemoglobin concentration is 4 g/dL (40 g/L) and the reticulocyte count is 0.1%. The **most** likely explanation for her clinical picture is:

a increased hemolysis due to hypersplenism
b aplastic crisis
c thrombotic crisis
d occult blood loss

39. The hypoproliferative red cell population in the bone marrow of uremic patients is caused by:

MLS ONLY

a infiltration of bone marrow by toxic waste products
b decreased levels of circulating erythropoietin
c defective globin synthesis
d overcrowding of bone marrow space by increased myeloid precursors

40. Which of the following characteristics are common to hereditary spherocytosis, hereditary elliptocytosis, hereditary stomatocytosis, and paroxysmal nocturnal hemoglobinuria?

a autosomal dominant inheritance
b red cell membrane defects
c positive direct antiglobulin test
d measured platelet count

41. An 89-year-old Caucasian female is transferred to the hospital from a nursing facility for treatment of chronic urinary tract infection with proteinuria. The patient presents with the laboratory results shown in this table:

test	result
WBC	$10.0 \times 10^3/\mu L$ ($10.0 \times 10^9/L$)
RBC	$3.1 \times 10^6/\mu L$ ($3.1 \times 10^{12}/L$)
HGB	7.2 g/dL (72 g/L)
HCT	24%
MCV	78 μm^3 (78 fL)
MCH	23 pg
MCHC	31%
serum iron	29 µg/dL (5.2 µmol/L)
TIBC	160 µg/dL (28.6 µmol/L)
serum ferritin	100 ng/mL (100 µg/L)

These data are most consistent with which of these clinical conditions?

a iron deficiency anemia
b anemia of chronic inflammation
c hemochromatosis
d acute blood loss

42. A patient is admitted with a history of chronic bleeding secondary to peptic ulcer. Hematology results reveal a severe microcytic, hypochromic anemia. Iron studies are requested. Which result set would be expected in this case?

result	serum iron	TIBC	storage iron
a	decreased	increased	increased
b	increased	decreased	increased
c	decreased	increased	decreased
d	increased	normal	decreased

a result a
b result b
c result c
d result d

43. Which of the following is **most** closely associated with iron deficiency anemia?

a iron overload in tissue
b macrocytes
c basophilic stippling
d chronic blood loss

44. Evidence indicates that the genetic defect in thalassemia usually results in:

a the production of abnormal globin chains
b a quantitative deficiency in RNA resulting in decreased globin chain production
c a structural change in the heme portion of the hemoglobin
d an abnormality in the alpha- or beta-chain binding or affinity

45. IALS ONLY A 20-year-old African-American man has peripheral blood changes suggesting thalassemia minor. The quantitative HbA_2 level is normal, but the HbF level is 5% (normal <2%). This is most consistent with:

a alpha thalassemia minor
b beta thalassemia minor
c delta-beta thalassemia minor
d hereditary persistence of fetal hemoglobin

46. IALS ONLY Anemia secondary to uremia and chronic renal disease characteristically is:

a microcytic, hypochromic
b hemolytic
c normocytic, normochromic
d macrocytic

47. Which of the following sets of laboratory findings is consistent with hemolytic anemia?

a decreased bilirubin; normal reticulocyte count
b increased serum lactate dehydrogenase; increased catabolism of heme
c decreased serum lactate dehydrogenase; normal catabolism of heme
d increased concentration of haptoglobin; marked hemoglobinuria

48. Deficiency of this enzyme is associated with a moderate to severe hemolytic anemia after the patient is exposed to certain drugs and characterized by red cell inclusions formed by denatured hemoglobin:

a lactate dehydrogenase
b G6PD
c pyruvate kinase
d hexokinase

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49. Patients with G6PD deficiency are least likely to have hemolytic episodes in which of the following situations?

a following the administration of oxidizing drugs
b following the ingestion of fava beans
c during infections
d spontaneously

50. A patient has a congenital nonspherocytic hemolytic anemia. After exposure to anti-malarial drugs, the patient experiences a severe hemolytic episode. This episode is characterized by red cell inclusions caused by hemoglobin denaturation. Which of the following conditions is **most** consistent with these findings?

a G6PD deficiency
b thalassemia major
c pyruvate kinase deficiency
d paroxysmal nocturnal hemoglobinuria

51. Which of the following is the most characteristic finding in autoimmune hemolytic anemia:

a increased reticulocyte count
b leukopenia and thrombocytopenia
c peripheral spherocytosis
d positive direct antiglobulin test

52. Peripheral blood smears from patients with untreated pernicious anemia are characterized by:

a pancytopenia and macrocytosis
b leukocytosis and elliptocytosis
c leukocytosis and ovalocytosis
d pancytopenia and microcytosis

53. Laboratory tests performed on a patient indicate macrocytosis and pancytopenia. Which of the following disorders is most likely?

a anemia of chronic inflammation
b vitamin B_{12} deficiency
c iron deficiency
d acute hemorrhage

54. A patient has the laboratory results shown in this table:

test	result
RBC	2.35×10^6/µL (2.35×10^{12}/L)
WBC	3.0×10^3/µL (3.0×10^9/L)
PLT	95.0×10^3/µL (95.0×10^9/L)
HGB	9.5 g/dL (95 g/L)
HCT	27%
MCV	115 μm^3 (115 fL)
MCHC	35%
MCH	40 pg

Which of the following tests would contribute toward the diagnosis?

a reticulocyte count
b platelet factor 3
c serum B_{12} and folate
d leukocyte alkaline phosphatase

55. The characteristic morphologic feature in folic acid deficiency is:

a macrocytosis
b target cells
c basophilic stippling
d rouleaux formation

56. A 50-year-old patient is found to have the laboratory results shown in this table:

test	result
HGB	7.0 g/dL (70 g/L)
HCT	20%
RBC	2.0×10^6/µL (2.0×10^{12}/L)

It is determined that the patient is suffering from pernicious anemia. Which of these sets of results is most likely to have been obtained from the same patient?

result	WBCs	platelets	reticulocytes
a	17,500	350,000	5.2%
b	7,500	80,000	4.1%
c	5,000	425,000	2.9%
d	3,500	80,000	0.8%

a result a
b result b
c result c
d result d

57. Megaloblastic asynchronous development in the bone marrow indicates which one of the following?

a proliferation of erythrocyte precursors
b impaired synthesis of DNA
c inadequate production of erythropoietin
d deficiency of G6PD

58. Which of the following are found in association with megaloblastic anemia?

a neutropenia and thrombocytopenia
b decreased LD activity
c increased erythrocyte folate levels
d decreased plasma bilirubin levels

59. Which of the results represents characteristic features of iron metabolism in patients with anemia of chronic inflammation?

result	serum iron	transferrin saturation	TIBC
a	normal	normal	normal
b	increased	increased	normal or slightly increased
c	normal	markedly increased	normal
d	decreased	decreased	normal or decreased

a result a
b result b
c result c
d result d

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60. A characteristic morphologic feature in hemoglobin C disease is:

a macrocytosis
b spherocytosis
c rouleaux formation
d target cells

61. Thalassemias are characterized by:

a structural abnormalities in the hemoglobin molecule
b absence of iron in hemoglobin
c decreased rate of heme synthesis
d decreased rate of globin synthesis

62. A patient has the laboratory results shown in this table:

test	result
RBC	$6.5 \times 10^6/\mu L$ ($6.5 \times 10^{12}/L$)
HGB	13.0 g/dL (130 g/L)
HCT	39.0%
MCV	65 μm^3 (65 fL)
MCH	21.5 pg
MCHC	33%

These results are compatible with:

a iron deficiency
b pregnancy
c thalassemia minor
d beta thalassemia major

63. Laboratory findings in hereditary spherocytosis include:

a decreased WBCs
b decreased RBC band 3 protein
c reticulocytopenia
d positive direct antiglobulin test

64. Which of the following types of polycythemia is a severely burned patient **most** likely to have?

a polycythemia vera
b polycythemia, secondary to hypoxia
c relative polycythemia associated with dehydration
d polycythemia associated with renal disease

65. The characteristic morphologic feature in lead poisoning is:

a macrocytosis
b target cells (codocytes)
c basophilic stippling
d rouleaux formation

66. The white cell feature most characteristic of pernicious anemia is:

a eosinophilia
b toxic granulation
c hypersegmentation
d reactive lymphocytes

67. Which parameter is most consistently abnormal in cases of hereditary spherocytosis?

a RBC count
b MCV
c Hemoglobin
d MCHC

68. What protein is commonly defective in hereditary elliptocytosis?

a ankyrin
b spectrin
c band 4.1
d elliptocin

69. What is the most common mechanism resulting in hereditary stomatocytosis?

a abnormal Na/K permeability
b deficient cytoskeletal structural proteins
c inability to repair oxidative stress damage
d ATP depletion due to glycolytic enzyme deficiency

70. The basic mechanism associated with the development of sideroblastic anemia is:

a enzymatic defect in heme synthesis causes iron accumulation
b quantitative decrease in the production of globin chains
c defective iron utilization
d ineffective erythropoietin production decreases RBC response

71. Individuals with Fanconi anemia characteristically show:

MLS ONLY

a increased HbF
b intravascular hemolysis
c ringed sideroblasts
d thrombocytosis

72. Which of the following tumors are associated with erythrocytosis due to excessive erythropoietin production?

MLS ONLY

a renal cell carcinoma
b sarcoma
c basal cell carcinoma
d squamous cell carcinoma of the lung

73. Which of the following features of G6PD deficiency are typically present on a Wright-Giemsa stained peripheral blood smear?

a Cabot rings
b microcytosis
c bite cells
d Heinz bodies

74. Which abnormal RBC morphology is associated with pyruvate kinase deficiency?

a acanthocytes
b dacryocytes
c echinocytes
d drepanocytes

75. Which of the following hemoglobinopathies is associated with rod shaped crystals?

a HbS
b HbC
c HbSC
d HbD

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76. Which of the following statements about hemoglobins D and G is true?

MLS ONLY

a they are clinically abnormal
b they both migrate with HbS on alkaline gel
c they are both caused by mutations in the beta-globin gene
d they cannot be separated by solubility testing

77. This hemoglobinopathy results from a fusion product of the delta and beta gene:

MLS ONLY

a HbD
b HbG
c HbLepore
d HbConstant Spring

78. Which of the following is consistent with the diagnosis of heterozygous beta-thalassemia?

a increased red blood cell count
b high MCV
c decreased HbA_2
d decreased iron stores

79. Hereditary persistence of fetal hemoglobin (HPFH) is due to a loss of expression of this globin chain:

MLS ONLY

a alpha
b beta
c gamma
d delta

80. What is the specificity of cold autoagglutinin disease?

a anti-i
b anti-H
c anti-Pr
d anti-I

81. What is the most common presentation of paroxysmal cold hemoglobinuria?

a older people with Raynaud syndrome
b children following a viral illness
c neonates with congenital syphilis
d alcoholics with advanced cirrhosis

82. Which of the following is the most common cause of anemia in hospitalized patients?

a inadequate iron intake
b inadequate folate intake
c hemolytic anemia
d anemia of chronic inflammation

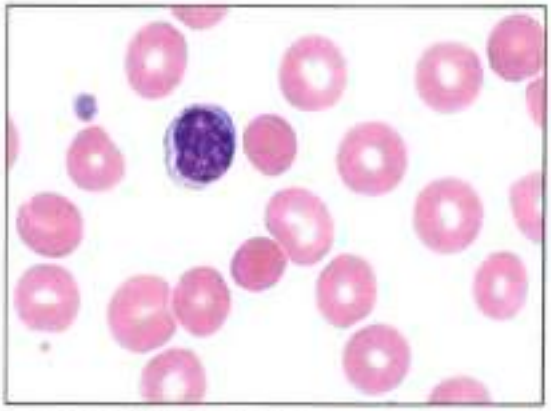
83. In a patient with an increased red cell mass into the 99th percentile and serum erythropoietin level below reference range for normal, which of the following criteria confirms a diagnosis of polycythemia vera?

a bone marrow panmyelosis
b inv(16) mutation
c *JAK2* V617F mutation
d *BCR/ABL1* translocation

84. A medical technologist is examining a peripheral smear and notices 7 large segmented neutrophils with between 5 and 7 lobes. Everything else about the CBC is otherwise normal. This observed morphologic change might develop months ahead of which of the following changes:

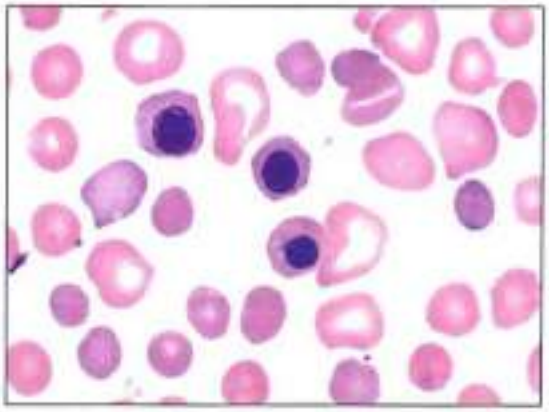
a an increase in MCV, MCH, and RDW
b a decrease in MCV, MCH, and RDW
c an increase in metamyelocytes and bands
d a bone marrow showing aplasia

85. The red blood cells in this image are representative of an anemia that is:



a microcytic, hypochromic
b nonmegaloblastic macrocytic
c normocytic, normochromic
d myelodysplastic

86. This image of the blood smear from a newborn most likely represents:



a severe G6PD deficiency
b hereditary spherocytosis
c untreated megaloblastic anemia
d HDN due to ABO incompatibility

87. Hemoglobin H disease results from:

a absence of 3 of 4 alpha genes
b absence of 2 of 4 alpha genes
c absence of 1 of 1 alpha genes
d absence of all 4 alpha genes

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88. The M:E ratio in polycythemia vera is usually:

a 4:1
b 10:1
c 50:1
d 75:15

89. In an uncomplicated case of severe iron deficiency anemia, which of these sets of laboratory data represents the typical pattern of results?

set	serum iron	serum TIBC	% saturation	marrow % sidero-blasts	marrow iron stores	serum ferritin	HbA_2
a	↓	↑	↓	↓	↑	↑	↑
b	↓	↓	↓	↓	↓	↓	↓
c	↓	↑	↓	↓	↓	↓	↓
d	↓	↓	↑	↑	↑	↑	↑

increased = ↑ decreased = ↓

a set a
b set b
c set c
d set d

90. A patient has a tumor that concentrates erythropoietin. He is most likely to have which of the following types of polycythemia?

a polycythemia vera
b polycythemia, secondary to hypoxia
c benign familial polycythemia
d polycythemia associated with renal disease

91. Which of the following types of polycythemia is most often associated with lung disease?

a polycythemia vera
b polycythemia, secondary to hypoxia
c relative polycythemia associated with dehydration
d polycythemia associated with renal disease

92. A patient has been treated for a malignant tumor for several years. His blood smear now shows:

oval macrocytes
Howell-Jolly bodies
hypersegmented neutrophils
large, agranular platelets

The most probable cause of this blood picture is:

a iron deficiency
b alcoholism
c dietary B_{12} deficiency
d chemotherapy

93. How does the bone marrow respond to anemic stress?

a expand production, release RBCs prematurely
b expand production, rush platelets into circulation
c diminish production, increase M:E
d diminish production, M:E remains normal

94. Which of the following conditions may contribute to lethargy, abdominal pain, and hemoglobinuria in some patients with a G6PD deficiency?

a dehydration
b excess iron
c ingesting fava beans
d increased glucose

B. Leukocytes

95. The most likely cause of the macrocytosis that often accompanies primary myelofibrosis is:

a folic acid deficiency
b increased reticulocyte count
c inadequate B_{12} absorption
d pyridoxine deficiency

96. Giant, vacuolated, multinucleated erythroid precursors are present in which of the following?

a chronic myelocytic leukemia
b primary myelofibrosis
c erythroleukemia
d acute myelocytic leukemia

97. Which of the following is a significant feature of dyserythropoiesis?

a persistently increased M:E ratio
b megaloblastoid erythropoiesis
c marked thrombocytosis
d decreased ferritin levels

98. The M:E ratio in erythroleukemia is usually:

a normal
b high
c low
d variable

99. Autoimmune hemolytic anemia is often a complication of:

a PV
b CML
c CLL
d HCL

100. Elevation of the total granulocyte count above $7.5 \times 10^3/\mu L$ ($7.5 \times 10^9/L$) is termed:

a relative lymphocytosis
b leukocytosis
c relative neutrophilic leukocytosis
d absolute neutrophilic leukocytosis

101. Elevation of the total white cell count above $11.0 \times 10^3/\mu L$ ($11 \times 10^9/L$) is termed:

a relative lymphocytosis
b absolute lymphocytosis
c leukocytosis
d relative neutrophilic leukocytosis

102. Elevation of the lymphocyte percentage above 45% is termed:

a relative lymphocytosis
b absolute lymphocytosis
c leukocytosis
d absolute neutrophilic leukocytosis

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103. The Philadelphia chromosome is formed by a translocation between:

- **a** chromosome 22 and chromosome 9
- **b** chromosome 21 and chromosome 9
- **c** chromosome 21 and chromosome 6
- **d** chromosome 22 and chromosome 6

104. The mechanism causing catecholamine-induced neutrophilia includes:

IELS ONLY

- **a** a shift in granulocytes from the marginating pool to the circulating pool
- **b** an increased exit of granulocytes from the circulation
- **c** a decreased exit of granulocytes from the bone marrow
- **d** granulocyte return from the tissues to the circulating pool

105. What accounts for the frequent smudge cells in CLL?

- **a** increased *in vivo* cell lysis
- **b** apoptosis related changes
- **c** artifact due to fragile cells
- **d** artifact due to heparin

106. A 14-year-old boy is seen in the ER complaining of a sore throat, swollen glands and fatigue. The CBC results are:

test	result
WBC	16.0×10^3/µL (16.0×10^9/L)
RBC	4.37×10^6/µL (4.37×10^{12}/L)
HGB	12.8 g/dL (128 g/L)
HCT	38.4%
PLT	180×10^3/µL (180×10^9/L)
absolute neutrophils	3.9×10^3/µL (3.9×10^9/L)
absolute lymphs	6.0×10^3/µL (6.0×10^9/L)
absolute monocytes	0.5×10^3/µL (0.5×10^9/L)
absolute reactive lymphs	3.2×10^3/µL (3.2×10^9/L)

What is the most likely diagnosis?

- **a** acute lymphocytic leukemia
- **b** chronic lymphocytic leukemia
- **c** viral hepatitis
- **d** infectious mononucleosis

107. The M:E ratio in chronic myelocytic leukemia is usually:

- **a** normal
- **b** high
- **c** low
- **d** variable

108. In the World Health Organization (WHO) classification, acute myelomonocytic leukemia would be named acute myeloid leukemia (AML):

- **a** with myelodysplastic-related changes
- **b** with recurrent cytogenetic changes
- **c** not otherwise specified
- **d** therapy-related

109. Abnormalities found in erythroleukemia include:

a rapid DNA synthesis
b marrow fibrosis
c megaloblastoid development
d increased erythrocyte survival

110. Neutropenia is usually associated with:

a mild bacterial infections
b viral infections
c bone marrow hyperplasia
d myeloproliferative neoplasms

111. Auer rods are **most** likely present in which of the following?

a chronic myelocytic leukemia
b primary myelofibrosis
c erythroleukemia
d acute myelocytic leukemia

112. The laboratory results shown in this table are obtained on a 45-year-old man complaining of chills and fever:

test	result	differential	%
WBC	$23.0 \times 10^3/\mu L$ ($23.0 \times 10^9/L$)	segmented neutrophils	60%
Philadelphia chromosome	negative	bands	21%
BCR/ABL fusion gene	negative	lymphocytes	11%
toxic granulation, Döhle bodies and vacuoles		monocytes	3%
		metamyelocytes	2%
		myelocytes	3%

These results are consistent with:

a neutrophilic leukemoid reaction
b polycythemia vera
c chronic myelocytic leukemia
d leukoerythroblastosis in myelofibrosis

113. In an uncomplicated case of infectious mononucleosis, which type of cells is affected?

a erythrocytes
b lymphocytes
c monocytes
d thrombocytes

114. The reactive lymphocyte seen in the peripheral blood smear of patients with infectious mononucleosis is probably derived from which of these cell types?

a T lymphocytes
b B lymphocytes
c monocytes
d mast cells

115. The disease most frequently present in patients with reactive lymphocytosis and persistently negative heterophile antibody tests is:

a toxoplasmosis
b cytomegalovirus (CMV) infection
c herpes virus infection
d viral hepatitis

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116. Dwarf or micromegakaryocytes may be found in the peripheral blood of patients with:

a pernicious anemia
b polycythemia
c primary myelofibrosis
d chronic lymphocytic leukemia

117. Which of the following is associated with pseudo-Pelger-Huët anomaly?

a aplastic anemia
b iron deficiency anemia
c chronic myeloid leukemia
d Chédiak-Higashi syndrome

118. Increased numbers of basophils are often seen in:

a acute infections
b chronic myelocytic leukemia
c chronic lymphocytic leukemia
d erythroblastosis fetalis (hemolytic disease of the newborn)

119. A hypercellular marrow with an M:E ratio of 6:1 is most commonly due to:

MLS ONLY

a lymphoid hyperplasia
b granulocytic hyperplasia
c normoblastic hyperplasia
d myeloid hypoplasia

120. The laboratory data shown in this table are obtained:

test	result
WBC	5.0×10^3/µL (5.0×10^9/L)
RBC	1.7×10^6/µL (1.7×10^{12}/L)
MCV	84.0 µm³ (84 fL)
PLT	89.0×10^3/µL (89×10^9/L)
Philadelphia chromosome	negative
BCR/ABL fusion gene	negative

1 megakaryoblast; 30 nucleated erythrocytes; teardrops; schistocytes; polychromasia; giant, bizarre platelets noted

differential	%
segmented neutrophils	16%
bands	22%
lymphocytes	28%
monocytes	16%
eosinophils	1%
basophils	1%
metamyelocytes	4%
myelocytes	3%
promyelocytes	4%
blasts	5%

These results are consistent with:

a essential thrombocythemia
b polycythemia vera
c chronic myelocytic leukemia
d primary myelofibrosis

121. A 50-year-old man is admitted into the hospital with acute leukemia. Laboratory findings include the results shown in this table:

test	result
myeloperoxidase stain	blast cells negative
terminal deoxynucleotidyl transferase (TdT)	blast cells positive
surface immunoglobulin	blast cells negative
CD2	blast cells negative
Philadelphia chromosome	positive
BCR/ABL fusion gene	positive

These results are most consistent with:

a acute myelogenous leukemia
b chronic lymphocytic leukemia in lymphoblastic transformation
c T-cell acute lymphocytic leukemia
d chronic myelogenous leukemia in lymphoblastic transformation

122. A 30-year-old man who had been diagnosed as having leukemia 2 years previously is readmitted because of cervical lymphadenopathy. Laboratory findings are shown in this table:

test	result	differential	%
WBC	39.6×10^3/μL (39.6×10^9/L)	polymorphonuclear leukocytes	7%
RBC	3.25×10^6/μL (3.25×10^{12}/L)	lymphocytes	4%
HGB	9.4 g/dL (94 g/L)	monocytes	2%
HCT	28.2%	eosinophils	3%
MCV	86.7 μm^3 (86.7 fL)	basophils	48%
MCH	29.0 pg	myelocytes	13%
MCHC	33.4%	promyelocytes	2%
PLT	53×10^3/μL (53×10^9/L)	metamyelocytes	8%
Philadelphia chromosome	positive	blasts	13%
BCR/ABL fusion gene	positive	NRBCs	11%
bone marrow	95% cellularity, 50% blast cells (some with myeloperoxidase positivity)		

These results are most consistent with:

a acute myeloid leukemia
b erythroleukemia
c chronic myelogenous leukemia (CML)
d CML in blast transformation

123. Biochemical abnormalities characteristic of polycythemia vera include:

a increased serum B_{12} binding capacity
b hypouricemia
c hypohistaminemia
d increased erythropoietin

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124. A differential cell count of 50-90% myeloblasts in a peripheral blood smear is typical of which of the following?

a chronic myelocytic leukemia
b primary myelofibrosis
c erythroleukemia
d acute myelocytic leukemia

125. The M:E ratio in acute myelocytic leukemia is usually:

a normal
b high
c low
d variable

126. Which of the following is most closely associated with acute promyelocytic leukemia?

a ringed sideroblasts
b disseminated intravascular coagulation
c micromegakaryocytes
d Philadelphia chromosome

127. Which of the following is most closely associated with chronic myelomonocytic leukemia?

a Philadelphia chromosome
b disseminated intravascular coagulation
c micromegakaryocytes
d lysozymuria

128. Which of the following is most closely associated with chronic myelogenous leukemia?

a ringed sideroblasts
b disseminated intravascular coagulation
c micromegakaryocytes
d *BCR/ABL* fusion gene

129. Acute (pure) erythroid leukemia is characterized by the presence of:

MLS ONLY
a more than 20% bone marrow myeloblasts
b less than 20% bone marrow proerythroblasts
c more than 30% bone marrow proerythroblasts
d less than 80% bone marrow erythroid precursors

130. The replacement of normal marrow precursor cells by an accumulation of blasts is a hallmark of:

a chronic lymphocytic leukemia
b myelodysplastic syndromes
c polycythemia vera
d acute myelocytic leukemia

131. All stages of neutrophils are most likely to be seen in the peripheral blood of a patient with:

a chronic myelocytic leukemia
b myelofibrosis with myeloid metaplasia
c erythroleukemia
d acute myelocytic leukemia

132. Which of the following conditions is a myeloproliferative neoplasm?

a refractory anemia
b secondary erythrocytosis
c myelomonocytic leukemia
d essential thrombocythemia

133. The laboratory results shown in this table are obtained on a 55-year-old man complaining of headaches and blurred vision:

test	result	differential	%
WBC	19.0 × 10^3/µL (19.0 × 10^9/L)	segmented neutrophils	84%
RBC	7.2 × 10^6/µL (7.2 × 10^{12}/L)	bands	10%
PLT	1056 × 10^3/µL (1056 × 10^9/L)	lymphocytes	3%
uric acid	13.0 mg/dL (0.76 mmol/L)	monocytes	2%
O_2 saturation	93%	eosinophils	1%
red cell volume	3911 mL (normal = 1600)		

These results are consistent with:

a neutrophilic leukemoid reaction
b polycythemia vera
c chronic myelocytic leukemia
d primary myelofibrosis

134. In comparison to malignant lymphocytes, reactive lymphocytes:

a have a denser nuclear chromatin
b have a regular cell shape
c have more cytoplasm and more mitochondria
d are morphologically more variable throughout the smear

135. Which pair of malignancies represent different clinical manifestations of the same disease?

a chronic lymphocytic leukemia and adult T-cell leukemia
b hairy cell leukemia and Hodgkin lymphoma
c chronic lymphocytic leukemia and small lymphocytic lymphoma
d Sézary syndrome and prolymphocytic leukemia

136. Increased levels of TdT activity are indicative of:

a Burkitt lymphoma
b acute promyelocytic leukemia
c acute lymphocytic leukemia
d eosinophilia

137. Which of the following is true of acute lymphoblastic leukemia (ALL)?

a occurs most commonly in children 1-2 years of age
b patient is asymptomatic
c primitive lymphoid-appearing cells accumulate in bone marrow
d children under 1 year of age have a good prognosis

138. The most common form of childhood leukemia is:

a acute lymphocytic
b acute granulocytic
c acute monocytic
d chronic granulocytic

139. Chronic lymphocytic leukemia is defined as a(n):

a malignancy of the thymus
b accumulation of prolymphocytes
c accumulation of hairy cells in the spleen
d accumulation of monoclonal B cells

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140. Hairy cell leukemia is a(an):

a acute myelocytic leukemia
b chronic leukemia of myelocytic origin
c chronic leukemia of lymphocytic origin
d acute leukemia of monocytic origin

141. Which of these characteristics is usually associated with hairy cell leukemia?

a neutrophilia
b mononuclear cells with ruffled edges
c positive for CD5
d increased resistance to infection

142. Morphologic variants of plasma cells include:

a flame cells
b Cabot cells
c Pelger-Huët anomaly
d Gaucher cells

143. Which of these following bone marrow findings favors the diagnosis of multiple myeloma?

IMLS ONLY
a presence of Reed Sternberg cells
b sheaths of immature plasma cells
c presence of occasional flame cell
d presence of plasmacytic satellitosis

144. Which of these clinical conditions has a B-cell origin?

MLS ONLY
a Sézary syndrome
b large granular lymphocytosis
c Sternberg sarcoma
d Waldenström macroglobulinemia

145. Which of these cell types is most likely identified in lesions of mycosis fungoides?

IMLS ONLY
a T lymphocytes
b B lymphocytes
c monocytes
d mast cells

146. The disease most closely associated with cytoplasmic granule fusion is:

a Chédiak-Higashi syndrome
b Pelger-Huët anomaly
c May-Hegglin anomaly
d Alder-Reilly anomaly

147. Which of these anomalies is an autosomal dominant disorder characterized by irregularly-sized inclusions in polymorphonuclear neutrophils, abnormal giant platelets and often thrombocytopenia?

a Pelger-Huët
b Chédiak-Higashi
c Alder-Reilly
d May-Hegglin

148. The disease most closely associated with granulocyte hyposegmentation is:

a May-Hegglin anomaly
b Pelger-Huët anomaly
c Chédiak-Higashi syndrome
d Gaucher disease

149. Which of the following is associated with Chédiak-Higashi syndrome?

a membrane defect of lysosomes
b Döhle bodies and giant platelets
c 2-lobed neutrophils
d mucopolysaccharidosis

150. Which of the following is associated with Alder-Reilly inclusions?

a membrane defect of lysosomes
b Döhle bodies and giant platelets
c 2-lobed neutrophils
d mucopolysaccharidosis

151. Which of the following is associated with May-Hegglin anomaly?

a membrane defect of lysosomes
b Döhle bodies and giant platelets
c chronic myelogenous leukemia
d mucopolysaccharidosis

152. A differential is performed on an asymptomatic patient. The differential included 60% neutrophils, 55 of which had 2 lobes and 5 had 3 lobes. There are no other abnormalities. This is consistent with which of these anomalies?

a Pelger-Huët
b May-Hegglin
c Alder-Reilly
d Chédiak-Higashi

153. MLS ONLY Patients with chronic granulomatous disease suffer from frequent pyogenic infections due to the inability of:

a lymphocytes to produce bacterial antibodies
b eosinophils to degranulate in the presence of bacteria
c neutrophils to kill phagocytized bacteria
d basophils to release histamine in the presence of bacteria

154. Which of the following can cause neutropenia?

a medications
b lymphoid neoplasms
c chronic myelogenous leukemia
d polycythemia vera

155. Which of the following immunohistochemical patterns is most consistent with CLL/SLL?

a CD5–/CD23–
b CD5–/CD23+
c CD5+/CD23–
d CD5+/CD23+

156. Which of the following morphologic characteristics is consistently associated with hairy cell leukemia?

a small cells
b clumped nuclear chromatin
c flocculent dark blue cytoplasm
d uneven cytoplasmic margins

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157. What is the minimum percentage of prolymphocytes that must be present for a diagnosis of prolymphocytic leukemia? MLS ONLY

a >10%
b >25%
c >55%
d >75%

158. Which of the following translocations is most commonly associated with Burkitt lymphoma? MLS ONLY

a t(8;14)
b t(11;18)
c t(15;17)
d t(18;22)

159. Which of the following genetic alterations is associated with a favorable prognosis in pre-B-ALL? MLS ONLY

a t(1;19)
b t(4;11)
c t(9;22)
d t(12;21)

160. What is the most common immunoglobulin present in plasma cell myeloma?

a IgG
b IgA
c light chains
d IgD

161. Which of the following is associated with mycosis fungoides? MLS ONLY

a Sézary cells
b hairy cells
c prolymphocytes
d large granular lymphocytes

162. Which of the following is a characteristic of Hodgkin lymphoma? MLS ONLY

a bimodal age distribution
b high incidence of peripheral blood infiltration
c unpredictable lymph node involvement
d uniformly fatal

163. Which of the following is suggestive of a myeloproliferative disorder?

a basopenia
b granulocytosis
c lack of toxic granulation
d cytopenias

164. Which of the following is characteristic of the myelodysplastic syndromes?

a neutrophilia
b unilineage or multilineage dyspoiesis
c effective hematopoiesis
d propensity to develop lymphoma

165. Which of the following is a WHO classification of myelodysplastic syndrome?

a chronic myelomonocytic leukemia (CMML)
b refractory thrombocytopenia
c refractory anemia with excessive blasts (RAEB)
d refractory neutrophilia

166. Which of the following cytogenetic abnormalities is associated with a more stable clinical course in myelodysplastic syndromes (MDSs)?

a monosomy 7
b del 5q
c loss of Y
d del 20q

167. Which of the following disorders is a category in the WHO classification of myelodysplastic syndromes/myeloproliferative neoplasms?

a acute myelomonocytic leukemia (AMML)
b chronic myeloid leukemia (CML)
c juvenile myelomonocytic leukemia (JMML)
d refractory anemia with excessive blasts (RAEB)

168. Which of the following features define chronic myelogenous leukemia?

a *JAK2* mutation
b myelocyte "bulge"
c basophilia, eosinophilia, and thrombocytosis
d t(9;22)

169. Which of the following are main groups of acute myeloid leukemia (AML) in the WHO classification system?

a AML with myeloproliferative related changes
b AML with lymphoid differentiation
c AML with biphenotypic morphology
d AML, NOS

170. What type of morphologically abnormal cells proliferate in acute myelogenous leukemia with inv(16)?

a lymphocytes
b erythrocytes
c platelets
d eosinophils

171. What defines acute myelogenous leukemia with minimal differentiation?

a >20% blasts positive for Sudan black B
b positivity for myeloperoxidase or nonspecific esterase in at least 50% of blasts
c positivity for myeloperoxidase or nonspecific esterase in at least 100% of blasts
d >20% blasts lacking cytochemical evidence of myeloid differentiation

172. What is the most common presentation of acute erythroid leukemia?

a >80% of nucleated marrow cells are erythroid precursors
b >50% of nucleated marrow cells are erythroid precursors and >20% of nonerythroid cells are myeloblasts
c >50% of nucleated marrow cells are erythroid precursors and <20% of nonerythroid cells are myeloblasts
d 80% blasts with 20% erythroid precursors, >20% myeloblasts

173. Which of the following is associated with acute megakaryoblastic leukemia?

a Down syndrome
b systemic mastocytosis
c isochromosome 17p
d <5% of blasts being megakaryoblastic

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174. Which of the following cell surface markers is associated with a more aggressive subtype of CLL/SLL? MLS ONLY

a CD13
b CD21
c CD38
d CD125

175. The leukemic phase of T cell lymphoma is marked by the following cellular abnormality: MLS ONLY

a Sézary cell
b Plasmacytoid lymphocyte
c Mantle cell
d Reed Sternberg cell

176. A bone marrow shows foam cells ranging from 20-100 µm, vacuolated cytoplasm containing sphingomyelin and faint PAS positivity. This cell type is most characteristic of: MLS ONLY

a Gaucher disease
b myeloma with Russell bodies
c DiGuglielmo disease
d Niemann-Pick disease

177. A useful chemical test for the diagnosis of hairy cell leukemia is the:

a peroxidase test
b Sudan black test
c periodic acid-Schiff test
d tartrate-resistant acid phosphatase test

178. A 30-year-old woman is admitted to the hospital for easy bruising and menorrhagia. Laboratory findings included the results shown in this table:

test	result	differential	%
WBC	3.5×10^3/µL (3.5×10^9/L)	polymorphonuclear leukocytes	3%
RBC	2.48×10^6/µL (2.48×10^{12}/L)	lymphocytes	1%
PLT	30×10^3/µL (30.0×10^9/L)	monocytes	2%
HGB	8.6 g/dL (86 g/L)	myelocytes	4%
HCT	25.0%	abnormal immature	58%
MCV	100.7 μm^3 (100.7 fL)	blasts	31%
MCH	34.7 pg	nRBC	1%
MCHC	34.3%		
PT	34.0 sec		
aPTT	62.5 sec		
TT	15.0 sec		
FDP	>40 µg/mL (>40 mg/L)		
fibrinogen	315 mg/dL (3.15 g/L) (control 200-400 mg/dL [2.0–4.0 g/L])		

Auer rods, 1+ macrocytes, 1+ polychromasia

The cells identified as "abnormal immature" are described as having lobulated nuclei with prominent nucleoli; the cytoplasm has intense azurophilic granulation over the nucleus, with some cells containing 1-20 Auer rods, frequently grouped in bundles. A t(15;17) chromosomal translocation is noted. Cells are myeloperoxidase positive. Which of these types of acute leukemia is most likely?

a myeloblastic
b promyelocytic
c myelomonocytic
d monocytic

179. Chronic lymphocytic leukemia cells are most likely to express which of the following cell surface markers?

a CD3, CD7, CD19, CD20
b CD4, CD5, CD19, CD20
c CD5, CD19, CD20, CD21
d CD13, CD33, CD107

180. Which of the following markers, typically detected in normal myeloid cells, are expressed on the surface of hairy cell leukemia lymphocytes?

a CD3
b CD8
c CD10
d CD11c

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C. Platelets

181. vWF antigen can be found in which of the following?

- **a** myeloblasts
- **b** monoblasts
- **c** lymphoblasts
- **d** megakaryocytes

182. The life span of a platelet is:

- **a** 5 days
- **b** 10 days
- **c** 20 days
- **d** 30 days

183. Aspirin affects platelet function by interfering with platelets' metabolism of:

MLS ONLY

- **a** cyclooxygenase
- **b** lipids
- **c** carbohydrates
- **d** nucleic acids

184. In patients who present with bleeding disorders caused by platelets, the most common type of bleeding is:

- **a** mucosal bleeding
- **b** hemarthrosis
- **c** delayed bleeding
- **d** deep hematomas

185. The anticoagulant that is best directed against platelets is:

MLS ONLY

- **a** argatroban
- **b** hirudin
- **c** tirofiban
- **d** dabigatran

186. How do ticlopidine and clopidogrel inhibit platelets?

MLS ONLY

- **a** binding von Willebrand factor
- **b** ADP mediated platelet aggregation
- **c** inhibit GPIIb/IIIa
- **d** depletion of platelet alpha granule content

187. von Willebrand factor mediates platelet adhesion by binding to platelet receptor:

- **a** GPIb/IIa
- **b** GPIb/GPIX/GPV
- **c** GPIIb/IIa
- **d** GPIb/GPIIIa/GP X

188. The disease state that presents with a quantitative platelet disorder is:

MLS ONLY

- **a** von Willebrand disease
- **b** hemophilia A
- **c** Glanzmann thrombasthenia
- **d** May-Hegglin anomaly

189. A patient presents with a very low platelet count and is diagnosed with acute idiopathic thrombocytopenic purpura (ITP). Which statement is associated with acute ITP?

MLS ONLY

- **a** it is found primarily in adults
- **b** spontaneous remission usually occurs within several weeks
- **c** women are more commonly affected
- **d** peripheral destruction of platelets is decreased

190. The most common cause of bleeding in patients is:

a qualitative platelet defect
b qualitative abnormality of fibrinogen
c quantitative abnormality of fibrinogen
d quantitative abnormality of platelets

191. MLS ONLY A 53-year-old man is in recovery following a triple bypass operation. Oozing is noted from his surgical wound. The laboratory data shown in this table are obtained:

test	result
hemoglobin	12.5 g/dL (125 g/L)
hematocrit	37%
prothrombin time	12.3 seconds
aPTT	34 seconds
platelet count	40.0 × 10^3/μL (40.0 × 10^9/L)
fibrinogen	250 mg/dL (2.5 g/L)

The most likely cause of bleeding would be:

a dilution of coagulation factors due to massive transfusion
b intravascular coagulation secondary to microaggregates
c hypofibrinogenemia
d dilutional thrombocytopenia

192. MLS ONLY ADAMTS13 deficiency is responsible for thrombocytopenia found in:

a TTP
b DIC
c HUS
d ITP

193. MLS ONLY Heparin induced thrombocytopenia (HIT) is an immune mediated complication associated with heparin therapy. Antibodies are produced against:

a ACLA
b PF4
c AT
d B2GP1

194. In polycythemia vera, the platelet count is:

a elevated
b normal
c decreased
d variable

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195. A 60-year-old man has a painful right knee and a slightly enlarged spleen. Hematology results include: MLS ONLY

test	result
hemoglobin	15 g/dL (150 g/L)
absolute neutrophil count	10.0×10^3/µL (10.0×10^9/L)
platelet count	900×10^3/µL (900×10^9/L)
uncorrected retic count	1%
morphology	normal red cell morphology and indices a slight increase in bands rare metamyelocyte and myelocyte giant and bizarre-shaped platelets

This is most compatible with:

a congenital spherocytosis
b rheumatoid arthritis with reactive thrombocytosis
c myelofibrosis
d idiopathic thrombocythemia (essential or primary)

196. Large blue inclusions bodies in WBC are found in patients with:

a Wiskott-Aldrich syndrome
b May-Hegglin Anomaly
c Ehlers-Danlos syndrome
d Hĕrmanský-Pudlák syndrome

197. A 42-year-old male presents with fatigue, difficulty breathing, tingling in the hands and feet. The laboratory results shown in this table are obtained:

test	result	reference range
PT	25.6 seconds	11.5-13.9 seconds
aPTT	77.5 seconds	25.0-35.7 seconds
HGB	23.5 g/dL	
HCT	76.2%	

The clinician questioned the results, as the patient had no bleeding symptoms. This result could be caused by:

a aplastic anemia
b polycythemia vera
c chronic lymphocytic leukemia
d sickle cell disorder

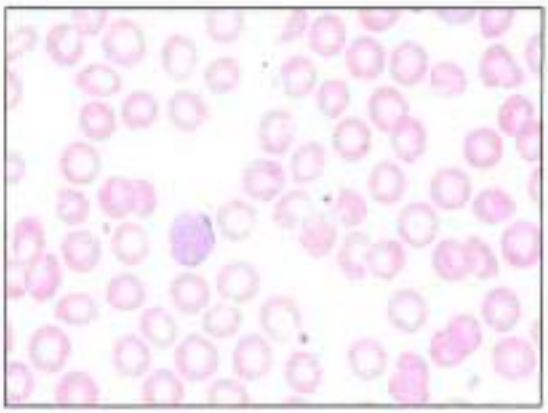
III. Laboratory Testing

A. Cell Counts

198. Which of the following is the formula to calculate an absolute cell count?

a number of cells counted/total count
b total count/number of cells counted
c 10× total count
d % of cells counted × total count

199. Using a supravital stain, the polychromatic red blood cells in the photomicrograph would probably be:



a rubricytes (polychromatophilic normoblast)
b reticulocytes
c sickle cells
d target cells

200. The mean value of a reticulocyte count on specimens of cord blood from healthy, full-term newborns is approximately:

a 0.5%
b 2.0%
c 5.0%
d 8.0%

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201. The values in the table are obtained using an automated hematology analyzer and are performed on a blood sample from a 25-year-old man:

test	patient	normal
WBC	5.1×10^3/µL (5.1×10^9/L)	$5.0\text{-}10.0 \times 10^3$/µL ($5.0\text{-}10.0 \times 10^9$/L)
RBC	2.94×10^6/µL (2.94×10^{12}/L)	$4.6\text{-}6.2 \times 10^6$/µL ($4.6\text{-}6.2 \times 10^{12}$/L)
HGB	13.8 g/dL (138 g/L)	14-18 g/dL (140-180 g/L)
HCT	35.4%	40-54%
MCV	128 µm^3 (128 fL)	82-90 µm^3 (82-90 fL)
MCH	46.7 pg	27-31 pg
MCHC	40%	32-36%

These results are most consistent with which of the following?

a megaloblastic anemia
b hereditary spherocytosis
c a high titer of cold agglutinins
d an elevated reticulocyte count

202. A patient has a high cold agglutinin titer. Automated hematology analyzer results reveal an elevated MCV, MCH and MCHC as well as a decreased RBC. Individual erythrocytes appear normal on a stained smear, but agglutinates are noted. The appropriate course of action would be to:

a perform the RBC, HGB, and HCT determinations using manual methods
b perform the RBC determination by a manual method; use the automated results for the HGB and HCT
c repeat the determinations using a microsample of diluted blood
d repeat the determinations using a prewarmed microsample of diluted blood

203. Supravital staining is important for reticulocytes since the cells must be living in order to stain the:

a remaining RNA in the cell
b iron before it precipitates
c cell membrane before it dries out
d denatured hemoglobin in the cell

204. Which of the following is used for staining reticulocytes?

a Giemsa stain
b Wright stain
c new methylene blue
d Prussian blue

205. Which of the following values is calculated from the red blood cell indices in an automated hematology analyzer?

a red blood cell count (RBC)
b hematocrit (HCT)
c mean corpuscular volume (MCV)
d red cell distribution width (RDW)

206. A hematology analyzer counts red blood cells by which method?

a impedance
b chromogenic
c photometric
d turbidimetric

207. The RDW is elevated on a CBC report. The smear would show:

a hypochromia
b anisocytosis
c poikilocytosis
d macrocytosis

208. An oncology patient has the results shown in this table:

test	day 1	day 3
WBC	8.0×10^3/µL (8.0×10^9/L)	2.0×10^3/µL (2.0×10^9/L)
RBC	3.50×10^6/µL (3.50×10^{12}/L)	3.45×10^6/µL (3.45×10^{12}/L)
HGB	10.0 g/dL (100 g/L)	9.9 g/dL (99 g/L)
HCT	29.8%	29.5%
PLT	180×10^3/µL (180×10^9/L)	150×10^3/µL (150×10^9/L)

The most probable explanation is:

a chemotherapy
b cold antibody
c clotted specimen
d inadequate mixing

209. A leukocyte count and differential on a 40-year-old Caucasian man revealed:

test	results
WBC	5.4×10^3/µL (5.4×10^9/L)
segmented neutrophils	20%
lymphocytes	58%
monocytes	20%
eosinophils	2%

This data represents:

a absolute lymphocytosis
b relative neutrophilia
c absolute neutropenia
d leukopenia

210. A leukocyte count and differential on a 40-year-old Caucasian man revealed:

test	results
WBC	5.4×10^3/µL (5.4×10^9/L)
segmented neutrophils	20%
lymphocytes	58%
monocytes	20%
eosinophils	2%

This represents:

a relative lymphocytosis
b absolute lymphocytosis
c relative neutrophilia
d leukopenia

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211. Given the data shown in this table:

test	results
WBC	8.5×10^3/µL (8.5×10^9/L)
segmented neutrophils	56%
bands	2%
lymphocytes	30%
monocytes	6%
eosinophils	6%

What is the absolute lymphocyte count?

a 170/µL (0.17×10^9/L)
b 510/µL (0.51×10^9/L)
c 2550/µL (2.55×10^9/L)
d 4760/µL (4.76×10^9/L)

212. Which of these is the formula to calculate a manual white cell count?

a (number of cells counted × dilution × 10)/number of squares counted
b (number of cells counted × dilution)/10 × number of squares counted
c number of cells counted × dilution
d number of cells counted × number of squares counted

213. If a WBC count is performed on a 1:100 dilution and the number of cells counted in 8 squares is 50, the total WBC count is:

a 5,000/µL (5.0×10^9/L)
b 6,250/µL (6.25×10^9/L)
c 50,000/µL (50.0×10^9/L)
d 62,500/µL (62.5×10^9/L)

214. A total leukocyte count is 10.0×10^3/µL (10.0×10^9/L) and 25 NRBCs are seen per 100 leukocytes on the differential. What is the corrected leukocyte count?

a 2,000/µL (2.0×10^9/L)
b 8,000/µL (8.0×10^9/L)
c 10,000/µL (10.0×10^9/L)
d 12,000/µL (12.0×10^9/L)

215. A flag of immature granulocytes (IG) is reported from a hematology analyzer. The next step is to perform a(an):

a auto verification
b smear review
c manual differential
d pathology review

216. In an automated hematology analyzer, the WBC printed result is +++. The next step is to:

a repeat after warming the sample to 37°C
b make an appropriate dilution of the sample
c recalibrate the machine from pooled samples
d request a new sample immediately

217. A specimen analyzed on an automated hematology instrument has a platelet count of 19×10^3/µL (19×10^9/L). The first procedure to follow is:

a report the count after the batch run is completed
b request a new specimen
c review the stained blood smear
d notify the laboratory manager

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218. The electrical resistance method of cell counting requires:

a equal-sized particles
b a conductive liquid
c 2 internal electrodes for current
d 3 apertures for counting

219. MLS ONLY An anemic patient has an RBC of 2.70 × 10^6/µL (2.70 × 10^{12}/L) and a hemoglobin of 13.5 g/dL (135 g/L) as determined by an automated hematology analyzer. Which of the following is the best explanation for these results?

a electrical interference
b specimen that is lipemic
c high anticoagulant to blood ratio
d a high coincidence rate

220. MLS ONLY The results shown in this table are obtained on an automated hematology analyzer:

test	result	test	result
WBC	6.5 × 10^3/µL (6.5 × 10^9/L)	MCV	90.1 µm^3 (90.1 fL)
RBC	4.55 × 10^6/µL (4.55 × 10^{12}/L)	MCH	39.6 pg
HGB	18.0 g/dL (180 g/L)	MCHC	43.4%
HCT	41.5%		

The first step in obtaining valid results is to:

a perform a microhematocrit
b correct the hemoglobin for lipemia
c dilute the blood
d replace the lysing agent

221. On an automated hematology analyzer, if the RBC is erroneously increased, how will other parameters be affected?

a increased MCHC
b increased hemoglobin
c decreased MCH
d increased MCV

222. On initial start-up of the automated hematology analyzer, one of the controls is slightly below the range for the MCV. Which of the following is indicated?

a call for service
b adjust the MCV up slightly
c shut down the instrument
d repeat the control

223. MLS ONLY In an electronic or laser particle cell counter, clumped platelets may interfere with which of the following parameters?

a white blood cell count
b red blood cell count
c hemoglobin
d hematocrit

224. Which of the following will cause erroneous results when using a phase optical system for enumerating platelets?

a lysed RBCs
b aggregated platelets
c diluted sample
d intact leukocytes

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225. The laboratory results shown in this table are obtained on an electronic particle counter:

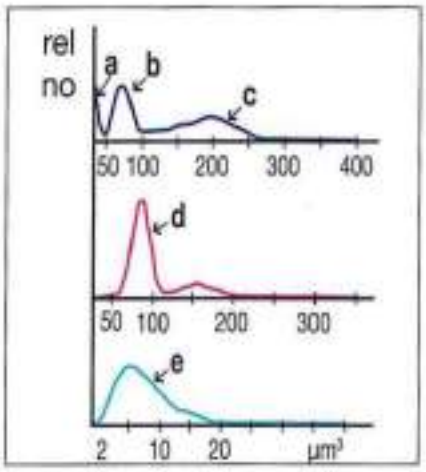
MLS ONLY

test	result	test	result
WBC	61.3×10^3/µL (61.3×10^9/L)	MCV	125 µm³ (125 fL)
RBC	1.19×10^6/µL (1.19×10^{12}/L)	MCHC	54.1%
HGB	9.9 g/dL (99 g/L)		
HCT	21%		

What action should be taken to obtain accurate results?

a dilute the specimen and recount
b warm the specimen and recount
c check the tube for clots
d clean the aperture tubes and recount

226. Which area in the histogram shown for an automated hematology analyzer represents the RBC distribution curve?



a area a
b area b
c area c
d area d

227. Blood collected in EDTA undergoes which of the following changes if kept at room temperature (22°C) for 6-24 hours?

a increased hematocrit and MCV
b increased ESR and MCV
c increased MCHC and MCV
d decreased reticulocyte count and hematocrit

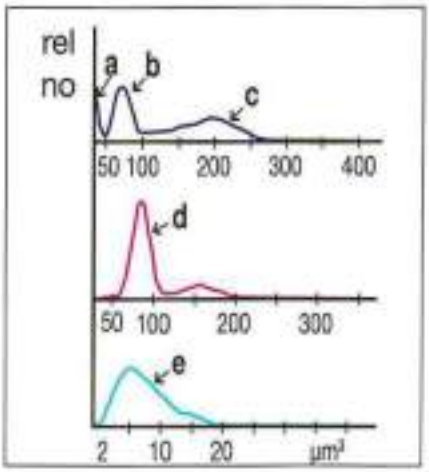
228. A platelet determination is performed on an automated instrument and a very low value is obtained. The platelets appear adequate when estimated from the stained blood film. The best explanation for this discrepancy is:

a many platelets are abnormally large
b blood sample is hemolyzed
c white cell fragments are present in the blood
d red cell fragments are present in the blood

229. MLS ONLY Two samples arrive from the emergency room (ER) on the same patient within 10 minutes of each other. Sample 1 has a WBC of 8.5×10^9/L and Sample 2 has a WBC of 20.5×10^9/L. What is the best course of action?

a report both results
b determine how the samples were collected
c determine if the samples are from the same patient
d consider a redraw of the specimen

230. MLS ONLY Which area in the histogram shown for an automated hematology analyzer represents the platelet distribution curve?



a area a
b area b
c area c
d area e

231. MLS ONLY The CBC results shown in this table are obtained from an automated hematology analyzer on a patient sample with lipemic plasma:

test	result
WBC	7.2×10^3/µL (7.2×10^9/L)
RBC	3.50×10^6/µL (3.50×10^{12}/L)
HGB	13.8 g/dL (138 g/L)
HCT	33.5%
MCV	92 µm³ (92 fL)
MCH	39.4 pg
MCHC	41.0%

Which of the following tests would probably be in error?

a WBC, RBC, MCV
b RBC, HCT, MCV
c RBC, HGB, HCT
d HGB, MCH, MCHC

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232. Blood is diluted 1:200, and a platelet count is performed. 180 platelets are counted in the center square millimeter on one side of the hemacytometer and 186 on the other side. The total platelet count is:

a 146,000/µL (1.46×10^{12}/L)
b 183,000/µL (1.83×10^{12}/L)
c 366,000/µL (3.66×10^{12}/L)
d 732,000/µL (7.32×10^{12}/L)

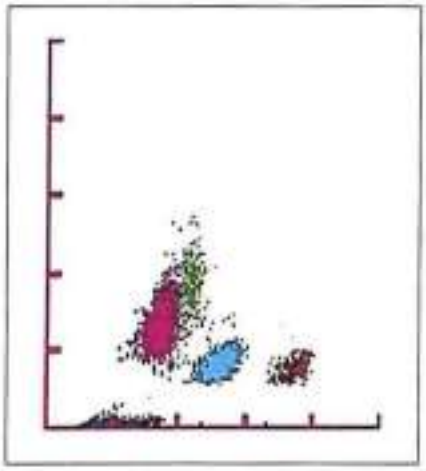
233. A phase-platelet count is performed and the total platelet count is 356,000/µL (3.56×10^{12}/L). Ten high power (100x) fields on the stained blood smear are examined for platelets and the results per field were:

16, 18, 15, 20, 19, 17, 19, 18, 20, 16

The next step would be to:

a report the phase-platelet count since it correlated well with the slide
b repeat the phase-platelet count on a recollected specimen and check for clumping
c check 10 additional fields on the blood smear
d repeat the platelet count using a different method

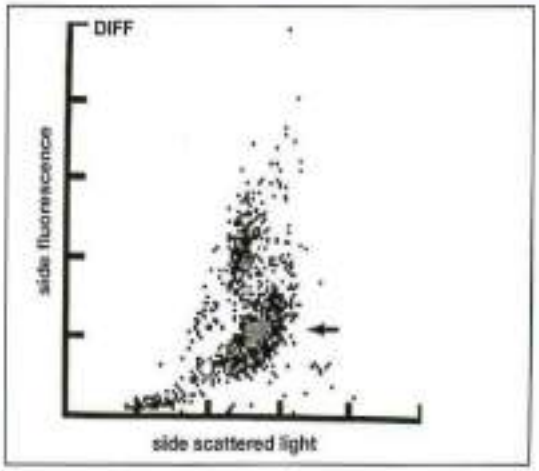
234. MLS ONLY The white cell count is flagged on a routine CBC. In reviewing the scatter plot shown from the analyzer, what cells will be present on the differential?



a reactive lymphocytes
b increased eosinophils
c increased lymphocytes
d nucleated red blood cells

235. What population of cells does the arrow indicate in the scattergram?

MLS ONLY



a lymphocytes
b monocytes
c basophils
d neutrophils

236. The CBC results shown in this table are obtained from an automated hematology analyzer.

MLS ONLY

test	result
WBC	$9.0 \times 10^3/\mu L$ ($9.0 \times 10^9/L$)
RBC	$4.62 \times 10^6/\mu L$ ($4.62 \times 10^{12}/L$)
HGB	14.7 g/dL (147 g/L)
HCT	43.4%
MCV	93.9 μm^3 (93.9 fL)
MCH	31.8 pg
MCHC	33.9%
PLT	$9.0 \times 10^3/\mu L$ ($230 \times 10^9/L$)
RDW	14.1

The white cell count is flagged. What laboratory operation would be followed?

a send the report
b scan and review the slide
c perform a platelet estimate
d ask for pathologist review

237. Which hematology manufacturer uses multi-angle polarized scatter separation (MAPSS) and fluorescence to perform a WBC differential?

MLS ONLY

a Beckman Coulter
b Sysmex XN
c Abbott Cell Dyne
d Siemens Advia

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238. Given this normal scattergram, the area indicated as a "6" represents which cell population?

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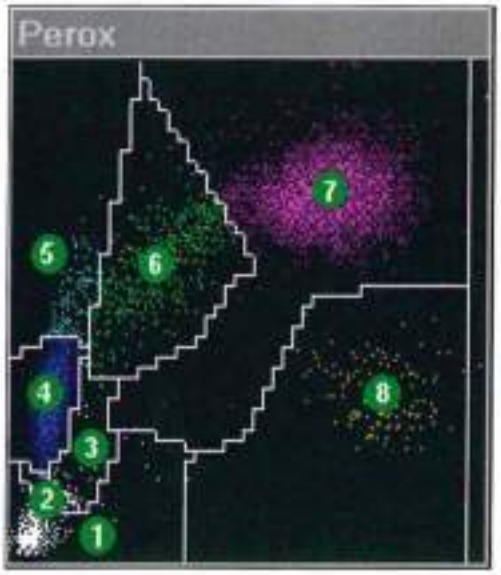

a monocytes
b large unstained cells
c platelet clumps
d nucleated RBCs

B. Differentials and Morphology Evaluation

239. In which of the following disease states are teardrop cells and abnormal platelets most characteristically seen?

a hemolytic anemia
b multiple myeloma
c G6PD deficiency
d primary myelofibrosis

240. Which of the following represent residual nuclear fragments?

a Pappenheimer bodies
b Cabot rings
c Heinz bodies
d target cells

241. Which of the following RBC inclusions are seen in sideroblastic anemia and contain high amounts of iron?

a Cabot rings
b Howell-Jolly bodies
c Heinz bodies
d Pappenheimer bodies

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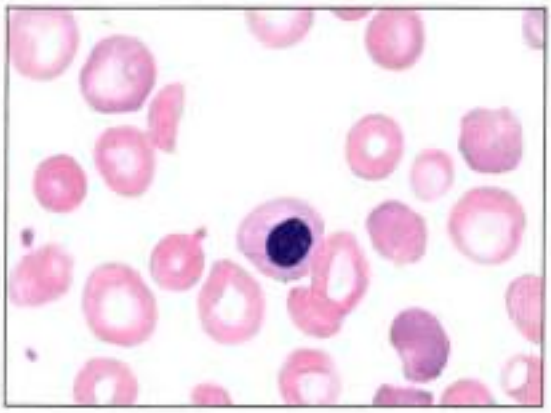
242. The nucleated cell in the photomicrograph may be seen in the peripheral blood of a normal newborn and is classified as a(an):



a basophilic normoblast
b polychromatophilic normoblast
c orthochromatic normoblast
d megaloblastic normoblast

243. A Wright-stained peripheral blood smear reveals blue, ring-shaped inclusions with red chromatin dots in several of the red blood cells. These inclusions are consistent with:

a Howell-Jolly bodies
b Cabot rings
c basophilic stippling
d malarial parasites

244. A red blood cell about 5 µm in diameter that stains bright red and shows no central pallor is a:

a spherocyte
b leptocyte
c microcyte
d macrocyte

245. The results shown in this table are obtained on a patient's blood:

test	result
HGB	11.5 g/dL (115 g/L)
HCT	34%
MCV	89 μm^3 (89 fL)
MCH	26 pg
MCHC	29%

Examination of a Wright-stained smear of the same sample would most likely show:

a macrocytic, normochromic erythrocytes
b microcytic, hypochromic erythrocytes
c normocytic, hypochromic erythrocytes
d normocytic, normochromic erythrocytes

246. Evidence of active red cell regeneration may be indicated on a blood smear by:

a nucleated red blood cells and polychromasia
b hypochromia and macrocytes
c basophilic stippling and nucleated red blood cells
d Howell-Jolly bodies and Cabot rings

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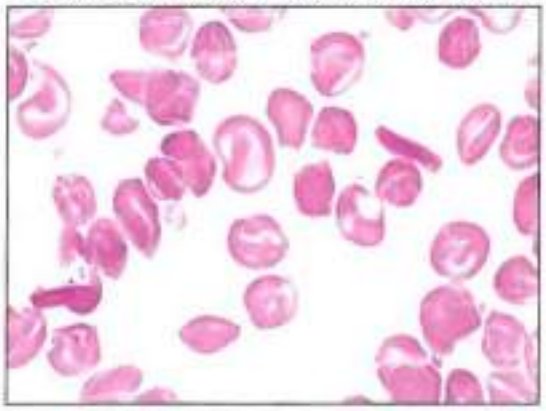
247. The smear represented in the photomicrograph displays:



a congenital ovalocytosis
b hemoglobin C disease
c poor RBC fixation
d delay in smear preparation

248. The presence of excessive rouleaux formation on a blood smear is often accompanied by an increased:

a reticulocyte count
b sedimentation rate
c hematocrit
d erythrocyte count

249. The characteristic peripheral blood morphologic feature in multiple myeloma is:

a cytotoxic T cells
b rouleaux formation
c spherocytosis
d macrocytosis

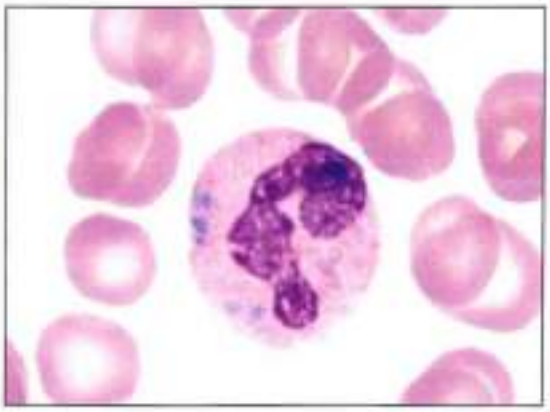
250. Many microspherocytes, schistocytes and spherocytes with budding cytoplasm can be seen on peripheral blood smears of patients with:

a hereditary spherocytosis
b pyruvate kinase deficiency
c thalassemia
d extensive burns

251. The light-colored zone adjacent to the nucleus in a plasma cell is the:

a ribosome
b chromatin
c mitochondria
d Golgi area

252. Inclusions in the cytoplasm of neutrophils as shown in the photomicrograph are known as:



a Auer bodies
b Howell-Jolly bodies
c Heinz bodies
d Döhle bodies

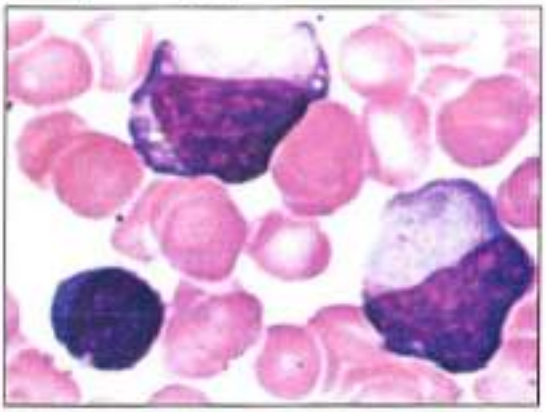
253. The term "shift to the left" refers to:

a a microscope adjustment
b immature cell forms in the peripheral blood
c a trend on a Levy-Jennings chart
d a calibration adjustment on an instrument

254. A term that means varying degrees of leukocytosis with a shift to the left and occasional nucleated red cells in the peripheral blood is:

a polycythemia vera
b erythroleukemia
c leukoerythroblastosis
d megaloblastoid

255. In the photomicrograph the small nucleated cell seen in the lower left corner is a:



a polychromatophilic normoblast (rubricyte)
b mature lymphocyte
c plasma cell
d lymphoblast

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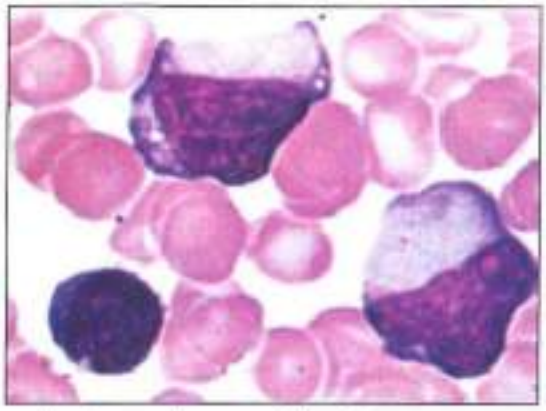
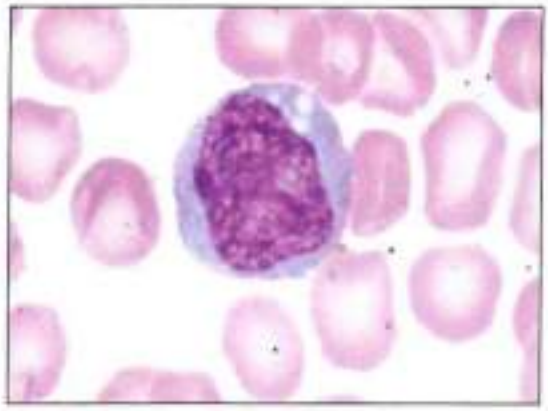
256. The large cells seen in the photomicrograph are most consistent with:



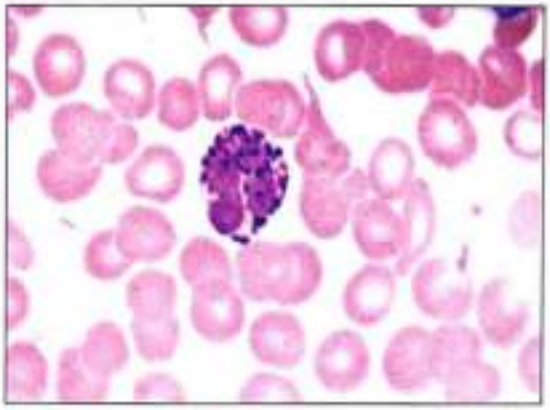
a chronic myelogenous leukemia
b infectious mononucleosis
c acute lymphocytic leukemia
d Sézary syndrome

257. The cell in the center of the photomicrograph is a:



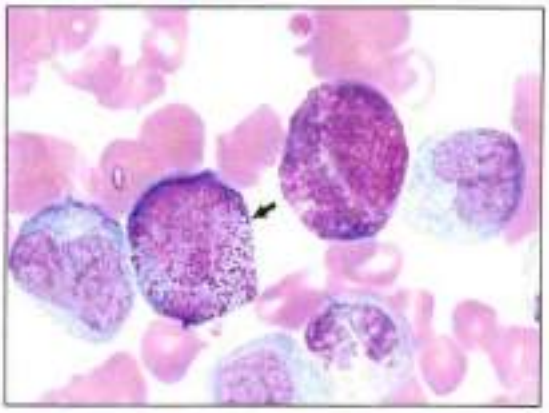
a promyelocyte
b lymphocyte
c neutrophil
d monocyte

258. The large cell in the center of the photomicrograph would be best described as a(n):



a neutrophil
b basophil
c eosinophil
d myelocyte

259. The large cell indicated by the arrow in the photomicrograph is a:



a myeloblast
b promyelocyte
c myelocyte
d metamyelocyte

260. A patient is diagnosed as having bacterial septicemia. Which of the following would best describe the expected change in his peripheral blood?

a granulocytic leukemoid reaction
b lymphocytic leukemoid reaction
c neutropenia
d eosinophilia

261. Characteristic morphologic features of reactive lymphocytes include:

a small size, clumped nuclear chromatin, scanty basophilic cytoplasm
b small size, nucleoli, granular cytoplasm
c large cell, oval nucleus, scanty blue-gray cytoplasm
d large size, irregularly shaped nucleus, abundant blue cytoplasm

262. If a blood smear is dried too slowly, the red blood cells are often:

a clumped
b crenated
c lysed
d destroyed

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263. The specimen of choice for preparation of blood films for manual differential leukocyte counts is whole blood collected in:

a EDTA
b oxalate
c citrate
d heparin

264. On a smear made directly from a finger stick, no platelets are found in the counting area. The first thing to do is:

a examine the slide for clumping
b obtain another smear
c perform a total platelet count
d request another finger stick

265. Evidence of a hemolytic event is present in the peripheral blood smear when which of the following is seen?

a basophilic stippling
b increased polychromasia
c increased target cells
d stomatocytes

C. Hemoglobin

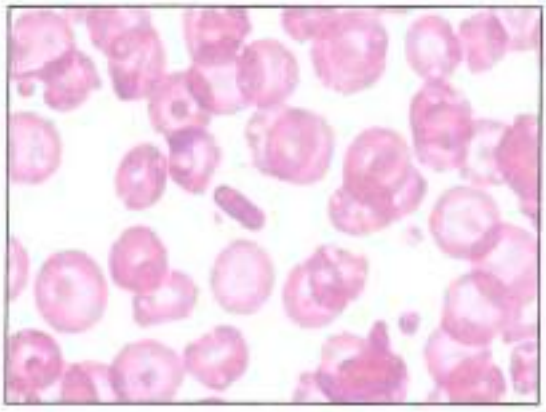
266. A common source of interference in measuring hemoglobin concentration is:

a hemolysis
b very high WBC count
c cold agglutinins
d clumped platelets

267. A patient with beta-thalassemia characteristically has a(n):

a elevated A_2 hemoglobin
b low fetal hemoglobin
c high serum iron
d normal red cell volume

268. An additional test indicated by the photomicrograph is:



a alkali denaturation
b alkaline phosphatase stain
c peroxidase stain
d hemoglobin electrophoresis

269. The most appropriate screening test for hemoglobin S is:

- **a** Kleihauer-Betke
- **b** dithionite solubility
- **c** heat instability
- **d** fluorescent spot

270. Which of the following is characteristic of hemoglobin H?

MLS ONLY

- **a** it is a tetramer of gamma chains
- **b** it is relatively stable
- **c** electrophoretically, it represents a "fast" hemoglobin
- **d** it has a lower oxygen affinity than hemoglobin A

271. In most cases of hereditary persistence of fetal hemoglobin (HPFH):

MLS ONLY

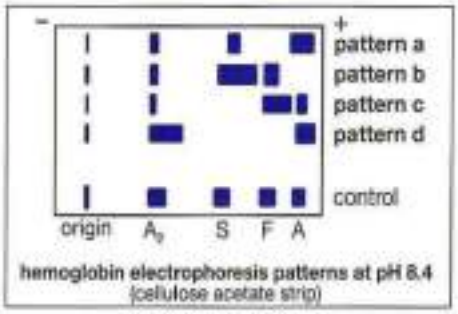
- **a** hemoglobin F is unevenly distributed throughout the erythrocytes
- **b** the black heterozygote has 75% hemoglobin F
- **c** beta- and gamma-chain synthesis is decreased
- **d** gamma-chain production equals alpha-chain production

272. When using the turbidity (solubility) method for detecting the presence of hemoglobin S, an incorrect interpretation may be made when there is a(n):

MLS ONLY

- **a** increased reticulocyte count
- **b** glucose concentration >150 mg/dL (8.3 mmol/L)
- **c** blood specimen >2 hours old
- **d** decreased hematocrit

273. Which pattern shown in this figure is consistent with homozygous beta-thalassemia?



hemoglobin electrophoresis patterns at pH 8.4
(cellulose acetate strip)

- **a** pattern a
- **b** pattern b
- **c** pattern c
- **d** pattern d

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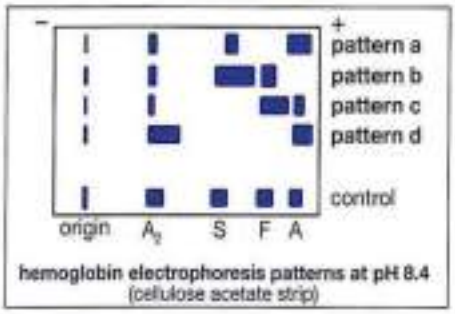
274. Which electrophoresis pattern shown in this figure is consistent with sickle cell trait?



hemoglobin electrophoresis patterns at pH 8.4
(cellulose acetate strip)

a pattern a
b pattern b
c pattern c
d pattern d

275. A native of Thailand has a normal hemoglobin level. Hemoglobin electrophoresis on cellulose acetate shows 70% HbA and approximately 30% of a hemoglobin with the mobility of HbA_2. This is most consistent with hemoglobin:

a C trait
b E trait
c O trait
d D trait

276. Automated methods of measuring hemoglobin cannot detect this form:

a methemoglobin
b carboxyhemoglobin
c deoxyhemoglobin
d sulfhemoglobin

277. In the hemoglobin solubility (dithionate) test, which type of hemoglobin causes turbidity (positive reaction)?

a HbD
b HbE
c HbS
d HbA

278. The presence of HbH may be demonstrated by:

a Prussian blue stain
b Wright stain
c Giemsa stain
d brilliant cresyl blue

279. MLS ONLY The hemoglobin variant that is seen frequently in the South East Asian population, demonstrates a microcytic blood smear, and migrates with HbC at pH 8.6 is:

a HbBarts
b HbF
c HbE
d HbH

D. Hematocrit

280. Automated hematology analyzers calculate the hematocrit using which of the following parameters?

a MCV and RBC
b MCH and RBC
c HGB and RBC
d MCV and MCHC

281. A false elevation in a manual hematocrit (microhematocrit) determination may result from:

a prolonged centrifugation
b *in vitro* hemolysis
c trapped plasma
d incomplete sealing of the hematocrit tube

E. Indices

282. What cell shape is **most** commonly associated with an increased MCHC?

a teardrop cells
b target cells
c spherocytes
d sickle cells

283. Which of the RBC indices is a measure of the amount of hemoglobin in individual red blood cells?

a MCHC
b MCV
c HCT
d MCH

284. The RDW-CV and RDW-SD performed by automated hematology analyzers are calculations that provide:

a an index of the distribution of RBC volumes
b a calculated mean RBC hemoglobin concentration
c a calculated mean cell hemoglobin
d the mean RBC volume

285. The laboratory findings on a patient are shown in this table:

test	result
MCV	55 μm^3 (55 fL)
MCHC	25%
MCH	17 pg

A stained blood film of this patient would most likely reveal a red cell picture that is:

a microcytic, hypochromic
b macrocytic, hypochromic
c normocytic, normochromic
d microcytic, normochromic

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286. A patient has the laboratory results shown in this table:

test	result
RBC	2.00 × 10^6/µL (2.00 × 10^{12}/L)
HCT	24%
HGB	6.8 g/dL (68 g/L)
reticulocytes	0.8%

The mean corpuscular volume (MCV) of the patient is:

a 35 fL
b 83 fL
c 120 fL
d 150 fL

287. Which of these formulas is used for calculating the mean corpuscular hemoglobin (MCH)?

a HCT/(RBC × 1000)
b HGB/HCT
c RBC/HCT
d (HGB × 10)/RBC

288. What is the MCH if the HCT is 20%, the RBC is 2.4 × 10^6/µL (2.4 × 10^{12}/L) and the HGB is 5 g/dL (50 g/L)?

a 21 pg
b 23 pg
c 25 pg
d 84 pg

289. Which of these formulas is used for calculating the MCHC?

a (HGB × 100)/HCT
b HGB/RBC
c RBC/HCT
d (HCT × 1000)/RBC

290. What is the MCHC if the HCT is 20%, the RBC is 2.4 × 10^6/µL (2.4 × 10^{12}/L) and the HGB is 5 g/dL (50 g/L)?

a 21%
b 25%
c 30%
d 34%

291. Which of these formulas is used for calculating the mean corpuscular volume (MCV)?

a (HGB × 10)/RBC
b HGB/HCT
c (HCT × 10)/RBC
d RBC/HCT

292. MLS ONLY The calculated erythrocyte indices on an adult man are MCV = 89 fL, MCH = 29 pg and MCHC = 38%. The calculations have been rechecked; erythrocytes on the peripheral blood smear appear normocytic and normochromic with no abnormal forms. The next step is to:

a report the results
b examine another smear
c repeat the hemoglobin and hematocrit
d repeat the erythrocyte count and hematocrit